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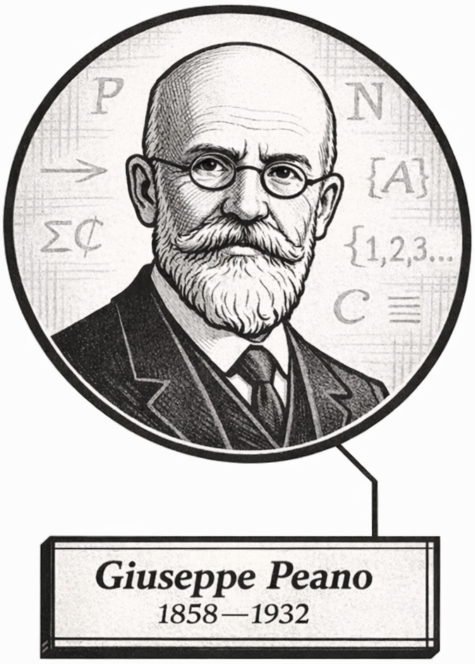

One, ah-ah-ah... Two, ah-ah-ah...

Guiseppe Peano (probably)

Volume II

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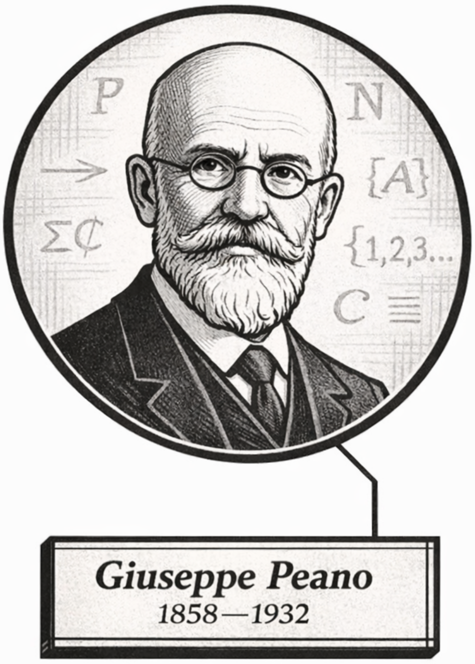

One, ah-ah-ah... Two, ah-ah-ah...
Guiseppe Peano (probably)

Volume II

Origins of the Numbers

Self-Study Notes

William Sollers

June 8, 2026

Part I

Foundations of Formal Number Systems

Chapter 1

Peano Systems

peano-systems

Peano Systems → Natural Numbers (ℕ)

Structural Roadmap

How we got here. Volume I established the logical background for axiomatic systems, relations, functions, and induction-style arguments. This chapter applies that apparatus to the abstract structure of counting, in the Dedekind–Peano tradition of describing the natural numbers by a first element, successor, and induction [Dedekind, 1901, Peano, 1889].

What this chapter develops. The chapter defines Peano systems, records their basic successor and predecessor theorems, proves the Peano Iterator Theorem and the general recursion theorem, and ends with categoricity up to isomorphism.

Where this leads. Once recursion is justified for every Peano system, the next chapter can work inside the canonical natural-number system and define arithmetic operations by recursion.

defining-peano-systems

Peano Systems → **Peano Axioms** → Recursion on Peano Systems

1.1 Peano Systems

The Peano Axioms

The Peano axioms make precise what it means for a system to behave like the natural numbers. They isolate the first element, the successor map, and the induction principle. These axioms do not define addition, multiplication, or order. They fix the structure that supports those later constructions. The formulation used here belongs to the Dedekind–Peano line of axiomatizing the natural-number structure [Dedekind, 1901, Peano, 1889].

Definition (Peano System)

Definition 1.1.1 (Peano System). A triple $(P, S, 1)$ is called a *Peano system* exactly when P is a set, $1 \in P$, $S : P \to P$ is a function, and the following axioms hold:

P1. $1 \in P$.

P2. For every $n \in P$, $S(n) \in P$.

P3. For every $n \in P$, $S(n) \neq 1$.

P4. For all $m, n \in P$, if $S(m) = S(n)$ then $m = n$.

P5. If $A \subseteq P$, $1 \in A$, and

$$\forall n \in P\ (n \in A \Longrightarrow S(n) \in A),$$

then $A = P$.

Remark (Standard quantified statement).

$$\begin{aligned} &(P, S, 1) \text{ is a Peano system} \\ \Longleftrightarrow\ &\Big(P \text{ is a set} \wedge 1 \in P \wedge S : P \to P \\ &\quad \wedge \forall n \in P\ (S(n) \in P) \\ &\quad \wedge \forall n \in P\ (S(n) \neq 1) \\ &\quad \wedge \forall m, n \in P\ (S(m) = S(n) \Longrightarrow m = n) \\ &\quad \wedge \forall A \subseteq P\ \big((1 \in A \wedge \forall n \in P\ (n \in A \Longrightarrow S(n) \in A)) \Longrightarrow A = P\big)\Big). \end{aligned}$$

Remark (Failure modes). A triple $(P, S, 1)$ fails to be a Peano system if the distinguished element does not lie in the underlying set, if the successor map leaves the set, if the successor hits 1, if the successor identifies two distinct elements, or if there exists a proper subset of P that contains 1 and is closed under successor.

Remark (Failure mode decomposition).

$$\begin{aligned} \neg\,(P, S, 1) \text{ is a Peano system} \Longrightarrow\ &1 \notin P \\ &\vee \exists n \in P\ (S(n) \notin P) \\ &\vee \exists n \in P\ (S(n) = 1) \\ &\vee \exists m, n \in P\ (m \neq n \wedge S(m) = S(n)) \\ &\vee \exists A \subseteq P\ \big(1 \in A \wedge \forall n \in P\ (n \in A \Longrightarrow S(n) \in A) \wedge A \neq P\big). \end{aligned}$$

Remark (Interpretation). A Peano system is the abstract data of counting: an underlying set, a first element, and a successor operation satisfying the usual non-collision and induction principles. The purpose of the definition is to isolate the ambient object before the individual axioms are recorded and used one by one.

Remark (Source alignment). The five displayed clauses are the house-style Peano-system presentation of Peano's axiomatization of arithmetic [Peano, 1889]. They are stated here in structural form so that later definitions and proofs can refer directly to the successor and induction clauses.

Remark (Comparison with Feferman). This definition corresponds closely to Definition 3.1 in Feferman [1964], where Peano systems are introduced as the abstract structural framework underlying the positive integers. The present notes expand the definition into explicit quantified, failure-mode, and decomposition forms in order to support later logical analysis and dependency extraction.

Axiom (One Is a Natural Number)

Axiom 1.1.2 (Distinguished Element Lies in the Underlying Set).

$$1 \in P.$$

Remark (Standard quantified statement).

$$1 \in P.$$

Remark (Interpretation). The counting system begins from a distinguished first element of the underlying set. This is the base point from which the remaining elements are generated by the successor operation.

Axiom (Closure Under Successor)

Axiom 1.1.3 (Closure Under Successor). If $n \in P$, then $S(n) \in P$.

Remark (Standard quantified statement).

$$\forall n \in P\ (S(n) \in P).$$

Remark (Interpretation). Applying the successor operation to an element of the underlying set never leaves that set. Successor is therefore an internal operation on the counting system.

Axiom (One Is Not a Successor)

Axiom 1.1.4 (Distinguished Element Is Not a Successor). For every $n \in P$, $S(n) \neq 1$.

Remark (Standard quantified statement).

$$\forall n \in P\ (S(n) \neq 1).$$

Remark (Interpretation). The first natural number does not arise by taking the successor of an earlier natural number. This keeps the counting system from folding backward onto its starting point.

Axiom (Injectivity of Successor)

Axiom 1.1.5 (Injectivity of Successor). If $m, n \in P$ and $S(m) = S(n)$, then $m = n$.

Remark (Standard quantified statement).

$$\forall m, n \in P\ (S(m) = S(n) \Longrightarrow m = n).$$

Remark (Interpretation). Different natural numbers do not collapse to the same successor. The successor operation preserves distinctness and lets predecessor reasoning work whenever a successor is known.

Axiom (Induction Axiom)

Axiom 1.1.6 (Induction Axiom). Let $A \subseteq P$. If $1 \in A$ and if

$$\forall n \in P\ (n \in A \Longrightarrow S(n) \in A),$$

then $A = P$.

Remark (Standard quantified statement).

$$\forall A \subseteq P\ \Big((1 \in A \wedge \forall n \in P\ (n \in A \Longrightarrow S(n) \in A)) \Longrightarrow A = P\Big).$$

Remark (Interpretation). Any subset of the underlying set that contains the distinguished element and is closed under successor must already contain the entire set. This axiom rules out rogue elements and is the logical source of mathematical induction.

Axiom (Existence of a Peano System)

Axiom 1.1.7 (Existence of a Peano System). There exists a Peano system.

Remark (Standard quantified statement).

$$\exists P\ \exists S\ \big((P, S, 1) \text{ is a Peano system}\big).$$

Remark (Interpretation). The Peano axioms are not merely a formal pattern; at least one structure satisfies them. This axiom guarantees that the abstract counting system is available before we give it the conventional name $\mathbb{N}$.

Remark (Historical note). This axiom corresponds to Axiom 3.2 of Feferman [1964], where the existence of at least one Peano system is taken as a foundational assumption.

Remark (Dependencies).

- Peano System

1.1.1 First Theorems on Peano Systems

Induction as a Working Principle

The induction axiom for a Peano system is formulated in terms of inductive subsets of the underlying set. Arithmetic arguments are usually written in terms of predicates rather than subsets. This theorem identifies the predicate-form induction principle derived from the Peano axioms. It is the standard form of induction used in the rest of the chapter. Historically, this is the working proof principle attached to Peano's axiomatization of the natural numbers [Peano, 1889].

Theorem (Induction Principle for a Peano System)

Theorem 1.1.8 (Induction Principle for a Peano System). *Let $(P, S, 1)$ be a Peano system, and let φ be a predicate on P. If*

$$\varphi(1) \qquad \textit{and} \qquad \forall n \in P\ (\varphi(n) \implies \varphi(S(n))),$$

then

$$\forall n \in P\ \varphi(n).$$

Go to proof.

Remark (Standard quantified statement).

Let $(P, S, 1)$ be a Peano system and let φ be a predicate on P.
$$\Big(\varphi(1) \wedge \forall n \in P\ (\varphi(n) \implies \varphi(S(n)))\Big) \implies \forall n \in P\ \varphi(n).$$

Remark (Contrapositive quantified statement).

Let $(P, S, 1)$ be a Peano system and let φ be a predicate on P.
$$\neg\forall n \in P\ \varphi(n) \implies \neg\Big(\varphi(1) \wedge \forall n \in P\ (\varphi(n) \implies \varphi(S(n)))\Big).$$

Remark (Interpretation). This theorem converts the subset form of induction into the predicate form used in ordinary arithmetic arguments. A property propagates through the entire Peano system once it holds at the distinguished first element and is preserved by successor. The theorem therefore turns the structural induction axiom into a proof principle for arbitrary predicates on P.

Remark (Comparison with Feferman). Peano's axiomatization includes induction as part of the natural-number structure [Peano, 1889]. Feferman derives the working form of mathematical induction from the subset formulation of the Peano axioms immediately after Theorem 3.3 in Chapter 3 of Feferman [1964]. The present notes formalize that induction principle as an explicit reusable theorem.

Remark (Dependencies).

- Peano System
- Induction Axiom

Inductive Subsets and Predecessors

Induction is expressed in terms of subsets of a Peano system. Two structural notions appear repeatedly in that setting: closure under successor and membership generated from the distinguished first element. Predecessor language is also convenient once it is known that every non-initial element arises as a successor.

Definition (Successor-Closed Subset)

Definition 1.1.9 (Successor-Closed Subset). Let $(P, S, 1)$ be a Peano system, and let $A \subseteq P$. The subset A is called *successor-closed* exactly when

$$\forall n \in P\ (n \in A \Longrightarrow S(n) \in A).$$

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } (P, S, 1) \text{ be a Peano system and let } A \subseteq P.\\ &A \text{ is successor-closed} \Longleftrightarrow \forall n \in P\ (n \in A \Longrightarrow S(n) \in A).\end{aligned}$$

Remark (Negated quantified statement).

$$\begin{aligned}&\text{Let } (P, S, 1) \text{ be a Peano system and let } A \subseteq P.\\ &A \text{ is not successor-closed} \Longleftrightarrow \exists n \in P\ (n \in A \wedge S(n) \notin A).\end{aligned}$$

Remark (Failure modes). A subset fails to be successor-closed exactly when it contains some element of P but fails to contain that element's successor.

Remark (Failure mode decomposition).

$$A \text{ is not successor-closed} \Longleftrightarrow \exists n \in P\ \underbrace{(n \in A \wedge S(n) \notin A)}_{\text{closure fails at } n}.$$

Remark (Interpretation). A successor-closed subset is stable under one step of counting. Once an element of A is present, its successor remains in A as well.

Remark (Comparison with Feferman). The notion of a successor-closed subset is implicit in condition (iii) of Feferman's Definition 3.1 [Feferman, 1964]. The present notes isolate it as a separate definition so that induction arguments can refer to the closure condition directly.

Remark (Dependencies).

- Peano System

Definition (Inductive Subset of a Peano System)

Definition 1.1.10 (Inductive Subset of a Peano System). Let $(P, S, 1)$ be a Peano system, and let $A \subseteq P$. The subset A is called *inductive* exactly when $1 \in A$ and A is successor-closed.

Remark (Standard quantified statement).

$$\text{Let } (P, S, 1) \text{ be a Peano system and let } A \subseteq P.$$
$$A \text{ is inductive} \iff \bigl(1 \in A \wedge \forall n \in P\ (n \in A \Longrightarrow S(n) \in A)\bigr).$$

Remark (Negated quantified statement).

$$\text{Let } (P, S, 1) \text{ be a Peano system and let } A \subseteq P.$$
$$A \text{ is not inductive} \iff \bigl(1 \notin A \vee \exists n \in P\ (n \in A \wedge S(n) \notin A)\bigr).$$

Remark (Failure modes). A subset fails to be inductive in exactly two ways: it may fail to contain the distinguished first element, or it may contain some element while failing to contain that element's successor.

Remark (Failure mode decomposition).

$$A \text{ is not inductive} \iff \underbrace{1 \notin A}_{\text{base point missing}} \vee \underbrace{\exists n \in P\ (n \in A \wedge S(n) \notin A)}_{\text{successor closure fails}}.$$

Remark (Interpretation). An inductive subset contains the distinguished first element and is closed under successor. Such a subset contains every stage obtained by starting at 1 and applying successor repeatedly.

Remark (Comparison with Feferman). The notion of an inductive subset packages the two hypotheses appearing in the induction clause of Feferman's Definition 3.1 [Feferman, 1964]. The present notes name this package so that the internal structure of induction proofs remains explicit.

Remark (Dependencies).

- Peano System
- Successor-Closed Subset

Definition (Predecessor in a Peano System)

Definition 1.1.11 (Predecessor in a Peano System). Let $(P, S, 1)$ be a Peano system, and let $x, u \in P$. The element u is called a *predecessor* of x exactly when

$$S(u) = x.$$

Remark (Standard quantified statement).

$$\text{Let } (P, S, 1) \text{ be a Peano system and let } x, u \in P.$$
$$u \text{ is a predecessor of } x \iff S(u) = x.$$

Remark (Negated quantified statement).

$$\text{Let } (P, S, 1) \text{ be a Peano system and let } x, u \in P.$$
$$u \text{ is not a predecessor of } x \iff S(u) \neq x.$$

Remark (Interpretation). A predecessor of x is an element whose successor is x. This language isolates the inverse-looking question attached to the successor map without assuming that successor is globally invertible.

Remark (Dependencies).

- Peano System

Generation by Successor

The Peano axioms describe a counting system generated from a distinguished first element by the successor operation. The next theorems make that generation principle explicit: every element is either the first element or a successor, every non-initial element has a unique predecessor, and no element can be its own successor.

Theorem (Every Element Is Either 1 or a Successor)

Theorem 1.1.12 (Every Element Is Either 1 or a Successor). *Let $(P, S, 1)$ be a Peano system. For every $x \in P$, either $x = 1$ or there exists $u \in P$ such that $S(u) = x$.* *Go to proof.*

Remark (Standard quantified statement).

$$\text{Let } (P, S, 1) \text{ be a Peano system.}$$
$$\forall x \in P \; \big(x = 1 \lor \exists u \in P \; (S(u) = x)\big).$$

Remark (Interpretation). Every element of a Peano system is reached by starting at the distinguished first element and moving forward by successor. The only possible exception to having a predecessor is the element 1 itself.

Remark (Historical note). This theorem is one of the basic successor consequences of Peano's axiomatization [Peano, 1889]. In the present organization it corresponds directly to Theorem 3.3 in Feferman [1964]. Feferman uses this result to show that the distinguished initial element is the only possible non-successor in a Peano system.

Remark (Dependencies).

- Peano System
- Distinguished Element Lies in the Underlying Set
- Closure Under Successor
- Induction Principle for a Peano System

Theorem (Successor Inequality Reflection)

Theorem 1.1.13 (Successor Inequality Reflection). *Let $(P, S, 1)$ be a Peano system. For all $a, b \in P$, if $a \neq b$, then $S(a) \neq S(b)$.* *Go to proof.*

Remark (Standard quantified statement).

$$\text{Let } (P, S, 1) \text{ be a Peano system.}$$
$$\forall a, b \in P \; (a \neq b \implies S(a) \neq S(b)).$$

Remark (Negated quantified statement).

$$\text{Let } (P, S, 1) \text{ be a Peano system.}$$
$$\exists a, b \in P \; (a \neq b \wedge S(a) = S(b)).$$

Remark (Failure modes). Successor inequality reflection fails exactly when two distinct elements have the same successor.

Remark (Failure mode decomposition).

$$\exists a, b \in P \qquad \underbrace{(a \neq b \wedge S(a) = S(b))}_{\text{distinct elements with a common successor}} \qquad .$$

Remark (Contrapositive quantified statement).

$$\text{Let } (P, S, 1) \text{ be a Peano system.}$$
$$\forall a, b \in P \; (S(a) = S(b) \implies a = b).$$

Remark (Interpretation). The successor operation reflects inequality: distinct inputs remain distinct after one successor step. This is the contrapositive form of successor injectivity, and it is the form used when an induction step must preserve a negative equality statement.

Remark (Dependencies).

- Peano System
- Injectivity of Successor

Corollary (Non-One Elements Have a Predecessor)

Corollary 1.1.14 (Non-One Elements Have a Predecessor). *Let $(P, S, 1)$ be a Peano system. For every $x \in P$, if $x \neq 1$, then there exists $u \in P$ such that $S(u) = x$.* *Go to proof.*

Remark (Standard quantified statement).

$$\text{Let } (P, S, 1) \text{ be a Peano system.}$$
$$\forall x \in P \; \big(x \neq 1 \implies \exists u \in P \; (S(u) = x)\big).$$

Remark (Negated quantified statement).

$$\text{Let } (P, S, 1) \text{ be a Peano system.}$$
$$\exists x \in P \; \big(x \neq 1 \wedge \forall u \in P \; (S(u) \neq x)\big).$$

Remark (Failure modes). The statement fails exactly when some element distinct from 1 has no predecessor.

Remark (Failure mode decomposition).

$$\exists x \in P \quad \underbrace{\big(x \neq 1 \wedge \forall u \in P\ (S(u) \neq x)\big)}_{\texttt{non-initial element with no predecessor}} \ .$$

Remark (Interpretation). Every non-initial element of a Peano system is reached by one successor step from an earlier element. This separates predecessor existence from predecessor uniqueness so that later arguments can cite only the part they need.

Remark (Comparison with Feferman). Feferman's Theorem 3.3 in Feferman [1964] contains this predecessor-existence fact as part of the structural analysis of Peano systems. The present notes record it separately before adding uniqueness.

Remark (Dependencies).

- Peano System
- Every Element Is Either 1 or a Successor

Corollary (Predecessor Existence and Uniqueness Away from 1)

Corollary 1.1.15 (Predecessor Existence and Uniqueness Away from 1). *Let $(P, S, 1)$ be a Peano system, and let $x \in P$ with $x \neq 1$. Then there exists a unique $u \in P$ such that $S(u) = x$.* *Go to proof.*

Remark (Standard quantified statement).

$$\text{Let } (P, S, 1) \text{ be a Peano system.}$$
$$\forall x \in P\ \big(x \neq 1 \Longrightarrow \exists! u \in P\ (S(u) = x)\big).$$

Remark (Interpretation). Every element distinct from 1 has exactly one predecessor. The successor map therefore organizes the non-initial part of a Peano system into one-step extensions of earlier elements.

Remark (Comparison with Feferman). Feferman proves predecessor existence and uniqueness together inside Theorem 3.3 of Feferman [1964]. The present notes separate the uniqueness statement into an independent corollary so that it can be cited directly in later proofs.

Remark (Dependencies).

- Peano System
- Non-One Elements Have a Predecessor
- Injectivity of Successor

Corollary (Unique Predecessor Characterization Away from 1)

Corollary 1.1.16 (Unique Predecessor Characterization Away from 1). *Let $(P, S, 1)$ be a Peano system. For every $x \in P$, $x \neq 1$ if and only if there exists a unique $u \in P$ such that $S(u) = x$.* *Go to proof.*

Remark (Standard quantified statement).

$$\text{Let } (P, S, 1) \text{ be a Peano system.}$$
$$\forall x \in P \left(x \neq 1 \iff \exists! u \in P \, (S(u) = x)\right).$$

Remark (Negated quantified statement).

$$\begin{aligned}
&\text{Let } (P, S, 1) \text{ be a Peano system.} \\
&\exists x \in P \Big(\big(x \neq 1 \land (\forall u \in P \, (S(u) \neq x) \\
&\qquad\qquad \lor \exists u, v \in P \, (S(u) = x \land S(v) = x \land u \neq v))\big) \\
&\qquad\quad \lor \left(x = 1 \land \exists u \in P \, (S(u) = x \land \forall v \in P \, (S(v) = x \implies v = u))\right)\Big).
\end{aligned}$$

Remark (Failure modes). The biconditional can fail in two ways: a non-initial element can fail to have a unique predecessor, or the distinguished first element can have a unique predecessor.

Remark (Failure mode decomposition).

$$\begin{aligned}
\exists x \in P \Big(&\underbrace{\big(x \neq 1 \land (\forall u \in P \, (S(u) \neq x) \lor \exists u, v \in P \, (S(u) = x \land S(v) = x \land u \neq v))\big)}_{\text{non-initial element lacks a unique predecessor}} \\
&\lor \underbrace{\left(x = 1 \land \exists u \in P \, (S(u) = x \land \forall v \in P \, (S(v) = x \implies v = u))\right)}_{\text{the first element has a unique predecessor}}\Big).
\end{aligned}$$

Remark (Interpretation). Being different from the distinguished first element is exactly the same as having a unique predecessor. This turns the informal phrase "away from 1" into an internal characterization of the successor structure.

Remark (Comparison with Feferman). This corollary packages the predecessor analysis surrounding Feferman's Theorem 3.3 in Feferman [1964]. The forward direction is the predecessor existence-and-uniqueness result, while the reverse direction uses the axiom that 1 is not a successor.

Remark (Dependencies).

- Peano System
- Distinguished Element Is Not a Successor
- Predecessor Existence and Uniqueness Away from 1

Corollary (The Distinguished Element Is the Unique Non-Successor)

Corollary 1.1.17 (The Distinguished Element Is the Unique Non-Successor). *Let $(P, S, 1)$ be a Peano system. An element $x \in P$ is not a successor of any element of P if and only if $x = 1$.* *Go to proof.*

Remark (Standard quantified statement).

$$\text{Let } (P, S, 1) \text{ be a Peano system.}$$
$$\forall x \in P \left(\neg \exists u \in P\ (S(u) = x) \iff x = 1\right).$$

Remark (Interpretation). The distinguished first element is characterized internally as the unique element with no predecessor. This gives a structural description of 1 that does not depend on external notation.

Remark (Comparison with Feferman). This corollary extracts a structural characterization of the distinguished initial element that is implicit in the discussion surrounding Theorem 3.3 of Feferman [1964].

Remark (Dependencies).

- Distinguished Element Is Not a Successor
- Predecessor Existence and Uniqueness Away from 1

Theorem (No Object Is Its Own Successor)

Theorem 1.1.18 (No Object Is Its Own Successor). *Let $(P, S, 1)$ be a Peano system. For every $n \in P$,*

$$S(n) \neq n.$$

Go to proof.

Remark (Standard quantified statement).

$$\text{Let } (P, S, 1) \text{ be a Peano system.}$$
$$\forall n \in P\ (S(n) \neq n).$$

Remark (Interpretation). Successor always moves forward in the counting structure. No element can be recovered by applying successor to itself.

Remark (Historical note). This theorem is a standard consequence of the Peano axioms in the classical development of the natural numbers [Peano, 1889]. Feferman includes the corresponding result as an exercise in Chapter 3 of Feferman [1964].

Remark (Dependencies).

- Distinguished Element Is Not a Successor
- Injectivity of Successor
- Induction Principle for a Peano System

1.1.2 Peano System Figure Plates

The following figure plates collect the visual diagrams for the Peano system and iterator material. Each plate is presented separately so that the diagram can be read as a full-page reference.

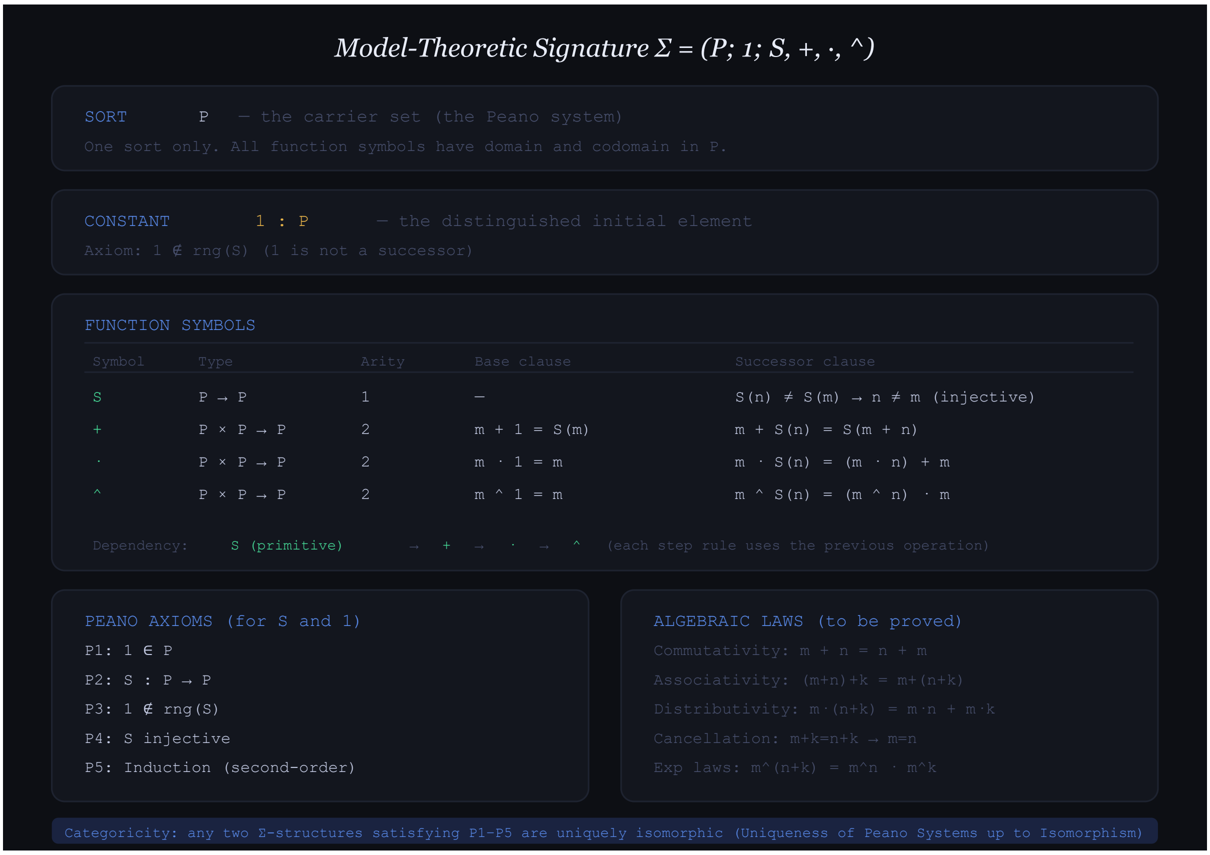

Figure 1.1: The model-theoretic signature for a Peano system and the later arithmetic operations generated from it.

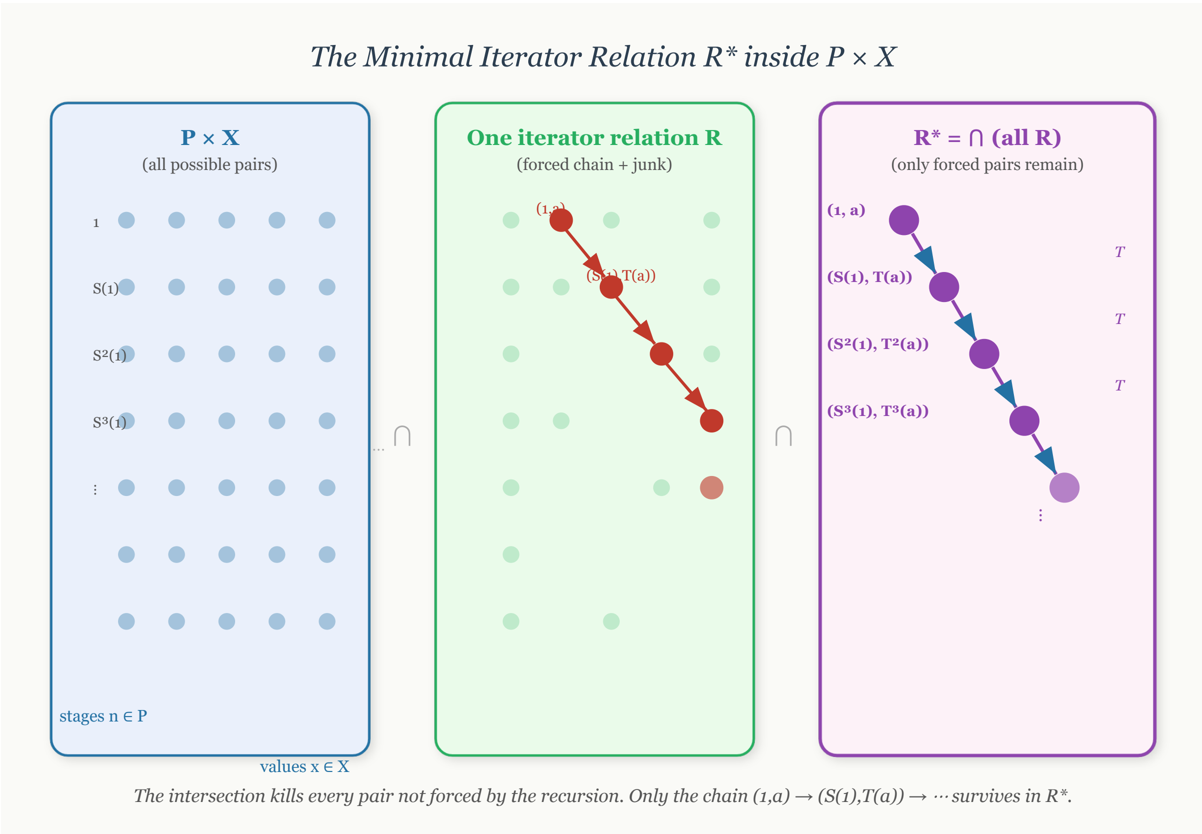

Figure 1.2: The intuitive construction of the minimal iterator relation R^* inside $P \times W$.

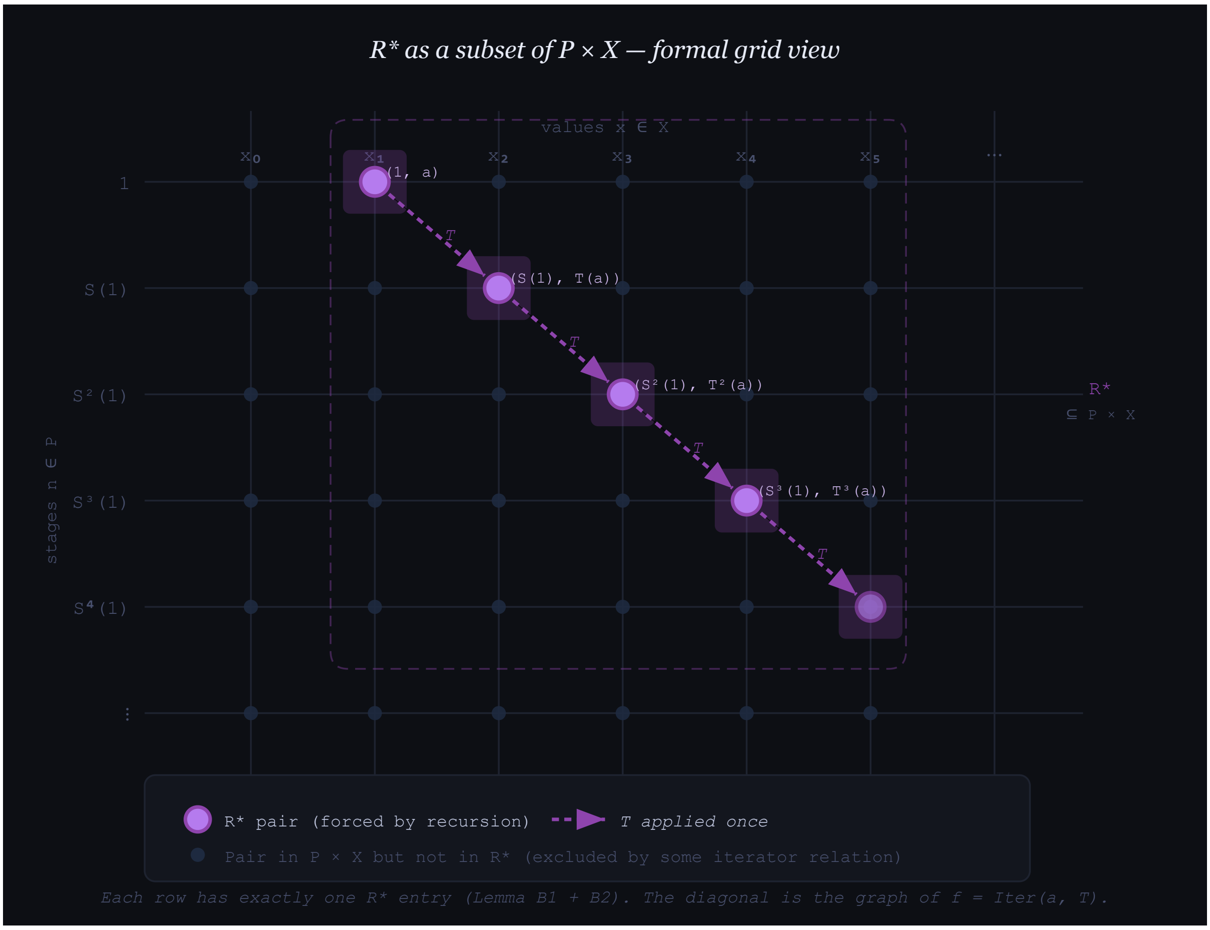

Figure 1.3: A formal grid view of R^* as the chain of forced ordered pairs in $P \times W$.

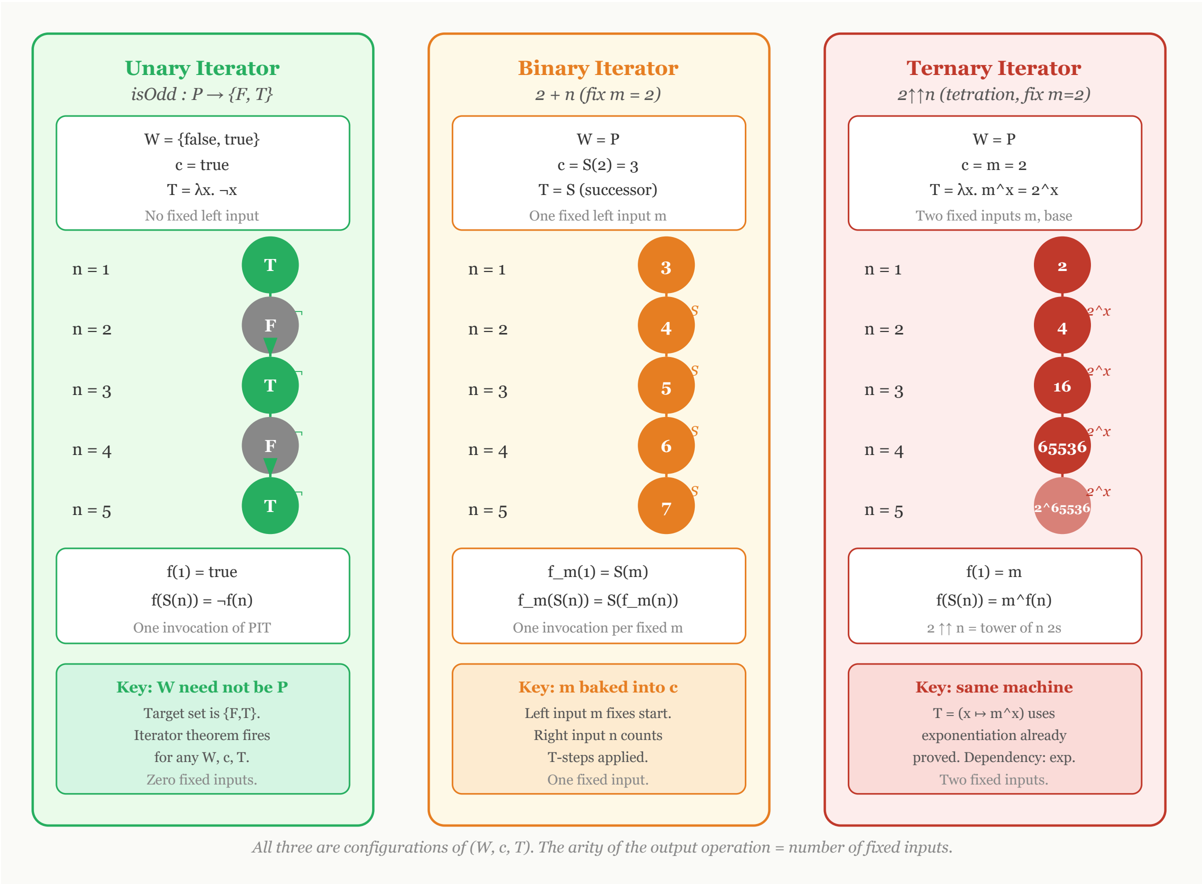

Figure 1.4: Unary, binary, and higher-arity iterator configurations after fixing the relevant parameters.

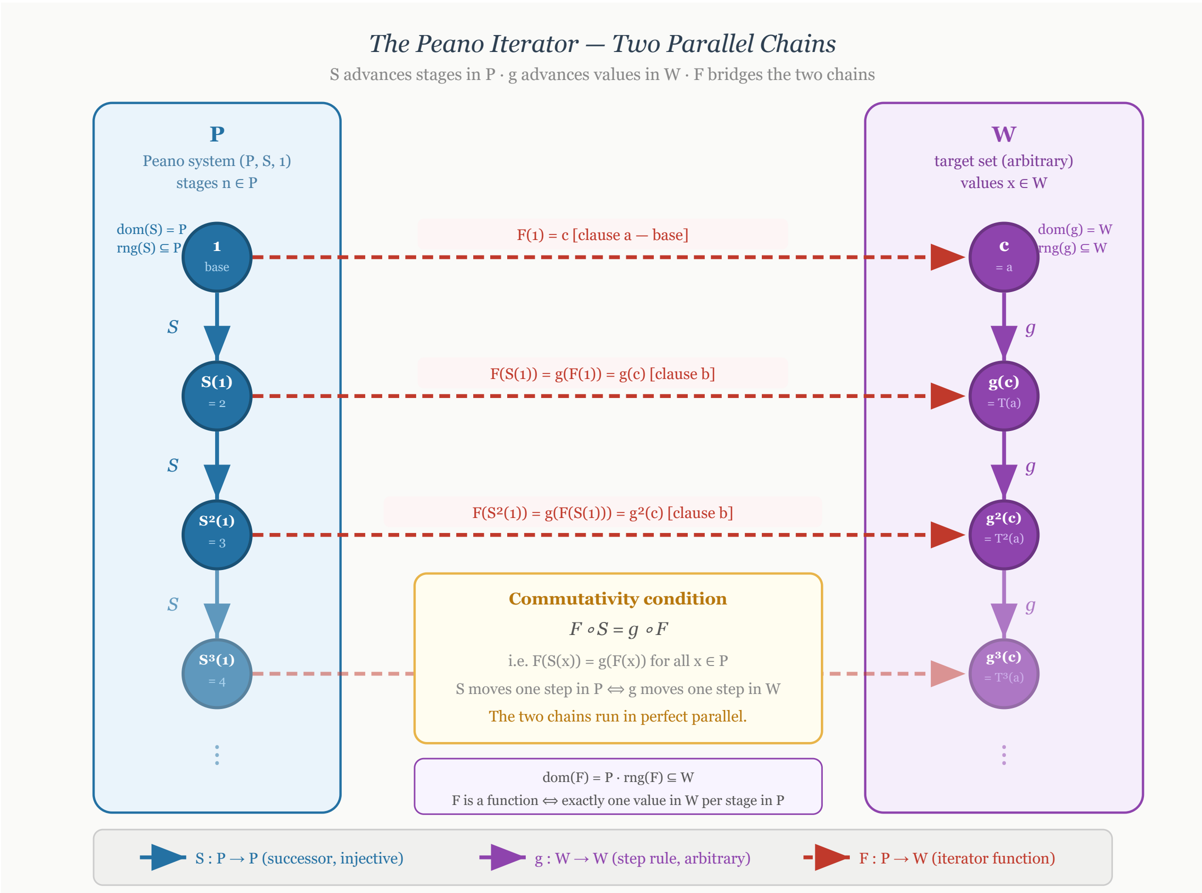

Figure 1.5: The Peano iterator as two parallel chains, one in P and one in W, connected by the iterator function.

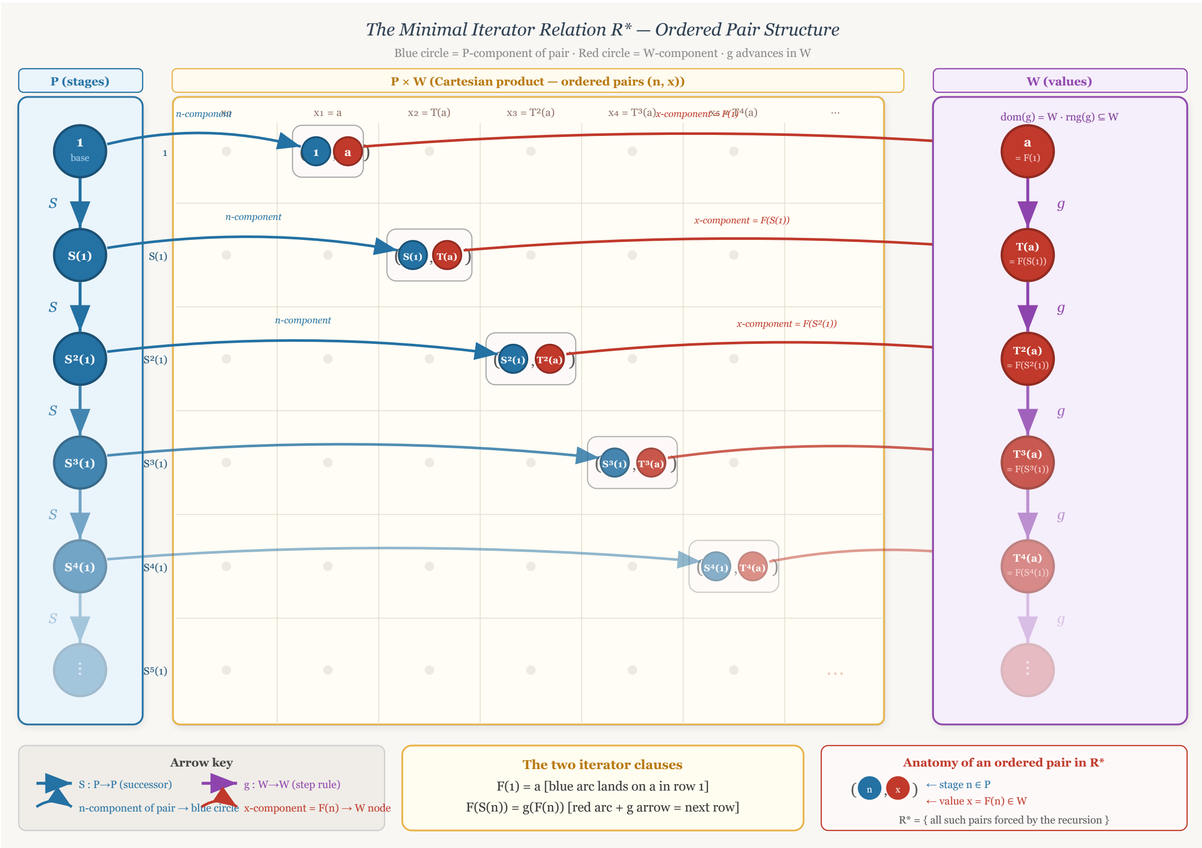

Figure 1.6: The three-zone view of R^*, separating stages in P, ordered pairs in $P \times W$, and values in W.

1.2 Examples of Peano Systems

Peano Systems as Abstract Structures

The examples in this section separate the abstract idea of a Peano system from any particular representation of the natural numbers. A Peano system is not a specific set such as $\mathbb{N}$. It is a structure consisting of a carrier set, a distinguished initial element, and a successor operation satisfying the Peano conditions. Different carriers can therefore realize the same abstract successor structure.

Remark (Formal reference). The formal definition used throughout this section is Peano System. In each example, the data are displayed as a triple (P, S, e) consisting of a carrier P, a successor map $S : P \to P$, and a distinguished first element $e \in P$.

Remark (Interpretation). The axioms say that the system has one starting point, the successor operation never merges two different elements, and induction reaches every element of the carrier. Thus the carrier is an infinite one-way chain generated from its initial element.

1.2.1 The Standard One-Based System

The Usual One-Based Carrier

The familiar natural-number model is the reference example, but it is only one realization of the Peano axioms. The carrier is the usual one-based set of natural numbers, the distinguished element is 1, and successor is the usual next-number operation.

Example (The Usual Natural Numbers). Let

$$P = \mathbb{N}, \qquad e = 1,$$

and define

$$S : P \to P, \qquad S(n) = n + 1.$$

Then $(P, S, e) = (\mathbb{N}, S, 1)$ is the usual one-based Peano system.

Remark (Structure). The Peano data are

$$\big(P, S, e\big) = \big(\mathbb{N},\, n \mapsto n + 1,\, 1\big).$$

Remark (Interpretation). This is the familiar model, but it is not the only possible carrier for a Peano system. The remaining examples use different elements while preserving the same abstract successor pattern.

1.2.2 Von Neumann Ordinals

Set-Theoretic Carriers

Von Neumann ordinals realize natural-number structure inside set theory. In this representation, each stage is a set, and successor is formed by adjoining the current stage as a new element.

Example (The Zero-Based Von Neumann Chain). Let

$$0 = \varnothing, \qquad 1 = \{0\}, \qquad 2 = \{0, 1\}, \qquad 3 = \{0, 1, 2\}, \quad \ldots$$

Let $P = \omega$, let $e = 0$, and define

$$S : P \to P, \qquad S(n) = n \cup \{n\}.$$

Then $(P, S, e) = (\omega, S, 0)$ is a zero-based Peano system.

Remark (Structure). The Peano data are

$$\big(P, S, e\big) = \big(\omega,\, n \mapsto n \cup \{n\},\, 0\big).$$

This example uses the zero-based convention common in set-theoretic foundations. The house convention for arithmetic in this volume remains one-based.

Example (The One-Based Von Neumann Chain). Let

$$P = \omega \setminus \{0\}, \qquad e = 1,$$

and define

$$S : P \to P, \qquad S(n) = n \cup \{n\}.$$

Then (P, S, e) is a one-based Peano system.

Remark (Structure). The Peano data are

$$\big(P, S, e\big) = \big(\omega \setminus \{0\},\, n \mapsto n \cup \{n\},\, 1\big).$$

Remark (Interpretation). In the Von Neumann model, each natural number is literally the set of all smaller natural numbers. This is not merely notation; it is a concrete set-theoretic realization of the abstract Peano structure.

1.2.3 Nonnumeric Peano Systems

Different Carriers, Same Successor Shape

A Peano carrier need not be the usual natural-number set, and its successor operation need not agree with the ambient order or arithmetic of a larger structure. What matters is the internally generated successor chain.

Example (The Powers of Ten). Let

$$P = \{10^{n-1} : n \in \mathbb{N}\}, \qquad e = 1,$$

and define

$$S : P \to P, \qquad S(10^{n-1}) = 10^n \quad (n \in \mathbb{N}).$$

Equivalently, for every $x \in P$,

$$S(x) = 10x.$$

Then (P, S, e) is a Peano system.

Remark (Structure). The Peano data are

$$\big(P, S, e\big) = \big(\{1, 10, 10^2, 10^3, \ldots\},\ x \mapsto 10x,\ 1\big).$$

Remark (Interpretation). The elements are not all natural numbers. They are only powers of ten. Nevertheless, starting from 1 and repeatedly multiplying by 10 produces a successor chain with exactly the Peano shape:

$$1 \mapsto 10 \mapsto 10^2 \mapsto 10^3 \mapsto \cdots.$$

Example (Strings of Repeated Letters). Let $\Sigma = \{a\}$ and let

$$P = \{a^n : n \in \mathbb{N}\}, \qquad e = a,$$

where a^n denotes the string consisting of n copies of a. Define

$$S : P \to P, \qquad S(a^n) = a^{n+1} \quad (n \in \mathbb{N}).$$

Then (P, S, e) is a Peano system.

Remark (Structure). The Peano data are

$$\big(P, S, e\big) = \big(\{a, aa, aaa, aaaa, \ldots\},\ a^n \mapsto a^{n+1},\ a\big).$$

Remark (Interpretation). This example is useful because the elements are not numbers at all. They are finite strings. The Peano structure is carried by the successor relation, not by the surface appearance of the elements.

Example (The Reciprocal Chain). Let

$$P = \left\{\frac{1}{n} : n \in \mathbb{N}\right\}, \qquad e = 1,$$

and define

$$S : P \to P, \qquad S\left(\frac{1}{n}\right) = \frac{1}{n+1} \quad (n \in \mathbb{N}).$$

Then (P, S, e) is a Peano system.

Remark (Structure). The Peano data are

$$\big(P, S, e\big) = \left(\left\{1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \ldots\right\},\ \frac{1}{n} \mapsto \frac{1}{n+1},\ 1\right).$$

Remark (Interpretation). This is a good example precisely because the carrier is a decreasing sequence in the usual order on $\mathbb{R}$:

$$1 > \frac{1}{2} > \frac{1}{3} > \frac{1}{4} > \cdots .$$

The Peano successor does not have to mean "larger than." It only means the next element in the generated successor chain. The order inherited from $\mathbb{R}$ is external structure, not part of the Peano-system data.

1.2.4 A Finite Alphabet Is Not a Peano System

A Failed Carrier

Finite chains help isolate what the Peano axioms forbid. A finite alphabet can support a partial next-letter operation, or it can support a cyclic successor operation, but neither choice gives a Peano system.

Example (The Finite Alphabet Fails). Let

$$P = \{A, B, C, \ldots, Z\}.$$

If one tries to define

$$S(A) = B, \quad S(B) = C, \quad \ldots,$$

then there is no value of $S(Z)$ inside the same finite alphabet that extends the one-way chain. If instead one defines $S(Z) = A$, then $S : P \to P$ is total but the successor operation forms a cycle.

Remark (Failure modes). The finite alphabet fails in one of two ways:

$$\begin{aligned}&\text{either } S \text{ is not a total map } P \to P,\\ &\text{or } S \text{ loops back into a cycle.}\end{aligned}$$

The first failure violates closure of successor. The second failure violates the one-way successor structure forced by the Peano axioms.

Remark (Interpretation). A finite alphabet cannot be a Peano system. Peano systems are necessarily infinite because induction and successor closure generate an unending chain from the distinguished first element.

1.2.5 Summary

The Core Lesson

The carrier of a Peano system can be numerical, set-theoretic, symbolic, or embedded in another ordered structure. The common content is the same abstract successor pattern.

Remark (Examples). The following are Peano systems once their successor operations are specified:

$$\begin{aligned}
&(\mathbb{N},\, n \mapsto n+1,\, 1),\\
&(\omega,\, n \mapsto n \cup \{n\},\, 0),\\
&(\omega \setminus \{0\},\, n \mapsto n \cup \{n\},\, 1),\\
&(\{1, 10, 10^2, 10^3, \ldots\},\, x \mapsto 10x,\, 1),\\
&(\{a, aa, aaa, aaaa, \ldots\},\, a^n \mapsto a^{n+1},\, a),\\
&\left(\left\{1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \ldots\right\},\, \frac{1}{n} \mapsto \frac{1}{n+1},\, 1\right).
\end{aligned}$$

They look different, but they have the same abstract Peano shape. Each is an infinite chain generated from a distinguished initial element by repeated successor.

1.3 Recursion on Peano Systems

Why Recursion Requires a Theorem

The Peano axioms describe the shape of a Peano system: there is a first element, a successor operation, no predecessor of the first element, no successor collisions, and an induction principle. They do not, by themselves, say that every recursive specification determines a function. A recursive display such as

$$f(1) = c \qquad \text{and} \qquad f(S(n)) = g(f(n))$$

is only a specification. It becomes a definition only after one proves that there exists a unique function satisfying it. This is the "specification is not existence" point emphasized in Feferman's treatment of recursive definition [Feferman, 1964, p. 67].

The proof used here follows the function-as-graph viewpoint. A function $f : P \to W$ is a subset of $P \times W$ whose elements are ordered pairs (n, w), one for each stage $n \in P$. The construction therefore moves through three zones: the stage set P, the Cartesian product grid $P \times W$, and the value set W. The recursion theorem is the result that identifies the right graph inside $P \times W$ and proves that it is the graph of exactly one function.

Remark (Prerequisite definition). An *inductive subset* of a Peano system is defined in Inductive Subset of a Peano System. In this section, induction is used only through the earlier Induction Principle for a Peano System.

1.3.1 Arity Examples Before the Theorem

Iterator Configurations

The iterator theorem has one formal shape but many uses. In every example below, the stage variable is the Peano stage $n \in P$, the value set is W, the first value is c, and the update rule is $g : W \to W$. These examples are motivating configurations only; the theorem establishing their legitimacy is proved afterward.

Remark (Example: isOdd).

component	choice
W	$\{\texttt{false}, \texttt{true}\}$
c	true
g	$x \mapsto \neg x$

n	1	2	3	4	5
$h(n)$	true	false	true	false	true

The defining clauses are

$$h(1) = \texttt{true} \qquad \text{and} \qquad h(S(n)) = \neg h(n).$$

This example shows that the value set W need not be the Peano system P. The stages are natural-number-like, but the values are truth values.

Remark (Example: addition with the left parameter fixed). Fix the left parameter at $m = 2$.

component	choice
W	P
c	$S(2) = 3$
g	S

n	1	2	3	4	5
$h(n)$	3	4	5	6	7

The defining clauses are

$$h(1) = S(2) \qquad \text{and} \qquad h(S(n)) = S(h(n)).$$

This is the iterator pattern behind $2 + n$ in the one-based convention. One parameter is fixed, and recursion runs in the remaining argument.

Remark (Example: tetration with the base fixed). Fix the base at $m = 2$.

component	choice
W	P
c	2
g	$x \mapsto 2^x$

n	1	2	3	4	5
$h(n)$	2	2^2	2^{2^2}	$2^{2^{2^2}}$	$2^{2^{2^{2^2}}}$

The defining clauses are

$$h(1) = 2 \qquad \text{and} \qquad h(S(n)) = 2^{h(n)}.$$

This example shows that the iterator theorem is not limited to the first arithmetic operations. Higher arithmetic still has the same formal configuration once the relevant operation on W is available.

Remark (Example: string repetition). Fix a string $s \in \Sigma^*$.

component	choice
W	Σ^*
c	s
g	$x \mapsto x \cdot s$

n	1	2	3	4	5
$h(n)$	s	$s \cdot s$	$s \cdot s \cdot s$	$s \cdot s \cdot s \cdot s$	$s \cdot s \cdot s \cdot s \cdot s$

The defining clauses are

$$h(1) = s \qquad \text{and} \qquad h(S(n)) = h(n) \cdot s.$$

This example is not arithmetic. It illustrates that recursion on P can generate values in any set equipped with an appropriate update rule.

1.3.2 Formal Development

Minimal Relations and Iterator Functions

The formal proof is organized in Feferman's graph style. First define the admissible relations in $P \times W$. Then take their intersection. The resulting minimal relation is shown to be consistent, deterministic, and complete, hence it is the graph of the required function. This follows the minimal-relation construction used in Feferman, Hrbacek–Jech, and related lecture-note presentations of the recursion theorem [Feferman, 1964, Hrbacek and Jech, 1999, Freiwald, 2009].

Definition (Iterator Data)

Definition 1.3.1 (Iterator Data). Let $(P, S, 1)$ be a Peano system, let W be a set, let $c \in W$, and let $g : W \to W$ be a function. The triple (W, c, g) is called ***iterator data*** for $(P, S, 1)$.

Remark (Standard quantified statement).

$$(W, c, g) \text{ is iterator data for } (P, S, 1) \iff \big(W \text{ is a set} \land c \in W \land g : W \to W\big).$$

Remark (Definition predicate reading).

$$\mathrm{IteratorData}(W, c, g) \iff \big(W \text{ is a set} \land c \in W \land g : W \to W\big).$$

Remark (Negated quantified statement).

$$(W, c, g) \text{ is not iterator data for } (P, S, 1) \iff \big(W \text{ is not a set} \lor c \notin W \lor g \text{ is not a functi}$$

Remark (Failure modes). Iterator data fails if the proposed value collection is not a set, if the initial value does not lie in the value set, or if the proposed update rule is not a self-map of the value set.

Remark (Failure mode decomposition).

$$\begin{aligned} & (W, c, g) \text{ is not iterator data for } (P, S, 1) \\ \iff & \underbrace{W \text{ is not a set}}_{\text{no value set}} \\ & \lor \underbrace{c \notin W}_{\text{invalid initial value}} \\ & \lor \underbrace{g \text{ is not a function } W \to W}_{\text{invalid update rule}}. \end{aligned}$$

Remark (Interpretation). Iterator data records the value set, the first value, and the fixed rule used to pass from each value to the next value.

Remark (Dependencies).

- Peano System

Definition (Iterator Relation)

Definition 1.3.2 (Iterator Relation). Let $(P, S, 1)$ be a Peano system, and let (W, c, g) be iterator data for it. A relation $R \subseteq P \times W$ is called an *iterator relation* for (W, c, g) exactly when

$$(1, c) \in R$$

and

$$\forall n \in P \, \forall w \in W \, \big((n, w) \in R \Longrightarrow (S(n), g(w)) \in R\big).$$

Remark (Standard quantified statement).

$$\begin{aligned} R \text{ is an iterator relation for } (W, c, g) \iff & R \subseteq P \times W \wedge (1, c) \in R \\ & \wedge \forall n \in P \, \forall w \in W \, \big((n, w) \in R \Longrightarrow (S(n), g(w)) \in \end{aligned}$$

Remark (Definition predicate reading).

$$\begin{aligned} \mathrm{IteratorRelation}(R, (P, S, 1), W, c, g) \iff & \mathrm{Relation}(R, P, W) \wedge (1, c) \in R \\ & \wedge \forall n \in P \, \forall w \in W \, \big((n, w) \in R \Longrightarrow (S(n), g(w)) \in R\big). \end{aligned}$$

Remark (Negated quantified statement).

$$\begin{aligned} R \text{ is not an iterator relation for } (W, c, g) \iff & R \not\subseteq P \times W \vee (1, c) \notin R \\ & \vee \exists n \in P \, \exists w \in W \, \big((n, w) \in R \wedge (S(n), g(w)) \end{aligned}$$

Remark (Failure modes). An iterator relation fails if it is not a relation inside $P \times W$, if it omits the required base pair, or if it is not closed under the iterator step.

Remark (Failure mode decomposition).

$$\begin{aligned} R \text{ is not an iterator relation for } (W, c, g) \iff & \underbrace{R \not\subseteq P \times W}_{\text{wrong ambient relation}} \\ & \vee \underbrace{(1, c) \notin R}_{\text{base pair missing}} \\ & \vee \underbrace{\exists n \in P \, \exists w \in W \, \big((n, w) \in R \wedge (S(n), g(w))}_{\text{step closure fails}} \end{aligned}$$

Remark (Interpretation). An iterator relation is any graph-like set of possible stage-value pairs that contains the required first pair and is closed under the recursive step.

Remark (Dependencies).

- Iterator Data

Definition (Minimal Iterator Relation)

Definition 1.3.3 (Minimal Iterator Relation). Let $(P, S, 1)$ be a Peano system, and let (W, c, g) be iterator data for it. The *minimal iterator relation* for (W, c, g) is

$$R^* = \bigcap \{R \subseteq P \times W : R \text{ is an iterator relation for } (W, c, g)\}.$$

Remark (Standard quantified statement).

$$(n, w) \in R^* \iff \forall R \subseteq P \times W \; \big(R \text{ is an iterator relation for } (W, c, g) \implies (n, w) \in R\big).$$

Remark (Definition predicate reading). Let $\mathcal{C}_{\text{it}}$ be the closure condition

$$\mathcal{C}_{\text{it}}(R) \iff \text{IteratorRelation}(R, (P, S, 1), W, c, g).$$

Then the definition says

$$\text{MinimalRelation}(R^*, \mathcal{C}_{\text{it}}).$$

Remark (Negated quantified statement).

$$(n, w) \notin R^* \iff \exists R \subseteq P \times W \; \big(R \text{ is an iterator relation for } (W, c, g) \wedge (n, w) \notin R\big).$$

Remark (Failure modes). A pair fails to belong to the minimal iterator relation exactly when some iterator relation omits that pair. Such a pair is not forced by the base and successor-closure clauses.

Remark (Interpretation). The minimal iterator relation consists exactly of the stage-value pairs forced by the base pair and the successor closure rule. This is the R^* construction used in the classical proof strategy [Feferman, 1964, Hrbacek and Jech, 1999, Freiwald, 2009].

Remark (Dependencies).

- Iterator Relation

Lemma (Consistency of the Minimal Iterator Relation)

Lemma 1.3.4 (Consistency of the Minimal Iterator Relation). *Let $(P, S, 1)$ be a Peano system, let (W, c, g) be iterator data for it, and let R^* be the minimal iterator relation for (W, c, g). Then R^* is an iterator relation for (W, c, g).* *Go to proof.*

Remark (Standard quantified statement).

$$R^* \text{ is an iterator relation for } (W, c, g).$$

Remark (Predicate reading).

$$\text{MinimalRelation}(R^*, \mathcal{C}_{\text{it}}) \implies \text{IteratorRelation}(R^*, (P, S, 1), W, c, g).$$

Remark (Negated quantified statement).

$$R^* \text{ is not an iterator relation for } (W, c, g).$$

Remark (Contrapositive quantified statement).

R^* is not an iterator relation for $(W, c, g) \implies R^*$ is not the minimal iterator relatio

Remark (Interpretation). The intersection construction does not lose the base pair or the successor closure condition. The minimal relation is therefore still one of the admissible iterator relations.

Remark (Dependencies).

- Minimal Iterator Relation

Lemma (Forced Values Are Unique)

Lemma 1.3.5 (Forced Values Are Unique). *Let $(P, S, 1)$ be a Peano system, let (W, c, g) be iterator data for it, and let R^* be the minimal iterator relation. If $(n, w_0) \in R^*$ and $(n, w_1) \in R^*$, then $w_0 = w_1$. Go to proof.*

Remark (Standard quantified statement).

$$\forall n \in P\ \forall w_0, w_1 \in W\ \big((n, w_0) \in R^* \wedge (n, w_1) \in R^* \implies w_0 = w_1\big).$$

Remark (Predicate reading).

$$\text{FunctionalRelation}(R^*, P, W).$$

Remark (Negated quantified statement).

$$\exists n \in P\ \exists w_0, w_1 \in W\ \big((n, w_0) \in R^* \wedge (n, w_1) \in R^* \wedge w_0 \neq w_1\big).$$

Remark (Failure modes). Forced value uniqueness fails exactly when one stage is assigned two distinct values by the minimal iterator relation.

Remark (Contrapositive quantified statement).

$\exists n \in P\ \exists w_0, w_1 \in W\ \big((n, w_0) \in R^* \wedge (n, w_1) \in R^* \wedge w_0 \neq w_1\big) \implies R^*$ is not functional at every s

Remark (Interpretation). No stage can be forced to carry two different values. This is the key determinacy step in the minimal-relation construction.

Remark (Dependencies).

- Consistency of the Minimal Iterator Relation

Lemma (Determinism of the Minimal Iterator Relation)

Lemma 1.3.6 (Determinism of the Minimal Iterator Relation). *Let $(P, S, 1)$ be a Peano system, let (W, c, g) be iterator data for it, and let R^* be the minimal iterator relation. For each $n \in P$, there is at most one $w \in W$ such that $(n, w) \in R^*$. Go to proof.*

Remark (Standard quantified statement).

$$\forall n \in P\ \forall w_0, w_1 \in W\ \big((n, w_0) \in R^* \wedge (n, w_1) \in R^* \implies w_0 = w_1\big).$$

Remark (Predicate reading).

$$\mathrm{FunctionalRelation}(R^*, P, W).$$

Remark (Negated quantified statement).

$$\exists n \in P\ \exists w_0, w_1 \in W\ \big((n, w_0) \in R^* \wedge (n, w_1) \in R^* \wedge w_0 \neq w_1\big).$$

Remark (Failure modes). Determinism fails exactly when a single stage has two distinct values in the minimal iterator relation.

Remark (Dependencies).

- Forced Values Are Unique

Lemma (Completeness of the Minimal Iterator Relation)

Lemma 1.3.7 (Completeness of the Minimal Iterator Relation). *Let $(P, S, 1)$ be a Peano system, let (W, c, g) be iterator data for it, and let R^* be the minimal iterator relation. For each $n \in P$, there exists $w \in W$ such that $(n, w) \in R^*$.* *Go to proof.*

Remark (Standard quantified statement).

$$\forall n \in P\ \exists w \in W\ ((n, w) \in R^*).$$

Remark (Predicate reading).

$$\mathrm{TotalRelation}(R^*, P, W).$$

Remark (Negated quantified statement).

$$\exists n \in P\ \forall w \in W\ ((n, w) \notin R^*).$$

Remark (Failure modes). Completeness fails exactly when some stage of P receives no value in the minimal iterator relation.

Remark (Dependencies).

- Consistency of the Minimal Iterator Relation
- Induction Principle for a Peano System

Lemma (Uniqueness of Iterator Functions)

Lemma 1.3.8 (Uniqueness of Iterator Functions). *Let $(P, S, 1)$ be a Peano system, and let (W, c, g) be iterator data for it. If $f, h : P \to W$ satisfy*

$$f(1) = c, \qquad h(1) = c,$$

and

$$\forall n \in P\ (f(S(n)) = g(f(n))) \qquad \textit{and} \qquad \forall n \in P\ (h(S(n)) = g(h(n))),$$

then $f = h$. *Go to proof.*

Remark (Standard quantified statement).

$$\begin{aligned}\forall f, h : P \to W \Big(f(1) = c \wedge h(1) = c \\ \wedge \forall n \in P \, (f(S(n)) = g(f(n))) \\ \wedge \forall n \in P \, (h(S(n)) = g(h(n))) \implies f = h \Big).\end{aligned}$$

Remark (Predicate reading).

$$\forall f, h : P \to W \ (\mathrm{IteratorFunction}(f, (P, S, 1), W, c, g) \wedge \mathrm{IteratorFunction}(h, (P, S, 1), W, c, g) \implies f = h)$$

Remark (Negated quantified statement).

$$\begin{aligned}\exists f, h : P \to W \Big(f(1) = c \wedge h(1) = c \\ \wedge \forall n \in P \, (f(S(n)) = g(f(n))) \\ \wedge \forall n \in P \, (h(S(n)) = g(h(n))) \wedge f \neq h \Big).\end{aligned}$$

Remark (Failure modes). Uniqueness fails exactly when two distinct functions satisfy the same base clause and the same successor clause.

Remark (Contrapositive quantified statement).

$$\begin{aligned}f \neq h \implies f(1) \neq c \vee h(1) \neq c \\ \vee \exists n \in P \, (f(S(n)) \neq g(f(n))) \\ \vee \exists n \in P \, (h(S(n)) \neq g(h(n))).\end{aligned}$$

Remark (Interpretation). The iterator clauses determine at most one function. Any difference between two candidate functions must show up as a failure of the base clause or a failure of one of the successor clauses.

Remark (Dependencies).

- Iterator Data
- Induction Principle for a Peano System

Lemma (Existence of an Iterator Function)

Lemma 1.3.9 (Existence of an Iterator Function). *Let $(P, S, 1)$ be a Peano system, and let (W, c, g) be iterator data for it. Then there exists a function $f : P \to W$ such that*

$$f(1) = c \qquad \text{and} \qquad \forall n \in P \, (f(S(n)) = g(f(n))).$$

Go to proof.

Remark (Standard quantified statement).

$$\exists f : P \to W \, \big(f(1) = c \wedge \forall n \in P \, (f(S(n)) = g(f(n))) \big).$$

Remark (Predicate reading).

$$\exists f : P \to W \ \ \mathrm{IteratorFunction}(f, (P, S, 1), W, c, g).$$

Remark (Negated quantified statement).

$$\forall f : P \to W \ \big(f(1) \neq c \vee \exists n \in P \ (f(S(n)) \neq g(f(n)))\big).$$

Remark (Failure modes). Existence fails exactly when every candidate function misses the base value or misses the recursive successor equation at some stage.

Remark (Contrapositive quantified statement).

$\neg\exists f : P \to W \ \big(f(1) = c \wedge \forall n \in P \ (f(S(n)) = g(f(n)))\big) \implies R^*$ is not both deterministic and co

Remark (Interpretation). Existence is the assertion that the minimal relation is not merely graph-like locally, but supplies a value at every stage and hence determines an actual function.

Remark (Dependencies).

- Determinism of the Minimal Iterator Relation
- Completeness of the Minimal Iterator Relation

Theorem (Peano Iterator Theorem)

Theorem 1.3.10 (Peano Iterator Theorem). *Let $(P, S, 1)$ be a Peano system, and let (W, c, g) be iterator data for it. Then there exists a unique function $f : P \to W$ such that*

$$f(1) = c \qquad \textit{and} \qquad \forall n \in P \ (f(S(n)) = g(f(n))).$$

Go to proof.

Remark (Standard quantified statement).

$$\exists! f : P \to W \ \big(f(1) = c \wedge \forall n \in P \ (f(S(n)) = g(f(n)))\big).$$

Remark (Predicate reading).

$$\exists! f : P \to W \ \ \mathrm{IteratorFunction}(f, (P, S, 1), W, c, g).$$

Remark (Negated quantified statement).

$$\neg\exists! f : P \to W \ \big(f(1) = c \wedge \forall n \in P \ (f(S(n)) = g(f(n)))\big).$$

Remark (Failure modes). The iterator theorem fails only if no function satisfies the iterator clauses or if more than one function satisfies them.

Remark (Failure mode decomposition).

$$
\begin{aligned}
&\neg\exists! f : P \to W \big(f(1) = c \land \forall n \in P\ (f(S(n)) = g(f(n)))\big) \\
\iff &\underbrace{\neg\exists f : P \to W \big(f(1) = c \land \forall n \in P\ (f(S(n)) = g(f(n)))\big)}_{\texttt{existence fails}} \\
&\lor \underbrace{\exists f, h : P \to W \big(f \neq h \land f(1) = c \land h(1) = c \land \forall n \in P\ (f(S(n)) = g(f(n))) \land \forall n \in P\ (h(S(n))}_{\texttt{uniqueness fails}}
\end{aligned}
$$

Remark (Interpretation). The theorem says that iterator data is sufficient to turn a recursive specification into a genuine definition: there is exactly one graph and hence exactly one function satisfying the base and successor clauses.

Remark (Historical note). This is the one-based Peano-system version of the iterator theorem in Feferman [1964], Mendelson [1982], Simpson [2009]. It is also the set-theoretic instance of the natural numbers object universal property [Lawvere, 1969]; Dedekind's recursion theorem is the classical ancestor [Dedekind, 1901].

Remark (Dependencies).

- Uniqueness of Iterator Functions
- Existence of an Iterator Function

Definition (Iterator-Generated Function)

Definition 1.3.11 (Iterator-Generated Function). Let $(P, S, 1)$ be a Peano system, and let (W, c, g) be iterator data for it. The unique function $f : P \to W$ satisfying

$$f(1) = c \qquad \text{and} \qquad \forall n \in P\ (f(S(n)) = g(f(n)))$$

is called the *iterator-generated function determined by* (W, c, g).

Remark (Standard quantified statement).

$$f \text{ is the iterator-generated function determined by } (W, c, g) \iff \big(f : P \to W \land f(1) = c \land \forall n$$

Remark (Definition predicate reading).

$$\mathrm{IteratorGeneratedFunction}(f, (P, S, 1), W, c, g) \iff \mathrm{IteratorFunction}(f, (P, S, 1), W, c, g) \land \exists! h : P \to W$$

Remark (Negated quantified statement).

$$
\begin{aligned}
f \text{ is not the iterator-generated function determined by } (W, c, g) \iff\ & f \text{ is not a functio} \\
& \lor f(1) \neq c \lor \exists n \in P
\end{aligned}
$$

Remark (Failure modes). A candidate fails to be the iterator-generated function if it is not a function from P to W, if it has the wrong base value, or if it violates the successor clause.

Remark (Interpretation). This definition packages the unique function supplied by the Peano Iterator Theorem as a named object available for later constructions.

Remark (Dependencies).

- Peano Iterator Theorem

Theorem (Iterator Base Clause)

Theorem 1.3.12 (Iterator Base Clause). *Let $(P, S, 1)$ be a Peano system, let (W, c, g) be iterator data for it, and let f be the iterator-generated function determined by (W, c, g). Then*

$$f(1) = c.$$

Go to proof.

Remark (Standard quantified statement).

$$f \text{ is the iterator-generated function determined by } (W, c, g) \Longrightarrow f(1) = c.$$

Remark (Predicate reading).

$$\text{IteratorGeneratedFunction}(f, (P, S, 1), W, c, g) \Longrightarrow f(1) = c.$$

Remark (Contrapositive quantified statement).

$$f(1) \neq c \Longrightarrow f \text{ is not the iterator-generated function determined by } (W, c, g).$$

Remark (Interpretation). The base clause is not an additional theorem about the generated function; it is one of the clauses built into the unique recursive specification.

Remark (Dependencies).

- Iterator-Generated Function

Theorem (Iterator Successor Clause)

Theorem 1.3.13 (Iterator Successor Clause). *Let $(P, S, 1)$ be a Peano system, let (W, c, g) be iterator data for it, and let f be the iterator-generated function determined by (W, c, g). Then*

$$\forall n \in P \ (f(S(n)) = g(f(n))).$$

Go to proof.

Remark (Standard quantified statement).

$$f \text{ is the iterator-generated function determined by } (W, c, g) \Longrightarrow \forall n \in P \ (f(S(n)) = g(f(n)))$$

Remark (Predicate reading).

$$\mathrm{IteratorGeneratedFunction}(f,(P,S,1),W,c,g) \implies \forall n \in P\ (f(S(n)) = g(f(n))).$$

Remark (Contrapositive quantified statement).

$\exists n \in P\ (f(S(n)) \neq g(f(n))) \implies f$ is not the iterator-generated function determined by (W

Remark (Interpretation). The successor clause says that the generated function advances by applying the update rule to the previous value at every Peano stage.

Remark (Dependencies).

- Iterator-Generated Function

Definition (Iteration of a Self-Map)

Definition 1.3.14 (Iteration of a Self-Map). Let $(P, S, 1)$ be a Peano system, let W be a set, let $c \in W$, and let $g : W \to W$. The iterator-generated function determined by (W, c, g) is called the *iteration of g from c along P*.

Remark (Standard quantified statement).

f is the iteration of g from c along $P \iff f$ is the iterator-generated function deter

Remark (Definition predicate reading).

f is the iteration of g from c along $P \iff \mathrm{IteratorGeneratedFunction}(f,(P,S,1),W,c,g)$.

Remark (Negated quantified statement).

f is not the iteration of g from c along $P \iff f$ is not the iterator-generated functi

Remark (Interpretation). Iteration is the special case of recursion in which the same self-map is used at every stage.

Remark (Dependencies).

- Iterator-Generated Function

Corollary (Uniqueness of Binary Iterator Operations)

Corollary 1.3.15 (Uniqueness of Binary Iterator Operations). *Let $(P, S, 1)$ be a Peano system, let A be a set, let W be a set, and suppose each $a \in A$ determines iterator data (W, c_a, g_a) for $(P, S, 1)$. If $F, G : A \times P \to W$ satisfy*

$$F(a,1) = c_a, \qquad G(a,1) = c_a$$

and

$$F(a,S(n)) = g_a(F(a,n)), \qquad G(a,S(n)) = g_a(G(a,n))$$

for all $a \in A$ and all $n \in P$, then $F = G$. Go to proof.

Remark (Standard quantified statement).

$$\forall a \in A\, \forall n \in P\, \Big(F(a,1) = c_a \wedge G(a,1) = c_a \\ \wedge F(a,S(n)) = g_a(F(a,n)) \wedge G(a,S(n)) = g_a(G(a,n)) \Big) \implies F = G.$$

Remark (Predicate reading). For each $a \in A$, write $F_a(n) = F(a,n)$ and $G_a(n) = G(a,n)$. Then

$$\forall a \in A\, \big(\mathrm{IteratorFunction}(F_a, (P,S,1), W, c_a, g_a) \wedge \mathrm{IteratorFunction}(G_a, (P,S,1), W, c_a, g_a) \implies F_a = G_a$$

Remark (Negated quantified statement).

$$\exists a \in A\, \exists n \in P\, \Big(F(a,1) = c_a \wedge G(a,1) = c_a \\ \wedge F(a,S(n)) = g_a(F(a,n)) \wedge G(a,S(n)) = g_a(G(a,n)) \wedge F \neq G \Big).$$

Remark (Failure modes). Binary iterator uniqueness fails if two different binary operations satisfy the same one-parameter iterator specification for every fixed parameter.

Remark (Interpretation). For each fixed first argument, the binary operation is an ordinary iterator in the second argument. Uniqueness for the binary operation is therefore pointwise uniqueness of iterator functions.

Remark (Historical note). This isolates the binary-operation uniqueness pattern used by Thurston before existence is invoked [Thurston, 1956].

Remark (Dependencies).

- Uniqueness of Iterator Functions

General Recursion

Pure iteration uses a fixed self-map $g : W \to W$. General recursion allows the successor rule to depend on the current stage as well as on the current value. The state-encoding theorem below reduces that stage-dependent case to the Peano Iterator Theorem.

Definition (Stage-Dependent Step Rule)

Definition 1.3.16 (Stage-Dependent Step Rule). Let $(P, S, 1)$ be a Peano system and let W be a set. A *stage-dependent step rule* on W over P is a function

$$\Phi : P \times W \to W.$$

Remark (Standard quantified statement).

$$\Phi \text{ is a stage-dependent step rule on } W \text{ over } P \iff \Phi : P \times W \to W.$$

Remark (Definition predicate reading).

$$\mathrm{StageDependentStepRule}(\Phi, P, W) \iff \Phi : P \times W \to W.$$

Remark (Negated quantified statement).

Φ is not a stage-dependent step rule on W over $P \iff \Phi$ is not a function $P \times W \to W$

Remark (Failure modes). A stage-dependent step rule fails precisely when it is not a well-defined function from stage-value pairs back into the value set.

Remark (Interpretation). A stage-dependent step rule allows the next value to depend on both the current Peano stage and the current value.

Remark (Dependencies).

- Peano System

Definition (General Recursive Function)

Definition 1.3.17 (General Recursive Function). Let $(P, S, 1)$ be a Peano system, let W be a set, let $c \in W$, and let $\Phi : P \times W \to W$ be a stage-dependent step rule. A function $f : P \to W$ is called the *general recursive function determined by* (W, c, Φ) exactly when

$$f(1) = c \qquad \text{and} \qquad \forall n \in P\ (f(S(n)) = \Phi(n, f(n))).$$

Remark (Standard quantified statement).

f is the general recursive function determined by $(W, c, \Phi) \iff f : P \to W \wedge f(1) = c$
$\wedge\, \forall n \in P\ (f(S(n)) = \Phi(n,$

Remark (Definition predicate reading).

$$\mathrm{GeneralRecursiveFunction}(f, (P, S, 1), W, c, \Phi) \iff f : P \to W \wedge f(1) = c$$
$$\wedge\, \forall n \in P\ (f(S(n)) = \Phi(n, f(n))).$$

Remark (Negated quantified statement).

f is not the general recursive function determined by $(W, c, \Phi) \iff f$ is not a function
$\vee\, f(1) \neq c \vee \exists n \in P\ ($

Remark (Failure modes). A candidate fails to be the general recursive function if it has the wrong domain or codomain, the wrong base value, or a failed stage-dependent successor equation.

Remark (Interpretation). General recursive functions are the stage-dependent analogues of iterator-generated functions.

Remark (Dependencies).

- Stage-Dependent Step Rule

Lemma (Uniqueness of General Recursive Functions)

Lemma 1.3.18 (Uniqueness of General Recursive Functions). *Let $(P, S, 1)$ be a Peano system, let W be a set, let $c \in W$, and let $\Phi : P \times W \to W$ be a stage-dependent step rule. If $f, h : P \to W$ satisfy*

$$f(1) = c, \qquad h(1) = c,$$

and

$$\forall n \in P\ (f(S(n)) = \Phi(n, f(n))) \qquad \text{and} \qquad \forall n \in P\ (h(S(n)) = \Phi(n, h(n))),$$

then $f = h$. *Go to proof.*

Remark (Standard quantified statement).

$$\begin{aligned}\forall f, h : P \to W\ \Big(f(1) = c \wedge h(1) = c \\ &\wedge \forall n \in P\ (f(S(n)) = \Phi(n, f(n))) \\ &\wedge \forall n \in P\ (h(S(n)) = \Phi(n, h(n))) \Longrightarrow f = h\Big).\end{aligned}$$

Remark (Predicate reading).

$\forall f, h : P \to W\ \big(\text{GeneralRecursiveFunction}(f, (P, S, 1), W, c, \Phi) \wedge \text{GeneralRecursiveFunction}(h, (P, S, 1),$

Remark (Negated quantified statement).

$$\begin{aligned}\exists f, h : P \to W\ \Big(f(1) = c \wedge h(1) = c \\ &\wedge \forall n \in P\ (f(S(n)) = \Phi(n, f(n))) \\ &\wedge \forall n \in P\ (h(S(n)) = \Phi(n, h(n))) \wedge f \neq h\Big).\end{aligned}$$

Remark (Failure modes). Uniqueness of general recursive functions fails if two distinct functions satisfy the same base clause and the same stage-dependent successor clause.

Remark (Contrapositive quantified statement).

$$\begin{aligned}f \neq h \Longrightarrow f(1) \neq c \vee h(1) \neq c \\ &\vee \exists n \in P\ (f(S(n)) \neq \Phi(n, f(n))) \\ &\vee \exists n \in P\ (h(S(n)) \neq \Phi(n, h(n))).\end{aligned}$$

Remark (Interpretation). Once the initial value and stage-dependent rule are fixed, two recursive candidates cannot diverge without violating one of the defining clauses.

Remark (Dependencies).

- General Recursive Function

- Induction Principle for a Peano System

Theorem (General Recursion by State Encoding)

Theorem 1.3.19 (General Recursion by State Encoding). *Let $(P, S, 1)$ be a Peano system, let W be a set, let $c \in W$, and let $\Phi : P \times W \to W$ be a stage-dependent step rule. There exists a function $H : P \to P \times W$ such that*

$$H(1) = (1, c)$$

and

$$\forall n \in P \ \big(H(S(n)) = (S(\pi_1(H(n))), \Phi(\pi_1(H(n)), \pi_2(H(n))))\big).$$

Go to proof.

Remark (Standard quantified statement).

$$\exists H : P \to P \times W \ \Big(H(1) = (1, c) \wedge \forall n \in P \ H(S(n)) = (S(\pi_1(H(n))), \Phi(\pi_1(H(n)), \pi_2(H(n))))\Big).$$

Remark (Predicate reading). Let $\Gamma_\Phi : P \times W \to P \times W$ be defined by

$$\Gamma_\Phi(p, w) = (S(p), \Phi(p, w)).$$

Then state encoding is the iterator assertion

$$\exists H : P \to P \times W \ \mathrm{IteratorFunction}(H, (P, S, 1), P \times W, (1, c), \Gamma_\Phi).$$

Remark (Negated quantified statement).

$$\forall H : P \to P \times W \ \Big(H(1) \neq (1, c) \\ \vee \exists n \in P \ H(S(n)) \neq (S(\pi_1(H(n))), \Phi(\pi_1(H(n)), \pi_2(H(n))))\Big).$$

Remark (Failure modes). State encoding fails if every candidate state function has the wrong initial state or violates the encoded successor step at some stage.

Remark (Interpretation). The theorem converts stage-dependent recursion into ordinary iteration by carrying the current stage together with the current value as the recursive state.

Remark (Dependencies).

- Peano Iterator Theorem
- Stage-Dependent Step Rule

Theorem (General Recursion Theorem)

Theorem 1.3.20 (General Recursion Theorem). *Let $(P, S, 1)$ be a Peano system, let W be a set, let $c \in W$, and let $\Phi : P \times W \to W$ be a stage-dependent step rule. Then there exists a unique function $f : P \to W$ such that*

$$f(1) = c \qquad \textit{and} \qquad \forall n \in P \ (f(S(n)) = \Phi(n, f(n))).$$

Go to proof.

Remark (Standard quantified statement).

$$\exists! f : P \to W \ \big(f(1) = c \wedge \forall n \in P \ (f(S(n)) = \Phi(n, f(n)))\big).$$

Remark (Predicate reading).

$$\exists! f : P \to W \ \ \mathrm{GeneralRecursiveFunction}(f, (P, S, 1), W, c, \Phi).$$

Remark (Negated quantified statement).

$$\neg\exists! f : P \to W \ \big(f(1) = c \wedge \forall n \in P \ (f(S(n)) = \Phi(n, f(n)))\big).$$

Remark (Failure modes). General recursion fails if no function satisfies the general recursive clauses or if two distinct functions satisfy them.

Remark (Failure mode decomposition).

$$\begin{aligned}
& \neg\exists! f : P \to W \ \big(f(1) = c \wedge \forall n \in P \ (f(S(n)) = \Phi(n, f(n)))\big) \\
\iff & \underbrace{\neg\exists f : P \to W \ \big(f(1) = c \wedge \forall n \in P \ (f(S(n)) = \Phi(n, f(n)))\big)}_{\texttt{existence fails}} \\
& \vee \underbrace{\exists f, h : P \to W \ \big(f \neq h \wedge f(1) = c \wedge h(1) = c \wedge \forall n \in P \ (f(S(n)) = \Phi(n, f(n))) \wedge \forall n \in P \ (h(S(}_{\texttt{uniqueness fails}}
\end{aligned}$$

Remark (Interpretation). General recursion justifies recursive definitions whose successor rule depends on the current stage as well as the current value.

Remark (Historical note). This is the stage-dependent form corresponding to Feferman's Theorem 3.4' [Feferman, 1964]. It is the form used later for arithmetic definitions; those definitions are deferred to their own sections [Feferman, 1964, Mendelson, 1982].

Remark (Dependencies).

- Uniqueness of General Recursive Functions
- General Recursion by State Encoding

Remark (Deferred arithmetic definitions). Definitions of addition, multiplication, exponentiation, and later arithmetic operations are not part of this section. They begin after the recursion principles have been established.

Theorem (Uniqueness of Peano Systems up to Isomorphism)

Theorem 1.3.21 (Uniqueness of Peano Systems up to Isomorphism). *Let $(P, S_P, 1_P)$ and $(Q, S_Q, 1_Q)$ be Peano systems. Then there exists a unique bijection $\theta : P \to Q$ such that*

$$\theta(1_P) = 1_Q$$

and

$$\forall n \in P \ (\theta(S_P(n)) = S_Q(\theta(n))).$$

Thus any two Peano systems are isomorphic. *Go to proof.*

Remark (Standard quantified statement).

$$\exists! \theta : P \to Q \left(\theta \texttt{ is bijective} \wedge \theta(1_P) = 1_Q \wedge \forall n \in P \left(\theta(S_P(n)) = S_Q(\theta(n)) \right) \right).$$

Remark (Predicate reading). Let $A = (P, S_P, 1_P)$ and $B = (Q, S_Q, 1_Q)$. The theorem asserts

$$\mathrm{Isomorphism}(A, B),$$

with unique witness $\theta : P \to Q$ satisfying the displayed base and successor-preservation clauses.

Remark (Negated quantified statement).

$$\neg \exists! \theta : P \to Q \left(\theta \texttt{ is bijective} \wedge \theta(1_P) = 1_Q \right.$$
$$\left. \wedge \forall n \in P \left(\theta(S_P(n)) = S_Q(\theta(n)) \right) \right).$$

Remark (Failure modes). Categoricity fails if no structure-preserving bijection exists between the two Peano systems or if more than one such bijection exists.

Remark (Interpretation). All Peano systems have the same structure up to a unique isomorphism. The elements may be named differently, but the distinguished first element and successor structure determine the translation.

Remark (Historical note). This is the categoricity theorem for Peano systems, following Feferman's Theorem 3.5 and Simpson's Peano-system presentation [Feferman, 1964, Simpson, 2009].

Remark (Dependencies).

- Peano System
- General Recursion Theorem

Proofs

Remark (Return). Return to Theorem

Theorem (Induction Principle for a Peano System). *Let $(P, S, 1)$ be a Peano system, and let φ be a predicate on P. If*

$$\varphi(1) \qquad \textit{and} \qquad \forall n \in P\ (\varphi(n) \Longrightarrow \varphi(S(n))),$$

then

$$\forall n \in P\ \varphi(n).$$

Proof. **Professional Standard Proof.**
Define $A := \{\, n \in P \mid \varphi(n) \,\}$. The hypothesis $\varphi(1)$ gives $1 \in A$. The hypothesis $\forall n \in P\,(\varphi(n) \Rightarrow \varphi(S(n)))$ gives $\forall n \in P\,(n \in A \Rightarrow S(n) \in A)$. Hence A is an inductive subset of P. By the induction axiom (P5), $A = P$. Therefore $\varphi(n)$ holds for every $n \in P$. $\square$

Proof. **Detailed Learning Proof.**

Step 1. Translate the predicate into a subset. Every predicate φ on a set P determines an extension: the set of elements for which the predicate holds. Define

$$A \;:=\; \{\, n \in P \mid \varphi(n) \,\}.$$

Since φ is a predicate on P, the subset A is well-defined as a subset of P, that is, $A \subseteq P$.

The goal $\forall n \in P\,\varphi(n)$ is now equivalent to $A = P$: every element of P satisfies φ if and only if every element of P belongs to A. The proof will establish $A = P$ by applying the induction axiom.

Step 2. Verify the base condition: $1 \in A$. The first hypothesis is $\varphi(1)$. By the definition of A, $\varphi(1)$ holds if and only if $1 \in A$. Since $\varphi(1)$ is given, we conclude $1 \in A$.

Step 3. Verify the successor-closure condition. The second hypothesis is

$$\forall n \in P\ (\varphi(n) \Longrightarrow \varphi(S(n))).$$

Let $n \in P$ be arbitrary and suppose $n \in A$. By the definition of A, $n \in A$ means precisely $\varphi(n)$. The second hypothesis then gives $\varphi(S(n))$. Since $S(n) \in P$ (by the closure axiom for a Peano system) and $\varphi(S(n))$ holds, the definition of A gives $S(n) \in A$.

Since n was arbitrary, this establishes

$$\forall n \in P\ (n \in A \Longrightarrow S(n) \in A).$$

The subset A is therefore successor-closed.

Step 4. Conclude that A is an inductive subset. Combining Steps 2 and 3:

$$1 \in A \quad \text{and} \quad \forall n \in P\ (n \in A \Longrightarrow S(n) \in A).$$

By the definition of an inductive subset (Definition: Inductive Subset of a Peano System), A is an inductive subset of the Peano system $(P, S, 1)$.

Step 5. Apply the induction axiom. The induction axiom (P5) states: any subset $A \subseteq P$ that contains 1 and is closed under successor satisfies $A =$

P. Steps 2 and 3 establish exactly these two conditions. Therefore $A = P$.

Step 6. Conclude the predicate holds universally. Since $A = P$, every element $n \in P$ belongs to A. By the definition of A, this means $\varphi(n)$ holds for every $n \in P$. That is,

$$\forall n \in P\ \varphi(n). \qquad \square$$

Remark (Proof structure). The proof is a direct application of the induction axiom (P5). The induction axiom is formulated in terms of subsets: any inductive subset of P must equal P itself. The predicate φ does not appear in (P5), so the first move is always to translate -- passing from the predicate form to the subset form via the extension $A := \{\, n \in P \mid \varphi(n) \,\}$. Once the translation is made, the two hypotheses map exactly onto the two conditions that (P5) requires: the base condition $1 \in A$ comes from $\varphi(1)$, and successor-closure of A comes from the step hypothesis on φ. The induction axiom then forces $A = P$, and translating back gives the conclusion.

This translation pattern -- predicate to subset, apply (P5), translate back -- is the invariant structure of every induction proof in the Peano system framework.

Remark (Dependencies).

- Peano System
- Induction Axiom (P5)
- Inductive Subset of a Peano System

Remark (Return). Return to Theorem

Theorem (Every Element Is Either 1 or a Successor). *Let $(P, S, 1)$ be a Peano system. For every $x \in P$, either $x = 1$ or there exists $u \in P$ such that $S(u) = x$.*

Proof. **Professional Standard Proof.**
Let

$$D := \{x \in P : x = 1 \text{ or } \exists u \in P\ (S(u) = x)\}.$$

We show that D is inductive. Since $1 = 1$, we have $1 \in D$.

Now let $x \in D$. By closure of successor, $S(x) \in P$. Moreover $x \in P$, so $S(x)$ is the successor of an element of P. Hence $S(x) \in D$. Therefore D contains 1 and is closed under successor. By the induction principle, $D = P$.

Thus every $x \in P$ lies in D, so every $x \in P$ satisfies either $x = 1$ or there exists $u \in P$ such that $S(u) = x$. □

Proof. **Detailed Learning Proof.**
Define a subset D of P by putting into D exactly those elements of P that already have the desired form:

$$D = \{x \in P : x = 1 \text{ or } \exists u \in P\ (S(u) = x)\}.$$

We prove that $D = P$ by induction.

For the base case, $1 \in D$ because $1 = 1$. Thus the distinguished first element has the required form.

For the successor step, assume $x \in D$. Since $D \subseteq P$, we have $x \in P$. The Peano closure axiom gives $S(x) \in P$. But then $S(x)$ is literally a successor, namely the successor of $x \in P$. Therefore

$$\exists u \in P\ (S(u) = S(x)),$$

for example by taking $u = x$. Hence $S(x) \in D$.

So D contains 1 and is closed under successor. By the induction principle for Peano systems, $D = P$. Since membership in D is exactly the assertion that an element is either 1 or a successor, the theorem follows. □

Remark (Proof structure). The proof forms the subset of all elements satisfying the desired dichotomy and then proves that this subset is inductive. The base case is immediate from $1 = 1$. The successor step is immediate because every successor is, by definition, a successor of an element of P.

Remark (Dependencies).

- Peano System
- Distinguished Element Lies in the Underlying Set
- Closure Under Successor
- Induction Principle for a Peano System

Remark (Return). Return to Theorem

Theorem (Successor Inequality Reflection). *Let $(P, S, 1)$ be a Peano system. For all $a, b \in P$, if $a \neq b$, then $S(a) \neq S(b)$.*

Professional Standard Proof.

Proof. TODO ☐

Detailed Learning Proof.

TODO: Expand the professional proof into a detailed learning proof.

Remark (Proof structure). This proof is intended to use the contrapositive form of injectivity of the successor map.

Remark (Dependencies).

- Peano System
- Injectivity of Successor

Remark (Return). Return to Corollary

Corollary (Non-One Elements Have a Predecessor). *Let $(P, S, 1)$ be a Peano system. For every $x \in P$, if $x \neq 1$, then there exists $u \in P$ such that $S(u) = x$.*

Professional Standard Proof.

Proof. TODO □

Detailed Learning Proof.

TODO: Expand the professional proof into a detailed learning proof.

Remark (Proof structure). This proof is intended to apply the dichotomy theorem and rule out the case $x = 1$ by the hypothesis $x \neq 1$.

Remark (Dependencies).

- Peano System
- Every Element Is Either 1 or a Successor

Remark (Return). Return to Corollary

Corollary (Predecessor Existence and Uniqueness Away from 1). *Let $(P, S, 1)$ be a Peano system, and let $x \in P$ with $x \neq 1$. Then there exists a unique $u \in P$ such that $S(u) = x$.*

Proof. **Professional Standard Proof.**
By Every Element Is Either 1 or a Successor, either $x = 1$ or there exists $u \in P$ such that $S(u) = x$. Since $x \neq 1$, the first alternative is impossible. Hence there exists $u \in P$ such that $S(u) = x$.

To prove uniqueness, let $v \in P$ and suppose $S(v) = x$. Since also $S(u) = x$, we have

$$S(v) = S(u).$$

By injectivity of the successor map, $v = u$. Therefore there exists a unique $u \in P$ such that $S(u) = x$. □

Proof. **Detailed Learning Proof.**
We first prove existence. Since $x \in P$, the dichotomy theorem applies to x. Thus either

$$x = 1$$

or there exists some $u \in P$ such that

$$S(u) = x.$$

The hypothesis says $x \neq 1$, so the first case cannot occur. Therefore the second case must occur, and a predecessor $u \in P$ of x exists.

Now we prove uniqueness. Fix one such $u \in P$ with $S(u) = x$. Let $v \in P$ be any other element satisfying

$$S(v) = x.$$

Then

$$S(v) = x = S(u),$$

so

$$S(v) = S(u).$$

The Peano successor map is injective, so $v = u$. Thus every predecessor of x is equal to the one already found. Hence the predecessor exists and is unique. □

Remark (Proof structure). Existence comes from the dichotomy theorem after ruling out the case $x = 1$. Uniqueness is a direct application of injectivity of the successor map: two elements with the same successor must be equal.

Remark (Dependencies).

- Peano System
- Every Element Is Either 1 or a Successor
- Injectivity of Successor

Remark (Return). Return to Corollary

Corollary (Unique Predecessor Characterization Away from 1). *Let $(P, S, 1)$ be a Peano system. For every $x \in P$, $x \neq 1$ if and only if there exists a unique $u \in P$ such that $S(u) = x$.*

Professional Standard Proof.

Proof. TODO □

Detailed Learning Proof.
TODO: Expand the professional proof into a detailed learning proof.

Remark (Proof structure). This proof is intended to prove both directions of the biconditional: use predecessor existence and uniqueness away from 1 for the forward direction, and use the axiom that 1 is not a successor for the reverse direction.

Remark (Dependencies).

- Peano System
- Distinguished Element Is Not a Successor
- Predecessor Existence and Uniqueness Away from 1

Remark (Return). Return to Corollary

Corollary (The Distinguished Element Is the Unique Non-Successor). *Let* $(P, S, 1)$ *be a Peano system. An element* $x \in P$ *is not a successor of any element of* P *if and only if* $x = 1$.

Proof. **Professional Standard Proof.**
Assume first that $x \in P$ is not a successor of any element of P. By Every Element Is Either 1 or a Successor, either $x = 1$ or there exists $u \in P$ such that $S(u) = x$. The second alternative contradicts the assumption that x is not a successor. Hence $x = 1$.

Conversely, suppose $x = 1$. By the Peano axiom that 1 is not a successor, there is no $u \in P$ such that $S(u) = 1$. Since $x = 1$, there is no $u \in P$ such that $S(u) = x$. Therefore x is not a successor of any element of P. □

Proof. **Detailed Learning Proof.**
We prove both implications.

First suppose x is not a successor of any element of P. Since $x \in P$, the dichotomy theorem says that x has one of two forms:

$$x = 1$$

or

$$\exists u \in P\ (S(u) = x).$$

The second possibility says exactly that x is a successor of an element of P, which is excluded by hypothesis. Therefore the only remaining possibility is $x = 1$.

Conversely suppose $x = 1$. The Peano axiom Distinguished Element Is Not a Successor says that no element of P has successor equal to 1. Hence no element of P has successor equal to x. Therefore x is not a successor of any element of P. □

Remark (Proof structure). The forward implication uses the dichotomy theorem and eliminates the successor case. The reverse implication is exactly the Peano axiom that the distinguished element is not a successor.

Remark (Dependencies).

- Distinguished Element Is Not a Successor
- Every Element Is Either 1 or a Successor

Remark (Return). Return to Theorem

Theorem (No Object Is Its Own Successor). *Let $(P, S, 1)$ be a Peano system. For every $n \in P$,*

$$S(n) \neq n.$$

Proof. **Professional Standard Proof.**
Let

$$D := \{x \in P : S(x) \neq x\}.$$

We prove that $D = P$ by induction.

First, $1 \in D$. Indeed, if $S(1) = 1$, then 1 would be the successor of the element $1 \in P$, contradicting the Peano axiom that 1 is not a successor. Thus $S(1) \neq 1$.

Now assume $n \in D$. Then $S(n) \neq n$. If $S(S(n)) = S(n)$, injectivity of S would imply $S(n) = n$, contradicting $n \in D$. Therefore $S(S(n)) \neq S(n)$, so $S(n) \in D$.

Thus D contains 1 and is closed under successor. By the induction principle, $D = P$. Hence for every $n \in P$, $S(n) \neq n$. $\square$

Proof. **Detailed Learning Proof.**
Define

$$D := \{x \in P : S(x) \neq x\}.$$

The goal is to prove that every element of P lies in D.

For the base case, we show $1 \in D$. This means showing

$$S(1) \neq 1.$$

If $S(1) = 1$, then 1 would be the successor of an element of P, namely 1 itself. This contradicts the Peano axiom that the distinguished element 1 is not a successor of any element of P. Therefore $S(1) \neq 1$, and so $1 \in D$.

For the inductive step, assume $n \in D$. Then

$$S(n) \neq n.$$

We must show that $S(n) \in D$, which means

$$S(S(n)) \neq S(n).$$

Suppose instead that $S(S(n)) = S(n)$. Since the successor map is injective, this equality of successors would imply

$$S(n) = n,$$

contradicting the inductive hypothesis. Hence $S(S(n)) \neq S(n)$, so $S(n) \in D$.

Therefore D is inductive. By the induction principle for Peano systems, $D = P$. Thus every $n \in P$ satisfies $S(n) \neq n$. $\square$

Remark (Proof structure). The proof uses induction on the subset of elements that are not equal to their own successors. The base case uses the axiom that 1 is not a successor. The successor step uses the contrapositive form of injectivity from $S(n) \neq n$, one obtains $S(S(n)) \neq S(n)$.

Remark (Dependencies).

- Distinguished Element Is Not a Successor
- Injectivity of Successor
- Induction Principle for a Peano System

Remark (Return). Return to Lemma

Remark (Proof vault). Handwritten proof record

Theorem (Consistency of the Minimal Iterator Relation). *Let $(P, S, 1)$ be a Peano system, let (W, c, g) be iterator data for it, and let R^* be the minimal iterator relation for (W, c, g). Then R^* is an iterator relation for (W, c, g).*

Proof. **Professional Standard Proof.**
Let

$$\mathcal{R} = \{R \subseteq P \times W : R \text{ is an iterator relation for } (W, c, g)\}.$$

By definition,

$$R^* = \bigcap \mathcal{R}.$$

We verify the three clauses in the definition of an iterator relation.

First, since every member of $\mathcal{R}$ is a subset of $P \times W$, its intersection is also a subset of $P \times W$. Hence

$$R^* \subseteq P \times W.$$

Second, let $R \in \mathcal{R}$. Because R is an iterator relation, $(1, c) \in R$. Since this holds for every $R \in \mathcal{R}$, the pair $(1, c)$ belongs to the intersection. Thus

$$(1, c) \in R^*.$$

Third, suppose $n \in P$, $w \in W$, and $(n, w) \in R^*$. Let $R \in \mathcal{R}$. Then $(n, w) \in R$, because R^* is the intersection of all relations in $\mathcal{R}$. Since R is an iterator relation, it is closed under the iterator step, so

$$(S(n), g(w)) \in R.$$

This holds for every $R \in \mathcal{R}$, and therefore

$$(S(n), g(w)) \in R^*.$$

We have shown that $R^* \subseteq P \times W$, that $(1, c) \in R^*$, and that R^* is closed under the iterator step. Hence R^* is an iterator relation for (W, c, g). □

Proof. **Detailed Learning Proof.**

Step 1. Identify the family being intersected. Let

$$\mathcal{R} = \{R \subseteq P \times W : R \text{ is an iterator relation for } (W, c, g)\}.$$

By definition, the minimal iterator relation is

$$R^* = \bigcap \mathcal{R}.$$

To prove that R^* is itself an iterator relation, each clause in the definition of iterator relation must be checked.

Step 2. Check that R^* is a relation from P to W. Every member of $\mathcal{R}$ is a subset of $P \times W$. An intersection of subsets of $P \times W$ is again a subset of $P \times W$. Therefore

$$R^* \subseteq P \times W.$$

Step 3. Check the base pair. Let $R \in \mathcal{R}$. Since R is an iterator relation for (W, c, g), it contains the base pair $(1, c)$. This is true for every $R \in \mathcal{R}$. Hence $(1, c)$ belongs to the intersection of all such relations, so

$$(1, c) \in R^*.$$

Step 4. Check closure under the iterator step. Suppose $n \in P$, $w \in W$, and $(n, w) \in R^*$. Let $R \in \mathcal{R}$ be arbitrary. Since R^* is the intersection of all relations in $\mathcal{R}$, the membership $(n, w) \in R^*$ implies $(n, w) \in R$.

Because R is an iterator relation, it is closed under the iterator step. Therefore

$$(S(n), g(w)) \in R.$$

This holds for every $R \in \mathcal{R}$. Hence $(S(n), g(w))$ belongs to the intersection of all relations in $\mathcal{R}$, and so

$$(S(n), g(w)) \in R^*.$$

Step 5. Conclude. The relation R^* is a subset of $P \times W$, contains the base pair $(1, c)$, and is closed under the iterator step. These are exactly the clauses in the definition of an iterator relation. Hence R^* is an iterator relation for (W, c, g). □

Remark (Proof structure). The proof uses the closure properties that define the class of iterator relations and verifies that those properties are preserved by intersection. The base-pair and step-closure clauses hold in every iterator relation in the family, so they hold in the intersection that defines R^*. The argument is therefore a direct verification from the definition of the minimal iterator relation.

Remark (Dependencies).

- Peano System
- Iterator Data on a Peano System
- Iterator Relation
- Minimal Iterator Relation

Remark (Return). Return to Lemma

Theorem (Forced Values Are Unique). *Let $(P, S, 1)$ be a Peano system, let (W, c, g) be iterator data for it, and let R^* be the minimal iterator relation. If $(n, w_0) \in R^*$ and $(n, w_1) \in R^*$, then $w_0 = w_1$.*

Remark (Load-bearing warning). The closure check is load-bearing: removing a forced pair must preserve the iterator-relation clauses.

Professional Standard Proof.

Proof. TODO ∎

Detailed Learning Proof.

TODO: Expand the professional proof into a detailed learning proof.

Remark (Proof structure). TODO: Record the proof strategy and major logical moves.

Remark (Dependencies).

- TODO: Add mathematical dependency links.

Remark (Return). Return to Lemma

Theorem (Determinism of the Minimal Iterator Relation). *Let $(P, S, 1)$ be a Peano system, let (W, c, g) be iterator data for it, and let R^* be the minimal iterator relation. For each $n \in P$, there is at most one $w \in W$ such that $(n, w) \in R^*$.*

Professional Standard Proof.

Proof. TODO ☐

Detailed Learning Proof.

TODO: Expand the professional proof into a detailed learning proof.

Remark (Proof structure). TODO: Record the proof strategy and major logical moves.

Remark (Dependencies).

- TODO: Add mathematical dependency links.

Remark (Return). Return to Lemma

Theorem (Completeness of the Minimal Iterator Relation). *Let $(P, S, 1)$ be a Peano system, let (W, c, g) be iterator data for it, and let R^* be the minimal iterator relation. For each $n \in P$, there exists $w \in W$ such that $(n, w) \in R^*$.*

Professional Standard Proof.

Proof. TODO □

Detailed Learning Proof.

TODO: Expand the professional proof into a detailed learning proof.

Remark (Proof structure). TODO: Record the proof strategy and major logical moves.

Remark (Dependencies).

- TODO: Add mathematical dependency links.

Remark (Return). Return to Lemma

Theorem (Uniqueness of Iterator Functions). *Let $(P, S, 1)$ be a Peano system, and let (W, c, g) be iterator data for it. If $f, h : P \to W$ satisfy*

$$f(1) = c, \qquad h(1) = c,$$

and

$$\forall n \in P\ (f(S(n)) = g(f(n))) \qquad \textit{and} \qquad \forall n \in P\ (h(S(n)) = g(h(n))),$$

then $f = h$.

Professional Standard Proof.

Proof. TODO □

Detailed Learning Proof.

TODO: Expand the professional proof into a detailed learning proof.

Remark (Proof structure). TODO: Record the proof strategy and major logical moves.

Remark (Dependencies).

- TODO: Add mathematical dependency links.

Remark (Return). Return to Lemma

Theorem (Existence of an Iterator Function). *Let $(P, S, 1)$ be a Peano system, and let (W, c, g) be iterator data for it. Then there exists a function $f : P \to W$ such that*

$$f(1) = c \qquad \textit{and} \qquad \forall n \in P\ (f(S(n)) = g(f(n))).$$

Professional Standard Proof.

Proof. TODO ∎

Detailed Learning Proof.

TODO: Expand the professional proof into a detailed learning proof.

Remark (Proof structure). TODO: Record the proof strategy and major logical moves.

Remark (Dependencies).

- TODO: Add mathematical dependency links.

Remark (Return). Return to Theorem

Theorem (Peano Iterator Theorem). *Let $(P, S, 1)$ be a Peano system, and let (W, c, g) be iterator data for it. Then there exists a unique function $f : P \to W$ such that*

$$f(1) = c \qquad \textit{and} \qquad \forall n \in P\ (f(S(n)) = g(f(n))).$$

Professional Standard Proof.

Proof. TODO □

Detailed Learning Proof.
TODO: Expand the professional proof into a detailed learning proof.

Remark (Proof structure). TODO: Record the proof strategy and major logical moves.

Remark (Dependencies).

- TODO: Add mathematical dependency links.

Remark (Return). Return to Theorem

Theorem (Iterator Base Clause). *Let $(P, S, 1)$ be a Peano system, let (W, c, g) be iterator data for it, and let f be the iterator-generated function determined by (W, c, g). Then*

$$f(1) = c.$$

Professional Standard Proof.

Proof. TODO ☐

Detailed Learning Proof.

TODO: Expand the professional proof into a detailed learning proof.

Remark (Proof structure). TODO: Record the proof strategy and major logical moves.

Remark (Dependencies).

- TODO: Add mathematical dependency links.

Remark (Return). Return to Theorem

Theorem (Iterator Successor Clause). *Let $(P, S, 1)$ be a Peano system, let (W, c, g) be iterator data for it, and let f be the iterator-generated function determined by (W, c, g). Then*

$$\forall n \in P \; (f(S(n)) = g(f(n))).$$

Professional Standard Proof.

Proof. TODO □

Detailed Learning Proof.

TODO: Expand the professional proof into a detailed learning proof.

Remark (Proof structure). TODO: Record the proof strategy and major logical moves.

Remark (Dependencies).

- TODO: Add mathematical dependency links.

Remark (Return). Return to Corollary

Theorem (Uniqueness of Binary Iterator Operations). *Let $(P, S, 1)$ be a Peano system, let A be a set, let W be a set, and suppose each $a \in A$ determines iterator data (W, c_a, g_a) for $(P, S, 1)$. If $F, G : A \times P \to W$ satisfy the same base and successor clauses for every $a \in A$, then $F = G$.*

Professional Standard Proof.

Proof. TODO □

Detailed Learning Proof.

TODO: Expand the professional proof into a detailed learning proof.

Remark (Proof structure). TODO: Record the proof strategy and major logical moves.

Remark (Dependencies).

- TODO: Add mathematical dependency links.

Remark (Return). Return to Lemma

Theorem (Uniqueness of General Recursive Functions). *Let $(P, S, 1)$ be a Peano system, let W be a set, let $c \in W$, and let $\Phi : P \times W \to W$ be a stage-dependent step rule. If $f, h : P \to W$ satisfy*

$$f(1) = c, \qquad h(1) = c,$$

and

$$\forall n \in P\ (f(S(n)) = \Phi(n, f(n))) \qquad \textit{and} \qquad \forall n \in P\ (h(S(n)) = \Phi(n, h(n))),$$

then $f = h$.

Professional Standard Proof.

Proof. TODO □

Detailed Learning Proof.

TODO: Expand the professional proof into a detailed learning proof.

Remark (Proof structure). TODO: Record the proof strategy and major logical moves.

Remark (Dependencies).

- TODO: Add mathematical dependency links.

Remark (Return). Return to Theorem

Theorem (General Recursion by State Encoding). *Let $(P, S, 1)$ be a Peano system, let W be a set, let $c \in W$, and let $\Phi : P \times W \to W$ be a stage-dependent step rule. There exists a function $H : P \to P \times W$ satisfying the stated state-encoding base and successor clauses.*

Professional Standard Proof.

Proof. TODO ☐

Detailed Learning Proof.

TODO: Expand the professional proof into a detailed learning proof.

Remark (Proof structure). TODO: Record the proof strategy and major logical moves.

Remark (Dependencies).

- TODO: Add mathematical dependency links.

Remark (Return). Return to Theorem

Theorem (General Recursion Theorem). *Let $(P, S, 1)$ be a Peano system, let W be a set, let $c \in W$, and let $\Phi : P \times W \to W$ be a stage-dependent step rule. Then there exists a unique function $f : P \to W$ such that*

$$f(1) = c \qquad \textit{and} \qquad \forall n \in P\ (f(S(n)) = \Phi(n, f(n))).$$

Professional Standard Proof.

Proof. TODO □

Detailed Learning Proof.

TODO: Expand the professional proof into a detailed learning proof.

Remark (Proof structure). TODO: Record the proof strategy and major logical moves.

Remark (Dependencies).

- TODO: Add mathematical dependency links.

Remark (Return). Return to Theorem

Theorem (Uniqueness of Peano Systems up to Isomorphism). *Let* $(P, S_P, 1_P)$ *and* $(Q, S_Q, 1_Q)$ *be Peano systems. Then there exists a unique bijection* $\theta : P \to Q$ *such that*

$$\theta(1_P) = 1_Q$$

and

$$\forall n \in P \ (\theta(S_P(n)) = S_Q(\theta(n))).$$

Thus any two Peano systems are isomorphic.

Proof. TODO $\square$

Capstone

Chapter 2

Natural Numbers ($\mathbb{N}$)

natural-numbers

Axiom Systems $\to$ Peano Systems $\to$ **Natural Numbers ($\mathbb{N}$)** $\to$ Whole Numbers ($\mathbb{W}$) $\to$ Integers ($\mathbb{Z}$)

Structural Roadmap

How we got here. The preceding chapter established Peano systems, induction, recursion, and categoricity. This chapter begins from that foundation and works in the canonical natural-number system.

What this chapter develops. The chapter defines addition and multiplication by recursion, proves the corresponding algebraic and order laws, and then records the resulting structural form of $\mathbb{N}$.

Where this leads. The natural numbers support counting and recursive arithmetic, but they do not contain an additive identity. That limitation first leads to the construction of the whole numbers, and then to the construction of the integers.

2.1 The Canonical Natural-Number System

Working in $\mathbb{N}$

The preceding chapter established the Peano-system facts needed for arithmetic: induction, successor analysis, recursion, and categoricity. From this point forward, $\mathbb{N}$ denotes the canonical natural-number Peano system, with distinguished first element 1 and successor map S. The arithmetic operations introduced in this chapter are defined inside that system by the recursion principles already proved for Peano systems.

Remark (Standing convention). Unless otherwise stated, variables m, n, k range over $\mathbb{N}$, the symbol 1 denotes the distinguished first natural number, and $S(n)$ denotes the successor of n.

2.2 Addition on $\mathbb{N}$

Recursive Definition of Addition

Addition on a Peano system is defined from the successor primitive by recursion. For each fixed left input m, the map $n \mapsto m + n$ is generated from the initial value $S(m)$ and the successor operator on the same Peano system. This definition makes addition a derived operation rather than an independent axiom.

Definition (Addition on a Peano System)

Definition 2.2.1 (Addition on a Peano System). Let $(P, S, 1)$ be a Peano system. For each $m \in P$, define the function

$$f_m : P \to P$$

to be the unique function satisfying

$$f_m(1) = S(m) \qquad \text{and} \qquad \forall n \in P\ (f_m(S(n)) = S(f_m(n))).$$

For $m, n \in P$, define

$$m + n := f_m(n).$$

Remark (Standard quantified statement).

Let $(P, S, 1)$ be a Peano system. For each $m \in P$, let $f_m : P \to P$ satisfy
$f_m(1) = S(m) \qquad \text{and} \qquad \forall n \in P\ (f_m(S(n)) = S(f_m(n)))$.
Then for all $m, n \in P$, $\quad m + n := f_m(n)$.

Remark (Negated quantified statement).

Let $(P, S, 1)$ be a Peano system. For each $m \in P$, let $f_m : P \to P$ satisfy
$f_m(1) = S(m) \qquad \text{and} \qquad \forall n \in P\ (f_m(S(n)) = S(f_m(n)))$.
Then for all $m, n, x \in P$, $\quad x \neq m + n \iff x \neq f_m(n)$.

Remark (Failure modes). The definition fails only when the designated value does not agree with the unique recursively generated function attached to the fixed left input m. Once the recursive map f_m is fixed, every incorrect candidate for $m + n$ is simply a value of P different from $f_m(n)$.

Remark (Failure mode decomposition).

Let $(P, S, 1)$ be a Peano system. For each $m \in P$, let $f_m : P \to P$ satisfy
$f_m(1) = S(m) \qquad \text{and} \qquad \forall n \in P\ (f_m(S(n)) = S(f_m(n)))$.

$$x \neq m + n \iff \underbrace{x \neq f_m(n)}_{\text{candidate value disagrees with the recursive output}}.$$

Remark (Interpretation). Addition is introduced by fixing the left input and recursively generating the right-input map. The value $m + 1$ is set to be the successor of m, and each further successor step in the right input advances the output by one more successor. This makes addition repeated succession.

Remark (Dependencies).

- Peano System
- Peano Iterator Theorem

Theorem (Addition Is Well-Defined on a Peano System)

Theorem 2.2.2 (Addition Is Well-Defined on a Peano System). *Let $(P, S, 1)$ be a Peano system. For every $m \in P$, there exists a unique function*

$$f_m : P \to P$$

such that

$$f_m(1) = S(m) \qquad \textit{and} \qquad \forall n \in P \ (f_m(S(n)) = S(f_m(n))).$$

Consequently the rule

$$(m, n) \longmapsto m + n := f_m(n)$$

defines a unique binary operation on P. *Go to proof.*

Remark (Standard quantified statement).

> Let $(P, S, 1)$ be a Peano system.
> $$\forall m \in P \ \exists! f_m : P \to P \ \Big(f_m(1) = S(m) \land \forall n \in P \ (f_m(S(n)) = S(f_m(n)))\Big).$$

Remark (Interpretation). The theorem shows that the recursive clauses for addition determine exactly one right-input map for each fixed left input. The addition operation is therefore not assumed; it is constructed from the successor map by recursion.

Remark (Dependencies).

- Addition on a Peano System
- Peano Iterator Theorem

Recursive Clauses as Reusable Equalities

The recursive definition of addition is given by the base value and successor step for the right-input map $n \mapsto m + n$. Those clauses are part of the definition, but they become substantially more useful once isolated as named theorems. Later algebraic arguments repeatedly cite these equalities rather than reopening the recursive construction each time.

Theorem (Addition With 1)

Theorem 2.2.3 (Addition With 1). *Let $(P, S, 1)$ be a Peano system. For every $m \in P$,*

$$m + 1 = S(m).$$

Go to proof.

Remark (Standard quantified statement).

Let $(P, S, 1)$ be a Peano system.
$\forall m \in P\ (m + 1 = S(m)).$

Remark (Interpretation). This theorem records the base clause of the recursive definition in the operation notation itself. Adding the distinguished first element applies one successor step to the left input.

Remark (Dependencies).

- Addition on a Peano System

Theorem (Addition Successor on the Right)

Theorem 2.2.4 (Addition Successor on the Right). *Let $(P, S, 1)$ be a Peano system. For every $m, n \in P$,*

$$m + S(n) = S(m + n).$$

Go to proof.

Remark (Standard quantified statement).

Let $(P, S, 1)$ be a Peano system.
$\forall m, n \in P\ (m + S(n) = S(m + n)).$

Remark (Interpretation). This theorem records the recursive step of addition in operation notation. Moving one successor step forward in the right input moves the sum one successor step forward as well.

Remark (Dependencies).

- Addition on a Peano System

Left Input Initialization

The recursive definition controls addition when the right input is advanced by successor. To use addition symmetrically, the first structural task is to understand what happens when the distinguished first element appears on the left. The next theorem shows that left addition by 1 also produces a single successor step.

Theorem (One Plus n)

Theorem 2.2.5 (One Plus n). *Let $(P, S, 1)$ be a Peano system. For every $n \in P$,*

$$1 + n = S(n).$$

Go to proof.

Remark (Standard quantified statement).

Let $(P, S, 1)$ be a Peano system.
$\forall n \in P\ (1 + n = S(n))$.

Remark (Interpretation). The theorem shows that the distinguished first element acts on either side as a single successor step. This is the first sign that the recursively defined operation is not tied permanently to the asymmetry of its definition.

Remark (Dependencies).

- Addition With 1
- Addition Successor on the Right
- Induction Principle for a Peano System

Associativity and Commutativity

Once the recursive clauses are available as reusable theorems, addition can be shown to satisfy the two fundamental algebraic symmetries of a binary operation. Associativity makes repeated addition unambiguous, and commutativity shows that the recursively defined operation does not depend on which input is treated as the evolving one.

Theorem (Addition Is Associative)

Theorem 2.2.6 (Addition Is Associative). *Let $(P, S, 1)$ be a Peano system. For every $a, b, c \in P$,*

$$(a + b) + c = a + (b + c).$$

Go to proof.

Remark (Standard quantified statement).

Let $(P, S, 1)$ be a Peano system.
$\forall a, b, c \in P\ ((a + b) + c = a + (b + c))$.

Remark (Interpretation). Associativity shows that successive additions can be regrouped without changing the value. This makes longer sums structurally coherent and prepares addition for later interaction with order and multiplication.

Remark (Dependencies).

- Addition With 1
- Addition Successor on the Right
- Induction Principle for a Peano System

Theorem (Addition Is Commutative)

Theorem 2.2.7 (Addition Is Commutative). *Let $(P, S, 1)$ be a Peano system. For every $a, b \in P$,*

$$a + b = b + a.$$

Go to proof.

Remark (Standard quantified statement).

Let $(P, S, 1)$ be a Peano system.
$\forall a, b \in P\ (a + b = b + a)$.

Remark (Interpretation). Although addition is defined by recursion on the right input, the resulting operation is symmetric in its two arguments. Commutativity shows that the apparent asymmetry of the defining clauses is an artifact of construction rather than a feature of the operation itself.

Remark (Dependencies).

- Addition With 1
- Addition Successor on the Right
- One Plus n
- Addition Is Associative
- Induction Principle for a Peano System

Cancellation Laws

The next structural property shows that addition preserves enough information to allow one to remove a common summand from an equation. These cancellation laws are the first substitute for subtraction inside a Peano system and are indispensable when order is later defined through additive comparison.

Theorem (Left Cancellation for Addition)

Theorem 2.2.8 (Left Cancellation for Addition). *Let $(P, S, 1)$ be a Peano system. For every $a, b, c \in P$,*

$$a + b = a + c \implies b = c.$$

Go to proof.

Remark (Standard quantified statement).

Let $(P, S, 1)$ be a Peano system.
$$\forall a, b, c \in P \ \big((a + b = a + c) \implies b = c\big).$$

Remark (Contrapositive quantified statement).

Let $(P, S, 1)$ be a Peano system.
$$\forall a, b, c \in P \ \big(b \neq c \implies a + b \neq a + c\big).$$

Remark (Interpretation). Left cancellation shows that adding the same element to two inputs cannot erase a difference between them. This is the algebraic core that later makes additive definitions of order behave well.

Remark (Dependencies).

- Injectivity of Successor
- Addition With 1
- Addition Successor on the Right
- One Plus n
- Addition Is Associative
- Addition Is Commutative
- Induction Principle for a Peano System

Theorem (Right Cancellation for Addition)

Theorem 2.2.9 (Right Cancellation for Addition). *Let $(P, S, 1)$ be a Peano system. For every $a, b, c \in P$,*

$$b + a = c + a \implies b = c.$$

Go to proof.

Remark (Standard quantified statement).

Let $(P, S, 1)$ be a Peano system.
$$\forall a, b, c \in P \ \big((b + a = c + a) \implies b = c\big).$$

Remark (Contrapositive quantified statement).

$$\text{Let } (P, S, 1) \text{ be a Peano system.}$$
$$\forall a, b, c \in P \; \big(b \neq c \Longrightarrow b + a \neq c + a\big).$$

Remark (Interpretation). Right cancellation is the companion to left cancellation. Once commutativity is known, the placement of the common summand no longer matters, so equality can be tested after adding the same element on either side.

Remark (Dependencies).

- Addition Is Commutative
- Left Cancellation for Addition

Additive Non-Return

Addition on $\mathbb{N}$ always moves forward by at least one successor step in its right input. The next theorem records the resulting non-return law: adding a natural number to m never returns the added natural number itself.

Theorem (No Natural Number Equals Itself Plus a Natural Number)

Theorem 2.2.10 (No Natural Number Equals Itself Plus a Natural Number). *Let* $(P, S, 1)$ *be a Peano system. For every* $m, n \in P$,

$$n \neq m + n.$$

Go to proof.

Remark (Standard quantified statement).

$$\text{Let } (P, S, 1) \text{ be a Peano system.}$$
$$\forall m, n \in P \; (n \neq m + n).$$

Remark (Interpretation). Because every element of P is at least one successor step, the sum $m + n$ lies strictly beyond n. The theorem rules out additive cycles and is a standard tool for order arguments on the natural numbers.

Remark (Source alignment). This is the house-style form of the additive non-return theorem in [Feferman, 1964].

Remark (Dependencies).

- Addition With 1
- Addition Successor on the Right
- No Object Is Its Own Successor
- Induction Principle for a Peano System

2.3 Multiplication on $\mathbb{N}$

Recursive Definition of Multiplication

Multiplication on a Peano system is defined from addition by recursion. For each fixed left input m, the map $n \mapsto m \cdot n$ is generated by starting at m and adding one further copy of m at each successor stage of the right input. This makes multiplication repeated addition rather than an independent primitive operation.

Definition (Multiplication on a Peano System)

Definition 2.3.1 (Multiplication on a Peano System). Let $(P, S, 1)$ be a Peano system. For each $m \in P$, define the function

$$g_m : P \to P$$

to be the unique function satisfying

$$g_m(1) = m \qquad \text{and} \qquad \forall n \in P\ (g_m(S(n)) = g_m(n) + m).$$

For $m, n \in P$, define

$$m \cdot n := g_m(n).$$

Remark (Standard quantified statement).

Let $(P, S, 1)$ be a Peano system. For each $m \in P$, let $g_m : P \to P$ satisfy
$g_m(1) = m \quad$ and $\quad \forall n \in P\ (g_m(S(n)) = g_m(n) + m)$.
Then for all $m, n \in P$, $\quad m \cdot n := g_m(n)$.

Remark (Negated quantified statement).

Let $(P, S, 1)$ be a Peano system. For each $m \in P$, let $g_m : P \to P$ satisfy
$g_m(1) = m \quad$ and $\quad \forall n \in P\ (g_m(S(n)) = g_m(n) + m)$.
Then for all $m, n, x \in P$, $\quad x \neq m \cdot n \iff x \neq g_m(n)$.

Remark (Failure modes). The definition fails only when a proposed value for $m \cdot n$ does not agree with the recursively generated function attached to the fixed left factor m. Once g_m is determined, every incorrect candidate product is simply a value different from $g_m(n)$.

Remark (Failure mode decomposition).

Let $(P, S, 1)$ be a Peano system. For each $m \in P$, let $g_m : P \to P$ satisfy
$g_m(1) = m \quad$ and $\quad \forall n \in P\ (g_m(S(n)) = g_m(n) + m)$.

$$x \neq m \cdot n \iff \underbrace{x \neq g_m(n)}_{\text{candidate value disagrees with the recursive output}}.$$

Remark (Interpretation). Multiplication is introduced by fixing the left input and recursively generating the right-input map. The value $m \cdot 1$ is initialized at m, and each successor step in the right input adds one further copy of m. Multiplication is therefore repeated addition.

Remark (Dependencies).

- Peano System
- General Recursion Theorem for a Peano System
- Addition on a Peano System

Theorem (Multiplication Is Well-Defined on a Peano System)

Theorem 2.3.2 (Multiplication Is Well-Defined on a Peano System)**.** *Let $(P, S, 1)$ be a Peano system. For every $m \in P$, there exists a unique function*

$$g_m : P \to P$$

such that

$$g_m(1) = m \qquad \text{and} \qquad \forall n \in P\ (g_m(S(n)) = g_m(n) + m).$$

Consequently the rule

$$(m, n) \longmapsto m \cdot n := g_m(n)$$

defines a unique binary operation on P. *Go to proof.*

Remark (Standard quantified statement).

Let $(P, S, 1)$ be a Peano system.
$$\forall m \in P\ \exists! g_m : P \to P\ \Big(g_m(1) = m \wedge \forall n \in P\ (g_m(S(n)) = g_m(n) + m)\Big).$$

Remark (Interpretation). The recursive clauses for multiplication determine exactly one right-input map for each fixed left input. The multiplication operation is therefore constructed from recursion and addition rather than assumed at the outset.

Remark (Dependencies).

- Multiplication on a Peano System
- General Recursion Theorem for a Peano System
- Addition on a Peano System

Recursive Clauses as Reusable Equalities

The recursive definition of multiplication is given by a base value and a successor step. As with addition, these clauses become more useful once they are isolated as named theorems that can be cited directly in later algebraic arguments.

Theorem (Multiplication With 1)

Theorem 2.3.3 (Multiplication With 1). *Let $(P, S, 1)$ be a Peano system. For every $m \in P$,*

$$m \cdot 1 = m.$$

Go to proof.

Remark (Standard quantified statement).

$$\text{Let } (P, S, 1) \text{ be a Peano system.}$$
$$\forall m \in P\ (m \cdot 1 = m).$$

Remark (Interpretation). This theorem records the base clause of the recursive definition in operation notation. Multiplying by the distinguished first element leaves the left input unchanged.

Remark (Dependencies).

- Multiplication on a Peano System

Theorem (Multiplication Successor on the Right)

Theorem 2.3.4 (Multiplication Successor on the Right). *Let $(P, S, 1)$ be a Peano system. For every $m, n \in P$,*

$$m \cdot S(n) = (m \cdot n) + m.$$

Go to proof.

Remark (Standard quantified statement).

$$\text{Let } (P, S, 1) \text{ be a Peano system.}$$
$$\forall m, n \in P\ (m \cdot S(n) = (m \cdot n) + m).$$

Remark (Interpretation). Moving one successor step forward in the right input adds one further copy of m to the product. This is the recursive step of multiplication written in operation notation.

Remark (Dependencies).

- Multiplication on a Peano System
- Addition on a Peano System

Left Identity

The recursive definition controls multiplication when the right input evolves. To use multiplication symmetrically, the first structural task is to understand what happens when the distinguished first element appears on the left. The next theorem identifies that left action explicitly.

Theorem (One Times n)

Theorem 2.3.5 (One Times n). *Let $(P, S, 1)$ be a Peano system. For every $n \in P$,*

$$1 \cdot n = n.$$

Go to proof.

Remark (Standard quantified statement).

Let $(P, S, 1)$ be a Peano system.
$\forall n \in P \ (1 \cdot n = n)$.

Remark (Interpretation). The distinguished first element acts as a multiplicative identity on the left. This is the multiplicative analogue of the theorem that $1 + n$ produces the successor of n in the additive theory.

Remark (Dependencies).

- Multiplication With 1
- Multiplication Successor on the Right
- Induction Principle for a Peano System

Distributivity and Associativity

Multiplication is built from repeated addition, so its first major structural law is distributivity over addition. Once distributivity is available, the operation can be shown to satisfy associativity, which makes iterated products well-defined without ambiguity of grouping.

Theorem (Multiplication Distributes Over Addition)

Theorem 2.3.6 (Multiplication Distributes Over Addition). *Let $(P, S, 1)$ be a Peano system. For every $a, b, c \in P$,*

$$a \cdot (b + c) = (a \cdot b) + (a \cdot c).$$

Go to proof.

Remark (Standard quantified statement).

Let $(P, S, 1)$ be a Peano system.
$\forall a, b, c \in P \ (a \cdot (b + c) = (a \cdot b) + (a \cdot c))$.

Remark (Interpretation). Distributivity expresses formally that multiplying by a means repeatedly adding copies of a, so a sum in the second factor splits into two separate blocks of repeated addition. This is the bridge between the additive and multiplicative structures.

Remark (Dependencies).

- Multiplication Successor on the Right
- Addition Is Associative
- Addition Is Commutative
- Induction Principle for a Peano System

Theorem (Left Distributivity of Multiplication Over Addition)

Theorem 2.3.7 (Left Distributivity of Multiplication Over Addition). *Let $(P, S, 1)$ be a Peano system. For every $a, b, c \in P$,*

$$(a+b) \cdot c = (a \cdot c) + (b \cdot c).$$

Go to proof.

Remark (Standard quantified statement).

$$\text{Let } (P, S, 1) \text{ be a Peano system.}$$
$$\forall a, b, c \in P \ \big((a+b) \cdot c = (a \cdot c) + (b \cdot c)\big).$$

Remark (Interpretation). The existing distributive law expands a sum in the right factor. This theorem expands a sum in the left factor. It follows from right distributivity together with commutativity of multiplication.

Remark (Source alignment). This is the house-style form of Feferman's left distributive law for multiplication [Feferman, 1964].

Remark (Dependencies).

- Multiplication Distributes Over Addition
- Multiplication Is Commutative
- Addition Is Commutative

Theorem (Multiplication Is Associative)

Theorem 2.3.8 (Multiplication Is Associative). *Let $(P, S, 1)$ be a Peano system. For every $a, b, c \in P$,*

$$(a \cdot b) \cdot c = a \cdot (b \cdot c).$$

Go to proof.

Remark (Standard quantified statement).

Let $(P, S, 1)$ be a Peano system.
$\forall a, b, c \in P\ ((a \cdot b) \cdot c = a \cdot (b \cdot c))$.

Remark (Interpretation). Associativity shows that repeated multiplication can be regrouped without changing the value. Once this law is established, longer products can be used without parenthetical ambiguity.

Remark (Dependencies).

- Multiplication Distributes Over Addition
- Multiplication With 1
- Multiplication Successor on the Right
- Induction Principle for a Peano System

Commutativity

The recursive definition of multiplication treats the right input as the evolving variable, but the resulting operation is not meant to privilege one factor over the other. The next theorem shows that the asymmetry of the construction disappears in the final operation.

Theorem (Multiplication Is Commutative)

Theorem 2.3.9 (Multiplication Is Commutative). *Let $(P, S, 1)$ be a Peano system. For every $a, b \in P$,*

$$a \cdot b = b \cdot a.$$

Go to proof.

Remark (Standard quantified statement).

Let $(P, S, 1)$ be a Peano system.
$\forall a, b \in P\ (a \cdot b = b \cdot a)$.

Remark (Interpretation). Although multiplication is defined recursively in one argument, the resulting operation is symmetric in the two factors. Commutativity shows that repeated addition of a to build $a \cdot b$ agrees exactly with repeated addition of b to build $b \cdot a$.

Remark (Dependencies).

- One Times n
- Multiplication Successor on the Right
- Multiplication Distributes Over Addition

- Addition Is Commutative
- Induction Principle for a Peano System

2.4 Order on $\mathbb{N}$

Strict and Non-Strict Order

Order on a Peano system is extracted from addition. Since the chapter is using the distinguished first element 1 rather than an additive zero, the strict order is the cleaner primitive notion: $a < b$ means that b is reached from a by adding a nontrivial amount. The non-strict order is then obtained by allowing equality in addition to strict increase.

Definition (Strict Less Than on a Peano System)

Definition 2.4.1 (Strict Less Than on a Peano System). Let $(P, S, 1)$ be a Peano system, and let $a, b \in P$. One writes

$$a < b$$

exactly when there exists $c \in P$ such that

$$b = a + c.$$

Remark (Standard quantified statement).

$$\text{Let } (P, S, 1) \text{ be a Peano system and let } a, b \in P.$$
$$a < b \iff \exists c \in P\ (b = a + c).$$

Remark (Negated quantified statement).

$$\text{Let } (P, S, 1) \text{ be a Peano system and let } a, b \in P.$$
$$a \not< b \iff \forall c \in P\ (b \neq a + c).$$

Remark (Failure modes). Strict inequality fails exactly when no additive witness in P carries a to b. In that case, every nontrivial additive step from a misses b.

Remark (Failure mode decomposition).

$$\text{Let } (P, S, 1) \text{ be a Peano system and let } a, b \in P.$$
$$a \not< b \iff \underbrace{\forall c \in P\ (b \neq a + c)}_{\text{every proposed additive witness fails}}.$$

Remark (Interpretation). The strict order records whether b lies genuinely ahead of a in the counting structure. The witness c is the amount of additive displacement needed to move from a to b.

Remark (Dependencies).

- Addition on a Peano System

Definition (Less Than or Equal on a Peano System)

Definition 2.4.2 (Less Than or Equal on a Peano System). Let $(P, S, 1)$ be a Peano system, and let $a, b \in P$. One writes

$$a \leq b$$

exactly when

$$a < b \qquad \text{or} \qquad a = b.$$

Remark (Standard quantified statement).

$$\text{Let } (P, S, 1) \text{ be a Peano system and let } a, b \in P.$$
$$a \leq b \iff \big(a < b \lor a = b\big).$$

Remark (Negated quantified statement).

$$\text{Let } (P, S, 1) \text{ be a Peano system and let } a, b \in P.$$
$$a \not\leq b \iff \big(a \not< b \land a \neq b\big).$$

Remark (Failure modes). The non-strict order fails in exactly two simultaneous ways: a is not strictly below b, and a is not equal to b. Thus b neither lies ahead of a nor coincides with it.

Remark (Failure mode decomposition).

$$\text{Let } (P, S, 1) \text{ be a Peano system and let } a, b \in P.$$
$$a \not\leq b \iff \underbrace{a \not< b}_{\text{no strict additive witness}} \land \underbrace{a \neq b}_{\text{equality also fails}}.$$

Remark (Interpretation). The relation $a \leq b$ allows either equality or a genuine forward step from a to b. It is the additive order relation derived from the strict witness definition.

Remark (Dependencies).

- Strict Less Than on a Peano System

Basic Order Laws

Once the order relations have been defined, the first task is to show that they satisfy the expected structural laws: reflexivity, transitivity, and antisymmetry. These are the properties that make the additive comparison relation behave as an honest order rather than a merely suggestive notation.

Theorem (Order Is Reflexive on a Peano System)

Theorem 2.4.3 (Order Is Reflexive on a Peano System). *Let $(P, S, 1)$ be a Peano system. For every $a \in P$,*

$$a \leq a.$$

Go to proof.

Remark (Standard quantified statement).

Let $(P, S, 1)$ be a Peano system.
$\forall a \in P\ (a \leq a)$.

Remark (Interpretation). Every element is comparable to itself. Reflexivity supplies the equality case inside the non-strict order.

Remark (Dependencies).

- Less Than or Equal on a Peano System

Theorem (Order Is Transitive on a Peano System)

Theorem 2.4.4 (Order Is Transitive on a Peano System). *Let $(P, S, 1)$ be a Peano system. For every $a, b, c \in P$,*

$$\big(a \leq b \wedge b \leq c\big) \implies a \leq c.$$

Go to proof.

Remark (Standard quantified statement).

Let $(P, S, 1)$ be a Peano system.
$\forall a, b, c \in P\ \big((a \leq b \wedge b \leq c) \implies a \leq c\big)$.

Remark (Interpretation). If b lies at or beyond a, and c lies at or beyond b, then c also lies at or beyond a. Transitivity expresses the consistency of additive comparison along a chain.

Remark (Dependencies).

- Strict Less Than on a Peano System
- Less Than or Equal on a Peano System
- Addition Is Associative

Theorem (Order Is Antisymmetric on a Peano System)

Theorem 2.4.5 (Order Is Antisymmetric on a Peano System). *Let $(P, S, 1)$ be a Peano system. For every $a, b \in P$,*

$$\big(a \leq b \wedge b \leq a\big) \implies a = b.$$

Go to proof.

Remark (Standard quantified statement).

$$\text{Let } (P,S,1) \text{ be a Peano system.}$$
$$\forall a,b \in P\ \big((a \le b \land b \le a) \implies a = b\big).$$

Remark (Contrapositive quantified statement).

$$\text{Let } (P,S,1) \text{ be a Peano system.}$$
$$\forall a,b \in P\ \big(a \ne b \implies \neg(a \le b \land b \le a)\big).$$

Remark (Interpretation). Two distinct elements cannot each lie at or beyond the other. Antisymmetry separates genuine equality from bidirectional comparability.

Remark (Dependencies).

- Strict Less Than on a Peano System
- Less Than or Equal on a Peano System
- Left Cancellation for Addition
- Addition Is Associative
- No Object Is Its Own Successor

Compatibility with Addition

The order on a Peano system is built from addition, so it must interact coherently with addition. The next theorems show that adding the same element to both sides preserves the comparison relation, regardless of whether the common summand is placed on the right or on the left.

Theorem (Addition Preserves Order on the Right)

Theorem 2.4.6 (Addition Preserves Order on the Right). *Let $(P,S,1)$ be a Peano system. For every $a,b,c \in P$,*

$$a \le b \implies a + c \le b + c.$$

Go to proof.

Remark (Standard quantified statement).

$$\text{Let } (P,S,1) \text{ be a Peano system.}$$
$$\forall a,b,c \in P\ \big(a \le b \implies a + c \le b + c\big).$$

Remark (Interpretation). Adding the same quantity to two comparable elements does not disturb their order. The additive witness for $a \le b$ survives after both sides are shifted by the same summand.

Remark (Dependencies).

- Strict Less Than on a Peano System
- Less Than or Equal on a Peano System
- Addition Is Associative

Theorem (Addition Preserves Order on the Left)

Theorem 2.4.7 (Addition Preserves Order on the Left). *Let $(P, S, 1)$ be a Peano system. For every $a, b, c \in P$,*

$$a \leq b \implies c + a \leq c + b.$$

Go to proof.

Remark (Standard quantified statement).

$$\begin{gathered}\text{Let } (P,S,1) \text{ be a Peano system.}\\ \forall a,b,c \in P\ \big(a \leq b \implies c + a \leq c + b\big).\end{gathered}$$

Remark (Interpretation). The same order preservation holds when the common summand is added on the left. This is the left-handed form of the additive compatibility law.

Remark (Dependencies).

- Addition Preserves Order on the Right
- Addition Is Commutative

Compatibility with Multiplication

Multiplication by a natural number is repeated addition by a positive amount. It therefore preserves strict comparisons, and because every multiplier lies at or beyond 1, it also reflects strict comparisons.

Theorem (Multiplication Preserves and Reflects Strict Order)

Theorem 2.4.8 (Multiplication Preserves and Reflects Strict Order). *Let $(P, S, 1)$ be a Peano system. For every $k, m, n \in P$,*

$$m < n \iff k \cdot m < k \cdot n.$$

Go to proof.

Remark (Standard quantified statement).

$$\begin{gathered}\text{Let } (P,S,1) \text{ be a Peano system.}\\ \forall k,m,n \in P\ \big(m < n \iff k \cdot m < k \cdot n\big).\end{gathered}$$

Remark (Interpretation). Multiplication by a fixed natural number is an order embedding of the natural numbers into themselves. It shifts strict inequalities forward without collapsing distinct values.

Remark (Source alignment). This is the house-style multiplication-order comparison theorem corresponding to Feferman's order laws for products [Feferman, 1964].

Remark (Dependencies).

- Strict Less Than on a Peano System
- Multiplication on a Peano System
- Multiplication Distributes Over Addition
- Multiplication Is Commutative
- Addition Preserves Order on the Right
- Left Cancellation for Addition

Successor Characterization and Trichotomy

Strict inequality can be recast in terms of stepping one successor beyond the smaller element. This successor reformulation sharpens the order relation and leads to trichotomy, which states that any two elements are linearly comparable.

Theorem (Strict Order Successor Characterization)

Theorem 2.4.9 (Strict Order Successor Characterization). *Let $(P, S, 1)$ be a Peano system. For every $a, b \in P$,*

$$a < b \iff S(a) \leq b.$$

Go to proof.

Remark (Standard quantified statement).

$$\text{Let } (P, S, 1) \text{ be a Peano system.}$$
$$\forall a, b \in P \; \big(a < b \iff S(a) \leq b\big).$$

Remark (Interpretation). To say that b is strictly larger than a is to say that b lies at or beyond the first successor step past a. This theorem converts strict order into a non-strict comparison involving successor.

Remark (Dependencies).

- Strict Less Than on a Peano System
- Less Than or Equal on a Peano System

- Addition With 1
- Addition Successor on the Right
- Addition Is Associative
- Addition Is Commutative
- Every Element Is Either 1 or a Successor

Theorem (Trichotomy on a Peano System)

Theorem 2.4.10 (Trichotomy on a Peano System). *Let $(P, S, 1)$ be a Peano system. For every $a, b \in P$, exactly one of the following holds:*

$$a < b, \qquad a = b, \qquad b < a.$$

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } (P, S, 1) \text{ be a Peano system.}\\&\forall a, b \in P \ \Big((a < b \lor a = b \lor b < a) \land\\&\qquad \neg(a < b \land a = b) \land \neg(a < b \land b < a) \land \neg(a = b \land b < a)\Big).\end{aligned}$$

Remark (Interpretation). Any two elements of a Peano system are linearly comparable: one lies below the other, or they coincide. Trichotomy is the theorem that upgrades the additive comparison relation from a partial-order structure to a total order.

Remark (Dependencies).

- Strict Less Than on a Peano System
- Less Than or Equal on a Peano System
- Order Is Reflexive on a Peano System
- Order Is Transitive on a Peano System
- Order Is Antisymmetric on a Peano System
- Strict Order Successor Characterization
- Induction Principle for a Peano System

2.5 Induction

Strong Induction

Ordinary induction propagates a property from each element to its immediate successor. Strong induction allows the inductive step to use the property at every predecessor, not just the immediate one. This broader hypothesis makes strong induction strictly more convenient for arguments in which the successor step requires information from elements earlier than the immediate predecessor.

Theorem (Strong Induction on a Peano System)

Theorem 2.5.1 (Strong Induction on a Peano System). *Let $(P, S, 1)$ be a Peano system, and let φ be a predicate on P. If*

$$\forall n \in P\ \big(\forall k \in P\ (k < n \Longrightarrow \varphi(k))\big) \Longrightarrow \varphi(n),$$

then

$$\forall n \in P\ \varphi(n).$$

Go to proof.

Remark (Standard quantified statement).

Let $(P, S, 1)$ be a Peano system and let φ be a predicate on P.
$\Big(\forall n \in P\ \big(\forall k \in P\ (k < n \Longrightarrow \varphi(k))\big) \Longrightarrow \varphi(n)\Big) \Longrightarrow \forall n \in P\ \varphi(n).$

Remark (Contrapositive quantified statement).

Let $(P, S, 1)$ be a Peano system and let φ be a predicate on P.
$\exists n \in P\ \neg\varphi(n) \Longrightarrow \exists n \in P\ \Big(\big(\forall k \in P\ (k < n \Longrightarrow \varphi(k))\big) \wedge \neg\varphi(n)\Big).$

Remark (Interpretation). Strong induction grants the inductive step access to the entire history below n, not just the single predecessor. The contrapositive gives the least counterexample principle: if any element fails φ, there is a least one, and that least one has φ holding at all its predecessors, which contradicts the strong-induction hypothesis. This makes strong induction and least-counterexample arguments two descriptions of the same logical structure.

Remark (Dependencies).

- Induction Principle for a Peano System
- Strict Less Than on a Peano System
- Trichotomy on a Peano System

Induction from an Arbitrary Base

Ordinary and strong induction each begin at the distinguished first element 1. Many arguments require starting at a later element. Induction from an arbitrary base m restricts the claim to elements at or above m and starts the inductive process there.

Theorem (Induction from an Arbitrary Base)

Theorem 2.5.2 (Induction from an Arbitrary Base). *Let $(P, S, 1)$ be a Peano system, let $m \in P$, and let φ be a predicate on P. If*

$$\varphi(m) \qquad \text{and} \qquad \forall n \in P\ \big(n \geq m \wedge \varphi(n) \implies \varphi(S(n))\big),$$

then

$$\forall n \in P\ \big(n \geq m \implies \varphi(n)\big).$$

Go to proof.

Remark (Standard quantified statement).

Let $(P, S, 1)$ be a Peano system, $m \in P$, and φ a predicate on P.
$$\Big(\varphi(m) \wedge \forall n \in P\ \big(n \geq m \wedge \varphi(n) \implies \varphi(S(n))\big)\Big) \implies \forall n \in P\ \big(n \geq m \implies \varphi(n)\big).$$

Remark (Interpretation). Apply ordinary induction to the shifted predicate $\psi(n) := (n \geq m \implies \varphi(n))$, which holds vacuously below m and reduces to φ above m. This theorem is the formal license for all arguments of the form "for all $n \geq m$, proceed by induction."

Remark (Dependencies).

- Induction Principle for a Peano System
- Less Than or Equal on a Peano System

2.6 Well-Ordering Principle

The Well-Ordering Principle

The well-ordering principle asserts that every nonempty subset of $\mathbb{N}$ contains a least element. This is not merely a consequence of the order axioms; it is a property that distinguishes $\mathbb{N}$ from dense ordered sets such as $\mathbb{Q}$, which have no smallest positive element. The well-ordering principle is equivalent to ordinary induction over $\mathbb{N}$, but it supports a distinct proof style: to establish a universal statement, assume it fails for some element and derive a contradiction by examining the least failure.

Theorem (Well-Ordering Principle)

Theorem 2.6.1 (Well-Ordering Principle). *Let $(P, S, 1)$ be a Peano system. Every nonempty subset of P has a least element with respect to the order $\leq$ on P.* *Go to proof.*

Remark (Standard quantified statement).

Let $(P, S, 1)$ be a Peano system.

$$\forall A \subseteq P \left(A \neq \varnothing \Longrightarrow \exists m \in A \; \forall a \in A \; (m \leq a) \right).$$

Remark (Predicate reading).

$$\text{WellOrderingPrinciple} := \forall A \left((A \subseteq P \wedge \exists a \, (a \in A)) \Rightarrow \exists m \, (m \in A \wedge \forall a \, (a \in A \Rightarrow m \leq a)) \right).$$

Remark (Negated quantified statement).

Let $(P, S, 1)$ be a Peano system.

$$\exists A \subseteq P \left(A \neq \varnothing \wedge \forall m \in A \; \exists a \in A \; (a < m) \right).$$

Remark (Failure modes). The well-ordering principle would fail if some nonempty subset of P had no least element. That means for every candidate minimum, a strictly smaller element of the subset exists, producing an infinite strictly descending chain in P. Induction rules this out.

Remark (Failure mode decomposition).

$$\neg \, \text{WellOrderingPrinciple} \Longleftrightarrow \exists A \subseteq P \left(\underbrace{A \neq \varnothing}_{\texttt{nonempty}} \wedge \underbrace{\forall m \in A \; \exists a \in A \; (a < m)}_{\texttt{no least element}} \right).$$

Remark (Interpretation). The well-ordering principle is the order-theoretic complement to induction. Induction says one can build up: start at 1 and keep stepping forward. Well-ordering says one can always descend to a minimum: any nonempty set has a floor. Proof strategies that use well-ordering typically assume a property fails somewhere, form the set of failures, extract its minimum by well-ordering, and derive a contradiction from the minimality of that element.

Remark (Dependencies).

- Peano System
- Less Than or Equal on a Peano System
- Induction Principle for a Peano System
- Trichotomy on a Peano System

Equivalence of Induction and Well-Ordering

The induction principle and the well-ordering principle are logically equivalent over a Peano system. Each implies the other without appealing to additional axioms. This equivalence matters structurally: it means that any argument written in the induction style can in principle be rewritten in the least-counterexample style, and vice versa.

Theorem (Induction and Well-Ordering Are Equivalent)

Theorem 2.6.2 (Induction and Well-Ordering Are Equivalent). *Let $(P, S, 1)$ be a Peano system equipped with the order $\leq$. The following are equivalent:*

1. *The induction principle: for every predicate φ on P, if $\varphi(1)$ holds and the successor step holds, then $\varphi(n)$ holds for all $n \in P$.*

2. *The well-ordering principle: every nonempty subset of P has a least element with respect to $\leq$.*

Go to proof.

Remark (Standard quantified statement).

Let $(P, S, 1)$ be a Peano system.

$$\text{WellOrderingPrinciple} \iff \Big(\forall \varphi \ (\varphi(1) \wedge \forall n \in P \ (\varphi(n) \implies \varphi(S(n))) \implies \forall n \in P \ \varphi(n))\Big).$$

Remark (Interpretation). Induction proves well-ordering by contrapositive: if a nonempty set A had no least element, the set of non-members of A would be inductive, forcing $A = \varnothing$, a contradiction. Well-ordering proves induction by forming the set of elements where φ fails and, if nonempty, extracting a least failure to derive a contradiction with the successor step.

Remark (Dependencies).

- Induction Principle for a Peano System
- Well-Ordering Principle

2.7 Algebraic Structure of $\mathbb{N}$

$\mathbb{N}$ as a Commutative Monoid Under Addition

The arithmetic established in the preceding sections can now be packaged into a named algebraic structure. Under addition, $\mathbb{N}$ is a commutative monoid with identity 0 — or, in the Peano-system setting that uses 1 as the distinguished element with no additive identity available internally, the correct statement is that $(\mathbb{N}, +)$ satisfies associativity and commutativity. This topic states the structural conclusion that follows from the individual arithmetic theorems proved above.

Theorem ($\mathbb{N}$ Is Closed, Associative, and Commutative Under Addition)

Theorem 2.7.1 ($\mathbb{N}$ Is Closed, Associative, and Commutative Under Addition). *Let $(P, S, 1)$ be a Peano system. The binary operation $+$ on P satisfies:*

1. ***Closure.*** *For all $a, b \in P$, $a + b \in P$.*
2. ***Associativity.*** *For all $a, b, c \in P$, $(a + b) + c = a + (b + c)$.*
3. ***Commutativity.*** *For all $a, b \in P$, $a + b = b + a$.*
4. ***Left cancellation.*** *For all $a, b, c \in P$, if $a + b = a + c$ then $b = c$.*
5. ***Right cancellation.*** *For all $a, b, c \in P$, if $b + a = c + a$ then $b = c$.*

Go to proof.

Remark (Standard quantified statement).

Let $(P, S, 1)$ be a Peano system.
$\forall a, b, c \in P : \ (a + b) + c = a + (b + c) \ \wedge \ a + b = b + a \ \wedge \ (a + b = a + c \Longrightarrow b = c).$

Remark (Interpretation). This theorem is a packaging result. It names the algebraic structure that has been proved piece by piece: the three individual theorems on associativity, commutativity, and cancellation together certify that $(P, +)$ is a cancellative commutative semigroup. There is no additive identity in P because P has no element 0 with $0 + n = n$ for all n; the smallest element under addition is 1, not an identity.

Remark (Dependencies).

- Addition Is Associative
- Addition Is Commutative
- Left Cancellation for Addition
- Right Cancellation for Addition

$\mathbb{N}$ as a Commutative Monoid Under Multiplication

Multiplication on P has a two-sided multiplicative identity, namely 1, the distinguished first element. Under multiplication, P is therefore a commutative monoid. This topic states the structural packaging for multiplication that parallels the additive packaging above.

Theorem ($\mathbb{N}$ Is a Commutative Monoid Under Multiplication)

Theorem 2.7.2 ($\mathbb{N}$ Is a Commutative Monoid Under Multiplication). *Let $(P, S, 1)$ be a Peano system. The binary operation $\cdot$ on P satisfies:*

1. ***Closure.*** *For all $a, b \in P$, $a \cdot b \in P$.*
2. ***Associativity.*** *For all $a, b, c \in P$, $(a \cdot b) \cdot c = a \cdot (b \cdot c)$.*
3. ***Commutativity.*** *For all $a, b \in P$, $a \cdot b = b \cdot a$.*
4. ***Identity.*** *For all $a \in P$, $1 \cdot a = a = a \cdot 1$.*

Consequently $(P, \cdot, 1)$ is a commutative monoid. *Go to proof.*

Remark (Standard quantified statement).

Let $(P, S, 1)$ be a Peano system.
$\forall a, b, c \in P : (a \cdot b) \cdot c = a \cdot (b \cdot c) \wedge a \cdot b = b \cdot a \wedge 1 \cdot a = a.$

Remark (Interpretation). The distinguished first element 1 is a multiplicative identity on both sides, so multiplication promotes P from a semigroup to a monoid. The absence of multiplicative inverses (other than $1^{-1} = 1$, which does not lie in P in the usual sense) is what prevents $(P, \cdot)$ from being a group. The structural limitation is intrinsic: no element of P other than 1 has a reciprocal in P.

Remark (Dependencies).

- Multiplication Is Associative
- Multiplication Is Commutative
- Multiplication With 1
- One Times n

$\mathbb{N}$ as a Semiring and the Absence of Subtraction

Together, addition and multiplication on P form a semiring: addition is a commutative semigroup, multiplication is a commutative monoid, and multiplication distributes over addition on both sides. The critical structural limitation is that subtraction is not a total operation on P. Since P has no additive identity and no additive inverses, the equation $a + x = b$ has a solution in P only when $b > a$; it has no solution when $b \leq a$. This failure of internal subtraction is the reason $\mathbb{Z}$ is introduced immediately after $\mathbb{N}$.

Theorem ($\mathbb{N}$ Is a Semiring)

Theorem 2.7.3 ($\mathbb{N}$ Is a Semiring). *Let $(P, S, 1)$ be a Peano system. Then $(P, +, \cdot)$ satisfies:*

1. *$(P, +)$ is a commutative semigroup.*
2. *$(P, \cdot, 1)$ is a commutative monoid.*
3. *Multiplication distributes over addition: for all $a, b, c \in P$,*

$$a \cdot (b + c) = (a \cdot b) + (a \cdot c).$$

Go to proof.

Remark (Standard quantified statement).

Let $(P, S, 1)$ be a Peano system.
$\forall a, b, c \in P : \ a \cdot (b + c) = (a \cdot b) + (a \cdot c) \ \wedge \ (a + b) + c = a + (b + c) \ \wedge \ a + b = b + a \ \wedge \ 1 \cdot a = a.$

Remark (Interpretation). A semiring packages the interaction of two operations without requiring additive inverses. The natural numbers are the canonical example of a semiring that is not a ring: they satisfy every ring axiom except the existence of additive inverses, which fails because there is no element -1 in P. The integers are the smallest ring extending P that repairs this failure.

Remark (Dependencies).

- $\mathbb{N}$ Is Closed, Associative, and Commutative Under Addition
- $\mathbb{N}$ Is a Commutative Monoid Under Multiplication
- Multiplication Distributes Over Addition

2.8 Powers and Growth

Exponentiation by Recursion

Exponentiation is defined on a Peano system by recursion on the exponent: $a^1 = a$ and $a^{S(n)} = a^n \cdot a$. This makes exponentiation a derived operation built from multiplication in exactly the way multiplication is built from addition. The recursive clauses are immediately isolated as named theorems so that later algebraic arguments can cite them directly.

Definition (Exponentiation on a Peano System)

Definition 2.8.1 (Exponentiation on a Peano System). Let $(P, S, 1)$ be a Peano system. For each $a \in P$, define the function

$$h_a : P \to P$$

to be the unique function satisfying

$$h_a(1) = a \qquad \text{and} \qquad \forall n \in P\ (h_a(S(n)) = h_a(n) \cdot a).$$

For $a, n \in P$, define

$$a^n := h_a(n).$$

Remark (Standard quantified statement).

Let $(P, S, 1)$ be a Peano system. For each $a \in P$, let $h_a : P \to P$ satisfy
$h_a(1) = a \qquad \text{and} \qquad \forall n \in P\ (h_a(S(n)) = h_a(n) \cdot a).$
Then for all $a, n \in P, \quad a^n := h_a(n).$

Remark (Negated quantified statement).

Let $(P, S, 1)$ be a Peano system. For each $a \in P$, let $h_a : P \to P$ satisfy
$h_a(1) = a \qquad \text{and} \qquad \forall n \in P\ (h_a(S(n)) = h_a(n) \cdot a).$
Then for all $a, n, x \in P, \quad x \neq a^n \iff x \neq h_a(n).$

Remark (Interpretation). Exponentiation is repeated multiplication in the same sense that multiplication is repeated addition. The base a is the fixed left factor; the exponent n counts how many times a is multiplied together. The recursive definition propagates rightward: $a^1 = a$, $a^2 = a^1 \cdot a = a \cdot a$, and so on.

Remark (Dependencies).

- Peano System
- General Recursion Theorem for a Peano System
- Multiplication on a Peano System

Theorem (Exponentiation Is Well-Defined on a Peano System)

Theorem 2.8.2 (Exponentiation Is Well-Defined on a Peano System). *Let $(P, S, 1)$ be a Peano system. For every $a \in P$, there exists a unique function*

$$h_a : P \to P$$

such that

$$h_a(1) = a \qquad \text{and} \qquad \forall n \in P\ (h_a(S(n)) = h_a(n) \cdot a).$$

Consequently the rule

$$(a, n) \longmapsto a^n := h_a(n)$$

defines a unique binary operation on P. *Go to proof.*

Remark (Standard quantified statement).

Let $(P, S, 1)$ be a Peano system.
$\forall a \in P\ \exists! h_a : P \to P\ \Big(h_a(1) = a \wedge \forall n \in P\ (h_a(S(n)) = h_a(n) \cdot a)\Big).$

Remark (Interpretation). The recursive clauses for exponentiation determine exactly one exponent map for each fixed base. Exponentiation is therefore constructed from recursion and multiplication rather than assumed as a primitive operation.

Remark (Source alignment). This is the house-style existence-and-uniqueness form of the recursive definition of exponentiation in [Feferman, 1964].

Remark (Dependencies).

- General Recursion Theorem for a Peano System
- Multiplication on a Peano System

Exponent Laws

The recursive definition immediately yields the standard exponent laws. These are not assumed; they are proved from the recursive clauses by induction. The three primary laws are: the sum law $a^{m+n} = a^m \cdot a^n$, the product law $(a \cdot b)^n = a^n \cdot b^n$, and the power law $(a^m)^n = a^{mn}$.

Theorem (Exponent Sum Law)

Theorem 2.8.3 (Exponent Sum Law). *Let $(P, S, 1)$ be a Peano system. For all $a, m, n \in P$,*

$$a^{m+n} = a^m \cdot a^n.$$

Go to proof.

Remark (Standard quantified statement).

$$\text{Let } (P, S, 1) \text{ be a Peano system. } \forall a, m, n \in P\ (a^{m+n} = a^m \cdot a^n).$$

Remark (Interpretation). Adding exponents corresponds to concatenating the two blocks of repeated multiplication. The proof is induction on n using the recursive clause $a^{S(n)} = a^n \cdot a$ and associativity of multiplication.

Remark (Dependencies).

- Exponentiation on a Peano System
- Multiplication Is Associative
- Induction Principle for a Peano System

Theorem (Exponent Product Law)

Theorem 2.8.4 (Exponent Product Law). *Let $(P, S, 1)$ be a Peano system. For all $a, b, n \in P$,*

$$(a \cdot b)^n = a^n \cdot b^n.$$

Go to proof.

Remark (Standard quantified statement).

$$\text{Let } (P, S, 1) \text{ be a Peano system. } \forall a, b, n \in P\ ((a \cdot b)^n = a^n \cdot b^n).$$

Remark (Interpretation). Raising a product to a power distributes over the factors. The proof is induction on n, using commutativity and associativity of multiplication to rearrange the interleaved factors at each successor step.

Remark (Dependencies).

- Exponentiation on a Peano System
- Multiplication Is Associative
- Multiplication Is Commutative
- Induction Principle for a Peano System

Theorem (Exponent Power Law)

Theorem 2.8.5 (Exponent Power Law). *Let $(P, S, 1)$ be a Peano system. For all $a, m, n \in P$,*

$$(a^m)^n = a^{m \cdot n}.$$

Go to proof.

Remark (Standard quantified statement).

$$\text{Let } (P, S, 1) \text{ be a Peano system.} \quad \forall a, m, n \in P\ ((a^m)^n = a^{m \cdot n}).$$

Remark (Interpretation). Raising a power to a further power multiplies the exponents. The proof is induction on n using the exponent sum law and the recursive definition of multiplication.

Remark (Dependencies).

- Exponentiation on a Peano System
- Exponent Sum Law
- Multiplication Successor on the Right
- Induction Principle for a Peano System

Order Compatibility of Powers

Exponentiation is order-sensitive in both its base and its exponent. For a fixed positive exponent, taking powers preserves and reflects strict order of bases. For a fixed base beyond 1, increasing the exponent strictly increases the value.

Theorem (Powers with Common Exponent Preserve Strict Order)

Theorem 2.8.6 (Powers with Common Exponent Preserve Strict Order). *Let $(P, S, 1)$ be a Peano system. For every $a, b, n \in P$,*

$$a < b \iff a^n < b^n.$$

Go to proof.

Remark (Standard quantified statement).

$$\text{Let } (P, S, 1) \text{ be a Peano system.}$$
$$\forall a, b, n \in P\ \left(a < b \iff a^n < b^n\right).$$

Remark (Interpretation). A positive exponent does not collapse the strict order between bases. Since the exponent is a natural number, exponentiation repeatedly multiplies by positive factors and preserves the order information.

Remark (Source alignment). This is the house-style common-exponent comparison theorem corresponding to Feferman's natural-number exponent laws [Feferman, 1964].

Remark (Dependencies).

- Exponentiation on a Peano System
- Multiplication Preserves and Reflects Strict Order

- Induction Principle for a Peano System

Theorem (Powers with Common Base Preserve Strict Order)

Theorem 2.8.7 (Powers with Common Base Preserve Strict Order). *Let $(P, S, 1)$ be a Peano system. For every $a, m, n \in P$,*

$$a^m < a^n \iff 1 < a \wedge m < n.$$

Go to proof.

Remark (Standard quantified statement).

Let $(P, S, 1)$ be a Peano system.
$\forall a, m, n \in P \left(a^m < a^n \iff (1 < a \wedge m < n)\right).$

Remark (Interpretation). Strict growth in the exponent occurs exactly when the base is beyond the multiplicative identity. If the base is 1, all powers are 1; if the base is larger than 1, each successor step in the exponent multiplies by a factor that strictly increases the value.

Remark (Source alignment). This is the house-style common-base comparison theorem corresponding to Feferman's exponent-order theorem [Feferman, 1964].

Remark (Dependencies).

- Exponentiation on a Peano System
- Multiplication Preserves and Reflects Strict Order
- Exponent Sum Law
- Induction Principle for a Peano System

Exponential Growth: $n < 2^n$

The most important growth comparison on $\mathbb{N}$ is that exponential growth outpaces linear growth: $n < 2^n$ for all $n \in P$. This single inequality captures why exponential towers dwarf polynomial bounds and is the prototypical example of an induction argument whose inductive step uses both the hypothesis and a multiplicative amplification.

Theorem ($n < 2^n$ for All $n \in \mathbb{N}$)

Theorem 2.8.8 ($n < 2^n$ for All $n \in \mathbb{N}$). *Let $(P, S, 1)$ be a Peano system, and let $2 := S(1)$. For every $n \in P$,*

$$n < 2^n.$$

Go to proof.

Remark (Standard quantified statement).

Let $(P, S, 1)$ be a Peano system and let $2 := S(1)$.
$\forall n \in P \ (n < 2^n)$.

Remark (Interpretation). The base case $1 < 2 = 2^1$ holds by definition of 2 and the strict order. For the inductive step, assume $n < 2^n$. Then $2^{S(n)} = 2^n{\cdot}2$. Since $2^n > n \geq 1$, we have $2^n{\cdot}2 > n{\cdot}2 = n{+}n > n{+}1 = S(n)$, where the last step uses that $n \geq 1$ so $n + n > n + 1$. Hence $S(n) < 2^{S(n)}$. The inequality $n < 2^n$ is the threshold that distinguishes the growth regimes and reappears in every comparison between polynomial and exponential bounds.

Remark (Dependencies).

- Exponentiation on a Peano System
- Strict Less Than on a Peano System
- Multiplication Distributes Over Addition
- Addition Preserves Order on the Right
- Induction Principle for a Peano System

Proofs

Remark (Return). Return to Theorem

Remark (Proof vault). Handwritten proof record

Theorem (Addition Is Well-Defined on a Peano System). *Let $(P, S, 1)$ be a Peano system. For every $m \in P$, there exists a unique function*

$$f_m : P \to P$$

such that

$$f_m(1) = S(m) \qquad \textit{and} \qquad \forall n \in P\ (f_m(S(n)) = S(f_m(n))).$$

Consequently the rule

$$(m, n) \longmapsto m + n := f_m(n)$$

defines a unique binary operation on P.

Proof. **Professional Standard Proof.**
Fix $m \in P$. Since $(P, S, 1)$ is a Peano system, $S : P \to P$ is a function. By closure under successor, $S(m) \in P$. Thus $(P, S(m), S)$ is iterator data for $(P, S, 1)$. By the Peano Iterator Theorem, there exists a unique function

$$f_m : P \to P$$

such that

$$f_m(1) = S(m) \qquad \text{and} \qquad \forall n \in P\ (f_m(S(n)) = S(f_m(n))).$$

This proves existence and uniqueness of the recursively specified function for the fixed element m.

Since $m \in P$ was arbitrary, each $m \in P$ determines exactly one such function f_m. Define

$$+ : P \times P \to P \qquad \text{by} \qquad (m, n) \longmapsto m + n := f_m(n).$$

For every pair $(m, n) \in P \times P$, the value $f_m(n)$ lies in P, because $f_m : P \to P$. The uniqueness of each f_m makes the value assigned to each pair uniquely determined. Hence the displayed rule defines a unique binary operation on P. □

Proof. **Detailed Learning Proof.**
Step 1. Fix the left input. Let $m \in P$ be fixed. Addition is constructed by holding this left input constant and recursively defining the right-input function.

Step 2. Build the iterator data. Because $(P, S, 1)$ is a Peano system, the successor map is a function

$$S : P \to P.$$

Also, since $m \in P$, closure under successor gives

$$S(m) \in P.$$

Therefore the value set P, the initial value $S(m)$, and the update map $S : P \to P$ form iterator data

$$(P, S(m), S)$$

for the Peano system $(P, S, 1)$.

Step 3. Apply the Peano Iterator Theorem. The Peano Iterator Theorem applied to the iterator data $(P, S(m), S)$ gives a unique function

$$f_m : P \to P$$

such that the base value is

$$f_m(1) = S(m)$$

and the successor step is

$$\forall n \in P\ (f_m(S(n)) = S(f_m(n))).$$

Thus, for this fixed m, the recursive clauses determine exactly one right-input function.

Step 4. Vary the left input. The argument did not use anything special about m except that $m \in P$. Hence every element $m \in P$ has its own uniquely determined function $f_m : P \to P$ satisfying the same form of recursive clauses.

Step 5. Define the operation pointwise. For a pair $(m, n) \in P \times P$, define

$$m + n := f_m(n).$$

This value belongs to P because f_m maps P into P and $n \in P$. Therefore the rule assigns an element of P to every ordered pair from $P \times P$.

Step 6. Check uniqueness of the operation. For each fixed m, the function f_m is unique by the iterator theorem. Therefore no other rule satisfying the same recursive specification can give a different value at any pair (m, n). The operation

$$(m, n) = m + n = f_m(n)$$

is consequently a uniquely determined binary operation on P. □

Remark (Proof structure). The proof fixes the left input m and packages the recursive data as the iterator data $(P, S(m), S)$. Closure under successor supplies the initial value $S(m) \in P$, and the Peano Iterator Theorem supplies the unique function $f_m : P \to P$. Varying m then defines the binary operation pointwise by $m+n = f_m(n)$; uniqueness is pointwise uniqueness of the iterator functions.

Remark (Dependencies).

- Peano System
- Closure Under Successor
- Iterator Data on a Peano System
- Addition on a Peano System
- Peano Iterator Theorem

Remark (Return). Return to Theorem

Theorem (Addition With 1). *Let $(P, S, 1)$ be a Peano system. For every $m \in P$,*

$$m + 1 = S(m).$$

Proof. **Professional Standard Proof.**
TODO. ∎

Proof. **Detailed Learning Proof.**
TODO. ∎

Remark (Proof structure). TODO.

Remark (Dependencies).

- Addition on a Peano System

Remark (Return). Return to Theorem

Theorem (Addition Successor on the Right). *Let* $(P, S, 1)$ *be a Peano system. For every* $m, n \in P$*,*

$$m + S(n) = S(m + n).$$

Proof. **Professional Standard Proof.**
TODO. □

Proof. **Detailed Learning Proof.**
TODO. □

Remark (Proof structure). TODO.

Remark (Dependencies).

- Addition on a Peano System

Remark (Return). Return to Theorem

Theorem (One Plus n). *Let* $(P, S, 1)$ *be a Peano system. For every* $n \in P$*,*

$$1 + n = S(n).$$

Proof. **Professional Standard Proof.**
TODO. □

Proof. **Detailed Learning Proof.**
TODO. □

Remark (Proof structure). TODO.

Remark (Dependencies).

- Addition With 1
- Addition Successor on the Right
- Induction Principle for a Peano System

Remark (Return). Return to Theorem

Theorem (Addition Is Associative). *Let $(P, S, 1)$ be a Peano system. For every $a, b, c \in P$,*

$$(a + b) + c = a + (b + c).$$

Proof. **Professional Standard Proof.**
TODO. ☐

Proof. **Detailed Learning Proof.**
TODO. ☐

Remark (Proof structure). TODO.

Remark (Dependencies).

- Addition With 1
- Addition Successor on the Right
- Induction Principle for a Peano System

Remark (Return). Return to Theorem

Theorem (Addition Is Commutative). *Let $(P, S, 1)$ be a Peano system. For every $a, b \in P$,*

$$a + b = b + a.$$

Proof. **Professional Standard Proof.**
TODO. ☐

Proof. **Detailed Learning Proof.**
TODO. ☐

Remark (Proof structure). TODO.

Remark (Dependencies).

- Addition With 1
- Addition Successor on the Right
- One Plus n
- Addition Is Associative
- Induction Principle for a Peano System

Remark (Return). Return to Theorem

Theorem (Left Cancellation for Addition). *Let* $(P, S, 1)$ *be a Peano system. For every* $a, b, c \in P$*,*

$$a + b = a + c \Longrightarrow b = c.$$

Proof. **Professional Standard Proof.**
TODO. □

Proof. **Detailed Learning Proof.**
TODO. □

Remark (Proof structure). TODO.

Remark (Dependencies).

- Injectivity of Successor
- Addition With 1
- Addition Successor on the Right
- One Plus n
- Addition Is Associative
- Addition Is Commutative
- Induction Principle for a Peano System

Remark (Return). Return to Theorem

Theorem (Right Cancellation for Addition). *Let $(P, S, 1)$ be a Peano system. For every $a, b, c \in P$,*

$$b + a = c + a \Longrightarrow b = c.$$

Proof. **Professional Standard Proof.**
TODO. □

Proof. **Detailed Learning Proof.**
TODO. □

Remark (Proof structure). TODO.

Remark (Dependencies).

- Addition Is Commutative
- Left Cancellation for Addition

Remark (Return). Return to Theorem

Theorem (No Natural Number Equals Itself Plus a Natural Number). *Let* $(P, S, 1)$ *be a Peano system. For every* $m, n \in P$,

$$n \neq m + n.$$

Proof. **Professional Standard Proof.**
TODO. □

Proof. **Detailed Learning Proof.**
TODO. □

Remark (Proof structure). TODO.

Remark (Dependencies).

- Addition With 1
- Addition Successor on the Right
- No Object Is Its Own Successor
- Induction Principle for a Peano System

Remark (Return). Return to Theorem

Theorem (Multiplication Is Well-Defined on a Peano System). *Let $(P, S, 1)$ be a Peano system. For every $m \in P$, there exists a unique function*

$$g_m : P \to P$$

such that

$$g_m(1) = m \qquad \textit{and} \qquad \forall n \in P\ (g_m(S(n)) = g_m(n) + m).$$

Consequently the rule

$$(m, n) \longmapsto m \cdot n := g_m(n)$$

defines a unique binary operation on P.

Proof. **Professional Standard Proof.**
TODO. ☐

Proof. **Detailed Learning Proof.**
TODO. ☐

Remark (Proof structure). TODO.

Remark (Dependencies).

- Multiplication on a Peano System
- General Recursion Theorem for a Peano System
- Addition on a Peano System

Remark (Return). Return to Theorem

Theorem (Multiplication With 1). *Let* $(P, S, 1)$ *be a Peano system. For every* $m \in P$*,*

$$m \cdot 1 = m.$$

Proof. **Professional Standard Proof.**
TODO. □

Proof. **Detailed Learning Proof.**
TODO. □

Remark (Proof structure). TODO.

Remark (Dependencies).

- Multiplication on a Peano System

Remark (Return). Return to Theorem

Theorem (Multiplication Successor on the Right). *Let $(P, S, 1)$ be a Peano system. For every $m, n \in P$,*

$$m \cdot S(n) = (m \cdot n) + m.$$

Proof. **Professional Standard Proof.**
TODO. ∎

Proof. **Detailed Learning Proof.**
TODO. ∎

Remark (Proof structure). TODO.

Remark (Dependencies).

- Multiplication on a Peano System
- Addition on a Peano System

Remark (Return). Return to Theorem

Theorem (One Times n). *Let $(P, S, 1)$ be a Peano system. For every $n \in P$,*

$$1 \cdot n = n.$$

Proof. **Professional Standard Proof.**
TODO. ☐

Proof. **Detailed Learning Proof.**
TODO. ☐

Remark (Proof structure). TODO.

Remark (Dependencies).

- Multiplication With 1
- Multiplication Successor on the Right
- Induction Principle for a Peano System

Remark (Return). Return to Theorem

Theorem (Multiplication Distributes Over Addition). *Let* $(P, S, 1)$ *be a Peano system. For every* $a, b, c \in P$*,*

$$a \cdot (b + c) = (a \cdot b) + (a \cdot c).$$

Proof. **Professional Standard Proof.**
TODO. □

Proof. **Detailed Learning Proof.**
TODO. □

Remark (Proof structure). TODO.

Remark (Dependencies).

- Multiplication Successor on the Right
- Addition Is Associative
- Addition Is Commutative
- Induction Principle for a Peano System

Remark (Return). Return to Theorem

Theorem (Left Distributivity of Multiplication Over Addition). *Let* $(P, S, 1)$ *be a Peano system. For every* $a, b, c \in P$*,*

$$(a + b) \cdot c = (a \cdot c) + (b \cdot c).$$

Proof. **Professional Standard Proof.**
TODO. □

Proof. **Detailed Learning Proof.**
TODO. □

Remark (Proof structure). TODO.

Remark (Dependencies).

- Multiplication Distributes Over Addition
- Multiplication Is Commutative
- Addition Is Commutative

Remark (Return). Return to Theorem

Theorem (Multiplication Is Associative). *Let $(P, S, 1)$ be a Peano system. For every $a, b, c \in P$,*

$$(a \cdot b) \cdot c = a \cdot (b \cdot c).$$

Proof. **Professional Standard Proof.**
TODO. □

Proof. **Detailed Learning Proof.**
TODO. □

Remark (Proof structure). TODO.

Remark (Dependencies).

- Multiplication Distributes Over Addition
- Multiplication With 1
- Multiplication Successor on the Right
- Induction Principle for a Peano System

Remark (Return). Return to Theorem

Theorem (Multiplication Is Commutative). *Let $(P, S, 1)$ be a Peano system. For every $a, b \in P$,*

$$a \cdot b = b \cdot a.$$

Proof. **Professional Standard Proof.**
TODO. □

Proof. **Detailed Learning Proof.**
TODO. □

Remark (Proof structure). TODO.

Remark (Dependencies).

- One Times n
- Multiplication Successor on the Right
- Multiplication Distributes Over Addition
- Addition Is Commutative
- Induction Principle for a Peano System

Remark (Return). Return to Theorem

Theorem (Order Is Reflexive on a Peano System). *Let $(P, S, 1)$ be a Peano system. For every $a \in P$,*

$$a \leq a.$$

Proof. **Professional Standard Proof.**
TODO. ☐

Proof. **Detailed Learning Proof.**
TODO. ☐

Remark (Proof structure). TODO.

Remark (Dependencies).

- Less Than or Equal on a Peano System

Remark (Return). Return to Theorem

Theorem (Order Is Transitive on a Peano System). *Let $(P, S, 1)$ be a Peano system. For every $a, b, c \in P$,*

$$\big(a \leq b \land b \leq c\big) \Longrightarrow a \leq c.$$

Proof. **Professional Standard Proof.**
TODO. □

Proof. **Detailed Learning Proof.**
TODO. □

Remark (Proof structure). TODO.

Remark (Dependencies).

- Strict Less Than on a Peano System
- Less Than or Equal on a Peano System
- Addition Is Associative

Remark (Return). Return to Theorem

Theorem (Order Is Antisymmetric on a Peano System). *Let* $(P, S, 1)$ *be a Peano system. For every* $a, b \in P$*,*

$$\big(a \leq b \wedge b \leq a\big) \implies a = b.$$

Proof. **Professional Standard Proof.**
TODO. ☐

Proof. **Detailed Learning Proof.**
TODO. ☐

Remark (Proof structure). TODO.

Remark (Dependencies).

- Strict Less Than on a Peano System
- Less Than or Equal on a Peano System
- Left Cancellation for Addition
- Addition Is Associative
- No Object Is Its Own Successor

Remark (Return). Return to Theorem

Theorem (Addition Preserves Order on the Right). *Let $(P, S, 1)$ be a Peano system. For every $a, b, c \in P$,*

$$a \leq b \Longrightarrow a + c \leq b + c.$$

Proof. **Professional Standard Proof.**
TODO. □

Proof. **Detailed Learning Proof.**
TODO. □

Remark (Proof structure). TODO.

Remark (Dependencies).

- Strict Less Than on a Peano System
- Less Than or Equal on a Peano System
- Addition Is Associative

Remark (Return). Return to Theorem

Theorem (Addition Preserves Order on the Left). *Let* $(P, S, 1)$ *be a Peano system. For every* $a, b, c \in P$*,*

$$a \leq b \Longrightarrow c + a \leq c + b.$$

Proof. **Professional Standard Proof.**
TODO. ∎

Proof. **Detailed Learning Proof.**
TODO. ∎

Remark (Proof structure). TODO.

Remark (Dependencies).

- Addition Preserves Order on the Right
- Addition Is Commutative

Remark (Return). Return to Theorem

Theorem (Multiplication Preserves and Reflects Strict Order). *Let $(P, S, 1)$ be a Peano system. For every $k, m, n \in P$,*

$$m < n \iff k \cdot m < k \cdot n.$$

Proof. **Professional Standard Proof.**
TODO. ☐

Proof. **Detailed Learning Proof.**
TODO. ☐

Remark (Proof structure). TODO.

Remark (Dependencies).

- Strict Less Than on a Peano System
- Multiplication on a Peano System
- Multiplication Distributes Over Addition
- Multiplication Is Commutative
- Addition Preserves Order on the Right
- Left Cancellation for Addition

Remark (Return). Return to Theorem

Theorem (Strict Order Successor Characterization). *Let $(P, S, 1)$ be a Peano system. For every $a, b \in P$,*

$$a < b \iff S(a) \leq b.$$

Proof. **Professional Standard Proof.**
TODO. ☐

Proof. **Detailed Learning Proof.**
TODO. ☐

Remark (Proof structure). TODO.

Remark (Dependencies).

- Strict Less Than on a Peano System
- Less Than or Equal on a Peano System
- Addition With 1
- Addition Successor on the Right
- Addition Is Associative
- Addition Is Commutative
- Every Element Is Either 1 or a Successor

Remark (Return). Return to Theorem

Theorem (Trichotomy on a Peano System). *Let $(P, S, 1)$ be a Peano system. For every $a, b \in P$, exactly one of the following holds:*

$$a < b, \qquad a = b, \qquad b < a.$$

Proof. **Professional Standard Proof.**
TODO. □

Proof. **Detailed Learning Proof.**
TODO. □

Remark (Proof structure). TODO.

Remark (Dependencies).

- Strict Less Than on a Peano System
- Less Than or Equal on a Peano System
- Order Is Reflexive on a Peano System
- Order Is Transitive on a Peano System
- Order Is Antisymmetric on a Peano System
- Strict Order Successor Characterization
- Induction Principle for a Peano System

Remark (Return). Return to Theorem

Theorem (Strong Induction on a Peano System). *Let $(P, S, 1)$ be a Peano system, and let φ be a predicate on P. If*

$$\forall n \in P \left(\forall k \in P \; (k < n \Longrightarrow \varphi(k))\right) \Longrightarrow \varphi(n),$$

then $\forall n \in P \; \varphi(n)$.

Proof. **Professional Standard Proof.**
TODO. □

Proof. **Detailed Learning Proof.**
TODO. □

Remark (Proof structure). TODO.

Remark (Dependencies).

- Induction Principle for a Peano System
- Strict Less Than on a Peano System
- Trichotomy on a Peano System

Remark (Return). Return to Theorem

Theorem (Induction from an Arbitrary Base). *Let $(P, S, 1)$ be a Peano system, let $m \in P$, and let φ be a predicate on P. If $\varphi(m)$ and $\forall n \in P\ (n \geq m \wedge \varphi(n) \Longrightarrow \varphi(S(n)))$, then $\forall n \in P\ (n \geq m \Longrightarrow \varphi(n))$.*

Proof. **Professional Standard Proof.**
TODO. □

Proof. **Detailed Learning Proof.**
TODO. □

Remark (Proof structure). TODO.

Remark (Dependencies).

- Induction Principle for a Peano System
- Less Than or Equal on a Peano System

Remark (Return). Return to Theorem

Theorem (Well-Ordering Principle). *Let $(P, S, 1)$ be a Peano system. Every nonempty subset of P has a least element with respect to the order $\leq$ on P.*

Proof. **Professional Standard Proof.**
TODO. ☐

Proof. **Detailed Learning Proof.**
TODO. ☐

Remark (Proof structure). TODO.

Remark (Dependencies).

- Peano System
- Less Than or Equal on a Peano System
- Induction Principle for a Peano System
- Trichotomy on a Peano System

Remark (Return). Return to Theorem

Theorem (Induction and Well-Ordering Are Equivalent). *Let $(P, S, 1)$ be a Peano system equipped with the order $\leq$. The induction principle and the well-ordering principle are equivalent over P.*

Proof. **Professional Standard Proof.**
TODO. □

Proof. **Detailed Learning Proof.**
TODO. □

Remark (Proof structure). TODO.

Remark (Dependencies).

- Induction Principle for a Peano System
- Well-Ordering Principle

Remark (Return). Return to Theorem

Theorem ($\mathbb{N}$ Is Closed, Associative, and Commutative Under Addition). *Let $(P, S, 1)$ be a Peano system. The operation $+$ on P is associative, commutative, and satisfies left and right cancellation.*

Proof. **Professional Standard Proof.**
TODO. $\square$

Proof. **Detailed Learning Proof.**
TODO. $\square$

Remark (Proof structure). TODO.

Remark (Dependencies).

- Addition Is Associative
- Addition Is Commutative
- Left Cancellation for Addition
- Right Cancellation for Addition

Remark (Return). Return to Theorem

Theorem ($\mathbb{N}$ Is a Commutative Monoid Under Multiplication). *Let $(P, S, 1)$ be a Peano system. The system $(P, \cdot, 1)$ is a commutative monoid: multiplication is associative and commutative, and 1 is a two-sided identity.*

Proof. **Professional Standard Proof.**
TODO. □

Proof. **Detailed Learning Proof.**
TODO. □

Remark (Proof structure). TODO.

Remark (Dependencies).

- Multiplication Is Associative
- Multiplication Is Commutative
- Multiplication With 1
- One Times n

Remark (Return). Return to Theorem

Theorem (ℕ Is a Semiring). *Let $(P, S, 1)$ be a Peano system. The system $(P, +, \cdot)$ is a semiring: $(P, +)$ is a commutative semigroup, $(P, \cdot, 1)$ is a commutative monoid, and multiplication distributes over addition on both sides.*

Proof. **Professional Standard Proof.**
TODO. □

Proof. **Detailed Learning Proof.**
TODO. □

Remark (Proof structure). TODO.

Remark (Dependencies).

- ℕ Is Closed, Associative, and Commutative Under Addition
- ℕ Is a Commutative Monoid Under Multiplication
- Multiplication Distributes Over Addition

Remark (Return). Return to Theorem

Theorem (Exponentiation Is Well-Defined on a Peano System). *Let $(P, S, 1)$ be a Peano system. For every $a \in P$, there exists a unique function*

$$h_a : P \to P$$

such that

$$h_a(1) = a \qquad \textit{and} \qquad \forall n \in P \ (h_a(S(n)) = h_a(n) \cdot a).$$

Consequently the rule

$$(a, n) \longmapsto a^n := h_a(n)$$

defines a unique binary operation on P.

Proof. **Professional Standard Proof.**
TODO. □

Proof. **Detailed Learning Proof.**
TODO. □

Remark (Proof structure). TODO.

Remark (Dependencies).

- General Recursion Theorem for a Peano System
- Multiplication on a Peano System

Remark (Return). Return to Theorem

Theorem (Exponent Sum Law). *Let $(P, S, 1)$ be a Peano system. For all $a, m, n \in P$,*

$$a^{m+n} = a^m \cdot a^n.$$

Proof. **Professional Standard Proof.**
TODO. ☐

Proof. **Detailed Learning Proof.**
TODO. ☐

Remark (Proof structure). TODO.

Remark (Dependencies).

- Exponentiation on a Peano System
- Multiplication Is Associative
- Induction Principle for a Peano System

Remark (Return). Return to Theorem

Theorem (Exponent Product Law). *Let $(P, S, 1)$ be a Peano system. For all $a, b, n \in P$,*

$$(a \cdot b)^n = a^n \cdot b^n.$$

Proof. **Professional Standard Proof.**
TODO. □

Proof. **Detailed Learning Proof.**
TODO. □

Remark (Proof structure). TODO.

Remark (Dependencies).

- Exponentiation on a Peano System
- Multiplication Is Associative
- Multiplication Is Commutative
- Induction Principle for a Peano System

Remark (Return). Return to Theorem

Theorem (Exponent Power Law). *Let* $(P, S, 1)$ *be a Peano system. For all* $a, m, n \in P$*,*

$$(a^m)^n = a^{m \cdot n}.$$

Proof. **Professional Standard Proof.**
TODO. ☐

Proof. **Detailed Learning Proof.**
TODO. ☐

Remark (Proof structure). TODO.

Remark (Dependencies).

- Exponentiation on a Peano System
- Exponent Sum Law
- Multiplication Successor on the Right
- Induction Principle for a Peano System

Remark (Return). Return to Theorem

Theorem (Powers with Common Exponent Preserve Strict Order). *Let $(P, S, 1)$ be a Peano system. For every $a, b, n \in P$,*

$$a < b \iff a^n < b^n.$$

Proof. **Professional Standard Proof.**
TODO. □

Proof. **Detailed Learning Proof.**
TODO. □

Remark (Proof structure). TODO.

Remark (Dependencies).

- Exponentiation on a Peano System
- Multiplication Preserves and Reflects Strict Order
- Induction Principle for a Peano System

Remark (Return). Return to Theorem

Theorem (Powers with Common Base Preserve Strict Order). *Let $(P, S, 1)$ be a Peano system. For every $a, m, n \in P$,*

$$a^m < a^n \iff 1 < a \wedge m < n.$$

Proof. **Professional Standard Proof.**
TODO. □

Proof. **Detailed Learning Proof.**
TODO. □

Remark (Proof structure). TODO.

Remark (Dependencies).

- Exponentiation on a Peano System
- Multiplication Preserves and Reflects Strict Order
- Exponent Sum Law
- Induction Principle for a Peano System

Remark (Return). Return to Theorem

Theorem ($n < 2^n$ for All $n \in \mathbb{N}$). *Let $(P, S, 1)$ be a Peano system, and let $2 := S(1)$. For every $n \in P$,*

$$n < 2^n.$$

Proof. **Professional Standard Proof.**
TODO. □

Proof. **Detailed Learning Proof.**
TODO. □

Remark (Proof structure). TODO.

Remark (Dependencies).

- Exponentiation on a Peano System
- Strict Less Than on a Peano System
- Multiplication Distributes Over Addition
- Addition Preserves Order on the Right
- Induction Principle for a Peano System

Capstone

Chapter 3

Constructing the Whole Numbers

whole-numbers

Peano Systems $\to$ Natural Numbers ($\mathbb{N}$) $\to$ **Whole Numbers ($\mathbb{W}$)** $\to$ Integers ($\mathbb{Z}$)

Structural Roadmap

How we got here. The preceding chapter developed arithmetic on the one-based natural numbers. That system has a first element 1, successor, induction, addition, multiplication, and order, but it does not contain an additive identity.

What this chapter develops. This chapter adjoins a new element 0, defines $\mathbb{W} = \mathbb{N} \cup \{0\}$, and extends successor, arithmetic, and order from $\mathbb{N}$ to $\mathbb{W}$.

Where this leads. The whole numbers repair the missing additive identity but still do not make subtraction internal. The next construction passes from $\mathbb{W}$ to $\mathbb{Z}$ by adjoining additive inverses formally.

3.1 Why Whole Numbers Are Introduced

Adjoining an Additive Identity

The natural-number system developed in the preceding chapter is one-based: its distinguished first element is 1. This is sufficient for counting and recursive arithmetic, but addition on $\mathbb{N}$ has no additive identity inside $\mathbb{N}$. The whole numbers are introduced by adjoining a new element before constructing the integers.

The Missing Identity

Remark (Exposition). In the one-based system, every element of $\mathbb{N}$ is positive. The operation $+$ is defined and behaves associatively and commutatively, but there is no element $e \in \mathbb{N}$ satisfying $e+n = n$ for every $n \in \mathbb{N}$. The element needed for that identity is not another successor-stage element; it is a new element placed before 1.

Remark (Interpretation). The passage from $\mathbb{N}$ to $\mathbb{W}$ is therefore not a change in the positive arithmetic already constructed. It is an extension that adds the missing additive identity and leaves the positive part intact.

Remark (Dependencies).

- Addition on a Peano System
- Addition Is Associative
- Addition Is Commutative

Remark (Exposition). Once 0 is available, addition can be packaged as a genuine commutative monoid. That structure is the natural input for the integer construction where ordered pairs are interpreted as formal differences and subtraction is made available by an equivalence-class construction.

3.2 Definition of the Whole Numbers

The Positive Part with a New Zero

The whole numbers are obtained by adjoining a single new element to the one-based natural numbers. The notation records two facts at once: the positive natural numbers remain present by inclusion, and the new element 0 is not one of those positive natural numbers.

The Underlying Set

Definition (Whole Numbers)

Definition 3.2.1 (Whole Numbers). Let $\mathbb{N}$ be the one-based natural-number system developed in the preceding chapter. Let 0 be a new element with $0 \notin \mathbb{N}$. Define

$$\mathbb{W} := \mathbb{N} \cup \{0\}.$$

Remark (Interpretation). Elements of $\mathbb{W}$ are either the new element 0 or positive natural numbers. The natural numbers embed into $\mathbb{W}$ by the inclusion map $n \mapsto n$. The element 0 is not a successor-stage element of the original one-based Peano system.

Remark (Dependencies).

- Peano System

3.3 Extending Successor

Successor After Adjoining Zero

The new successor map on $\mathbb{W}$ agrees with the old successor on the positive part and sends the new initial element 0 to 1. This makes $\mathbb{W}$ a zero-based successor chain while preserving the one-based successor behavior already proved on $\mathbb{N}$.

The Extended Successor

Definition (Successor on $\mathbb{W}$)

Definition 3.3.1 (Successor on $\mathbb{W}$). Define $S_{\mathbb{W}} : \mathbb{W} \to \mathbb{W}$ by

$$S_{\mathbb{W}}(0) = 1 \qquad \text{and} \qquad S_{\mathbb{W}}(n) = S_{\mathbb{N}}(n) \quad (n \in \mathbb{N}).$$

Remark (Interpretation). The definition inserts 0 immediately before the positive chain. After the first step, successor is exactly the successor already defined on $\mathbb{N}$.

Remark (Dependencies).

- Whole Numbers
- Peano System

Theorem (Successor on $\mathbb{W}$ Is Well-Defined)

Theorem 3.3.2 (Successor on $\mathbb{W}$ Is Well-Defined). *The rule defining $S_{\mathbb{W}}$ determines a function $S_{\mathbb{W}} : \mathbb{W} \to \mathbb{W}$.* *Go to proof.*

Remark (Interpretation). The cases are exclusive because $0 \notin \mathbb{N}$, and the positive case lands in $\mathbb{N} \subseteq \mathbb{W}$.

Remark (Dependencies).

- Whole Numbers
- Successor on $\mathbb{W}$

Theorem (0 Is Not a Successor in $\mathbb{W}$)

Theorem 3.3.3 (0 Is Not a Successor in $\mathbb{W}$). *There is no $a \in \mathbb{W}$ such that $S_{\mathbb{W}}(a) = 0$.* *Go to proof.*

Remark (Interpretation). The new element 0 is the first element of the extended chain. It is placed before 1, not obtained by applying successor to some earlier element.

Remark (Dependencies).

- Successor on $\mathbb{W}$
- 1 Is Not a Successor

Theorem (Every Nonzero Element of $\mathbb{W}$ Is a Successor)

Theorem 3.3.4 (Every Nonzero Element of $\mathbb{W}$ Is a Successor). *If $a \in \mathbb{W}$ and $a \neq 0$, then there exists $b \in \mathbb{W}$ such that $a = S_{\mathbb{W}}(b)$.* Go to proof.

Remark (Interpretation). The element 1 is the successor of 0, and every positive natural number beyond 1 remains a successor in the original Peano chain.

Remark (Dependencies).

- Whole Numbers
- Successor on $\mathbb{W}$
- Induction Axiom

Theorem (Successor on $\mathbb{W}$ Is Injective)

Theorem 3.3.5 (Successor on $\mathbb{W}$ Is Injective). *For all $a, b \in \mathbb{W}$,*

$$S_{\mathbb{W}}(a) = S_{\mathbb{W}}(b) \implies a = b.$$

Go to proof.

Remark (Interpretation). The extended successor still never merges two inputs. The only new case is the comparison between 0 and a positive natural number, and that case is ruled out by the original non-return condition for the first element.

Remark (Dependencies).

- Successor on $\mathbb{W}$
- Injectivity of Successor
- 1 Is Not a Successor

3.4 Extending Addition to $\mathbb{W}$

Addition with Zero

Addition on $\mathbb{W}$ keeps the old natural-number addition whenever both inputs are positive and adds identity clauses for the new element 0. The case split is exclusive because $0 \notin \mathbb{N}$.

The Extended Addition Operation

Definition (Addition on $\mathbb{W}$)

Definition 3.4.1 (Addition on $\mathbb{W}$). Define $+_{\mathbb{W}} : \mathbb{W} \times \mathbb{W} \to \mathbb{W}$ by

$$0 +_{\mathbb{W}} a = a, \qquad a +_{\mathbb{W}} 0 = a,$$

and, for $a, b \in \mathbb{N}$,

$$a +_{\mathbb{W}} b = a +_{\mathbb{N}} b.$$

Remark (Interpretation). If either input is 0, the new identity clause applies. If both inputs lie in $\mathbb{N}$, the old operation is used. Since $0 \notin \mathbb{N}$, the positive-positive clause does not overlap with the zero clauses.

Remark (Dependencies).

- Whole Numbers
- Addition on a Peano System

Theorem (Addition on $\mathbb{W}$ Is Well-Defined)

Theorem 3.4.2 (Addition on $\mathbb{W}$ Is Well-Defined). *The case definition of $+_{\mathbb{W}}$ determines a binary operation on $\mathbb{W}$.* *Go to proof.*

Remark (Interpretation). Well-definedness is a case check. The zero cases are new, the positive-positive case is inherited from $\mathbb{N}$, and compatibility is automatic because the positive-positive case cannot contain 0.

Remark (Dependencies).

- Whole Numbers
- Addition on $\mathbb{W}$
- Addition Is Well-Defined on a Peano System

Theorem (Zero Is an Additive Identity on $\mathbb{W}$)

Theorem 3.4.3 (Zero Is an Additive Identity on $\mathbb{W}$). *For every $a \in \mathbb{W}$,*

$$0 +_{\mathbb{W}} a = a \qquad \textit{and} \qquad a +_{\mathbb{W}} 0 = a.$$

Go to proof.

Remark (Interpretation). This is the structural feature missing from the one-based natural numbers. After adjoining 0, addition has a genuine identity element.

Remark (Dependencies).

- Addition on $\mathbb{W}$

Theorem (Addition on $\mathbb{W}$ Extends Addition on $\mathbb{N}$)

Theorem 3.4.4 (Addition on $\mathbb{W}$ Extends Addition on $\mathbb{N}$). *For all* $a, b \in \mathbb{N}$,

$$a +_{\mathbb{W}} b = a +_{\mathbb{N}} b.$$

Go to proof.

Remark (Interpretation). The extension does not alter sums of positive natural numbers.

Remark (Dependencies).

- Addition on $\mathbb{W}$
- Addition on a Peano System

Theorem (Addition on $\mathbb{W}$ Is Associative)

Theorem 3.4.5 (Addition on $\mathbb{W}$ Is Associative). *For all* $a, b, c \in \mathbb{W}$,

$$(a +_{\mathbb{W}} b) +_{\mathbb{W}} c = a +_{\mathbb{W}} (b +_{\mathbb{W}} c).$$

Go to proof.

Remark (Interpretation). Associativity follows by reducing zero cases to the identity law and reducing the positive-positive-positive case to associativity on $\mathbb{N}$.

Remark (Dependencies).

- Zero Is an Additive Identity on $\mathbb{W}$
- Addition Is Associative

Theorem (Addition on $\mathbb{W}$ Is Commutative)

Theorem 3.4.6 (Addition on $\mathbb{W}$ Is Commutative). *For all* $a, b \in \mathbb{W}$,

$$a +_{\mathbb{W}} b = b +_{\mathbb{W}} a.$$

Go to proof.

Remark (Interpretation). The zero cases are symmetric by definition, and the positive-positive case is the commutativity theorem already proved for $\mathbb{N}$.

Remark (Dependencies).

- Addition on $\mathbb{W}$
- Addition Is Commutative

Theorem (Addition on $\mathbb{W}$ Satisfies Cancellation)

Theorem 3.4.7 (Addition on $\mathbb{W}$ Satisfies Cancellation). *For all $a, b, c \in \mathbb{W}$,*

$$a +_{\mathbb{W}} b = a +_{\mathbb{W}} c \Longrightarrow b = c.$$

Go to proof.

Remark (Interpretation). When $a = 0$, cancellation is the identity law. When all relevant entries are positive, it is the natural-number cancellation theorem.

Remark (Dependencies).

- Zero Is an Additive Identity on $\mathbb{W}$
- Left Cancellation for Addition

3.5 Extending Multiplication to $\mathbb{W}$

Multiplication with Zero

Multiplication on $\mathbb{W}$ keeps the old natural-number multiplication on positive inputs and adds the absorbing behavior of the new element 0. The cases are compatible because $0 \notin \mathbb{N}$.

The Extended Multiplication Operation

Definition (Multiplication on $\mathbb{W}$)

Definition 3.5.1 (Multiplication on $\mathbb{W}$). Define $\cdot_{\mathbb{W}} : \mathbb{W} \times \mathbb{W} \to \mathbb{W}$ by

$$0 \cdot_{\mathbb{W}} a = 0, \qquad a \cdot_{\mathbb{W}} 0 = 0,$$

and, for $a, b \in \mathbb{N}$,

$$a \cdot_{\mathbb{W}} b = a \cdot_{\mathbb{N}} b.$$

Remark (Interpretation). The zero cases are new. The positive-positive case is inherited from $\mathbb{N}$. Since $0 \notin \mathbb{N}$, these clauses do not compete.

Remark (Dependencies).

- Whole Numbers
- Multiplication on a Peano System

Theorem (Multiplication on $\mathbb{W}$ Is Well-Defined)

Theorem 3.5.2 (Multiplication on $\mathbb{W}$ Is Well-Defined). *The case definition of $\cdot_{\mathbb{W}}$ determines a binary operation on $\mathbb{W}$. Go to proof.*

Remark (Interpretation). The inherited positive case is already well-defined on $\mathbb{N}$, and the new zero cases land in the distinguished element $0 \in \mathbb{W}$.

Remark (Dependencies).

- Multiplication on $\mathbb{W}$
- Multiplication Is Well-Defined on a Peano System

Theorem (Zero Is Absorbing for Multiplication on $\mathbb{W}$)

Theorem 3.5.3 (Zero Is Absorbing for Multiplication on $\mathbb{W}$). *For every $a \in \mathbb{W}$,*

$$0 \cdot_{\mathbb{W}} a = 0 \qquad \textit{and} \qquad a \cdot_{\mathbb{W}} 0 = 0.$$

Go to proof.

Remark (Dependencies).

- Multiplication on $\mathbb{W}$

Theorem (One Is a Multiplicative Identity on $\mathbb{W}$)

Theorem 3.5.4 (One Is a Multiplicative Identity on $\mathbb{W}$). *For every $a \in \mathbb{W}$,*

$$1 \cdot_{\mathbb{W}} a = a \qquad \textit{and} \qquad a \cdot_{\mathbb{W}} 1 = a.$$

Go to proof.

Remark (Dependencies).

- Multiplication on $\mathbb{W}$
- Multiplication With 1
- One Times n

Theorem (Multiplication on $\mathbb{W}$ Extends Multiplication on $\mathbb{N}$)

Theorem 3.5.5 (Multiplication on $\mathbb{W}$ Extends Multiplication on $\mathbb{N}$). *For all $a, b \in \mathbb{N}$,*

$$a \cdot_{\mathbb{W}} b = a \cdot_{\mathbb{N}} b.$$

Go to proof.

Remark (Dependencies).

- Multiplication on $\mathbb{W}$
- Multiplication on a Peano System

Theorem (Multiplication on $\mathbb{W}$ Is Associative)

Theorem 3.5.6 (Multiplication on $\mathbb{W}$ Is Associative). *For all $a, b, c \in \mathbb{W}$,*

$$(a \cdot_{\mathbb{W}} b) \cdot_{\mathbb{W}} c = a \cdot_{\mathbb{W}} (b \cdot_{\mathbb{W}} c).$$

Go to proof.

Remark (Dependencies).

- Zero Is Absorbing for Multiplication on $\mathbb{W}$
- Multiplication Is Associative

Theorem (Multiplication on $\mathbb{W}$ Is Commutative)

Theorem 3.5.7 (Multiplication on $\mathbb{W}$ Is Commutative). *For all $a, b \in \mathbb{W}$,*

$$a \cdot_{\mathbb{W}} b = b \cdot_{\mathbb{W}} a.$$

Go to proof.

Remark (Dependencies).

- Multiplication on $\mathbb{W}$
- Multiplication Is Commutative

Theorem (Multiplication Distributes over Addition on $\mathbb{W}$)

Theorem 3.5.8 (Multiplication Distributes over Addition on $\mathbb{W}$). *For all $a, b, c \in \mathbb{W}$,*

$$a \cdot_{\mathbb{W}} (b +_{\mathbb{W}} c) = (a \cdot_{\mathbb{W}} b) +_{\mathbb{W}} (a \cdot_{\mathbb{W}} c).$$

Go to proof.

Remark (Dependencies).

- Addition on $\mathbb{W}$
- Multiplication on $\mathbb{W}$
- Multiplication Distributes Over Addition

3.6 Order on $\mathbb{W}$

Zero as the Least Element

The order on $\mathbb{W}$ extends the order on $\mathbb{N}$ and places the new element 0 below every whole number. Thus the positive part keeps its old order, while the extended system gains a least element.

The Extended Order

Definition (Order on $\mathbb{W}$)

Definition 3.6.1 (Order on $\mathbb{W}$). Define $\leq_{\mathbb{W}}$ on $\mathbb{W}$ by declaring

$$0 \leq_{\mathbb{W}} a \quad (a \in \mathbb{W}),$$

and, for $a, b \in \mathbb{N}$,

$$a \leq_{\mathbb{W}} b \iff a \leq_{\mathbb{N}} b.$$

Remark (Dependencies).

- Whole Numbers
- Less Than or Equal on a Peano System

Definition (Strict Order on $\mathbb{W}$)

Definition 3.6.2 (Strict Order on $\mathbb{W}$). Define $<_{\mathbb{W}}$ on $\mathbb{W}$ by declaring

$$0 <_{\mathbb{W}} n \quad (n \in \mathbb{N}),$$

and, for $a, b \in \mathbb{N}$,

$$a <_{\mathbb{W}} b \quad \iff \quad a <_{\mathbb{N}} b.$$

No element of $\mathbb{W}$ is strictly less than 0.

Remark (Interpretation). The non-strict order makes 0 least. The strict order says exactly that the positive natural numbers lie after 0 and keep their old strict ordering.

Remark (Dependencies).

- Order on $\mathbb{W}$
- Strict Less Than on a Peano System

Theorem (Order on $\mathbb{W}$ Extends Order on $\mathbb{N}$)

Theorem 3.6.3 (Order on $\mathbb{W}$ Extends Order on $\mathbb{N}$). *For all $a, b \in \mathbb{N}$,*

$$a \leq_{\mathbb{W}} b \iff a \leq_{\mathbb{N}} b, \qquad a <_{\mathbb{W}} b \iff a <_{\mathbb{N}} b.$$

Go to proof.

Remark (Dependencies).

- Order on $\mathbb{W}$
- Strict Order on $\mathbb{W}$

Theorem (Zero Is the Least Element of $\mathbb{W}$)

Theorem 3.6.4 (Zero Is the Least Element of $\mathbb{W}$). *For every $a \in \mathbb{W}$,*

$$0 \leq_{\mathbb{W}} a.$$

Go to proof.

Remark (Dependencies).

- Order on $\mathbb{W}$

Theorem (Order on $\mathbb{W}$ Is Reflexive)

Theorem 3.6.5 (Order on $\mathbb{W}$ Is Reflexive). *For every $a \in \mathbb{W}$, $a \leq_{\mathbb{W}} a$.*
Go to proof.

Remark (Dependencies).

- Order on $\mathbb{W}$
- Order Is Reflexive on a Peano System

Theorem (Order on $\mathbb{W}$ Is Antisymmetric)

Theorem 3.6.6 (Order on $\mathbb{W}$ Is Antisymmetric). *For all $a, b \in \mathbb{W}$,*

$$a \leq_{\mathbb{W}} b \text{ and } b \leq_{\mathbb{W}} a \implies a = b.$$

Go to proof.

Remark (Dependencies).

- Order on $\mathbb{W}$
- Order Is Antisymmetric on a Peano System

Theorem (Order on $\mathbb{W}$ Is Transitive)

Theorem 3.6.7 (Order on $\mathbb{W}$ Is Transitive). *For all $a, b, c \in \mathbb{W}$,*

$$a \leq_{\mathbb{W}} b \text{ and } b \leq_{\mathbb{W}} c \implies a \leq_{\mathbb{W}} c.$$

Go to proof.

Remark (Dependencies).

- Order on $\mathbb{W}$
- Order Is Transitive on a Peano System

Theorem (Order on $\mathbb{W}$ Is Total)

Theorem 3.6.8 (Order on $\mathbb{W}$ Is Total). *For all $a, b \in \mathbb{W}$,*

$$a \leq_{\mathbb{W}} b \quad \text{or} \quad b \leq_{\mathbb{W}} a.$$

Go to proof.

Remark (Dependencies).

- Order on $\mathbb{W}$
- Trichotomy on a Peano System

Theorem (Well-Ordering Principle for $\mathbb{W}$)

Theorem 3.6.9 (Well-Ordering Principle for $\mathbb{W}$). *Every nonempty subset of $\mathbb{W}$ has a least element with respect to $\leq_{\mathbb{W}}$. Go to proof.*

Remark (Dependencies).

- Order on $\mathbb{W}$
- Zero Is the Least Element of $\mathbb{W}$
- Well-Ordering Principle

3.7 Algebraic Structure of $\mathbb{W}$

Packaging the Extended Operations

After addition, multiplication, and order have been extended, the main point of $\mathbb{W}$ can be recorded structurally. The whole numbers form the zero-based arithmetic system used as the bridge from positive natural numbers to the integer construction.

Algebraic Packages

Theorem ($(\mathbb{W}, +, 0)$ Is a Commutative Monoid)

Theorem 3.7.1 ($(\mathbb{W}, +, 0)$ Is a Commutative Monoid). *The structure $(\mathbb{W}, +_{\mathbb{W}}, 0)$ is a commutative monoid. Go to proof.*

Remark (Dependencies).

- Zero Is an Additive Identity on $\mathbb{W}$
- Addition on $\mathbb{W}$ Is Associative
- Addition on $\mathbb{W}$ Is Commutative

Theorem ($(\mathbb{W}, \cdot, 1)$ Is a Commutative Monoid)

Theorem 3.7.2 ($(\mathbb{W}, \cdot, 1)$ Is a Commutative Monoid). *The structure $(\mathbb{W}, \cdot_{\mathbb{W}}, 1)$ is a commutative monoid. Go to proof.*

Remark (Dependencies).

- One Is a Multiplicative Identity on $\mathbb{W}$
- Multiplication on $\mathbb{W}$ Is Associative
- Multiplication on $\mathbb{W}$ Is Commutative

Theorem ($(\mathbb{W}, +, \cdot, 0, 1)$ Is a Commutative Semiring)

Theorem 3.7.3 ($(\mathbb{W}, +, \cdot, 0, 1)$ Is a Commutative Semiring). *The structure $(\mathbb{W}, +_{\mathbb{W}}, \cdot_{\mathbb{W}}, 0, 1)$ is a commutative semiring. Go to proof.*

Remark (Dependencies).

- $(\mathbb{W}, +, 0)$ Is a Commutative Monoid
- $(\mathbb{W}, \cdot, 1)$ Is a Commutative Monoid
- Multiplication Distributes over Addition on $\mathbb{W}$
- Zero Is Absorbing for Multiplication on $\mathbb{W}$

Theorem ($\mathbb{N}$ Embeds into $\mathbb{W}$ as the Positive Part)

Theorem 3.7.4 ($\mathbb{N}$ Embeds into $\mathbb{W}$ as the Positive Part). *The inclusion map $\mathbb{N} \hookrightarrow \mathbb{W}$ preserves successor, addition, multiplication, and order on positive inputs.* *Go to proof.*

Remark (Dependencies).

- Whole Numbers
- Addition on $\mathbb{W}$ Extends Addition on $\mathbb{N}$
- Multiplication on $\mathbb{W}$ Extends Multiplication on $\mathbb{N}$
- Order on $\mathbb{W}$ Extends Order on $\mathbb{N}$

Theorem ($\mathbb{W}$ Has No Additive Inverses Except 0)

Theorem 3.7.5 ($\mathbb{W}$ Has No Additive Inverses Except 0). *If $a, b \in \mathbb{W}$ and*

$$a +_{\mathbb{W}} b = 0,$$

then $a = 0$ and $b = 0$. *Go to proof.*

Remark (Dependencies).

- Addition on $\mathbb{W}$
- Zero Is the Least Element of $\mathbb{W}$
- No Natural Number Equals Itself Plus a Natural Number

Theorem ($\mathbb{W}$ Is Not a Ring)

Theorem 3.7.6 ($\mathbb{W}$ Is Not a Ring). *With the operations $+_{\mathbb{W}}$ and $\cdot_{\mathbb{W}}$, the whole numbers do not form a ring.* *Go to proof.*

Remark (Interpretation). The obstruction is not associativity, commutativity, or distributivity. The obstruction is the absence of additive inverses for nonzero whole numbers.

Remark (Dependencies).

- $(\mathbb{W}, +, \cdot, 0, 1)$ Is a Commutative Semiring
- $\mathbb{W}$ Has No Additive Inverses Except 0

3.8 Transition to $\mathbb{Z}$

From Zero to Additive Inverses

The whole numbers fix the missing additive identity, but they do not make subtraction available in general. The integer construction repairs this by adjoining additive inverses formally.

The Next Construction

Remark (Exposition). The passage from $\mathbb{N}$ to $\mathbb{W}$ adds the element needed for $0 + a = a$. It does not add solutions to equations such as $a + x = b$ when $b < a$. Thus $\mathbb{W}$ is still not closed under subtraction.

Remark (Exposition). The construction of $\mathbb{Z}$ repairs this by adjoining additive inverses formally. One standard route uses ordered pairs from $\mathbb{W}\times\mathbb{W}$, with a pair (a, b) interpreted as a formal difference $a-b$. The equivalence relation then identifies different pairs that represent the same intended difference.

Remark (Dependencies).

- $(\mathbb{W}, +, 0)$ Is a Commutative Monoid
- $\mathbb{W}$ Has No Additive Inverses Except 0
- $\mathbb{W}$ Is Not a Ring

Proofs

Remark (Return). Return to Theorem

Theorem (Successor on $\mathbb{W}$ Is Well-Defined). *The rule defining $S_{\mathbb{W}}$ determines a function $S_{\mathbb{W}} : \mathbb{W} \to \mathbb{W}$.*

Proof. **Professional Standard Proof.**
TODO. ☐

Proof. **Detailed Learning Proof.**
TODO. ☐

Remark (Proof structure). TODO.

Remark (Dependencies).

- Successor on $\mathbb{W}$

Remark (Return). Return to Theorem

Theorem (0 Is Not a Successor in $\mathbb{W}$). *There is no $a \in \mathbb{W}$ such that $S_{\mathbb{W}}(a) = 0$.*

Proof. **Professional Standard Proof.**
TODO. ☐

Proof. **Detailed Learning Proof.**
TODO. ☐

Remark (Proof structure). TODO.

Remark (Dependencies).

- Successor on $\mathbb{W}$
- 1 Is Not a Successor

Remark (Return). Return to Theorem

Theorem (Every Nonzero Element of $\mathbb{W}$ Is a Successor). *If $a \in \mathbb{W}$ and $a \neq 0$, then there exists $b \in \mathbb{W}$ such that $a = S_{\mathbb{W}}(b)$.*

Proof. **Professional Standard Proof.**
TODO. □

Proof. **Detailed Learning Proof.**
TODO. □

Remark (Proof structure). TODO.

Remark (Dependencies).

- Whole Numbers
- Successor on $\mathbb{W}$

Remark (Return). Return to Theorem

Theorem (Successor on $\mathbb{W}$ Is Injective). *For all* $a, b \in \mathbb{W}$, $S_{\mathbb{W}}(a) = S_{\mathbb{W}}(b) \implies a = b$.

Proof. **Professional Standard Proof.**
TODO. □

Proof. **Detailed Learning Proof.**
TODO. □

Remark (Proof structure). TODO.

Remark (Dependencies).

- Successor on $\mathbb{W}$
- Injectivity of Successor

Remark (Return). Return to Theorem

Theorem (Addition on $\mathbb{W}$ Is Well-Defined). *The case definition of $+_{\mathbb{W}}$ determines a binary operation on $\mathbb{W}$.*

Proof. **Professional Standard Proof.**
TODO. □

Proof. **Detailed Learning Proof.**
TODO. □

Remark (Proof structure). TODO.

Remark (Dependencies).

- Addition on $\mathbb{W}$
- Addition Is Well-Defined on a Peano System

Remark (Return). Return to Theorem

Theorem (Zero Is an Additive Identity on $\mathbb{W}$). *For every $a \in \mathbb{W}$, $0 +_{\mathbb{W}} a = a$ and $a +_{\mathbb{W}} 0 = a$.*

Proof. **Professional Standard Proof.**
TODO. □

Proof. **Detailed Learning Proof.**
TODO. □

Remark (Proof structure). TODO.

Remark (Dependencies).

- Addition on $\mathbb{W}$

Remark (Return). Return to Theorem

Theorem (Addition on $\mathbb{W}$ Extends Addition on $\mathbb{N}$). *For all* $a, b \in \mathbb{N}$, $a +_{\mathbb{W}} b = a +_{\mathbb{N}} b$.

Proof. **Professional Standard Proof.**
TODO. ☐

Proof. **Detailed Learning Proof.**
TODO. ☐

Remark (Proof structure). TODO.

Remark (Dependencies).

- Addition on $\mathbb{W}$

Remark (Return). Return to Theorem

Theorem (Addition on $\mathbb{W}$ Is Associative). *For all* $a, b, c \in \mathbb{W}$, $(a +_{\mathbb{W}} b) +_{\mathbb{W}} c = a +_{\mathbb{W}} (b +_{\mathbb{W}} c)$.

Proof. **Professional Standard Proof.**
TODO. □

Proof. **Detailed Learning Proof.**
TODO. □

Remark (Proof structure). TODO.

Remark (Dependencies).

- Zero Is an Additive Identity on $\mathbb{W}$
- Addition Is Associative

Remark (Return). Return to Theorem

Theorem (Addition on $\mathbb{W}$ Is Commutative). *For all* $a, b \in \mathbb{W}$*,* $a +_{\mathbb{W}} b = b +_{\mathbb{W}} a$*.*

Proof. **Professional Standard Proof.**
TODO. □

Proof. **Detailed Learning Proof.**
TODO. □

Remark (Proof structure). TODO.

Remark (Dependencies).

- Addition on $\mathbb{W}$
- Addition Is Commutative

Remark (Return). Return to Theorem

Theorem (Addition on $\mathbb{W}$ Satisfies Cancellation). *For all* $a, b, c \in \mathbb{W}$, $a +_{\mathbb{W}} b = a +_{\mathbb{W}} c \implies b = c$.

Proof. **Professional Standard Proof.**
TODO. □

Proof. **Detailed Learning Proof.**
TODO. □

Remark (Proof structure). TODO.

Remark (Dependencies).

- Zero Is an Additive Identity on $\mathbb{W}$
- Left Cancellation for Addition

Remark (Return). Return to Theorem

Theorem (Multiplication on $\mathbb{W}$ Is Well-Defined). *The case definition of* $\cdot_{\mathbb{W}}$ *determines a binary operation on* $\mathbb{W}$.

Proof. **Professional Standard Proof.**
TODO. □

Proof. **Detailed Learning Proof.**
TODO. □

Remark (Proof structure). TODO.

Remark (Dependencies).

- Multiplication on $\mathbb{W}$
- Multiplication Is Well-Defined on a Peano System

Remark (Return). Return to Theorem

Theorem (Zero Is Absorbing for Multiplication on $\mathbb{W}$). *For every $a \in \mathbb{W}$, $0 \cdot_{\mathbb{W}} a = 0$ and $a \cdot_{\mathbb{W}} 0 = 0$.*

Proof. **Professional Standard Proof.**
TODO. ☐

Proof. **Detailed Learning Proof.**
TODO. ☐

Remark (Proof structure). TODO.

Remark (Dependencies).

- Multiplication on $\mathbb{W}$

Remark (Return). Return to Theorem

Theorem (One Is a Multiplicative Identity on $\mathbb{W}$). *For every $a \in \mathbb{W}$, $1 \cdot_{\mathbb{W}} a = a$ and $a \cdot_{\mathbb{W}} 1 = a$.*

Proof. **Professional Standard Proof.**
TODO. □

Proof. **Detailed Learning Proof.**
TODO. □

Remark (Proof structure). TODO.

Remark (Dependencies).

- Multiplication on $\mathbb{W}$
- Multiplication With 1
- One Times n

Remark (Return). Return to Theorem

Theorem (Multiplication on $\mathbb{W}$ Extends Multiplication on $\mathbb{N}$). *For all* $a, b \in \mathbb{N}$, $a \cdot_{\mathbb{W}} b = a \cdot_{\mathbb{N}} b$.

Proof. **Professional Standard Proof.**
TODO. ☐

Proof. **Detailed Learning Proof.**
TODO. ☐

Remark (Proof structure). TODO.

Remark (Dependencies).

- Multiplication on $\mathbb{W}$

Remark (Return). Return to Theorem

Theorem (Multiplication on $\mathbb{W}$ Is Associative). *For all* $a, b, c \in \mathbb{W}$, $(a \cdot_{\mathbb{W}} b) \cdot_{\mathbb{W}} c = a \cdot_{\mathbb{W}} (b \cdot_{\mathbb{W}} c)$.

Proof. **Professional Standard Proof.**
TODO. □

Proof. **Detailed Learning Proof.**
TODO. □

Remark (Proof structure). TODO.

Remark (Dependencies).

- Zero Is Absorbing for Multiplication on $\mathbb{W}$
- Multiplication Is Associative

Remark (Return). Return to Theorem

Theorem (Multiplication on $\mathbb{W}$ Is Commutative). *For all $a, b \in \mathbb{W}$, $a \cdot_{\mathbb{W}} b = b \cdot_{\mathbb{W}} a$.*

Proof. **Professional Standard Proof.**
TODO. □

Proof. **Detailed Learning Proof.**
TODO. □

Remark (Proof structure). TODO.

Remark (Dependencies).

- Multiplication on $\mathbb{W}$
- Multiplication Is Commutative

Remark (Return). Return to Theorem

Theorem (Multiplication Distributes over Addition on $\mathbb{W}$). *For all* $a, b, c \in \mathbb{W}$, $a \cdot_{\mathbb{W}} (b +_{\mathbb{W}} c) = (a \cdot_{\mathbb{W}} b) +_{\mathbb{W}} (a \cdot_{\mathbb{W}} c)$.

Proof. **Professional Standard Proof.**
TODO. □

Proof. **Detailed Learning Proof.**
TODO. □

Remark (Proof structure). TODO.

Remark (Dependencies).

- Addition on $\mathbb{W}$
- Multiplication on $\mathbb{W}$
- Multiplication Distributes Over Addition

Remark (Return). Return to Theorem

Theorem (Order on $\mathbb{W}$ Extends Order on $\mathbb{N}$). *For all $a, b \in \mathbb{N}$, the non-strict and strict orders on $\mathbb{W}$ agree with the corresponding orders on $\mathbb{N}$.*

Proof. **Professional Standard Proof.**
TODO. □

Proof. **Detailed Learning Proof.**
TODO. □

Remark (Proof structure). TODO.

Remark (Dependencies).

- Order on $\mathbb{W}$
- Strict Order on $\mathbb{W}$

Remark (Return). Return to Theorem

Theorem (Zero Is the Least Element of $\mathbb{W}$). *For every* $a \in \mathbb{W}$*,* $0 \leq_{\mathbb{W}} a$*.*

Proof. **Professional Standard Proof.**
TODO. ∎

Proof. **Detailed Learning Proof.**
TODO. ∎

Remark (Proof structure). TODO.

Remark (Dependencies).

- Order on $\mathbb{W}$

Remark (Return). Return to Theorem

Theorem (Order on $\mathbb{W}$ Is Reflexive). *For every* $a \in \mathbb{W}$, $a \leq_{\mathbb{W}} a$.

Proof. **Professional Standard Proof.**
TODO. ☐

Proof. **Detailed Learning Proof.**
TODO. ☐

Remark (Proof structure). TODO.

Remark (Dependencies).

- Order on $\mathbb{W}$
- Order Is Reflexive on a Peano System

Remark (Return). Return to Theorem

Theorem (Order on $\mathbb{W}$ Is Antisymmetric). *For all* $a, b \in \mathbb{W}$*,* $a \leq_{\mathbb{W}} b$ *and* $b \leq_{\mathbb{W}} a$ *imply* $a = b$*.*

Proof. **Professional Standard Proof.**
TODO. □

Proof. **Detailed Learning Proof.**
TODO. □

Remark (Proof structure). TODO.

Remark (Dependencies).

- Order on $\mathbb{W}$
- Order Is Antisymmetric on a Peano System

Remark (Return). Return to Theorem

Theorem (Order on $\mathbb{W}$ Is Transitive). *For all $a, b, c \in \mathbb{W}$, if $a \leq_{\mathbb{W}} b$ and $b \leq_{\mathbb{W}} c$, then $a \leq_{\mathbb{W}} c$.*

Proof. **Professional Standard Proof.**
TODO. □

Proof. **Detailed Learning Proof.**
TODO. □

Remark (Proof structure). TODO.

Remark (Dependencies).

- Order on $\mathbb{W}$
- Order Is Transitive on a Peano System

Remark (Return). Return to Theorem

Theorem (Order on $\mathbb{W}$ Is Total). *For all $a, b \in \mathbb{W}$, $a \leq_{\mathbb{W}} b$ or $b \leq_{\mathbb{W}} a$.*

Proof. **Professional Standard Proof.**
TODO. ∎

Proof. **Detailed Learning Proof.**
TODO. ∎

Remark (Proof structure). TODO.

Remark (Dependencies).

- Order on $\mathbb{W}$
- Trichotomy on a Peano System

Remark (Return). Return to Theorem

Theorem (Well-Ordering Principle for $\mathbb{W}$). *Every nonempty subset of* $\mathbb{W}$ *has a least element with respect to* $\leq_{\mathbb{W}}$.

Proof. **Professional Standard Proof.**
TODO. □

Proof. **Detailed Learning Proof.**
TODO. □

Remark (Proof structure). TODO.

Remark (Dependencies).

- Order on $\mathbb{W}$
- Well-Ordering Principle

Remark (Return). Return to Theorem

Theorem ($(\mathbb{W}, +, 0)$ Is a Commutative Monoid). *The structure $(\mathbb{W}, +_{\mathbb{W}}, 0)$ is a commutative monoid.*

Proof. **Professional Standard Proof.**
TODO. ☐

Proof. **Detailed Learning Proof.**
TODO. ☐

Remark (Proof structure). TODO.

Remark (Dependencies).

- Zero Is an Additive Identity on $\mathbb{W}$
- Addition on $\mathbb{W}$ Is Associative
- Addition on $\mathbb{W}$ Is Commutative

Remark (Return). Return to Theorem

Theorem (($\mathbb{W}, \cdot, 1$) Is a Commutative Monoid). *The structure* $(\mathbb{W}, \cdot_{\mathbb{W}}, 1)$ *is a commutative monoid.*

Proof. **Professional Standard Proof.**
TODO. ☐

Proof. **Detailed Learning Proof.**
TODO. ☐

Remark (Proof structure). TODO.

Remark (Dependencies).

- One Is a Multiplicative Identity on $\mathbb{W}$
- Multiplication on $\mathbb{W}$ Is Associative
- Multiplication on $\mathbb{W}$ Is Commutative

Remark (Return). Return to Theorem

Theorem ($(\mathbb{W}, +, \cdot, 0, 1)$ Is a Commutative Semiring). *The structure* $(\mathbb{W}, +_{\mathbb{W}}, \cdot_{\mathbb{W}}, 0, 1)$ *is a commutative semiring.*

Proof. **Professional Standard Proof.**
TODO. □

Proof. **Detailed Learning Proof.**
TODO. □

Remark (Proof structure). TODO.

Remark (Dependencies).

- $(\mathbb{W}, +, 0)$ Is a Commutative Monoid
- $(\mathbb{W}, \cdot, 1)$ Is a Commutative Monoid
- Multiplication Distributes over Addition on $\mathbb{W}$

Remark (Return). Return to Theorem

Theorem ($\mathbb{N}$ Embeds into $\mathbb{W}$ as the Positive Part). *The inclusion map $\mathbb{N} \hookrightarrow \mathbb{W}$ preserves successor, addition, multiplication, and order on positive inputs.*

Proof. **Professional Standard Proof.**
TODO. ☐

Proof. **Detailed Learning Proof.**
TODO. ☐

Remark (Proof structure). TODO.

Remark (Dependencies).

- Whole Numbers
- Addition on $\mathbb{W}$ Extends Addition on $\mathbb{N}$
- Multiplication on $\mathbb{W}$ Extends Multiplication on $\mathbb{N}$
- Order on $\mathbb{W}$ Extends Order on $\mathbb{N}$

Remark (Return). Return to Theorem

Theorem ($\mathbb{W}$ Has No Additive Inverses Except 0). *If $a, b \in \mathbb{W}$ and $a +_{\mathbb{W}} b = 0$, then $a = 0$ and $b = 0$.*

Proof. **Professional Standard Proof.**
TODO. ☐

Proof. **Detailed Learning Proof.**
TODO. ☐

Remark (Proof structure). TODO.

Remark (Dependencies).

- Addition on $\mathbb{W}$
- Zero Is the Least Element of $\mathbb{W}$

Remark (Return). Return to Theorem

Theorem ($\mathbb{W}$ Is Not a Ring). *With the operations $+_{\mathbb{W}}$ and $\cdot_{\mathbb{W}}$, the whole numbers do not form a ring.*

Proof. **Professional Standard Proof.**
TODO. □

Proof. **Detailed Learning Proof.**
TODO. □

Remark (Proof structure). TODO.

Remark (Dependencies).

- $\mathbb{W}$ Has No Additive Inverses Except 0

Chapter 4

Integers ($\mathbb{Z}$)

Where You Are in the Journey

Propositional Logic → Predicate Calculus → Sets & Functions → Proof Techniques → Axiom Systems & Natural Numbers → Whole Numbers → **The Integers** → Rationals & Ordered Fields → Real Line Foundations → ⋯

How we got here. The whole numbers gave us an additive identity, addition, multiplication, and a well-founded order, but subtraction is not available in general: for $a < b$ in $\mathbb{W}$, the expression $a - b$ is not represented as a whole number. Every subsequent number system depends on fixing this, so we pause here to do it rigorously.

What this chapter builds. We follow two complementary sources. Tao's *Analysis I*, Chapter 4 constructs $\mathbb{Z}$ from $\mathbb{N} \times \mathbb{N}$ via formal pairs and proves all arithmetic properties directly. Mendelson's *Number Systems and the Foundations of Analysis*, Chapters 3–4 performs the same construction but frames the result in the language of abstract algebra: rings, integral domains, and ordered integral domains. Together they give both the computational fluency and the structural vocabulary that real analysis and abstract algebra require.

Where this leads. The ring and integral-domain structure established here is not special to $\mathbb{Z}$: the same axioms, the same order properties, and the same absolute-value toolkit reappear verbatim for $\mathbb{Q}$ and $\mathbb{R}$. Everything algebraic built later inherits the patterns established here.

Structural Roadmap

Both sources follow the same logical spine:

Equivalence Relation ⟶ Definitions ⟶ Well-Definedness ⟶ Ring Laws ⟶ Order ⟶ Absolute Value

Nothing is assumed beyond what has been proved for $\mathbb{N}$. The available toolkit at each step is exactly what precedes it — most critically, the cancellation law for natural-number addition, which carries the entire construction forward.

The global progression is:

1. **The equivalence relation.** Integers are equivalence classes of pairs $(a, b) \in \mathbb{N} \times \mathbb{N}$ under $(a, b) \sim (c, d) \iff a + d = c + b$. Transitivity requires cancellation in $\mathbb{N}$.

2. **Addition and multiplication.** Defined on representatives; well-definedness verified separately for each operation (Tao L4.1.3; Mendelson Lemmas 2.1, 2.3).

3. **Ring laws.** The nine identities of Tao's P4.1.6 (= Mendelson Thms 2.2, 2.4) are exactly the axioms of a commutative ring with unit. All proofs expand pairs and reduce to arithmetic in $\mathbb{N}$.

4. **Integral domain.** No zero divisors (P4.1.8 / Thm 2.4(v)) and the cancellation law (C4.1.9 / Thm 3.3) elevate $\mathbb{Z}$ from a ring to an integral domain. Mendelson's §3.3 names and defines these structures abstractly.

5. **Order.** Tao defines order directly via $\mathbb{N}$-differences (Def 4.1.10). Mendelson axiomatises an ordered integral domain (Def 4.1, axioms O1–O5) and instantiates it for $\mathbb{Z}$ via a positivity set (Thm 4.3, Cor 4.6). The same five axioms recur unchanged for $\mathbb{R}$.

6. **Absolute value.** Mendelson's Theorem 4.8 (11 parts, for any ordered integral domain) provides a permanent toolkit — triangle inequality, reverse triangle inequality, multiplicativity — that requires no re-proof when we reach $\mathbb{Q}$ and $\mathbb{R}$. Tao defers absolute value to the rational chapter.

7. **Induction boundary.** Induction fails for $\mathbb{Z}$ (Tao Ex 4.1.8): there is no smallest element to anchor a descent below 0. Strong induction on $\mathbb{N}$ remains available for the positive integers inside $\mathbb{Z}$; Mendelson Thm 4.7 recovers a Peano system from $\mathcal{P}_{\mathbb{Z}}$ to make this precise.

4.1 Tao Construction

Remark 4.1.1 (Why the integers are needed)*.* The natural number system $\mathbb{N}$ with addition and multiplication has reached its limits: subtraction is not available. Given $a, b \in \mathbb{N}$ with $a < b$, the expression $a - b$ has no value in $\mathbb{N}$. To fix this, we pass to a larger system $\mathbb{Z}$ in which every natural number has an additive inverse.

Remark 4.1.2 (The circularity problem)*.* The naive approach define an integer as a difference $a - b$ of two natural numbers is circular, because the symbol $-$ is exactly what we are trying to construct. Tao resolves this by introducing a **placeholder symbol** $--$ with no arithmetic meaning, using $a{—}b$ purely as notation for a formal pair (a, b). Once subtraction is defined at the end of the construction, we verify that $a{—}b$ coincides with $a - b$ and discard the scaffolding.

Remark 4.1.3 (Why not define integers as signed naturals?)*.* One might define an integer as either a positive natural number, zero, or the negation of a positive natural number. Lemma 4.1.5 (trichotomy) shows this classification is correct, but *using it as the definition* forces case-splits for every arithmetic operation: negative $\times$ negative, positive $+$ negative of different sizes, etc. The verification of the ring laws becomes enormously messy. The equivalence-class construction pays one upfront cost and avoids all subsequent case-explosion.

Definition 4.1.1 (Integers)

An **integer** is an expression of the form $a—b$, where a and b are natural numbers. Two integers are **equal**:

$$a—b \;=\; c—d \quad\iff\quad a+d \;=\; c+b.$$

The set of all integers is denoted $\mathbb{Z}$.

Remark 4.1.4 (Reading the equality condition). The condition $a+d=c+b$ is the cross-addition criterion that would hold for genuine differences: $a-b=c-d \iff a+d=c+b$. It uses only addition in $\mathbb{N}$, which is already available, and avoids any reference to subtraction.

Remark 4.1.5 (Equality must be verified as a legitimate equivalence relation). Four axioms must be checked: reflexivity, symmetry, transitivity, and substitution. Reflexivity and symmetry are immediate from the definition. Transitivity requires the cancellation law for natural numbers (Proposition 2.2.6): the key load-bearing result from Chapter 2. Substitution cannot be verified until operations on $\mathbb{Z}$ are defined, and must be re-checked for each operation.

Definition 4.1.2 (Addition and Multiplication)

$$(a—b)+(c—d) \;:=\; (a+c)—(b+d)$$
$$(a—b)\times(c—d) \;:=\; (ac+bd)—(ad+bc)$$

Remark 4.1.6 (Motivation for the multiplication formula). Expanding the product of genuine differences: $(a-b)(c-d)=ac-ad-bc+bd=(ac+bd)-(ad+bc)$. The formula is forced by this foreknowledge.

Lemma 4.1.7 (Tao 4.1.3 Addition and multiplication are well-defined). *Let* $a,b,a',b',c,d\in\mathbb{N}$. *If* $(a—b)=(a'—b')$, *then:*

$$(a—b)+(c—d)=(a'—b')+(c—d),$$
$$(a—b)\times(c—d)=(a'—b')\times(c—d),$$

and similarly when replacing $(c—d)$ *by an equal integer.*

Remark 4.1.8 (Why well-definedness must be checked). The operations are defined in terms of representatives (a,b) of an equivalence class. If equal inputs (i.e., equal integers with different representatives) gave different outputs, the operation would not be a function on $\mathbb{Z}$ it would be a function on representations which is meaningless. This check is the substitution axiom for each operation.

Remark 4.1.9 (Embedding $\mathbb{N}$ in $\mathbb{Z}$). The integers of the form $n—0$ behave identically to the natural numbers:

$$(n—0)+(m—0)=(n+m)—0, \qquad (n—0)\times(m—0)=nm—0.$$

Furthermore $(n—0)=(m—0)$ if and only if $n=m$. We therefore identify $n\equiv n—0$, embedding $\mathbb{N}$ into $\mathbb{Z}$. In particular $0=0—0$ and $1=1—0$.

Definition 4.1.4 (Negation)

$-(a—b) \ := \ b—a.$
In particular, for a positive natural number $n = n—0$:

$$-n \ := \ 0—n.$$

Remark 4.1.10 (Well-definedness of negation). Negation must also be checked as well-defined: if $(a—b) = (a'—b')$ then $-(a—b) = -(a'—b')$. This is Exercise 4.1.2.

Lemma 4.1.11 (Tao 4.1.5 Trichotomy of integers). *Let x be an integer. Then exactly one of the following holds:*

(a) $x = 0$.

(b) $x = n$ *for some positive natural number* n.

(c) $x = -n$ *for some positive natural number* n.

*If (b) holds we call x a **positive integer**; if (c) holds we call x a **negative integer**.*

Remark 4.1.12 (Proof sketch). Existence of one case: $x = a—b$. By trichotomy of $\mathbb{N}$ (P2.2.13), either $a > b$, $a = b$, or $a < b$. Each case yields (b), (a), or (c) respectively. Mutual exclusion uses P2.2.8 and Proposition 2.2.6.

Proposition 4.1.6 (Laws of Algebra for $\mathbb{Z}$)

Let $x, y, z \in \mathbb{Z}$. Then:

$x + y = y + x$	(commutativity of addition)
$(x + y) + z = x + (y + z)$	(associativity of addition)
$x + 0 = 0 + x = x$	(additive identity)
$x + (-x) = (-x) + x = 0$	(additive inverse)
$xy = yx$	(commutativity of multiplication)
$(xy)z = x(yz)$	(associativity of multiplication)
$x \cdot 1 = 1 \cdot x = x$	(multiplicative identity)
$x(y + z) = xy + xz$	(left distributive law)
$(y + z)x = yx + zx$	(right distributive law)

Remark 4.1.13 (This is the commutative ring axioms). These nine identities are exactly the definition of a **commutative ring**. Without the identity $xy = yx$ the remaining eight would give a ring. Note: these properties were proved for $\mathbb{N}$, but $\mathbb{Z} \supsetneq \mathbb{N}$, so the proofs must be redone.

Remark 4.1.14 (Proof strategy). Write $x = (a—b)$, $y = (c—d)$, $z = (e—f)$ and expand both sides in terms of natural number arithmetic. This is far cleaner than case-splitting on sign via Lemma 4.1.5. Tao demonstrates the method on associativity of multiplication.

Definition 4.1.15 (Subtraction). For integers x, y:

$$x - y \ := \ x + (-y).$$

Remark 4.1.16 (Recovering the —— notation)*.* Once subtraction is defined, one checks that for natural numbers a, b:

$$a - b = (a—0) + (0—b) = a—b.$$

The placeholder —— is now equal to genuine subtraction; the scaffolding is removed and —— is discarded.

Proposition 4.1.17 (Tao 4.1.8 No zero divisors)**.** *Let $a, b \in \mathbb{Z}$ with $ab = 0$. Then $a = 0$ or $b = 0$ (or both).*

Corollary 4.1.18 (Tao 4.1.9 Cancellation law for $\mathbb{Z}$)**.** *Let $a, b, c \in \mathbb{Z}$ with $ac = bc$ and $c \neq 0$. Then $a = b$.*

Definition 4.1.10 (Ordering of $\mathbb{Z}$)

Let $n, m \in \mathbb{Z}$.

$$n \geq m \iff n = m + a \text{ for some } a \in \mathbb{N}.$$

$$n > m \iff n \geq m \text{ and } n \neq m.$$

Remark 4.1.19 (Consistency with $\mathbb{N}$)*.* This definition is verbatim the same as Definition 2.2.11 for $\mathbb{N}$. Since the embedding $n \equiv n—0$ is consistent with addition, the two orderings agree on natural numbers.

Lemma 4.1.20 (Tao 4.1.11 Properties of order on $\mathbb{Z}$)**.** *Let $a, b, c \in \mathbb{Z}$. Then:*

(a) $a > b$ if and only if $a - b$ is a positive natural number.

*(b) (**Addition preserves order.**) If $a > b$, then $a + c > b + c$.*

*(c) (**Positive multiplication preserves order.**) If $a > b$ and c is a positive integer, then $ac > bc$.*

*(d) (**Negation reverses order.**) If $a > b$, then $-a < -b$.*

*(e) (**Transitivity.**) If $a > b$ and $b > c$, then $a > c$.*

*(f) (**Trichotomy.**) Exactly one of $a > b$, $a < b$, $a = b$ holds.*

Remark 4.1.21 (Strategy: derive from part (a))*.* Part (a) reformulates $>$ in terms of positivity. Parts (b)-(f) all follow from (a) by translating into statements about positive natural numbers, where the analogous results are already known.

Remark 4.1.22 (Exercise 4.1.8 Induction does not apply to $\mathbb{Z}$)*.* Axiom P5 (induction) does not carry over to the integers. Specifically: there exists a property $P(n)$ of integers such that $P(0)$ is true and $P(n) \Rightarrow P(n{+}{+})$ for all $n \in \mathbb{Z}$, yet $P(n)$ fails for some $n \in \mathbb{Z}$. This is because $\mathbb{Z}$ has no smallest element there is nothing to anchor a descent below 0. The situation becomes worse for $\mathbb{Q}$ and $\mathbb{R}$.

Integer Toolkit Tao ğ4.1 Quick Reference

Label	Statement	Proof method
Def 4.1.1	$a—b = c—d \iff a+d = c+b$	Definition
Def 4.1.2	Addition and multiplication formulas	Definition
L4.1.3	Operations are well-defined	Expand; use $a+b' = a'+b$
Def 4.1.4	$-(a—b) := b—a$	Definition
L4.1.5	Trichotomy of $\mathbb{Z}$	Cases $a > b$, $a = b$, $a < b$
P4.1.6	Nine ring laws	Expand via representatives
Def 4.1.7	$x - y := x + (-y)$	Definition
P4.1.8	No zero divisors	Uses L2.3.3
C4.1.9	Cancellation: $ac = bc, c \neq 0 \Rightarrow a = b$	Uses P4.1.8 or C2.3.7+L4.1.5
Def 4.1.10	Order: $n \geq m \iff n = m + a,\ a \in \mathbb{N}$	Definition
L4.1.11	Six order properties	Use part (a) as bridge

4.2 Mendelson Construction

ğ3.1 The Equivalence Relation on $P \times P$

Remark 4.2.1 (Starting point). Mendelson begins from $P \times P$, the set of all ordered pairs of positive integers (his notation for $\mathbb{N} \setminus \{0\}$). Tao uses $\mathbb{N} \times \mathbb{N}$ (including zero). The idea is identical: a pair (n, j) represents the "formal difference" $n - j$.

Definition 3.1 (Equivalence relation on $P \times P$)

For natural numbers n, k, j, i, define a relation $\sim$ on ordered pairs by

$$(n, j) \sim (k, i) \iff n + i = k + j.$$

Theorem 4.2.2 (Mendelson 1.1 -- $\sim$ is an equivalence relation). *For all natural numbers h, i, j, k, m, n:*

(R) $(h, i) \sim (h, i)$ *(Reflexivity)*

(S) $(h, i) \sim (j, k) \Rightarrow (j, k) \sim (h, i)$ *(Symmetry)*

(T) $\big[(h, i) \sim (j, k) \wedge (j, k) \sim (m, n)\big] \Rightarrow (h, i) \sim (m, n)$ *(Transitivity)*

Remark 4.2.3 (Transitivity uses cancellation). As in Tao, transitivity of $\sim$ requires the cancellation law for addition in $\mathbb{N}$. This is the same load-bearing step in both constructions.

Definition 3.2 (The integers $\mathbb{Z}$)

$\mathbb{Z}$ is the set of all equivalence classes of $P \times P$ under $\sim$. Elements of $\mathbb{Z}$ are called **integers**.
Distinguished elements: $0_{\mathbb{Z}} = [(1, 1)]$, $\quad 1_{\mathbb{Z}} = [(2, 1)]$.

Remark 4.2.4 (Notation comparison). Mendelson writes $[(n, j)]$ for an equivalence class; Tao writes $a—b$ for the same object. Mendelson's notation makes the set-theoretic content explicit. Tao's notation is cleaner for computation.

ğ3.2 Addition and Multiplication

Lemma 4.2.5 (Mendelson 2.1 -- Addition well-defined). *If* $(n, j) \sim (n_1, j_1)$ *and* $(k, i) \sim (k_1, i_1)$*, then* $(n + k,\ j + i) \sim (n_1 + k_1,\ j_1 + i_1)$*.*

Definition 3.3 (Addition)

For $\alpha, \beta \in \mathbb{Z}$ with representatives $(n, j) \in \alpha$ and $(k, i) \in \beta$:

$$\alpha +_{\mathbb{Z}} \beta = [(n + k,\ j + i)].$$

Lemma 2.1 guarantees independence of representatives.

Theorem 4.2.6 (Mendelson 2.2 -- Properties of addition). *For all* $\alpha, \beta, \gamma \in \mathbb{Z}$*:*

(i) Commutativity: $\alpha +_{\mathbb{Z}} \beta = \beta +_{\mathbb{Z}} \alpha$*.*

(ii) Associativity: $\alpha +_{\mathbb{Z}} (\beta +_{\mathbb{Z}} \gamma) = (\alpha +_{\mathbb{Z}} \beta) +_{\mathbb{Z}} \gamma$*.*

(iii) Additive identity: $\alpha +_{\mathbb{Z}} 0_{\mathbb{Z}} = \alpha$*.*

(iv) Unique additive inverse: $\exists!\, \delta \in \mathbb{Z}$ *such that* $\alpha +_{\mathbb{Z}} \delta = 0_{\mathbb{Z}}$*.*

Remark 4.2.7 (Uniqueness of the inverse). Part (iv) asserts both existence *and* uniqueness of $-\alpha$. Tao's P4.1.6 asserts existence only; uniqueness follows later from C4.1.9. Mendelson builds uniqueness into the statement.

Lemma 4.2.8 (Mendelson 2.3 -- Multiplication well-defined). *If* $(n, j) \sim (n_1, j_1)$ *and* $(k, i) \sim (k_1, i_1)$*, then* $(nk + ji,\ jk + ni) \sim (n_1k_1 + j_1i_1,\ j_1k_1 + n_1i_1)$*.*

Definition 3.4 (Multiplication)

For $\alpha, \beta \in \mathbb{Z}$ with representatives $(n, j) \in \alpha$ and $(k, i) \in \beta$:

$$\alpha \times_{\mathbb{Z}} \beta = [(nk + ji,\ jk + ni)].$$

Lemma 2.3 guarantees well-definedness.

Theorem 4.2.9 (Mendelson 2.4 -- Properties of multiplication). *For all* $\alpha, \beta, \gamma \in \mathbb{Z}$*:*

(i) Commutativity: $\alpha \times_{\mathbb{Z}} \beta = \beta \times_{\mathbb{Z}} \alpha$*.*

(ii) Associativity: $\alpha \times_{\mathbb{Z}} (\beta \times_{\mathbb{Z}} \gamma) = (\alpha \times_{\mathbb{Z}} \beta) \times_{\mathbb{Z}} \gamma$*.*

(iii) Distributivity: $\alpha \times_{\mathbb{Z}} (\beta +_{\mathbb{Z}} \gamma) = (\alpha \times_{\mathbb{Z}} \beta) +_{\mathbb{Z}} (\alpha \times_{\mathbb{Z}} \gamma)$*.*

(iv) Multiplicative identity: $\alpha \times_{\mathbb{Z}} 1_{\mathbb{Z}} = \alpha$*.*

(v) No zero divisors: $\alpha \neq 0_{\mathbb{Z}} \wedge \beta \neq 0_{\mathbb{Z}} \Rightarrow \alpha \times_{\mathbb{Z}} \beta \neq 0_{\mathbb{Z}}$*.*

Remark 4.2.10 (Structural consequence). Theorems 2.2 and 2.4 establish that $(\mathbb{Z}, +_{\mathbb{Z}}, \times_{\mathbb{Z}})$ is an **integral domain**: a commutative ring with unit and no zero divisors. Mendelson makes this explicit in §3.3 via abstract definitions.

ğ3.3 Rings and Integral Domains (Abstract Theory)

Remark 4.2.11 (Abstract vs concrete). Mendelson develops ring theory abstractly before applying it to $\mathbb{Z}$. Tao works concretely throughout, naming the ring structure only in Remark 4.1.7.

Definition 3.5 (Ring; integral domain)

A **ring** $(R, +, \times)$: $(R, +)$ is an abelian group; $\times$ is associative; $\times$ distributes over $+$ on both sides. A ring is **commutative** if $xy = yx$; has a **unit** if $\exists 1$ with $x \cdot 1 = 1 \cdot x = x$. A nonzero x is a **zero divisor** if $\exists$ nonzero y with $xy = 0$. An **integral domain** is a commutative ring with unit, $0 \neq 1$, and no zero divisors.

Theorem 4.2.12 (Mendelson 3.3 -- Cancellation $\Leftrightarrow$ no zero divisors). *In a commutative ring with unit:*

$$(\forall x, y, z : \ xy = xz \wedge x \neq 0 \Rightarrow y = z) \iff \textit{no zero divisors.}$$

Theorem 4.2.13 (Mendelson 3.4 -- Trivial ring). *In a ring with unit:* $0 = 1 \iff R$ *is a singleton.*

ğ4 Ordered Integral Domains and Order on $\mathbb{Z}$

Remark 4.2.14 (Strategy). Mendelson proves order theory for any integral domain first, then instantiates it for $\mathbb{Z}$. This gives order on $\mathbb{Q}$ and $\mathbb{R}$ for free later.

Definition 4.1 (Ordered integral domain)

$(R, +, \times, <)$ is an **ordered integral domain** if $(R, +, \times)$ is an integral domain $(0 \neq 1)$ and $<$ satisfies:

(O1) $x \not< x$ *Irreflexivity*

(O2) $x < y \wedge y < z \Rightarrow x < z$ *Transitivity*

(O3) $x < y \vee x = y \vee y < x$ *Trichotomy*

(O4) $x < y \Rightarrow x + z < y + z$ *Addition-monotone*

(O5) $x < y \wedge 0 < z \Rightarrow xz < yz$ *Positive-mult-monotone*

x is **positive** if $0 < x$; **negative** if $x < 0$.

Theorem 4.2.15 (Mendelson 4.1 -- Consequences of order axioms). *In any ordered integral domain: (i) exactly one of* $x < y$*,* $x = y$*,* $y < x$ *holds; (ii)* $x < y \wedge u < v \Rightarrow x + u < y + v$*; (iii)* $0 < z \wedge xz < yz \Rightarrow x < y$*.*

Theorem 4.2.16 (Mendelson 4.2 -- Order and positivity). *In any ordered integral domain: (i)* $x < y \iff y - x$ *positive; (ii)* $x < y \iff x - y$ *negative; (iii)* $x < y \iff -y < -x$*; (iv) sum and product of positives are positive; (v) product of two negatives is positive.*

Theorem 4.2.17 (Mendelson 4.3 -- Positivity set construction). *Let $(R, +, \times)$ be an integral domain, $0 \neq 1$. If $\mathcal{P} \subseteq R$ satisfies: $0 \notin \mathcal{P}$; $\forall x : x \in \mathcal{P} \vee x = 0 \vee -x \in \mathcal{P}$; $\mathcal{P}$ closed under $+$ and $\times$; then $(R, +, \times, <)$ with $x < y \iff y - x \in \mathcal{P}$ is an ordered integral domain and $\mathcal{P}$ is its positive set.*

Remark 4.2.18 (Why this matters). Theorem 4.3 reduces verifying five order axioms to four positivity conditions. Mendelson uses it to order $\mathbb{Z}$ via $\mathcal{P}_{\mathbb{Z}}$ without checking (O1)-(O5) directly.

Lemma 4.2.19 (Mendelson 4.4 -- Positivity is class-invariant). *For $\alpha \in \mathbb{Z}$ and any two representatives $(n, j), (k, i) \in \alpha$: $j < n \iff i < k$.*

Definition 4.5 (Positivity set $\mathcal{P}_{\mathbb{Z}}$)

$$\mathcal{P}_{\mathbb{Z}} = \{\alpha \in \mathbb{Z} : \forall (n, j) \in \alpha,\ j < n\}.$$

Lemma 4.4 ensures this is well-defined. Intuitively: α is positive iff its representative has second coordinate strictly less than first, i.e. $n - j > 0$.

Lemma 4.2.20 (Mendelson 4.5). *$\mathcal{P}_{\mathbb{Z}}$ satisfies the four conditions of Theorem 4.3.*

Corollary 4.2.21 (Mendelson 4.6). *$(\mathbb{Z}, +_{\mathbb{Z}}, \times_{\mathbb{Z}}, <_{\mathbb{Z}})$ is an ordered integral domain with positive set $\mathcal{P}_{\mathbb{Z}}$.*

Theorem 4.2.22 (Mendelson 4.7 -- Recovery of Peano system). *Let $T(x) = x +_{\mathbb{Z}} 1_{\mathbb{Z}}$ for $x \in \mathcal{P}_{\mathbb{Z}}$. Then $(\mathcal{P}_{\mathbb{Z}}, T, 1_{\mathbb{Z}})$ is a Peano system.*

Remark 4.2.23 (Significance). Starting from a Peano system for $\mathbb{N}$, we construct $\mathbb{Z}$, and then prove the positive integers *inside* $\mathbb{Z}$ form a new Peano system consistent with the original. Tao does not prove this explicitly.

ğ4 Absolute Value

Definition 4.6 (Absolute value)

In any ordered integral domain: $|x| = x$ if $0 \leq x$; $|x| = -x$ if $x < 0$.

Theorem 4.2.24 (Mendelson 4.8 -- Properties of absolute value). *In any ordered integral domain, $|\cdot|$ satisfies: (1) $|x| \geq 0$; (2) $|x| = 0 \iff x = 0$; (3) $|-x| = |x|$; (4) $|x - y| = |y - x|$; (5) $|xy| = |x||y|$; (6) $-|x| \leq x \leq |x|$; (7) $|z| < u \iff -u < z < u$; (8) $|z| \leq u \iff -u \leq z \leq u$; (9) $u \geq v \wedge u \geq -v \Rightarrow u \geq |v|$; (10) $|x + y| \leq |x| + |y|$ (triangle inequality); (11) $|x - y| \geq \big||x| - |y|\big|$ (reverse triangle inequality).*

ğ5 Division, Divisibility, Primes

Remark 4.2.25 (Scope). This material is not in Tao Ch 4 and is not a prerequisite for real analysis. It is valuable background for number theory and abstract algebra. See the comparison table for coverage decisions.

Theorem 4.2.26 (Mendelson 5.1 -- Euclidean division). *For any integer $\alpha > 1$ and any integer β, there exist unique integers q, r with $\beta = q\alpha + r$ and $0 \le r < \alpha$.*

Definition 4.2.27 (Divisibility). $\alpha \mid \beta \iff \exists \gamma \in \mathbb{Z} : \beta = \alpha\gamma$.

Theorem 4.2.28 (Mendelson 5.2-5.9). *Standard divisibility properties, existence and Bézout form of gcd, characterisation of relative primeness, infinitude of primes (Euclid), Euclid's lemma ($\rho \mid \alpha\beta \Rightarrow \rho \mid \alpha$ or $\rho \mid \beta$), and the fundamental theorem of arithmetic (unique prime factorisation up to order and sign).*

ğ6 Integers Modulo n (Optional)

Remark 4.2.29 (Scope). Congruence mod n as an equivalence relation, $\mathbb{Z}_n$ as a commutative ring with unit; integral domain iff n is prime. Not required for the real analysis track.

ğ7-9 Integer Action, Embedding Theory, Uniqueness of $\mathbb{Z}$

Remark 4.2.30 (Scope). Chapters 7-9 are graduate-level algebra. Key results: **Thm 8.8**: every characteristic-0 integral domain contains a copy of $\mathbb{Z}$. **Thm 9.5**: $\mathcal{D} \cong \mathbb{Z}$ iff characteristic 0 and no proper subdomains. **Thm 9.8**: any two well-ordered integral domains are isomorphic -- $\mathbb{Z}$ is unique up to isomorphism. These results are beyond what is needed for real analysis but are worth knowing as statements.

Definition 4.2.31 (Well-ordered integral domain). An ordered integral domain is **well-ordered** if every nonempty subset of its positive elements contains a least element.

Theorem 4.2.32 (Mendelson 9.8 -- Uniqueness of $\mathbb{Z}$). *Any two well-ordered integral domains are isomorphic. In particular, $\mathbb{Z}$ is, up to isomorphism, the unique well-ordered integral domain.*

4.3 Tao vs. Mendelson: Comparison Table

Tao vs. Mendelson: Integer Theory Statement-by-Statement Comparison

Remark (How to read this table). Each row is one mathematical concept. **Coverage** recommendations: **Core** = do both; **Tao only** = Tao suffices, Mendelson adds abstraction only; **Men. only** = Mendelson covers this, Tao does not; **Skip** = beyond real analysis scope for now.

Block 1: The Construction of $\mathbb{Z}$

Concept	Tao	Mendelson	Same idea?	Coverage
Formal pairs as integers	Def 4.1.1: $a—b$, $a—b = c—d \iff a+d = c+b$	Def 3.1–3.2: $(n,j) \sim (k,i) \iff n+i = k+j$; $\mathbb{Z}$ = equiv. classes	Yes identical relation, different notation	**Core**
Equality is an equivalence relation	Ex 4.1.1 (reflexivity, symmetry); transitivity proved in text	Thm 1.1 (all three parts)	Yes	**Core**
Distinguished elements $0, 1$	$0 = 0—0$, $1 = 1—0$ (implicit in Def 4.1.1)	$0_{\mathbb{Z}} = [(1,1)]$, $1_{\mathbb{Z}} = [(2,1)]$ (explicit definition)	Yes same objects	**Core**
Embedding $\mathbb{N} \hookrightarrow \mathbb{Z}$	Remark after Def 4.1.2: $n \equiv n—0$	Thm 4.7 + Cor 9.7 (recovered as Peano system)	Tao informal; Mendelson proves formally	**Core**

Block 2: Addition

Concept	Tao	Mendelson	Same?	Coverage
Addition formula	Def 4.1.2: $(a—b)+(c—d) := (a+c)—(b+d)$	Def 3.3 (after Lem 2.1): $\alpha +_{\mathbb{Z}} \beta = [(n+k, j+i)]$	Yes	**Core**
Well-definedness of addition	L4.1.3 (combines add. and mult.)	Lem 2.1 (addition separately)	Yes Mendelson separates the two	**Core**
Commutativity	P4.1.6 (i)	Thm 2.2 (i)	Yes	**Core**
Associativity	P4.1.6 (ii)	Thm 2.2 (ii)	Yes	**Core**
Additive identity	P4.1.6 (iii): $x+0 = 0+x = x$	Thm 2.2 (iii): $\alpha + 0_{\mathbb{Z}} = \alpha$	Yes	**Core**
Additive inverse (negation)	Def 4.1.4: $-(a—b) := b—a$; P4.1.6 (iv) existence	Thm 2.2 (iv): unique δ with $\alpha + \delta = 0$	Tao: existence; Men.: existence + uniqueness in one statement	**Core**
Negation well-defined	Ex 4.1.2	(Implicit in Lem 2.1 / Thm 2.2)	Yes Tao makes it an explicit exercise	**Core**

Block 3: Multiplication

Concept	Tao	Mendelson	Same?	Coverage
Multiplication formula	Def 4.1.2: $(a—b)(c—d) := (ac+bd)—(ad+bc)$	Def 3.4 (after Lem 2.3): $\alpha \times_{\mathbb{Z}} \beta = [(nk + ji, jk + ni)]$	Yes	**Core**
Well-definedness of multiplication	L4.1.3	Lem 2.3	Yes	**Core**
Commutativity	P4.1.6 (v)	Thm 2.4 (i)	Yes	**Core**
Associativity	P4.1.6 (vi); proved in text as model calculation	Thm 2.4 (ii)	Yes	**Core**
Multiplicative identity	P4.1.6 (vii)	Thm 2.4 (iv)	Yes	**Core**
Distributive law	P4.1.6 (viii, ix)	Thm 2.4 (iii)	Yes	**Core**
$(-1) \times a = -a$	Ex 4.1.3	Not stated separately (follows from ring axioms)	Tao only as exercise	**Core (Tao)**
No zero divisors	P4.1.8	Thm 2.4 (v)	Yes	**Core**
Cancellation law	C4.1.9	Thm 3.3 (as equivalence, in any comm. ring with unit)	Mendelson more general	**Core**
$\mathbb{Z}$ is a commutative ring / integral domain	Rem 4.1.7 (named but not defined abstractly)	Def 3.5 + Thms 2.2, 2.4 (built up formally)	Mendelson makes it explicit	**Core**

Block 4: Order

Concept	Tao	Mendelson	Same?	Coverage
Order definition	Def 4.1.10: $n \geq m \iff n = m + a$, $a \in \mathbb{N}$	Def 4.1 (abstract OID axioms) + $<_{\mathbb{Z}}$ via $\mathcal{P}_{\mathbb{Z}}$ (Cor 4.6)	Same result; Mendelson builds via positivity set	**Core**
Order irreflexivity	(Implicit in Def 4.1.10 and L4.1.11(f))	Explicitly O1 in Def 4.1	Men. more explicit	**Core**
Trichotomy of integers	L4.1.5: every integer is positive, zero, or negative (one only)	Thm 4.1(i) + Def 4.1 O3	Yes	**Core**
$a > b \iff a - b$ positive	L4.1.11(a)	Thm 4.2(i)	Yes	**Core**
Addition preserves order	L4.1.11(b)	Def 4.1 O4 (axiom); Thm 4.1(ii) (addition of inequalities)	Yes	**Core**
Positive mult. preserves order	L4.1.11(c)	Def 4.1 O5 (axiom)	Yes	**Core**
Negation reverses order	L4.1.11(d)	Thm 4.2(iii)	Yes	**Core**
Transitivity	L4.1.11(e)	Def 4.1 O2 (axiom)	Yes	**Core**
Order trichotomy (strict)	L4.1.11(f)	Thm 4.1(i)	Yes	**Core**
Positivity set construction (Thm 4.3)	—	Thm 4.3: $\mathcal{P}$ satisfying 4 conditions $\Rightarrow$ OID	Men. only	**Men. only** (worth knowing)
Product of negatives is positive	(Derivable from P4.1.6 + L4.1.11)	Thm 4.2(v) (stated explicitly)	Tao implicit; Men. explicit	**Core**
Recovery of Peano system in $\mathbb{Z}$	—	Thm 4.7: $(\mathcal{P}_{\mathbb{Z}}, T, 1_{\mathbb{Z}})$ is a Peano system	Men. only	**Men. only** (important conceptually)

Block 5: Absolute Value

Concept	Tao	Mendelson	Same?	Coverage
Definition of $\lvert x\rvert$	(Defined for rationals in Def 4.3.1, not integers separately)	Def 4.6 (for any ordered integral domain)	Men. gives it for $\mathbb{Z}$ directly; Tao defers to $\mathbb{Q}$	**Men. only** (do it here)
Triangle inequality $\lvert x+y\rvert \leq \lvert x\rvert + \lvert y\rvert$	P4.3.3(b) (for rationals)	Thm 4.8(10)	Yes Mendelson proves it for $\mathbb{Z}$	**Core**
Multiplicativity $\lvert xy\rvert = \lvert x\rvert\lvert y\rvert$	P4.3.3(d) (for rationals)	Thm 4.8(5)	Yes	**Core**
Reverse triangle inequality	(Implicit)	Thm 4.8(11) explicitly	Men. explicit	**Core**
Full 11-part absolute value theorem	Spread across P4.3.3 (for $\mathbb{Q}$)	Thm 4.8 (complete, for any OID)	Men. more systematic	**Men. only** (do once, use forever)

Block 6: Induction and Structural Observations

Concept	Tao	Mendelson	Same?	Coverage
Induction fails for $\mathbb{Z}$	Ex 4.1.8: explicit counterexample	Implied by lack of least element; Thm 9.6 addresses positives only	Tao makes it an explicit exercise	**Core (Tao)**
Subtraction definition	$x - y := x + (-y)$ (after Def 4.1.4)	Standard (follows from additive inverse)	Yes	**Core**
$0 \neq 1$ in $\mathbb{Z}$	Implicit (P2.2.8 + L4.1.5)	Thm 3.4 (explicit: $0 = 1 \iff$ trivial ring)	Men. explicit	**Core**

Block 7: Material in Mendelson Only — Coverage Decision Required

Concept	Mendelson ref.	Pre-req for RA?	Recommendation
Abstract ring and integral domain definitions	ğ3.3, Def 3.5	No (but clarifying)	Include notes; no proof sheets needed
Cancellation ⇔ no zero divisors (abstract)	Thm 3.3	No	Notes only; proof optional
Trivial ring theorem ($0 = 1 \Rightarrow$ singleton)	Thm 3.4	No	Skip proof; note the statement
Abstract ordered integral domain axioms	Def 4.1	Indirectly (same axioms used for $\mathbb{R}$)	**Include** these axioms recur throughout real analysis
Positivity set construction (Thm 4.3)	Thm 4.3	No (used internally)	Notes only elegant but not exercised directly
Peano recovery (Thm 4.7)	Thm 4.7	No	Notes only; important for logical completeness
Euclidean division (ğ5, Thm 5.1)	Thm 5.1	No	**Proof sheet** classical and useful
Divisibility properties (ğ5, Thm 5.2)	Thm 5.2 (11 parts)	No	**Proof sheet** good algebraic practice
GCD existence and Bézout (Thm 5.3)	Thm 5.3	No	**Proof sheet** important
Infinitude of primes (Thm 5.6)	Thm 5.6	No	**Proof sheet** Euclid's classic argument
Fundamental theorem of arithmetic (Thm 5.9)	Thm 5.9	No	**Proof sheet** core number theory
Modular arithmetic (ğ6)	Thms 6.1–6.6	No	**Skip** for now; revisit in algebra track
Integer action on domains (ğ7, Thms 7.1–7.4)	Thms 7.1–7.4	No	**Skip** graduate algebra
$N_{\mathcal{D}}$, $Z_{\mathcal{D}}$ embedding theory (ğ8)	Thms 8.1–8.9	No	**Skip** graduate algebra
Subdomain theory (ğ9, Thms 9.1–9.4)	Thms 9.1–9.4	No	**Skip** graduate algebra
Characterisation of $\mathbb{Z}$ (Thm 9.5)	Thm 9.5	No	Notes only worth knowing as a statement
Well-ordered integral domain (Def, Cor 9.7)	Def + Cor 9.7	No	Notes only
Uniqueness of $\mathbb{Z}$ up to isomorphism (Thm 9.8)	Thm 9.8	No	**Note the statement** philosophically important

Summary: Recommended Coverage

Core (both sources): Equality, addition, multiplication, well-definedness, ring laws, no zero divisors, cancellation, order, trichotomy, six order properties.

Mendelson adds (do notes, consider proof sheets): Abstract ring/integral domain definitions (§3.3); abstract OID axioms (Def 4.1) — these recur in real analysis; absolute value on $\mathbb{Z}$ (Thm 4.8) — Tao defers to $\mathbb{Q}$; positivity set construction (Thm 4.3) — elegant technique; Peano recovery (Thm 4.7) — logical closure; number theory: Euclidean division, divisibility, GCD, primes (ğ5).

Skip for now (graduate algebra): Modular arithmetic (ğ6), integer action on domains (ğ7), $N_{\mathcal{D}}/Z_{\mathcal{D}}$ embedding theory (ğ8–9). Know the statements of Thms 9.5 and 9.8 (uniqueness of $\mathbb{Z}$).

4.4 Proofs

Chapter 5

Rational Numbers ($\mathbb{Q}$)

rationals

Integers ($\mathbb{Z}$) $\rightarrow$ **Rationals ($\mathbb{Q}$)** $\rightarrow$ Real Numbers ($\mathbb{R}$)

Structural Roadmap

How we got here. The integers gave us additive inverses, so subtraction became internal to the number system. Division still fails: for example, $1 \div 2$ is not an integer. The rational numbers adjoin multiplicative inverses for all nonzero integers.

What this chapter develops. We construct $\mathbb{Q}$ from $\mathbb{Z}$ using equivalence classes of ordered pairs, prove that $\mathbb{Q}$ is an ordered field, study its density and approximation properties, identify the problems that $\mathbb{Q}$ cannot resolve, and then see why $\mathbb{Q}$ is still incomplete.

Where this leads. The rational numbers are already rich enough to support most of arithmetic and much of algebra, but not rich enough to capture all limits. That failure motivates the construction of $\mathbb{R}$.

Chapter Roadmap -- The Rational Numbers

This chapter develops the rational numbers in stages. At each stage, the new structure is introduced and then the theorems proving that the structure works are placed immediately after it.

Toolkit — Rational Number Development

equivalence classes $\longrightarrow$ arithmetic $\longrightarrow$ field $\longrightarrow$ order $\longrightarrow$ ordered field $\longrightarrow$ approximation $\longrightarrow$ gaps

1. Construct rational numbers as equivalence classes of integer pairs.
2. Define arithmetic using representatives and prove well-definedness.
3. Prove the algebraic laws of rational arithmetic.
4. Package those laws by proving that $\mathbb{Q}$ is a field.

5. Define positivity and order on $\mathbb{Q}$.
6. Prove that $\mathbb{Q}$ is an ordered field.
7. Develop linear approximation: density, dyadics, small denominators, mediants, Farey sequences, and Stern–Brocot.
8. Develop nonlinear approximation: square bracketing and rational square approximations.
9. Convert repeated approximation into rational sequences and Cauchy behavior.
10. Exhibit rational gaps, especially the square-root gap at 2.
11. Transition to the real numbers as the completion of rational arithmetic and order.

5.1 Constructing Rational Numbers as Equivalence Classes

5.1.1 Definitions and Theorems

The rational numbers are built from the integers in essentially the same way that the integers were built from the natural numbers: one begins with concrete representatives, identifies those that ought to represent the same object, and then passes to equivalence classes.

Definition 5.1.1 (Fraction pairs). Let $\mathbb{Z}$ denote the set of integers and define

$$W = \mathbb{Z} \times (\mathbb{Z} \setminus \{0\}).$$

Thus

$$W = \{(a, b) : a, b \in \mathbb{Z},\ \ b \neq 0\}.$$

Each ordered pair (a, b) is intended to represent the formal fraction a/b.

Remark 5.1.2 (Numerator and denominator). If $(a, b) \in W$, then a is called the *numerator* and b is called the *denominator*. At this stage (a, b) is still only an ordered pair; it is not yet a rational number.

Remark 5.1.3. Different pairs should represent the same rational number. For example,

$$(1, 2), \quad (2, 4), \quad (3, 6)$$

all ought to describe the same quantity. To make this precise we introduce an equivalence relation.

Definition 5.1.4 (Equality of fraction pairs). For $(a, b), (c, d) \in W$, define

$$(a, b) \sim (c, d) \quad \Longleftrightarrow \quad ad = bc.$$

In fully quantified form,

$$\forall (a, b), (c, d) \in W, \qquad (a, b) \sim (c, d) \Longleftrightarrow ad = bc.$$

Remark 5.1.5. This is the familiar cross-multiplication test for equality of fractions. For example,

$$(1, 2) \sim (2, 4)$$

because

$$1 \cdot 4 = 2 \cdot 2.$$

Theorem 5.1.6 (The Fraction-Pair Relation Is an Equivalence Relation). *The relation $\sim$ is an equivalence relation on W.*

Remark (Proof). Go to proof.

Remark 5.1.7. Once this is known, the equivalence classes of $\sim$ partition W. Each class is what we shall call a rational number.

Definition 5.1.8 (Rational numbers). The set of *rational numbers* is the quotient

$$\mathbb{Q} = W/\sim.$$

If $(a, b) \in W$, its equivalence class is denoted by

$$[(a, b)].$$

We interpret $[(a, b)]$ as the rational number represented informally by a/b.

Definition 5.1.9 (Distinguished elements). Define

$$0 := [(0, 1)] \qquad \text{and} \qquad 1 := [(1, 1)].$$

Remark 5.1.10. These definitions are well-defined because

$$(0, 1) \sim (0, b) \quad (b \neq 0) \qquad \text{and} \qquad (1, 1) \sim (a, a) \quad (a \neq 0).$$

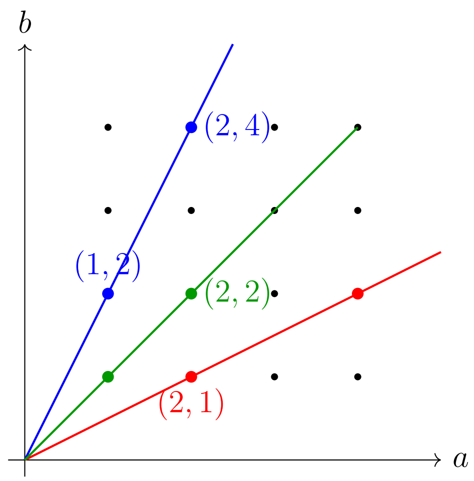

Figure 5.1: Integer pairs representing the same rational number lie on the same line through the origin.

5.2 Defining Rational Arithmetic from Representatives

This section begins where the quotient construction leaves off: operations are first written using representatives, and then their well-definedness is proved so that they descend to genuine operations on rational numbers.

Definition 5.2.1 (Addition and multiplication). Let $[(a,b)], [(c,d)] \in \mathbb{Q}$, where $(a,b), (c,d) \in W$. Define

$$[(a,b)] + [(c,d)] := [(ad+bc, bd)]$$

and

$$[(a,b)] \cdot [(c,d)] := [(ac, bd)].$$

Lemma 5.2.2 (Addition Is Well-Defined on Rational Classes). *Let $a,b,c,d,e,f,g,h \in \mathbb{Z}$ with $bdfh \neq 0$. If*

$$(a,b) \sim (e,f) \qquad \textit{and} \qquad (c,d) \sim (g,h),$$

then

$$(ad+bc, bd) \sim (eh+fg, fh).$$

Remark (Proof). Go to proof.

Lemma 5.2.3 (Multiplication Is Well-Defined on Rational Classes). *Let $a,b,c,d,e,f,g,h \in \mathbb{Z}$ with $bdfh \neq 0$. If*

$$(a,b) \sim (e,f) \qquad \textit{and} \qquad (c,d) \sim (g,h),$$

then

$$(ac, bd) \sim (eg, fh).$$

Remark (Proof). Go to proof.

Remark 5.2.4 (Why well-definedness matters). These formulas are written in terms of representatives. Because one rational number has many representatives, it must be proved that choosing different representatives leads to the same resulting equivalence class. This is the first serious instance of a very common pattern in mathematics:

representatives $\longrightarrow$ operations $\longrightarrow$ well-defined operations on equivalence classes.

Remark 5.2.5 (Three levels of representation). When working with rational numbers, it is essential to distinguish among the following three objects.

Object	**Lives in**	**Meaning**
(a,b)	W	raw integer pair
$[(a,b)]$	$\mathbb{Q}$	rational number
$\frac{a}{b}$	notation	informal representation of $[(a,b)]$

The pair (a,b) is not itself a rational number. The rational number is the entire equivalence class

$$[(a,b)] = \{(c,d) \in W : ad = bc\}.$$

Remark 5.2.6 (A rational number is an infinite set). For every nonzero integer k,

$$(ka, kb) \sim (a, b).$$

Hence the class $[(a, b)]$ contains infinitely many representatives:

$$(a, b), (2a, 2b), (3a, 3b), \ldots, (-a, -b), (-2a, -2b), \ldots$$

Thus every rational number is an infinite set.

Remark 5.2.7 (Canonical representative). Although a rational number has infinitely many representatives, each nonzero rational number can be written uniquely in the form

$$\frac{a}{b} \qquad \text{with} \qquad \gcd(a, b) = 1, \quad b > 0.$$

This is the *reduced form* of the rational number.

5.2.1 Consequences

Remark 5.2.8 (Quotient structures). The construction of $\mathbb{Q}$ is the first major example in these notes of the general pattern

$$\text{set} + \text{equivalence relation} \longrightarrow \text{quotient structure}.$$

The same pattern later appears in modular arithmetic, quotient groups, quotient rings, quotient spaces, and projective geometry.

5.2.2 Algebra of Equivalence Classes

Remark 5.2.9 (Role of the equivalence-class arithmetic theorems). The theorems in this subsection work entirely at the level of equivalence classes and integer arithmetic. They require nothing beyond Definition 5.2.1, the well-definedness lemmas, and facts about $\mathbb{Z}$.

In particular, Theorem 5.2.15 and Theorem 5.2.17 do not appeal to field axioms. They *establish*, by direct computation, that $[(-a, b)]$ and $[(b, a)]$ satisfy the defining conditions for additive and multiplicative inverses respectively. These facts are therefore *inputs to* the proof that $\mathbb{Q}$ is a field, not consequences of it. The field proof can cite them rather than repeat the computation.

The later rational algebra section re-expresses the same identities using fraction notation a/b inside the already-established field. Those re-derivations are about *notation and usability*, not about new mathematical content.

Theorem 5.2.10 (Integer embedding preserves addition). *For all $m, n \in \mathbb{Z}$,*

$$[(m, 1)] + [(n, 1)] = [(m + n,\ 1)].$$

Remark (Proof). Go to proof.

Theorem 5.2.11 (Integer embedding preserves multiplication). *For all $m, n \in \mathbb{Z}$,*

$$[(m, 1)] \cdot [(n, 1)] = [(mn,\ 1)].$$

Remark (Proof). Go to proof.

Remark 5.2.12 (The integer embedding). These two theorems establish that the map $n \mapsto [(n,1)]$ is a ring homomorphism from $\mathbb{Z}$ into $\mathbb{Q}$: it sends the integer operations to the rational operations. Moreover $[(m,1)] = [(n,1)]$ if and only if $m = n$, so the map is injective and $\mathbb{Z}$ sits inside $\mathbb{Q}$ as a subring. We therefore identify each integer n with the rational class $[(n,1)]$ and write n for $[(n,1)]$ without further comment.

Theorem 5.2.13 (Zero class criterion). *For every $[(a,b)] \in \mathbb{Q}$,*

$$[(a,b)] = [(0,1)] \quad \Longleftrightarrow \quad a = 0.$$

Remark (Proof). Go to proof.

Theorem 5.2.14 (Scaling of equivalence classes). *For every $[(a,b)] \in \mathbb{Q}$ and every nonzero integer $e \neq 0$,*

$$[(a,b)] = [(ae, be)].$$

Remark (Proof). Go to proof.

Theorem 5.2.15 (Negation of an equivalence class). *For every $[(a,b)] \in \mathbb{Q}$,*

$$-[(a,b)] = [(-a,b)] = [(a,-b)].$$

Remark (Proof). Go to proof.

Theorem 5.2.16 (Subtraction of equivalence classes). *For every $[(a,b)], [(c,d)] \in \mathbb{Q}$,*

$$[(a,b)] - [(c,d)] = [(ad - bc,\, bd)].$$

Remark (Proof). Go to proof.

Theorem 5.2.17 (Reciprocal of a nonzero equivalence class). *For every $[(a,b)] \in \mathbb{Q}$ with $a \neq 0$,*

$$[(a,b)]^{-1} = [(b,a)].$$

Remark (Proof). Go to proof.

5.2.3 Fraction Arithmetic Formulas in $\mathbb{Q}$

After representative-level addition, multiplication, negation, and reciprocals have been defined, the usual fraction formulas can be stated as algebraic rules inside $\mathbb{Q}$. These results should be read as the familiar symbolic calculus that is justified by the preceding well-definedness theorems.

Definition 5.2.18 (Division and reciprocal in $\mathbb{Q}$). Let $a, b \in \mathbb{Q}$ with $b \neq 0$. We define

$$\frac{a}{b}$$

to be the unique rational number x such that

$$b \cdot x = a.$$

We also define the reciprocal of b by

$$b^{-1} := \frac{1}{b}.$$

Remark 5.2.19. This is not a new primitive operation. It is shorthand for solving the equation $b \cdot x = a$ inside the field $\mathbb{Q}$.

Theorem 5.2.20 (Division by One in $\mathbb{Q}$). *For every $a \in \mathbb{Q}$,*

$$\frac{a}{1} = a.$$

Remark (Proof). Go to proof.

Theorem 5.2.21 (A Rational Quotient Is Zero Exactly When Its Numerator Is Zero). *For every $a, b \in \mathbb{Q}$ with $b \neq 0$,*

$$\frac{a}{b} = 0 \iff a = 0.$$

Remark (Proof). Go to proof.

Theorem 5.2.22 (Cross-Multiplication Criterion for Rational Equality). *For every $a, b, c, d \in \mathbb{Q}$ with $b \neq 0$ and $d \neq 0$,*

$$\frac{a}{b} = \frac{c}{d} \iff a \cdot d = b \cdot c.$$

Remark (Proof). Go to proof.

Theorem 5.2.23 (Scaling Numerator and Denominator by the Same Nonzero Rational). *For every $a, b, e \in \mathbb{Q}$ with $b \neq 0$ and $e \neq 0$,*

$$\frac{a}{b} = \frac{a \cdot e}{b \cdot e}.$$

Remark (Proof). Go to proof.

Theorem 5.2.24 (Fraction Addition Rule in $\mathbb{Q}$). *For every $a, b, c, d \in \mathbb{Q}$ with $b \neq 0$ and $d \neq 0$,*

$$\frac{a}{b} + \frac{c}{d} = \frac{a \cdot d + b \cdot c}{b \cdot d}.$$

Remark (Proof). Go to proof.

Theorem 5.2.25 (Fraction Subtraction Rule in $\mathbb{Q}$). *For every $a, b, c, d \in \mathbb{Q}$ with $b \neq 0$ and $d \neq 0$,*

$$\frac{a}{b} - \frac{c}{d} = \frac{a \cdot d - b \cdot c}{b \cdot d}.$$

Remark (Proof). Go to proof.

Theorem 5.2.26 (Fraction Multiplication Rule in $\mathbb{Q}$). *For every $a, b, c, d \in \mathbb{Q}$ with $b \neq 0$ and $d \neq 0$,*

$$\frac{a}{b} \cdot \frac{c}{d} = \frac{a \cdot c}{b \cdot d}.$$

Remark (Proof). Go to proof.

Theorem 5.2.27 (Negation Rule for Rational Fractions). *For every $a, b \in \mathbb{Q}$ with $b \neq 0$,*

$$-\left(\frac{a}{b}\right) = \frac{-a}{b} = \frac{a}{-b}.$$

Remark (Proof). Go to proof.

Theorem 5.2.28 (Reciprocal of a Nonzero Rational Fraction). *For every* $a, b \in \mathbb{Q}$ *with* $a \neq 0$ *and* $b \neq 0$*,*

$$\left(\frac{a}{b}\right)^{-1} = \frac{b}{a}.$$

Remark (Proof). Go to proof.

Remark 5.2.29 (What this package gives). These identities are the algebraic rules that make fraction notation usable. They turn the abstract field structure of $\mathbb{Q}$ into the familiar symbolic calculus of rational expressions.

5.3 Algebraic Laws of Rational Arithmetic

The previous section proves that the operations on rational classes are well-defined. The next step is to prove that these operations obey the ordinary algebraic laws expected of fractions.

Remark (Existing proof inventory). The detailed algebraic law proofs are currently packaged in Field Structure. This section is the intended home for the representativ level proofs of the individual laws before they are bundled into the theorem that $\mathbb{Q}$ is a field.

Theorem (Rational Addition Is Commutative)

Theorem 5.3.1 (Rational Addition Is Commutative). *For all* $x, y \in \mathbb{Q}$*,*

$$x + y = y + x.$$

Remark (Status). Missing as an individual representative-level proof. It is included implicitly in Theorem 5.4.7.

Theorem (Rational Addition Is Associative)

Theorem 5.3.2 (Rational Addition Is Associative). *For all* $x, y, z \in \mathbb{Q}$*,*

$$(x + y) + z = x + (y + z).$$

Remark (Status). Missing as an individual representative-level proof. It is included implicitly in Theorem 5.4.7.

Theorem (Rational Additive Identity)

Theorem 5.3.3 (Rational Additive Identity). *For every* $x \in \mathbb{Q}$*,*

$$x + 0 = x = 0 + x.$$

Remark (Status). Missing as an individual representative-level proof. Expected representative: $0 = [(0, 1)]$.

Theorem (Rational Multiplication Is Commutative)

Theorem 5.3.4 (Rational Multiplication Is Commutative). *For all* $x, y \in \mathbb{Q}$,

$$xy = yx.$$

Remark (Status). Missing as an individual representative-level proof. It is included implicitly in Theorem 5.4.7.

Theorem (Rational Multiplication Is Associative)

Theorem 5.3.5 (Rational Multiplication Is Associative). *For all* $x, y, z \in \mathbb{Q}$,

$$(xy)z = x(yz).$$

Remark (Status). Missing as an individual representative-level proof. It is included implicitly in Theorem 5.4.7.

Theorem (Rational Multiplicative Identity)

Theorem 5.3.6 (Rational Multiplicative Identity). *For every* $x \in \mathbb{Q}$,

$$x \cdot 1 = x = 1 \cdot x.$$

Remark (Status). Missing as an individual representative-level proof. Expected representative: $1 = [(1, 1)]$.

Theorem (Rational Distributivity)

Theorem 5.3.7 (Rational Distributivity). *For all* $x, y, z \in \mathbb{Q}$,

$$x(y + z) = xy + xz.$$

Remark (Status). Missing as an individual representative-level proof. It is included implicitly in Theorem 5.4.7.

5.4 Field Structure

Once addition and multiplication are shown to be well-defined on equivalence classes of fraction pairs, the rational numbers become much more than a construction. They become an algebraic number system in their own right.

Definition 5.4.1 (Integral domain). A system

$$(F, +, \cdot, 0, 1)$$

is called an *integral domain* if:

1. $(F, +, 0)$ is a commutative group;
2. multiplication on F is associative and commutative;
3. $1 \neq 0$ and
$$\forall a \in F, \qquad 1 \cdot a = a = a \cdot 1;$$

4. multiplication distributes over addition;

5. F has no zero divisors, that is,

$$\forall a, b \in F, \qquad ab = 0 \Longrightarrow a = 0 \text{ or } b = 0.$$

Definition 5.4.2 (Field). A system

$$(F, +, \cdot, 0, 1)$$

is called a *field* if it is an integral domain and satisfies:

$$\forall a, b \in F, \qquad b \neq 0 \Longrightarrow \exists x \in F \ (b \cdot x = a).$$

Equivalently, every nonzero element of F has a multiplicative inverse.

Definition 5.4.3 (Subfield). Let

$$(E, +, \cdot, 0, 1) \qquad \text{and} \qquad (F, +, \cdot, 0, 1)$$

be fields with $E \subseteq F$. We say that E is a *subfield* of F if the operations and distinguished elements on E are the restrictions of those on F.

Definition 5.4.4 (Nonnegative integer exponents in a field). Let F be a field, let $x \in F$, and let $n \in \mathbb{N}$. Define

$$x^0 := 1 \qquad \text{and} \qquad x^{n+1} := x^n \cdot x.$$

In particular $x^1 = x$ and $x^2 = x \cdot x$.

Remark 5.4.5. This is the same recursive definition used for natural-number exponentiation in $\mathbb{N}$, now stated inside an arbitrary field. No inversion is required and the definition is valid for all $x \in F$, including $x = 0$.

Definition 5.4.6 (Negative integer exponents in a field). Let F be a field, let $x \in F$ with $x \neq 0$, and let $m \in \mathbb{N}$. Define

$$x^{-m} := (x^{-1})^m.$$

Theorem 5.4.7 ($\mathbb{Q}$ Is an Integral Domain). *With the operations of Definition 5.2.1, the system*

$$(\mathbb{Q}, +, \cdot, 0, 1)$$

is an integral domain.

Remark (Proof). Go to proof.

Remark 5.4.8. This theorem packages the algebraic facts that follow from the quotient construction: addition and multiplication are well-defined, the field laws hold except for multiplicative inverses of nonzero elements, and there are no zero divisors.

Theorem 5.4.9 ($\mathbb{Q}$ Is a Field). *With the operations of Definition 5.2.1, the system*

$$(\mathbb{Q}, +, \cdot, 0, 1)$$

is a field.

Remark (Proof). Go to proof.

Remark 5.4.10 (What structure is gained). To say that $\mathbb{Q}$ is a field means, in particular, that:

- addition and multiplication are closed on $\mathbb{Q}$;
- both operations are associative and commutative;
- 0 and 1 serve as additive and multiplicative identities;
- every rational has an additive inverse;
- every nonzero rational has a multiplicative inverse;
- multiplication distributes over addition.

Remark 5.4.11 (Why this matters). This is the point where division becomes internal to the number system. The integers allowed subtraction, but not division by arbitrary nonzero elements. The rationals repair exactly that failure.

Theorem 5.4.12 (Negative Exponents Agree with Inverse Powers). *Let F be a field, let $x \in F$ with $x \neq 0$, and let $m, n \in \mathbb{Z}$. Then*

$$x^{-m} = (x^{-1})^m = (x^m)^{-1}.$$

Moreover, the usual laws of exponents continue to hold for integer exponents.

Remark (Proof). Go to proof.

Theorem 5.4.13 (Power of a Product in a Field). *Let $(F, +, \cdot, 0, 1)$ be a field. Let $q, r \in F \setminus \{0\}$ and let $a \in \mathbb{Z}$. Then*

$$(qr)^a = q^a r^a.$$

Remark (Proof). Go to proof.

Theorem 5.4.14 (Integer Exponent Addition Law). *Let $(F, +, \cdot, 0, 1)$ be a field, let $q \in F \setminus \{0\}$, and let $a, b \in \mathbb{Z}$. Then*

$$q^a q^b = q^{a+b}.$$

Remark (Proof). Go to proof.

Theorem 5.4.15 (Power of a Power in a Field). *Let $(F, +, \cdot, 0, 1)$ be a field, let $q \in F \setminus \{0\}$, and let $a, b \in \mathbb{Z}$. Then*

$$(q^a)^b = q^{ab}.$$

Remark (Proof). Go to proof.

Remark 5.4.16 (Elementary consequences of field axioms). The following identities hold in any field. They are consequences of the field axioms alone and require no order structure. They are stated here because they are invoked silently throughout the rational algebra proofs; naming them gives those proofs clean citation targets.

Theorem 5.4.17 (Zero product). *Let F be a field and let $a \in F$. Then*
$$a \cdot 0 = 0.$$

Remark (Proof). Go to proof.

Theorem 5.4.18 (Double negation). *Let F be a field and let $a \in F$. Then*
$$-(-a) = a.$$

Remark (Proof). Go to proof.

Theorem 5.4.19 (Negation of a product). *Let F be a field and let $a, b \in F$. Then*
$$-(a \cdot b) = (-a) \cdot b = a \cdot (-b).$$

Remark (Proof). Go to proof.

Corollary 5.4.20 (Product of negatives). *Let F be a field and let $a, b \in F$. Then*
$$(-a) \cdot (-b) = a \cdot b.$$

Remark (Proof). Go to proof.

Theorem 5.4.21 (Additive cancellation). *Let F be a field and let $a, b, c \in F$. If*
$$a + c = b + c,$$
then $a = b$.

Remark (Proof). Go to proof.

Theorem 5.4.22 (Multiplicative cancellation). *Let F be a field and let $a, b, c \in F$ with $c \neq 0$. If*
$$a \cdot c = b \cdot c,$$
then $a = b$.

Remark (Proof). Go to proof.

Theorem 5.4.23 (Uniqueness of multiplicative inverse). *Let F be a field and let $a \in F$ with $a \neq 0$. If $b \in F$ satisfies $a \cdot b = 1$, then $b = a^{-1}$.*

Remark (Proof). Go to proof.

Theorem 5.4.24 (The inverse of one is one). *In any field,*
$$1^{-1} = 1.$$

Remark (Proof). Go to proof.

Remark 5.4.25 (Absolute value). An ordered field carries a natural notion of absolute value: for $x \in F$ define $|x| = x$ if $x \geq 0$ and $|x| = -x$ if $x < 0$. The full definition and its standard properties -- non-negativity, multiplicativity, and the triangle inequality -- are developed in the modulus chapter of Volume III once the order machinery is fully in place. All uses of $|\cdot|$ in the sequence and Cauchy sections below anticipate that treatment; the reader should treat them as shorthand for the piecewise definition above.

Remark 5.4.26 (What comes next). Field structure alone does not yet explain inequalities, positivity, or approximation. Those belong to the order story, which is why the next step is to endow $\mathbb{Q}$ with order axioms compatible with the field operations.

5.5 Defining the Order on $\mathbb{Q}$

The rational numbers are not merely a field. They also carry an order relation compatible with the algebraic operations, so that $\mathbb{Q}$ becomes an ordered field.

Remark 5.5.1. For many arguments it is cleaner to encode order using the set of positive elements rather than to take $<$ as primitive.

Definition 5.5.2 (Positive cone). Let F be a field. A subset $P \subset F$ is called a *positive cone* if it satisfies:

1. if $a, b \in P$, then $a + b \in P$;
2. if $a, b \in P$, then $ab \in P$;
3. for every $a \in F$, exactly one of the following holds:
$$a \in P, \qquad a = 0, \qquad -a \in P.$$

Define the order relation by
$$a < b \iff b - a \in P.$$

Remark 5.5.3. This viewpoint makes the interaction between algebra and order transparent. Inequalities can be translated into statements about positive elements and then handled algebraically.

Definition 5.5.4 (Ordered integral domain). An *ordered integral domain* is an integral domain
$$(F, +, \cdot, 0, 1)$$
together with a relation $<$ on F such that:

1. for all $a, b \in F$, exactly one of the relations
$$a < b, \qquad a = b, \qquad b < a$$
holds;
2. if $a < b$, then for every $c \in F$,
$$a + c < b + c;$$
3. if $a < b$ and $0 < c$, then
$$ac < bc.$$

Definition 5.5.5 (Ordered field). An *ordered field* is a field
$$(F, +, \cdot, 0, 1)$$
together with a relation $<$ on F making F into an ordered integral domain.

Definition 5.5.6 (Ordered subfield). Let E and F be ordered fields with $E \subseteq F$. We say that E is an *ordered subfield* of F if E is a subfield of F and the order on E is the restriction of the order on F.

Definition 5.5.7 (Densely ordered system). Let S be a set with a relation $<$. We say that $(S, <)$ is *densely ordered* if:

1. $(S, <)$ is a simply ordered system; and
2. for every $x, y \in S$ with $x < y$, there exists $z \in S$ such that
$$x < z < y.$$

5.6 The Ordered Field Structure of $\mathbb{Q}$

$\mathbb{Q}$ as an Ordered Field

Once the order has been defined on $\mathbb{Q}$, the main task is to prove that it is compatible with the field operations. This compatibility is captured by the ordered field axioms and then deployed to derive the basic sign-arithmetic laws that govern inequalities in $\mathbb{Q}$.

Theorem ($\mathbb{Q}$ Is an Ordered Field)

Theorem 5.6.1 ($\mathbb{Q}$ Is an Ordered Field). *With its usual order, the system*

$$(\mathbb{Q}, +, \cdot, 0, 1, <)$$

is an ordered field. *Go to proof.*

Remark (Standard quantified statement).

$$\begin{aligned}&(\mathbb{Q}, +, \cdot, 0, 1) \text{ is a field and } < \text{ is an order on } \mathbb{Q}\\&\text{such that for all } a, b, c \in \mathbb{Q}:\\&\quad (a < b \Longrightarrow a + c < b + c)\\&\quad \wedge\ (a < b \wedge 0 < c \Longrightarrow ac < bc).\end{aligned}$$

Remark (Interpretation). This theorem records the compatibility of the rational order with the field operations. It is this compatibility that makes inequalities in $\mathbb{Q}$ behave as expected under addition, subtraction, multiplication, and division by positive elements.

Remark (Dependencies).
- $\mathbb{Q}$ Is a Field
- Ordered Field
- Positive Cone

Sign-Arithmetic Consequences

From the ordered field axioms one derives the fundamental sign-arithmetic laws: squares are nonnegative, positive elements have positive inverses, multiplication by a positive element preserves order, and multiplication by a negative element reverses it. These four facts are the working toolkit for rational inequalities.

Theorem 5.6.2 (Squares Are Nonnegative). *In any ordered field F, for every $a \in F$,*

$$a^2 \geq 0.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall a \in F, \quad a^2 \geq 0.$$

Remark (Contrapositive quantified statement).

$$\forall a \in F, \quad a^2 < 0 \Longrightarrow \perp.$$

Remark (Interpretation). No element of an ordered field has a negative square. This immediately rules out the existence of $\sqrt{-1}$ inside any ordered field and explains why complex numbers require abandoning the order structure.

Remark (Dependencies).
- Ordered Field
- $\mathbb{Q}$ Is an Ordered Field

Theorem 5.6.3 (Positive Elements Have Positive Inverses). *In any ordered field, if $a > 0$, then $a^{-1} > 0$. Go to proof.*

Remark (Standard quantified statement).

$$\forall a \in F, \quad a > 0 \Longrightarrow a^{-1} > 0.$$

Remark (Contrapositive quantified statement).

$$\forall a \in F, \quad a^{-1} \leq 0 \Longrightarrow a \leq 0.$$

Remark (Interpretation). Dividing by a positive number cannot change the direction of an inequality. This is the formal basis for the rule that multiplying both sides of an inequality by a positive denominator preserves the direction.

Remark (Dependencies).
- Ordered Field
- Squares Are Nonnegative
- 1 Is Positive

Theorem 5.6.4 (Multiplication by a Positive Element Preserves Order). *In any ordered field, if $a < b$ and $c > 0$, then*

$$ac < bc.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall a, b, c \in F, \quad a < b \land c > 0 \Longrightarrow ac < bc.$$

Remark (Interpretation). Multiplying both sides of an inequality by a positive element leaves the direction unchanged. This is the order-field axiom itself, stated as an independent theorem for citation.

Remark (Dependencies).
- Ordered Field

Theorem 5.6.5 (Multiplication by a Negative Element Reverses Order). *In any ordered field, if $a < b$ and $c < 0$, then*

$$ac > bc.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall a, b, c \in F, \quad a < b \land c < 0 \implies ac > bc.$$

Remark (Interpretation). Multiplying both sides of an inequality by a negative element reverses the direction. This is the formal explanation of the familiar rule that one "flips the inequality sign" when multiplying by a negative number.

Remark (Dependencies).

- Ordered Field
- Multiplication by a Positive Element Preserves Order

Positivity of 1 and Its Consequences

The positivity of 1 is not an axiom but a theorem derivable from the ordered field structure. It then propagates to the natural numbers embedded in any ordered field.

Theorem (1 Is Positive)

Theorem 5.6.6 (1 Is Positive). *In any ordered field,*

$$1 > 0.$$

Go to proof.

Remark (Standard quantified statement).

$$1 > 0.$$

Remark (Interpretation). The multiplicative identity of any ordered field is positive. This anchors the order structure: every natural number n embedded in the field via repeated addition of 1 is also positive.

Remark (Dependencies).

- Ordered Field
- Squares Are Nonnegative

Corollary 5.6.7 (Natural Numbers Are Positive). *In any ordered field, for every natural number n,*

$$n > 0.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall n \in \mathbb{N}, \quad n > 0.$$

Remark (Interpretation). Every natural number, when embedded in an ordered field by repeated addition of the identity 1, is a positive element of the field. This is the foundational fact behind the Archimedean property.

Remark (Dependencies).

- 1 Is Positive
- Ordered Field

Fraction Comparison in an Ordered Field

The order on an ordered field extends to fractions via sign-arithmetic. The results below give explicit necessary and sufficient conditions for comparing two fractions and for determining the order of their reciprocals.

Theorem 5.6.8 (Positivity of a Quotient via the Product ab). *For any $a, b \in F$ with $b \neq 0$,*

$$0 < \frac{a}{b} \iff 0 < a \cdot b.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall a, b \in F, \quad b \neq 0 \implies \left(0 < \tfrac{a}{b} \iff 0 < ab\right).$$

Remark (Interpretation). A fraction is positive exactly when numerator and denominator have the same sign equivalently, when their product is positive. This reduces sign-testing of fractions to sign-testing of products.

Remark (Dependencies).

- Ordered Field
- Positive Elements Have Positive Inverses

Theorem 5.6.9 (General Fraction Comparison in an Ordered Field). *For any $a, b, c, d \in F$ with $b \neq 0$ and $d \neq 0$,*

$$\frac{a}{b} < \frac{c}{d} \iff (a \cdot d) \cdot (b \cdot d) < (b \cdot c) \cdot (b \cdot d).$$

Go to proof.

Remark (Standard quantified statement).

$$\forall a, b, c, d \in F, \quad b \neq 0 \wedge d \neq 0 \implies \left(\tfrac{a}{b} < \tfrac{c}{d} \iff (ad)(bd) < (bc)(bd)\right).$$

Remark (Interpretation). This cross-multiplication criterion for fraction comparison is valid for arbitrary nonzero denominators, regardless of sign. It reduces fraction order to the order of products, which is controlled by the sign-arithmetic theorems.

Remark (Dependencies).

- Ordered Field
- Multiplication by a Positive Element Preserves Order
- Multiplication by a Negative Element Reverses Order

Theorem 5.6.10 (Fraction Comparison When the Denominator Product Is Positive). *For any $a, b, c, d \in F$, if $b \neq 0$, $d \neq 0$, and $0 < b \cdot d$, then*

$$\frac{a}{b} < \frac{c}{d} \quad \Longleftrightarrow \quad a \cdot d < b \cdot c.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall a, b, c, d \in F, \quad b \neq 0 \wedge d \neq 0 \wedge 0 < bd \Longrightarrow \left(\tfrac{a}{b} < \tfrac{c}{d} \Longleftrightarrow ad < bc\right).$$

Remark (Interpretation). When the denominator product is positive in particular when both denominators are positive fraction comparison reduces to the familiar cross-multiplication rule: $a/b < c/d$ iff $ad < bc$.

Remark (Dependencies).

- General Fraction Comparison in an Ordered Field
- Multiplication by a Positive Element Preserves Order

Theorem 5.6.11 (Reciprocal Order for Positive Elements). *In any ordered field, if $0 < d < b$, then*

$$0 < \frac{1}{b} < \frac{1}{d}.$$

Conversely, if $0 < 1/b < 1/d$, then $0 < d < b$. Go to proof.

Remark (Standard quantified statement).

$$\forall b, d \in F, \quad 0 < d < b \Longleftrightarrow 0 < \tfrac{1}{b} < \tfrac{1}{d}.$$

Remark (Interpretation). Taking reciprocals reverses the strict order among positive elements. Larger positive denominators produce smaller positive reciprocals. This is the formal basis for the rule that $1/n$ decreases as n increases.

Remark (Dependencies).

- Positive Elements Have Positive Inverses
- Multiplication by a Positive Element Preserves Order

Theorem (Ordered Fields Are Densely Ordered)

Theorem 5.6.12 (Ordered Fields Are Densely Ordered). *Every ordered field is densely ordered by its order relation. Go to proof.*

Remark (Standard quantified statement).

$$\forall a, b \in F, \quad a < b \Longrightarrow \exists c \in F \ (a < c < b).$$

Remark (Interpretation). In any ordered field, between any two elements there is always a third. The witness is $(a + b)/2$: it is rational (the field is closed under addition and division by 2, which equals $1 + 1$) and lies strictly between a and b by the sign-arithmetic laws. This is the abstract theorem that specializes to the density of $\mathbb{Q}$ in itself.

Remark (Dependencies).

- Ordered Field
- Densely Ordered System
- Multiplication by a Positive Element Preserves Order
- 1 Is Positive

5.7 Linear Approximation in $\mathbb{Q}$

5.7.1 Rational Density and Order Facts

The rational numbers are not merely numerous; they are tightly packed. Between any two distinct rationals one can find another, and small rational perturbations can always be made without leaving $\mathbb{Q}$.

Theorem (Between Any Two Rational Numbers Lies Another Rational)

Theorem 5.7.1 (Between Any Two Rational Numbers Lies Another Rational). *If $r, s \in \mathbb{Q}$ and $r < s$, then there exists $t \in \mathbb{Q}$ such that*

$$r < t < s.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall r, s \in \mathbb{Q}\ \big(r < s \Longrightarrow \exists t \in \mathbb{Q}\ (r < t < s)\big).$$

Remark (Predicate reading).

$$\mathrm{Rational}(r) \land \mathrm{Rational}(s) \land r < s \Longrightarrow \exists t\ \big(\mathrm{Rational}(t) \land r < t < s\big).$$

Remark (Negated quantified statement).

$$\exists r, s \in \mathbb{Q}\ \big(r < s \land \forall t \in \mathbb{Q}\ (t \le r \lor s \le t)\big).$$

Remark (Negation predicate reading).

$$\mathrm{Rational}(r) \land \mathrm{Rational}(s) \land r < s \land \neg\exists t\ \big(\mathrm{Rational}(t) \land r < t < s\big).$$

Remark (Failure modes). The theorem could fail only if some pair of distinct rationals had no room for another rational between them. The midpoint construction shows this never happens in $\mathbb{Q}$.

Remark (Failure mode decomposition).

$$r < s \Longrightarrow r < \frac{r+s}{2} < s, \qquad \frac{r+s}{2} \in \mathbb{Q}.$$

Remark (Interpretation). This is the order-density of $\mathbb{Q}$ in itself. Rational numbers have no adjacent neighbors.

Remark (Dependencies).
- $\mathbb{Q}$ Is an Ordered Field

Corollary 5.7.2 ($\mathbb{Q}$ Is Dense in Itself). *The subset* $\mathbb{Q}$ *is order dense in itself.* *Go to proof.*

Remark (Standard quantified statement).

$$\forall r, s \in \mathbb{Q} \left(r < s \implies \exists t \in \mathbb{Q} \left(r < t < s\right)\right).$$

Remark (Predicate reading).

$$\mathrm{DenseIn}(\mathbb{Q}, \mathbb{Q}).$$

Remark (Negated quantified statement).

$$\exists r, s \in \mathbb{Q} \left(r < s \land \forall t \in \mathbb{Q} \left(t \leq r \lor s \leq t\right)\right).$$

Remark (Negation predicate reading).

$$\neg \mathrm{DenseIn}(\mathbb{Q}, \mathbb{Q}).$$

Remark (Failure modes). This could fail only if $\mathbb{Q}$ contained a genuine gap when viewed as an ordered set in its own right. The previous theorem rules that out.

Remark (Dependencies).
- Between Any Two Rational Numbers Lies Another Rational

Corollary 5.7.3 (The Rational Numbers Have No Adjacent Points). *There do not exist rational numbers $r < s$ such that no rational number lies strictly between them.* *Go to proof.*

Remark (Standard quantified statement).

$$\neg \exists r, s \in \mathbb{Q} \left(r < s \land \forall t \in \mathbb{Q} \left(t \leq r \lor s \leq t\right)\right).$$

Remark (Predicate reading).

$$\forall r, s \in \mathbb{Q} \left(r < s \implies \exists t \in \mathbb{Q} \left(r < t < s\right)\right).$$

Remark (Negated quantified statement).

$$\exists r, s \in \mathbb{Q} \left(r < s \land \neg \exists t \in \mathbb{Q} \left(r < t < s\right)\right).$$

Remark (Failure modes). Adjacent points would mean that some rational interval had no internal rational point. The midpoint argument prevents that.

Remark (Dependencies).
- Between Any Two Rational Numbers Lies Another Rational

Theorem ($\mathbb{Q}$ Has No Minimum Positive Element)

Theorem 5.7.4 ($\mathbb{Q}$ Has No Minimum Positive Element). *For every $q \in \mathbb{Q}$ with $q > 0$, there exists $r \in \mathbb{Q}$ such that*

$$0 < r < q.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall q \in \mathbb{Q}\ \big(q > 0 \Longrightarrow \exists r \in \mathbb{Q}\ (0 < r < q)\big).$$

Remark (Predicate reading).

$$\text{Rational}(q) \wedge q > 0 \Longrightarrow \exists r\ \big(\text{Rational}(r) \wedge 0 < r < q\big).$$

Remark (Negated quantified statement).

$$\exists q \in \mathbb{Q}\ \big(q > 0 \wedge \forall r \in \mathbb{Q}\ (0 < r \Longrightarrow q \leq r)\big).$$

Remark (Failure modes). This would fail only if some positive rational were the smallest positive rational. Halving it produces a strictly smaller positive rational, so no such minimum can exist.

Remark (Interpretation). Positive rationals can always be halved. There is no smallest positive rational number.

Remark (Dependencies).
- $\mathbb{Q}$ Is an Ordered Field

Lemma 5.7.5 (Upper/Lower Perturbation Lemma, Rational Version). *If $q \in \mathbb{Q}$ and $\varepsilon \in \mathbb{Q}$ with $\varepsilon > 0$, then there exist $r, s \in \mathbb{Q}$ such that*

$$q - \varepsilon < r < q < s < q + \varepsilon.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall q, \varepsilon \in \mathbb{Q},\ \varepsilon > 0\ \ \exists r, s \in \mathbb{Q}\ \big(q - \varepsilon < r < q < s < q + \varepsilon\big).$$

Remark (Negated quantified statement).

$$\exists q, \varepsilon \in \mathbb{Q},\ \varepsilon > 0\ \ \forall r, s \in \mathbb{Q}\ \neg\big(q - \varepsilon < r < q < s < q + \varepsilon\big).$$

Remark (Interpretation). Every rational point has rational points strictly on both sides within any prescribed positive distance. This is the working tool for punctured-ball arguments in $\mathbb{Q}$.

Remark (Dependencies).
- $\mathbb{Q}$ Has No Minimum Positive Element

Lemma 5.7.6 (Rational Perturbation Lemma). *If $q \in \mathbb{Q}$ and $\varepsilon > 0$, then there exists $r \in \mathbb{Q}$ such that*

$$0 < |r - q| < \varepsilon.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall q \in \mathbb{Q},\ \forall \varepsilon > 0\ \ \exists r \in \mathbb{Q}\ \big(0 < |r - q| < \varepsilon\big).$$

Remark (Negated quantified statement).

$$\exists q \in \mathbb{Q},\ \exists \varepsilon > 0\ \ \forall r \in \mathbb{Q}\ \big(r = q \vee |r - q| \geq \varepsilon\big).$$

Remark (Interpretation). Every rational point is a limit point of $\mathbb{Q}$: it can be approached arbitrarily closely by other rational numbers.

Remark (Dependencies).
- Upper/Lower Perturbation Lemma

Theorem (Rational Chain with Uniformly Small Steps)

Theorem 5.7.7 (Rational Chain with Uniformly Small Steps). *Let $r, s \in \mathbb{Q}$ with $r < s$, and let $p \in \mathbb{Q}$ with $p > 0$. Then there exist $m \in \mathbb{N}$ and rational numbers $q_1, q_2, \ldots, q_m$ such that*

$$q_1 = r, \qquad q_m = s,$$

and

$$0 < q_k - q_{k-1} < p \qquad \textit{for every } k = 2, \ldots, m.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall r, s, p \in \mathbb{Q} \Big(r < s \wedge p > 0 \implies \exists m \in \mathbb{N}\, \exists q_1, \ldots, q_m \in \mathbb{Q}\, \big(q_1 = r \wedge q_m = s \wedge \forall k \in \{2, \ldots, m\}\, (0 < q_k - q_{k-1}$$

Remark (Interpretation). Any rational interval can be traversed by finitely many rational steps each smaller than a prescribed positive bound. This is the discrete counterpart of the Archimedean property applied to step traversal.

Remark (Dependencies).
- Archimedean Property of $\mathbb{Q}$

5.7.2 Dyadic Rational Approximation

Dyadic rationals are the rational numbers obtained by repeated halving. They are the simplest systematic approximation grid and foreshadow binary expansions, nested brackets, and Cauchy constructions.

Definition (Dyadic Rational)

Definition 5.7.8 (Dyadic rational). A rational number d is called *dyadic* if there exist $m \in \mathbb{Z}$ and $n \in \mathbb{N}$ such that

$$d = \frac{m}{2^n}.$$

Remark (Standard quantified statement).

$$\mathrm{Dyadic}(d) \;:=\; \exists m \in \mathbb{Z}\, \exists n \in \mathbb{N}\, \big(d = m/2^n\big).$$

Remark (Negated quantified statement).

$$\neg\,\mathrm{Dyadic}(d) \;:=\; \forall m \in \mathbb{Z}\, \forall n \in \mathbb{N}\, \big(d \neq m/2^n\big).$$

Remark (Interpretation). Dyadic rationals are the grid points produced by subdividing the real line into intervals of width $1/2^n$. They are dense in $\mathbb{Q}$ and in $\mathbb{R}$ and are closed under addition, subtraction, and halving.

Remark (Dependencies).
- Exponentiation on a Peano System

- Rational Numbers

Theorem 5.7.9 (Dyadic Rationals Are Rational). *Every dyadic rational is an element of* $\mathbb{Q}$. *Go to proof.*

Remark (Standard quantified statement).

$$\forall d, \quad \mathrm{Dyadic}(d) \implies d \in \mathbb{Q}.$$

Remark (Interpretation). This is immediate from the definition: $m/2^n$ is a ratio of integers with nonzero denominator, hence rational. The theorem isolates the fact for explicit citation.

Remark (Dependencies).
- Dyadic Rational
- Rational Numbers

Theorem 5.7.10 (Dyadic Midpoint Closure). *If a and b are dyadic rationals, then $(a+b)/2$ is dyadic. Go to proof.*

Remark (Standard quantified statement).

$$\forall a, b, \quad \mathrm{Dyadic}(a) \wedge \mathrm{Dyadic}(b) \implies \mathrm{Dyadic}\Big(\frac{a+b}{2}\Big).$$

Remark (Interpretation). The dyadic grid is closed under midpoint formation. This makes dyadic rationals the natural substrate for bisection algorithms.

Remark (Dependencies).
- Dyadic Rational
- Dyadic Rationals Are Rational

Theorem (Dyadic Approximation Error Bound)

Theorem 5.7.11 (Dyadic Approximation Error Bound). *For every $x \in \mathbb{Q}$ and every $n \in \mathbb{N}$, there exists $m \in \mathbb{Z}$ such that*

$$\frac{m}{2^n} \leq x < \frac{m+1}{2^n}.$$

Consequently,

$$0 \leq x - \frac{m}{2^n} < \frac{1}{2^n}.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall x \in \mathbb{Q}\ \forall n \in \mathbb{N}\ \exists m \in \mathbb{Z}\ \frac{m}{2^n} \leq x < \frac{m+1}{2^n}.$$

Remark (Interpretation). Every rational can be trapped between two consecutive dyadic grid points at resolution $1/2^n$. The error of approximating x by the lower grid point $m/2^n$ is less than $1/2^n$, which by the Archimedean property can be made arbitrarily small.

Remark (Dependencies).
- Well-Ordering Principle
- Archimedean Property of $\mathbb{Q}$
- Dyadic Rational

5.7.3 The Archimedean Property

Two different ideas are often confused when one first studies the rational numbers. The *Archimedean property* says that the natural numbers grow beyond every fixed rational. The *density of* $\mathbb{Q}$ says that between any two distinct rationals there is another rational. The first controls large-scale size; the second controls local refinement of intervals.

Theorem (Archimedean Property of $\mathbb{Q}$)

Theorem 5.7.12 (Archimedean property of $\mathbb{Q}$). *For every $x \in \mathbb{Q}$ there exists $n \in \mathbb{N}$ such that*

$$n > x.$$

Equivalently, for every $x > 0$ there exists $n \in \mathbb{N}$ such that

$$\frac{1}{n} < x.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall x \in \mathbb{Q}\, \exists n \in \mathbb{N}\, (n > x).$$

Remark (Interpretation). The reciprocals of natural numbers become arbitrarily small. This theorem rules out the existence of positive infinitesimals in $\mathbb{Q}$ and is the foundation of all the approximation results in the chapter.

Remark (Dependencies).

- $\mathbb{Q}$ Is an Ordered Field
- Natural Numbers Are Positive

Remark 5.7.13 (Forward reference). The statement that $\mathbb{Q}$ is dense in $\mathbb{R}$ is best treated as a theorem about how the rationals sit inside the completed real line, not as an internal theorem about $\mathbb{Q}$ alone. Its canonical home is therefore in the reals chapter.

5.7.4 The Mediant Property

Toolkit: Mediants, Farey Sequences, and Stern–Brocot

Once density is known abstractly, it is natural to ask for a *constructive* way to produce rationals between two given ones, and to understand the precise arithmetic of the gaps between them. The atomic notions developed here are:

- the **mediant**, and its order-theoretic property;
- **Farey sequences**, the canonical enumeration of rationals by denominator bound;
- the **Farey Neighbor Theorem** and the exact gap formula;
- the **Mediant Gap Lemma**, quantifying what happens when the mediant is inserted;
- the **Stern–Brocot tree**, which enumerates all positive rationals without repetition.

5.7.4.1 Mediants

Definition (Mediant)

Definition 5.7.14 (Mediant). Let $a/b, c/d \in \mathbb{Q}$ with $b, d > 0$. The *mediant* of these two fractions is

$$\frac{a+c}{b+d}.$$

Remark (Standard quantified statement).

$$\forall a, c \in \mathbb{Z},\ \forall b, d \in \mathbb{Z}_{>0}, \quad \mathrm{Mediant}\left(\tfrac{a}{b}, \tfrac{c}{d}\right) = \frac{a+c}{b+d}.$$

Remark (Interpretation). The mediant is formed by adding numerators and denominators separately. This is not fraction addition it is a distinct operation that always produces a value strictly between its two inputs when they are in order. Its strength is constructive: given any two rationals, the mediant produces a specific third rational between them with full arithmetic control over where it lands and what gap it leaves.

Remark (Dependencies). Order and arithmetic of $\mathbb{Q}$, rational operations.

Theorem 5.7.15 (Mediant inequality). *If $a/b < c/d$ with $b, d > 0$, then*

$$\frac{a}{b} < \frac{a+c}{b+d} < \frac{c}{d}.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall a, c \in \mathbb{Z},\ \forall b, d \in \mathbb{Z}_{>0}, \quad \frac{a}{b} < \frac{c}{d} \implies \frac{a}{b} < \frac{a+c}{b+d} < \frac{c}{d}.$$

Remark (Interpretation). The mediant always lands strictly inside the interval. This makes it a concrete construction proof of density: given any two rationals, produce a third one between them without appeal to existence arguments.

Remark (Dependencies).

- Mediant
- $\mathbb{Q}$ Is an Ordered Field

Remark 5.7.16 (Local versus global density). The mediant property is *local*: it refines a given interval by inserting a specific rational between two given rationals. The Archimedean property is *global*: it says the natural numbers eventually exceed any fixed rational. Together they describe both the fine structure and the unboundedness of $\mathbb{Q}$.

5.7.4.2 Farey Sequences

Definition (Farey Sequence)

Definition 5.7.17 (Farey sequence). For $n \in \mathbb{N}$, $n \geq 1$, the *Farey sequence of order* n, denoted F_n, is the finite sequence of all rational numbers a/b satisfying

$$0 \leq \frac{a}{b} \leq 1, \qquad b \leq n, \qquad \gcd(a, b) = 1,$$

arranged in strictly increasing order.

Remark (Standard quantified statement).

$$F_n = \left\{ \frac{a}{b} \in \mathbb{Q} \;\middle|\; 0 \leq a \leq b \leq n,\ \gcd(a, b) = 1 \right\} \text{ ordered increasingly.}$$

Remark (Interpretation). F_n is the canonical list of all fractions between 0 and 1 with denominator at most n, each in reduced form. As n increases, new fractions are inserted between existing neighbors. The first few sequences are:

$$\begin{aligned}
F_1 &= \left(\tfrac{0}{1}, \tfrac{1}{1}\right), \\
F_2 &= \left(\tfrac{0}{1}, \tfrac{1}{2}, \tfrac{1}{1}\right), \\
F_3 &= \left(\tfrac{0}{1}, \tfrac{1}{3}, \tfrac{1}{2}, \tfrac{2}{3}, \tfrac{1}{1}\right), \\
F_4 &= \left(\tfrac{0}{1}, \tfrac{1}{4}, \tfrac{1}{3}, \tfrac{1}{2}, \tfrac{2}{3}, \tfrac{3}{4}, \tfrac{1}{1}\right).
\end{aligned}$$

Each new fraction inserted between two Farey neighbors is their mediant, and the neighbors of a newly inserted fraction always satisfy the Farey determinant condition.

Remark (Dependencies). Mediant, Rational numbers, reduced-form representatives.

Theorem (Farey Neighbor Theorem)

Theorem 5.7.18 (Farey Neighbor Theorem). *Let* $a/b < c/d$ *be two consecutive fractions in the Farey sequence* F_n*, with* $\gcd(a,b) = \gcd(c,d) = 1$ *and* $b, d > 0$*. Then*

$$bc - ad = 1.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall \tfrac{a}{b}, \tfrac{c}{d} \in F_n, \quad \tfrac{a}{b} \text{ and } \tfrac{c}{d} \text{ consecutive} \implies bc - ad = 1.$$

Remark (Interpretation). The determinant $bc - ad$ measures the "arithmetic distance" between two reduced fractions. For consecutive Farey neighbors this quantity is always exactly 1 the smallest it can be for distinct reduced fractions with positive denominators. This is why Farey neighbors are best approximations within their denominator range: no reduced fraction with smaller denominator can fit between them.

Remark (Negated quantified statement).

$$\exists \tfrac{a}{b}, \tfrac{c}{d} \in F_n, \quad \tfrac{a}{b} \text{ and } \tfrac{c}{d} \text{ consecutive} \wedge bc - ad \neq 1.$$

This negation is false: the theorem asserts the determinant is always 1.

Remark (Dependencies).

- Farey Sequence
- Mediant

Theorem 5.7.19 (Gap between Farey neighbors). *Let* $a/b < c/d$ *be consecutive fractions in a Farey sequence. Then*

$$\frac{c}{d} - \frac{a}{b} = \frac{1}{bd}.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall \tfrac{a}{b}, \tfrac{c}{d} \text{ consecutive in some } F_n, \quad \frac{c}{d} - \frac{a}{b} = \frac{1}{bd}.$$

Remark (Interpretation). The gap between consecutive Farey neighbors is determined entirely by their denominators. Since bd grows as n increases, the gaps shrink this is the arithmetic explanation of why $\mathbb{Q}$ becomes arbitrarily dense. The identity also connects directly to Dirichlet approximation: if x lies between two Farey neighbors, it is within $1/(bd)$ of each, giving a denominator-controlled approximation error.

Remark (Dependencies).

- Farey Neighbor Theorem

Theorem 5.7.20 (Mediant property of Farey neighbors). *If* a/b *and* c/d *are consecutive fractions in a Farey sequence, then their mediant* $(a+c)/(b+d)$ *lies strictly between them:*

$$\frac{a}{b} < \frac{a+c}{b+d} < \frac{c}{d}.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall \tfrac{a}{b}, \tfrac{c}{d} \text{ consecutive Farey neighbors}, \quad \frac{a}{b} < \frac{a+c}{b+d} < \frac{c}{d}.$$

Remark (Interpretation). This is the Mediant Inequality applied in the Farey context. It guarantees that the mediant insertion operation always produces a fraction that fits strictly between consecutive neighbors, which is the mechanism by which F_n grows into F_{n+1} (and by which the Stern-Brocot tree is generated).

Remark (Dependencies).
- Mediant Inequality
- Farey Sequence

Lemma 5.7.21 (Mediant Gap Lemma). *If $a/b < c/d$ with $b, d > 0$, then the mediant $(a+c)/(b+d)$ lies strictly between them, and the two subgaps satisfy*

$$\frac{a+c}{b+d} - \frac{a}{b} = \frac{1}{b(b+d)}, \qquad \frac{c}{d} - \frac{a+c}{b+d} = \frac{1}{d(b+d)}.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall a, c \in \mathbb{Z},\ \forall b, d \in \mathbb{Z}_{>0}, \quad \frac{a}{b} < \frac{c}{d} \implies \frac{a+c}{b+d} - \frac{a}{b} = \frac{1}{b(b+d)} \ \wedge\ \frac{c}{d} - \frac{a+c}{b+d} = \frac{1}{d(b+d)}.$$

Remark (Interpretation). Inserting the mediant divides the interval $[a/b, c/d]$ into two subintervals of lengths $1/(b(b+d))$ and $1/(d(b+d))$. Both are strictly smaller than the original gap $1/(bd)$ (since $b + d > b$ and $b + d > d$). This exact shrinkage formula is the engine behind the Stern-Brocot limit theory: repeated mediant insertion forces interval lengths to zero at a controlled rate determined entirely by the denominators.

Remark (Dependencies).
- Mediant Inequality

5.7.4.3 The Stern-Brocot Tree

Theorem 5.7.22 (Stern-Brocot enumeration). *Every positive rational number appears exactly once in reduced form in the Stern-Brocot tree.* *Go to proof.*

Remark (Standard quantified statement).

$\forall q \in \mathbb{Q},\ q > 0,\ \exists!$ node in the Stern-Brocot tree whose reduced fraction equals q.

Remark (Interpretation). The Stern-Brocot tree is a complete, duplication-free enumeration of all positive rationals. Every positive rational is reachable by following a unique finite left-right path from the root: go left if the target is less than the current node, right if greater, and stop when equality holds. The fact that this process always terminates (for rational targets) and never revisits a number is a theorem not a construction and its proof uses the Farey determinant condition together with a descent argument analogous to the Euclidean algorithm.

Remark (Dependencies).
- Mediant

- Farey Neighbor Theorem
- Mediant Gap Lemma

Remark 5.7.23 (The Farey-Stern-Brocot connection). The determinant condition $bc-ad = 1$ is the arithmetic shadow of the Stern-Brocot tree. Adjacent fractions in the tree always satisfy this condition, and the mediant inserts the unique next fraction between them. This is why the Farey gap formula $c/d - a/b = 1/(bd)$ and the Mediant Gap Lemma control approximation errors so precisely: the entire arithmetic structure of the tree is encoded in a single determinant identity.

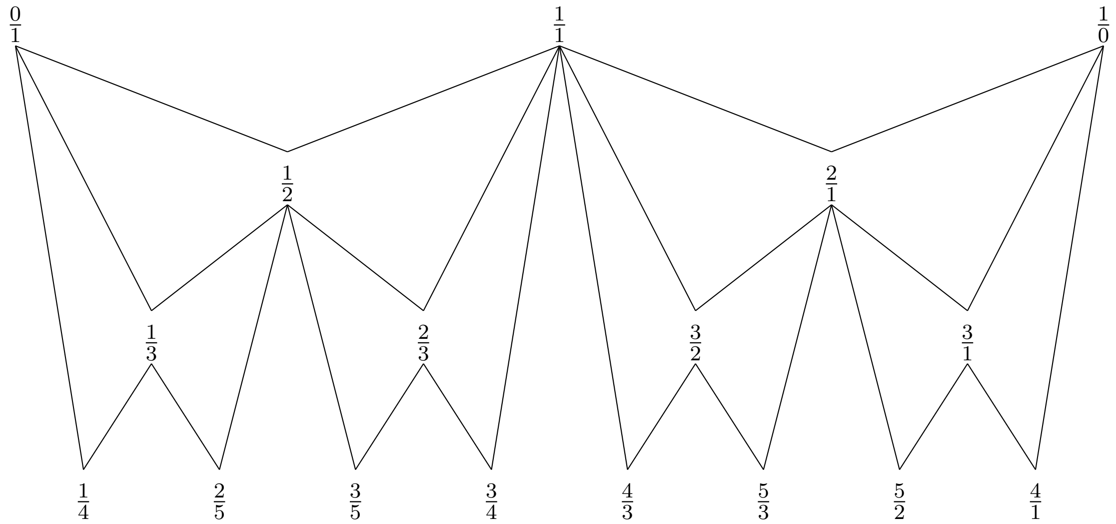

Figure 5.2: The Stern-Brocot tree with sentinel fractions $0/1$ and $1/0$. Each new fraction is the mediant of its two parent bounds. Every positive rational number appears exactly once in reduced form.

5.7.4.4 Stern-Brocot Limit Theory

Remark 5.7.24. The Stern-Brocot tree does more than enumerate the positive rationals. Its infinite paths encode the positive irrationals as limits of rational sequences. The four results below build this theory from the ground up, each depending on the previous one.

Lemma 5.7.25 (Denominator growth on infinite Stern-Brocot paths). *Let $a_0/b_0, a_1/b_1, a_2/b_2,$ be the sequence of endpoint fractions encountered along any infinite path in the Stern-Brocot tree. Then the denominators are strictly increasing:*

$$b_0 < b_1 < b_2 < \cdots ,$$

and in particular $b_n \to \infty$. Go to proof.

Remark (Standard quantified statement).

$$\forall n \in \mathbb{N}, \quad b_n < b_{n+1}. \qquad \forall M > 0 \ \exists N \in \mathbb{N} \ \forall n \geq N, \quad b_n > M.$$

Remark (Interpretation). At each step in the tree the new fraction produced by the mediant has denominator equal to the sum of the two parent denominators, which strictly exceeds each of them. A path that turns left or right at every level therefore visits fractions with strictly increasing denominators, and since the denominators are positive integers they must diverge to infinity. This is the key lemma that makes the limit theory work: it guarantees that the Stern-Brocot intervals shrink to zero length.

Remark (Dependencies).

- Farey Neighbor Theorem
- Induction Principle for a Peano System

Lemma 5.7.26 (Stern-Brocot interval lengths shrink to zero). *Let $(I_n)_{n\geq 0}$ be the nested rational intervals generated by the Stern-Brocot refinement process targeting a real number $x > 0$, where $I_n = [a_n/b_n,\ c_n/d_n]$. Then the interval lengths $\ell_n = c_n/d_n - a_n/b_n$ satisfy*

$$\lim_{n\to\infty} \ell_n = 0.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall n \geq N, \quad \ell_n < \varepsilon.$$

Remark (Interpretation). By the Farey gap formula, $\ell_n = 1/(b_n d_n)$. By the Denominator Growth Lemma, both b_n and d_n tend to infinity. Therefore $1/(b_n d_n) \to 0$. The Archimedean property of $\mathbb{Q}$ converts this into an explicit N: given $\varepsilon > 0$, find N such that $b_N d_N > 1/\varepsilon$.

Remark (Dependencies).

- Gap between Farey Neighbors
- Denominator Growth on Infinite SB Paths
- Archimedean Property of $\mathbb{Q}$

Proposition 5.7.27 (Nested Stern-Brocot intervals contain a unique limit point). *Let $(I_n)_{n\geq 0}$ be the nested sequence of closed rational intervals generated by the Stern-Brocot refinement process targeting $x > 0$, with $I_n = [a_n/b_n,\ c_n/d_n]$. Then there exists a unique real number $\xi \in \bigcap_{n=0}^{\infty} I_n$. Go to proof.*

Remark (Standard quantified statement).

$$\exists!\, \xi \in \mathbb{R}\ \ \forall n \in \mathbb{N}, \quad \frac{a_n}{b_n} \leq \xi \leq \frac{c_n}{d_n}.$$

Remark (Interpretation). The left endpoints (a_n/b_n) form a monotone increasing sequence bounded above; the right endpoints (c_n/d_n) form a monotone decreasing sequence bounded below. Both converge in $\mathbb{R}$ by the Monotone Convergence Theorem, and since the interval lengths tend to zero, the two limits must coincide. Uniqueness follows immediately: any two points in $\bigcap I_n$ differ by at most ℓ_n for every n, hence by zero.

This is where completeness of $\mathbb{R}$ enters: the Monotone Convergence Theorem requires the least-upper-bound property, which $\mathbb{Q}$ does not have. This is precisely why ξ need not be rational.

Remark (Dependencies). • Stern-Brocot Interval Lengths Shrink to Zero

Corollary 5.7.28 (Every infinite Stern-Brocot path converges to its target). *Let $x > 0$ be a real number and let (I_n) be the Stern-Brocot refinement sequence targeting x. The sequences of left and right endpoints both converge to x:*

$$\lim_{n\to\infty} \frac{a_n}{b_n} = \lim_{n\to\infty} \frac{c_n}{d_n} = x.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall n \geq N, \quad \left| x - \frac{a_n}{b_n} \right| < \varepsilon\ \wedge\ \left| x - \frac{c_n}{d_n} \right| < \varepsilon.$$

Remark (Interpretation). By construction, $x \in I_n$ for every n: the refinement rule always replaces the endpoint that would push x outside the interval, so x is retained in $\bigcap I_n$. But the proposition says this intersection contains exactly one point, namely ξ. Therefore $\xi = x$, and the endpoint sequences converge to x.

Remark (Dependencies). • Nested SB Intervals Contain a Unique Point

Corollary 5.7.29 (Infinite Stern-Brocot paths converge to irrationals). *A positive real number x corresponds to a* finite *path in the Stern-Brocot tree if and only if x is rational. Equivalently, x corresponds to an* infinite *path if and only if x is irrational. Go to proof.*

Remark (Standard quantified statement).

$x \in \mathbb{Q}_{>0} \iff$ the SB refinement process for x terminates in finitely many steps.

Remark (Interpretation). The refinement process terminates at the first step where x equals the mediant. Since every mediant is rational, termination forces $x \in \mathbb{Q}$. Conversely, if $x = p/q$ is rational, the Denominator Growth Lemma forces the denominators past q in finitely many steps, at which point x must have already been identified as a mediant. So the rational numbers are exactly those reached by finite paths, and the irrationals are exactly those requiring infinite paths. This is the cleanest formulation of the incompleteness of $\mathbb{Q}$ via the Stern-Brocot tree.

Remark (Dependencies). • Farey Neighbor Theorem

- Denominator Growth on Infinite SB Paths
- No Rational Number Has Square 2

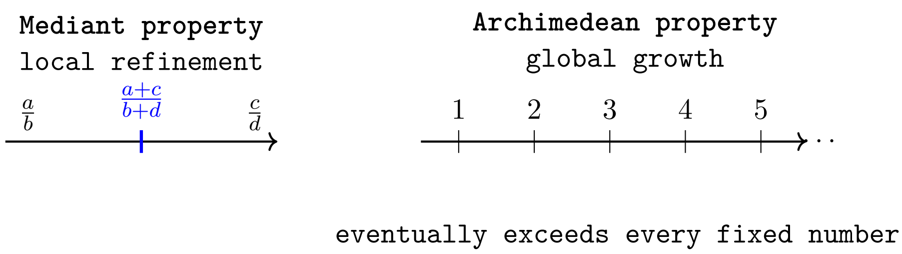

Figure 5.3: The mediant property and the Archimedean property govern different features of the rational numbers. The mediant inserts new rationals between nearby rationals, whereas the Archimedean property controls the large-scale behavior of the number system.

5.8 Square Bracketing and Nonlinear Approximation

Linear approximation controls distances on the number line. Nonlinear approximation asks for rational inputs whose images under nonlinear operations, such as squaring, bracket or approximate a target value.

Square Approximation Toolkit

The basic square-approximation strategy is to trap a target positive rational r between two rational squares, $a^2 < r < b^2$, and then refine the upper bracket using a fine dyadic grid. The least grid point whose square crosses r gives an approximation from above with controlled overshoot.

Theorem (Rational Square Bracketing)

Theorem 5.8.1 (Rational Square Bracketing). *For every $r \in \mathbb{Q}$ with $r > 0$, there exist $a, b \in \mathbb{Q}$ with $a > 0$ and $b > 0$ such that*

$$a^2 < r < b^2.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall r \in \mathbb{Q},\ r > 0\ \ \exists a, b \in \mathbb{Q},\ a > 0,\ b > 0\ \ a^2 < r < b^2.$$

Remark (Negated quantified statement).

$$\exists r \in \mathbb{Q},\ r > 0\ \ \forall a, b \in \mathbb{Q},\ a > 0,\ b > 0\ \ r \leq a^2 \vee b^2 \leq r.$$

Remark (Interpretation). Every positive rational lies strictly between the squares of two positive rationals. The witnesses are $a = r/(r+1)$ and $b = r+1$: one checks $a^2 < r$ and $r < b^2$ directly from the ordered field arithmetic of $\mathbb{Q}$.

Remark (Dependencies).

- $\mathbb{Q}$ Is an Ordered Field
- Archimedean Property of $\mathbb{Q}$

Grid Crossing and Approximation from Above

Given a rational target $r > 0$ and a grid of step size $1/q$, the well-ordering principle guarantees a least integer p such that $(p/q)^2 > r$. The predecessor $(p-1)/q$ has square at most r, and the overshoot $(p/q)^2 - r$ is controlled by $2p/q^2$, which can be made smaller than any prescribed $1/n$ by taking q large enough.

Lemma 5.8.2 (First Grid Point Crossing). *Let $r \in \mathbb{Q}$ with $r > 0$ and let $q \in \mathbb{N}$ with $q \geq 1$. If $K \in \mathbb{N}$ satisfies $K^2 > r$, then the set*

$$A = \{m \in \mathbb{N} : m^2 > rq^2\}$$

is nonempty and therefore has a least element. *Go to proof.*

Remark (Standard quantified statement).

$$\forall r \in \mathbb{Q},\ r > 0\ \ \forall q \in \mathbb{N},\ q \geq 1\ \ \big(\exists K \in \mathbb{N}\ (K^2 > r)\big) \implies \exists p_0 \in \mathbb{N}\ \big(p_0 \in A \wedge \forall m \in A\ (p_0 \leq m)\big).$$

Remark (Interpretation). The set A is nonempty because $Kq \in A$. By the Well-Ordering Principle, every nonempty subset of $\mathbb{N}$ has a least element. That least element p is the first grid point p/q whose square exceeds r.

Remark (Dependencies).

- Well-Ordering Principle
- Rational Square Bracketing

Theorem (Rational Square Approximation from Above)

Theorem 5.8.3 (Rational Square Approximation from Above). *Let $r \in \mathbb{Q}$ with $r > 0$, and let $n \in \mathbb{N}$ with $n \geq 1$. Then there exists $s \in \mathbb{Q}$ with $s > 0$ such that*

$$r < s^2 < r + \frac{1}{n}.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall r \in \mathbb{Q},\ r > 0\ \ \forall n \in \mathbb{N},\ n \geq 1\ \ \exists s \in \mathbb{Q},\ s > 0\ \ r < s^2 < r + \tfrac{1}{n}.$$

Remark (Interpretation). Choose grid denominator q large enough that $2Kq^{-1} < 1/n$ where K is a rational upper bound for $\sqrt{r}$. Let p be the least natural number with $p^2 > rq^2$ and set $s = p/q$. Then $s^2 > r$ by construction and $s^2 - r = (p^2 - rq^2)/q^2 \leq (2p-1)/q^2 < 2K/q < 1/n$.

Remark (Dependencies).

- First Grid Point Crossing
- Archimedean Property of $\mathbb{Q}$
- Rational Square Bracketing

Theorem (Rational Square Approximation from Below)

Theorem 5.8.4 (Rational Square Approximation from Below). *Let $r \in \mathbb{Q}$ with $r > 0$, and let $n \in \mathbb{N}$ with $n \geq 1$. If $r - 1/n > 0$, then there exists $s \in \mathbb{Q}$ with $s > 0$ such that*

$$r - \frac{1}{n} < s^2 < r.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall r \in \mathbb{Q},\ r > 0\ \ \forall n \in \mathbb{N},\ n \geq 1,\ r - \tfrac{1}{n} > 0\ \ \exists s \in \mathbb{Q},\ s > 0\ \ r - \tfrac{1}{n} < s^2 < r.$$

Remark (Interpretation). The rational $r - 1/n$ is positive by hypothesis. By the Approximation from Above theorem applied to $r-1/n$, there exists s with $(r - 1/n) < s^2 < (r - 1/n) + 1/n = r$.

Remark (Dependencies).

- Rational Square Approximation from Above

5.8.1 Problems in $\mathbb{Q}$

Toolkit: Problems in $\mathbb{Q}$

This section assembles the formal infrastructure of rational sequences and their limits, then uses that infrastructure to make precise exactly what goes wrong inside $\mathbb{Q}$. The atomic notions are:

- convergence of a rational sequence to a rational limit;
- uniqueness of limits;
- subsequences and their convergence;
- boundedness of convergent and Cauchy sequences;
- closure of the Cauchy property under addition and multiplication;
- an equivalence relation on Cauchy sequences, which is the raw material for the construction of $\mathbb{R}$.

Remark 5.8.5. The rational numbers are ordered, Archimedean, and dense. It is tempting to believe they should already suffice for analysis. The first serious obstruction appears when one asks which sequences have limits, and where those limits live.

Definition (Convergence of a Rational Sequence)

Definition 5.8.6 (Convergence of a rational sequence). Let (x_n) be a sequence of rational numbers and let $a \in \mathbb{Q}$. The sequence (x_n) *converges to* a if

$$\forall \varepsilon \in \mathbb{Q},\ \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall n \geq N, \quad |x_n - a| < \varepsilon.$$

We write $\lim_{n\to\infty} x_n = a$, or $x_n \to a$ as $n \to \infty$. The rational number a is called the *limit* of (x_n). A sequence that does not converge to any element of $\mathbb{Q}$ is called *divergent in* $\mathbb{Q}$.

Remark (Standard quantified statement).

$$\text{ConvergesTo}(x_n, a) \;:=\; \forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall n \geq N, \quad |x_n - a| < \varepsilon.$$

Remark (Negated quantified statement).

$$\neg\,\text{ConvergesTo}(x_n, a) \;:=\; \exists \varepsilon > 0\ \forall N \in \mathbb{N}\ \exists n \geq N, \quad |x_n - a| \geq \varepsilon.$$

Remark (Failure modes). A sequence fails to converge to a in one of two ways:

1. **Oscillation:** terms return repeatedly to both sides of a with spacing bounded away from zero. Example: $((-1)^n)$.

2. **Unbounded drift:** terms recede without bound and approach no rational number at all. Example: (n).

Remark (Interpretation). Think of ε as a challenge radius: the definition demands that the sequence eventually stays inside every ε-ball around a, no matter how small the radius. A sequence converges when it can meet every challenge, and diverges as soon as some challenge cannot be met.

Remark (Dependencies).

- Sequence of Rational Numbers
- Archimedean Property of $\mathbb{Q}$

Theorem 5.8.7 (Limits of rational sequences are unique). *If a sequence of rational numbers converges, then its limit is unique.* *Go to proof.*

Remark (Standard quantified statement).

$$\forall (x_n) \subseteq \mathbb{Q},\ \big(\text{ConvergesTo}(x_n, a) \wedge \text{ConvergesTo}(x_n, b)\big) \implies a = b.$$

Remark (Interpretation). A sequence cannot approach two distinct rational numbers simultaneously. If it did, the two alleged limits would need to be both arbitrarily close to the same terms, forcing them to be arbitrarily close to each other -- hence equal. This is the 2ε triangle-inequality squeeze.

Remark (Dependencies).

- Convergence of a Rational Sequence
- $\mathbb{Q}$ Is an Ordered Field

Definition 5.8.8 (Subsequence of a rational sequence). Let (x_n) be a sequence of rational numbers and let (k_n) be a strictly increasing sequence of natural numbers: $k_1 < k_2 < k_3 < \cdots$. The sequence (x_{k_n}) is called a ***subsequence*** of (x_n).

Remark (Standard quantified statement).

$$\big(\forall n \in \mathbb{N},\ k_{n+1} > k_n\big) \implies (x_{k_n}) \text{ is a subsequence of } (x_n).$$

Remark (Interpretation). A subsequence selects an infinite subcollection of terms in their original order. The key elementary fact is that $k_n \geq n$ for all n, so subsequence indices grow at least as fast as n itself.

Remark (Dependencies).

- Sequence of Rational Numbers

Theorem 5.8.9 (Subsequences of convergent sequences converge to the same limit). *Let (x_n) be a sequence of rational numbers with $x_n \to a$. Then every subsequence (x_{k_n}) of (x_n) also converges to a.* *Go to proof.*

Remark (Standard quantified statement).

$$\mathrm{ConvergesTo}(x_n, a) \implies \forall (k_n) \text{ strictly increasing}, \mathrm{ConvergesTo}(x_{k_n}, a).$$

Remark (Contrapositive quantified statement).

$$\exists (k_n) \text{ strictly increasing}, \neg\,\mathrm{ConvergesTo}(x_{k_n}, a) \implies \neg\,\mathrm{ConvergesTo}(x_n, a).$$

Remark (Interpretation). The contrapositive is the working tool: to prove (x_n) diverges it suffices to find two subsequences converging to distinct limits, or one subsequence that diverges.

Remark (Dependencies).

- Convergence of a Rational Sequence
- Subsequence of a Rational Sequence

Definition 5.8.10 (Bounded sequence of rational numbers). A sequence (x_n) of rational numbers is:

- ***bounded above*** if $\exists \Lambda \in \mathbb{Q}\ \forall n \in \mathbb{N},\ x_n \leq \Lambda$;
- ***bounded below*** if $\exists \lambda \in \mathbb{Q}\ \forall n \in \mathbb{N},\ x_n \geq \lambda$;
- ***bounded*** if $\exists \Gamma \in \mathbb{Q},\ \Gamma > 0\ \ \forall n \in \mathbb{N},\ |x_n| \leq \Gamma$.

Remark (Standard quantified statement).

$$\mathrm{Bounded}(x_n) \;:=\; \exists \Gamma \in \mathbb{Q},\ \Gamma > 0,\ \ \forall n \in \mathbb{N},\ \ |x_n| \leq \Gamma.$$

Remark (Negated quantified statement).

$$\neg\,\mathrm{Bounded}(x_n) \;:=\; \forall \Gamma > 0\ \ \exists n \in \mathbb{N},\ \ |x_n| > \Gamma.$$

Remark (Interpretation). Boundedness is a global envelope condition: one single rational bound must contain every term.

Remark (Dependencies).

- Sequence of Rational Numbers

Theorem 5.8.11 (Convergent sequences are bounded). *Every convergent sequence of rational numbers is bounded. Go to proof.*

Remark (Standard quantified statement).

$$\text{ConvergesTo}(x_n, a) \implies \text{Bounded}(x_n).$$

Remark (Interpretation). Convergence forces the tail of the sequence into a bounded neighborhood of the limit; the finite initial segment is bounded by the maximum of finitely many terms. The converse fails: $((-1)^n)$ is bounded but does not converge.

Remark (Dependencies).

- Convergence of a Rational Sequence
- Bounded Sequence of Rational Numbers

Theorem 5.8.12 (Cauchy sequences are bounded). *Every Cauchy sequence of rational numbers is bounded. Go to proof.*

Remark (Standard quantified statement).

$$\text{CauchySequence}(x_n) \implies \text{Bounded}(x_n).$$

Remark (Interpretation). The Cauchy condition forces the tail -- all terms beyond index N -- to lie within distance 1 of x_{N+1}. The finite initial segment contributes finitely many terms bounded by their maximum absolute value.

Remark (Dependencies).

- Cauchy Sequence in $\mathbb{Q}$
- Bounded Sequence of Rational Numbers

Theorem 5.8.13 (Cauchy sequences are closed under addition and multiplication). *If (x_n) and (y_n) are Cauchy sequences of rational numbers, then so are (x_n+y_n) and (x_ny_n). Go to proof.*

Remark (Standard quantified statement).

$$\text{CauchySequence}(x_n)\wedge\text{CauchySequence}(y_n) \implies \text{CauchySequence}(x_n+y_n) \wedge \text{CauchySequence}(x_ny_n).$$

Remark (Interpretation). For addition: an $\varepsilon/2$ split suffices. For multiplication: Cauchy sequences are bounded, so cross-terms $|y_n||x_n-x_m|$ and $|x_m||y_n-y_m|$ are controlled by a common bound Λ, and $\varepsilon/(2\Lambda)$ serves as the input tolerance for each factor.

Remark (Dependencies).

- Cauchy Sequence in $\mathbb{Q}$
- Cauchy Sequences Are Bounded

Definition (Equivalence of Cauchy Sequences in $\mathbb{Q}$)

Definition 5.8.14 (Equivalence of Cauchy sequences in $\mathbb{Q}$). Let $\mathbb{Q}^c$ denote the set of all Cauchy sequences of rational numbers. Two sequences $(x_n), (y_n) \in \mathbb{Q}^c$ are ***equivalent***, written $(x_n) \sim (y_n)$, if

$$\forall \varepsilon > 0 \;\exists N \in \mathbb{N} \;\forall n \geq N, \quad |x_n - y_n| < \varepsilon.$$

Equivalently, $(x_n) \sim (y_n)$ if and only if $(x_n - y_n)$ converges to 0.

Remark (Standard quantified statement).

$$\mathrm{CauchyEquiv}(x_n, y_n) \;\coloneqq\; \mathrm{ConvergesTo}(x_n - y_n,\, 0).$$

Remark (Negated quantified statement).

$$\neg\,\mathrm{CauchyEquiv}(x_n, y_n) \;\coloneqq\; \exists \varepsilon > 0 \;\; \forall N \in \mathbb{N} \;\; \exists n \geq N, \quad |x_n - y_n| \geq \varepsilon.$$

Remark (Interpretation). Two Cauchy sequences are equivalent when their difference is a null sequence. The equivalence classes under $\sim$ are exactly the real numbers in the Cauchy-sequence construction of $\mathbb{R}$.

Remark (Dependencies).

- Cauchy Sequence in $\mathbb{Q}$
- Convergence of a Rational Sequence

Theorem (The Cauchy Equivalence Relation on $\mathbb{Q}^c$)

Theorem 5.8.15 (The relation $\sim$ is an equivalence relation on $\mathbb{Q}^c$). *The relation $\sim$ defined on $\mathbb{Q}^c$ by $(x_n) \sim (y_n) \iff \mathrm{ConvergesTo}(x_n - y_n, 0)$ is an equivalence relation.* *Go to proof.*

Remark (Standard quantified statement). The relation $\sim$ on $\mathbb{Q}^c$ satisfies:

$$\begin{aligned}
&\forall (x_n) \in \mathbb{Q}^c, \quad (x_n) \sim (x_n); \\
&\forall (x_n), (y_n) \in \mathbb{Q}^c, \quad (x_n) \sim (y_n) \Longrightarrow (y_n) \sim (x_n); \\
&\forall (x_n), (y_n), (z_n) \in \mathbb{Q}^c, \quad (x_n) \sim (y_n) \wedge (y_n) \sim (z_n) \Longrightarrow (x_n) \sim (z_n).
\end{aligned}$$

Remark (Interpretation). Reflexivity: $|x_n - x_n| = 0 < \varepsilon$ always. Symmetry: $|x_n - y_n| = |y_n - x_n|$. Transitivity: the triangle inequality with an $\varepsilon/2$ split.

Remark (Dependencies).

- Equivalence of Cauchy Sequences in $\mathbb{Q}$
- Cauchy Sequences Are Closed Under Addition

Theorem 5.8.16 (Cauchy equivalence is preserved under addition and multiplication). *Let $(x_n), (y_n), (u_n), (v_n) \in \mathbb{Q}^c$. Suppose $(x_n) \sim (u_n)$ and $(y_n) \sim (v_n)$. Then:*

$$(x_n + y_n) \sim (u_n + v_n) \qquad \textit{and} \qquad (x_n y_n) \sim (u_n v_n).$$

Go to proof.

Remark (Standard quantified statement).

$$(x_n) \sim (u_n) \wedge (y_n) \sim (v_n) \implies (x_n + y_n) \sim (u_n + v_n) \;\wedge\; (x_n y_n) \sim (u_n v_n).$$

Remark (Interpretation). This is the well-definedness result for arithmetic on equivalence classes. Without it, adding or multiplying representatives from two classes could land in different classes, making the operations on $\mathbb{R} = \mathbb{Q}^c/\sim$ ill-defined. The proof uses boundedness of Cauchy sequences in the same way as the analogous well-definedness result for multiplication on $\mathbb{Q}$.

Remark (Dependencies).

- Equivalence of Cauchy Sequences in $\mathbb{Q}$

- Cauchy Sequences Are Closed Under Operations
- Cauchy Sequences Are Bounded

Remark 5.8.17 (Diagram of implications in $\mathbb{Q}$).

$$\text{convergent in } \mathbb{Q} \implies \text{Cauchy in } \mathbb{Q} \implies \text{bounded in } \mathbb{Q},$$

$$\text{bounded} \not\Longrightarrow \text{Cauchy}, \qquad \text{Cauchy} \not\Longrightarrow \text{convergent in } \mathbb{Q}.$$

Each failure is witnessed by an explicit counterexample. The last implication holds in $\mathbb{R}$ but not in $\mathbb{Q}$: that is exactly the completeness axiom.

5.9 Sequential Approximation in $\mathbb{Q}$

5.9.1 Rational Cauchy Sequences

The rational numbers form an ordered field and they approximate real numbers densely. At this stage it is tempting to believe that they should already be adequate for analysis. The first serious obstruction appears when one studies limits of rational sequences.

Definition (Sequence of Rational Numbers)

Definition 5.9.1 (Sequence of rational numbers). A *sequence of rational numbers* is a function

$$x : \mathbb{N} \to \mathbb{Q}.$$

It is usually written $(x_n)_{n\in\mathbb{N}}$ or simply $x_1, x_2, x_3, \ldots$.

Remark (Standard quantified statement).

$$\text{RationalSequence}(x) \;:=\; x : \mathbb{N} \to \mathbb{Q}.$$

Remark (Interpretation). Thinking of a sequence as a function emphasizes that a sequence is not merely a list but an indexed family of values. This viewpoint makes it natural to add, subtract, and compare sequences term by term.

Remark (Dependencies).
- Rational Numbers

Definition (Cauchy Sequence in $\mathbb{Q}$)

Definition 5.9.2 (Cauchy sequence in $\mathbb{Q}$). A sequence (x_n) of rational numbers is called a *Cauchy sequence* if

$$\forall \varepsilon > 0 \;\; \exists N \in \mathbb{N} \;\; \forall m, n \geq N, \qquad |x_m - x_n| < \varepsilon.$$

Remark (Standard quantified statement).

$$\text{CauchySequence}(x_n) \;:=\; \forall \varepsilon > 0 \;\; \exists N \in \mathbb{N} \;\; \forall m, n \geq N, \quad |x_m - x_n| < \varepsilon.$$

Remark (Negated quantified statement).

$$\neg\,\text{CauchySequence}(x_n) \;:=\; \exists \varepsilon > 0 \;\; \forall N \in \mathbb{N} \;\; \exists m, n \geq N, \quad |x_m - x_n| \geq \varepsilon.$$

Remark (Interpretation). A Cauchy sequence is one whose terms eventually cluster together as tightly as one wishes. This definition does not mention a limit; it speaks only about the internal behavior of the sequence.

Remark (Dependencies).
- Sequence of Rational Numbers
- $\mathbb{Q}$ Is an Ordered Field

Theorem (Every Convergent Rational Sequence Is Cauchy)

Theorem 5.9.3 (Every Convergent Rational Sequence Is Cauchy). *Every convergent sequence in $\mathbb{Q}$ is a Cauchy sequence.* *Go to proof.*

Remark (Standard quantified statement).

$$\text{ConvergesTo}(x_n, a) \implies \text{CauchySequence}(x_n).$$

Remark (Contrapositive quantified statement).

$$\neg\,\text{CauchySequence}(x_n) \implies \forall a \in \mathbb{Q}\ \neg\,\text{ConvergesTo}(x_n, a).$$

Remark (Interpretation). The triangle inequality gives $|x_m - x_n| \le |x_m - a| + |a - x_n|$. Taking N large enough that both terms are less than $\varepsilon/2$ gives the Cauchy condition. The converse is the issue: in $\mathbb{Q}$ it can fail.

Remark (Dependencies).
- Convergence of a Rational Sequence
- Cauchy Sequence in $\mathbb{Q}$

Theorem (There Exist Cauchy Sequences in $\mathbb{Q}$ That Do Not Converge to a Rational Number)

Theorem 5.9.4 (There Exist Cauchy Sequences in $\mathbb{Q}$ That Do Not Converge to a Rational Number). *There exist Cauchy sequences in $\mathbb{Q}$ that do not converge to a rational number.* *Go to proof.*

Remark (Standard quantified statement).

$$\exists (x_n) \subseteq \mathbb{Q}\ \Big(\text{CauchySequence}(x_n) \land \forall q \in \mathbb{Q}\ \neg\,\text{ConvergesTo}(x_n, q)\Big).$$

Remark (Negated quantified statement).

$$\forall (x_n) \subseteq \mathbb{Q}\ \big(\text{CauchySequence}(x_n) \implies \exists q \in \mathbb{Q}\ \text{ConvergesTo}(x_n, q)\big).$$

This negation is the (false) completeness claim for $\mathbb{Q}$.

Remark (Failure modes). The theorem would fail only if every Cauchy sequence of rationals converged to some rational limit. The decimal approximations to $\sqrt{2}$ show this is not so.

Remark (Failure mode decomposition).

$$1,\ 1.4,\ 1.41,\ 1.414,\ 1.4142, \ldots$$

is Cauchy in $\mathbb{Q}$, but its limit in $\mathbb{R}$ is $\sqrt{2}$, and the irrationality theorem gives $\sqrt{2} \notin \mathbb{Q}$.

Remark (Interpretation). Internal coherence is not enough. A rational sequence can have terms that settle down relative to one another and still fail to land on a rational number. This is the structural incompleteness of $\mathbb{Q}$.

Remark (Dependencies).
- Cauchy Sequence in $\mathbb{Q}$
- No Rational Number Has Square 2

5.9.2 Farey Sequences, Stern-Brocot, and Limiting Processes

The mediant operation, Farey sequences, and the Stern–Brocot tree are often introduced as combinatorial or number-theoretic constructions on the rational numbers. However, they also encode genuine limiting processes. In this sense they form a bridge between number theory and analysis.

The key idea is that repeated rational refinement produces sequences of rationals converging to real numbers. Thus these constructions do not merely organize $\mathbb{Q}$; they also point beyond $\mathbb{Q}$ toward $\mathbb{R}$.

Remark 5.9.5. There are at least three major ways in which Farey and Stern-Brocot structures enter limiting processes:

1. they generate convergent sequences of rational approximations;
2. they control approximation errors through exact gap formulas;
3. they encode irrational numbers as limits of infinite refinement paths.

Theorem 5.9.6 (Stern-Brocot interval refinement). *Let $x > 0$ be a real number. Suppose one begins with two rational numbers*

$$\frac{a}{b} < x < \frac{c}{d}, \qquad b, d > 0,$$

and repeatedly inserts the mediant

$$\frac{a+c}{b+d}.$$

At each step, replace one endpoint by the mediant so that the new interval still contains x. Then this process generates a nested sequence of rational intervals containing x. *Go to proof.*

Remark 5.9.7. This is a rational form of interval bisection. Instead of dividing an interval at its midpoint, one divides it at its mediant.

Remark 5.9.8. If the process is continued indefinitely, one obtains a sequence of nested intervals

$$\left[\frac{a_1}{b_1}, \frac{c_1}{d_1}\right] \supseteq \left[\frac{a_2}{b_2}, \frac{c_2}{d_2}\right] \supseteq \left[\frac{a_3}{b_3}, \frac{c_3}{d_3}\right] \supseteq \cdots$$

whose endpoints are rational numbers and whose lengths shrink. This is the first point at which the Stern-Brocot picture begins to look like an analytic limiting process rather than a purely arithmetic construction.

Remark 5.9.9 (Mediants generate convergent sequences). Starting with two rational bounds and repeatedly inserting mediants often produces monotone sequences of rational numbers converging to a real number.

For example, to approximate $\sqrt{2}$ one may begin with

$$1 < \sqrt{2} < 2.$$

Repeated refinement yields rational approximations such as

$$\frac{3}{2}, \quad \frac{7}{5}, \quad \frac{17}{12}, \quad \frac{41}{29}, \ldots$$

which converge to $\sqrt{2}$.

Thus the mediant process is not only a way to find intermediate rationals; it is also a mechanism for constructing convergent rational sequences.

Remark 5.9.10 (Farey neighbors control approximation error). If

$$\frac{a}{b} < \frac{c}{d}$$

are Farey neighbors, then

$$bc - ad = 1$$

and hence

$$\frac{c}{d} - \frac{a}{b} = \frac{1}{bd}.$$

Therefore, if a real number x lies between these two rationals, then its distance from either endpoint is controlled by a quantity of order

$$\frac{1}{bd}.$$

As the denominators grow, these gaps shrink. Thus the Farey gap formula gives a precise arithmetic explanation of why rational approximations can become arbitrarily accurate.

Remark 5.9.11 (Connection with Dirichlet approximation). Dirichlet's approximation theorem states that for every real number x there exist infinitely many rational numbers $\frac{p}{q}$ such that

$$\left| x - \frac{p}{q} \right| < \frac{1}{q^2}.$$

This bound is much stronger than mere density. Its geometric content is that one can find rational approximations inside intervals whose lengths shrink roughly like reciprocal quadratic functions of the denominator.

Farey gaps and Stern-Brocot refinement provide a concrete geometric picture of how such small intervals arise.

Remark 5.9.12 (Continued fractions and Stern-Brocot). The Stern-Brocot tree and continued fractions encode the same basic approximation process.

Every irrational number has an infinite continued fraction expansion

$$x = a_0 + \cfrac{1}{a_1 + \cfrac{1}{a_2 + \cfrac{1}{a_3 + \cdots}}}.$$

Its convergents

$$\frac{p_0}{q_0}, \frac{p_1}{q_1}, \frac{p_2}{q_2}, \ldots$$

satisfy

$$\lim_{n \to \infty} \frac{p_n}{q_n} = x.$$

These same approximants can be interpreted as arising from a path in the Stern-Brocot tree. Thus the Stern-Brocot tree gives a combinatorial model of continued-fraction convergence.

Theorem 5.9.13 (Stern-Brocot Limit Theorem). *Every irrational number $x > 0$ determines a unique infinite path in the Stern-Brocot tree. The sequence of rational numbers encountered along this path converges to x.*

Conversely, every infinite path in the Stern-Brocot tree determines a unique positive irrational number as its limit. *Go to proof.*

Remark 5.9.14 (Interpretation). This theorem says that the Stern-Brocot tree does not merely enumerate the rational numbers. Its infinite paths encode all positive irrational numbers as limits of rational approximations.

Thus:

- finite Stern-Brocot paths correspond to rational numbers;
- infinite Stern-Brocot paths correspond to irrational numbers.

In this way the Stern-Brocot tree reaches beyond $\mathbb{Q}$ and touches the full continuum of positive real numbers.

Remark 5.9.15 (Why this matters in analysis). This is one of the first places in elementary analysis where every irrational number appears through a single systematic construction based entirely on rational numbers.

The slogan is:

$$\text{infinite rational refinement} \longrightarrow \text{irrational limit}.$$

That makes the theorem especially important conceptually. It shows that irrational numbers are not mysterious isolated objects added from outside; they arise naturally as limits of coherent rational processes.

Remark 5.9.16 (Nested interval viewpoint). The Stern-Brocot limit process can also be understood through nested intervals.

Each step produces a smaller rational interval containing the target number. Thus one obtains

$$I_1 \supseteq I_2 \supseteq I_3 \supseteq \cdots$$

with

$$x \in I_n \quad \text{for all } n,$$

and with lengths tending to 0.

Hence the limit point is singled out by an infinite nested system of rational intervals. This is closely related in spirit to the nested interval principle and to Dedekind's idea of locating a real number by order cuts.

5.9.2.1 Consequences

Remark 5.9.17 (Why the chapter forces the issue). The theorem flow of this chapter does not merely show that rational numbers are useful. It shows that rational methods inevitably generate processes whose natural endpoints are not always rational.

That pressure builds in a very specific order:

1. mediant refinement and Stern-Brocot paths generate nested rational intervals;
2. Farey gaps explain why those intervals become arbitrarily small;
3. rational approximation theorems show that the target can be approached as closely as one likes;
4. Cauchy-style limiting behavior appears naturally inside $\mathbb{Q}$;
5. yet some of the targets produced by these processes are missing from $\mathbb{Q}$ itself.

Remark 5.9.18 (The first visible obstruction: the gap at $\sqrt{2}$). The most concrete example is Theorem 5.10.1: there is no rational number whose square is 2.

This is not an isolated curiosity. By Lemma 5.10.3 and Theorem 5.10.4, positive rational numbers can be trapped between rational squares and approached by rational squares with arbitrarily small error. So the arithmetic of $\mathbb{Q}$ naturally produces sequences that home in on locations that $\mathbb{Q}$ itself does not occupy.

Remark 5.9.19 (Refinement without completion). The mediant and Stern-Brocot constructions sharpen this picture. By Theorem 5.9.6, one obtains nested rational intervals containing a target real number. By Theorem 5.7.7, rational intervals can also be traversed by arbitrarily small rational increments. These are exactly the kinds of structures that suggest limit formation.

But the chapter's point is that repeated rational refinement does not guarantee a rational endpoint. The process is internally coherent while its limit may lie outside $\mathbb{Q}$.

Remark 5.9.20 (Cauchy pressure). This is the same tension that appears in the study of rational Cauchy sequences. A sequence can become arbitrarily tightly packed in the internal metric sense and still fail to converge to a rational number. The problem with $\mathbb{Q}$ is therefore not lack of density or lack of approximation power. The problem is failure of completion.

Remark 5.9.21 (The inescapable conclusion). Taken together, the theorem flow of this chapter leads to a single conclusion:

$$\text{rational refinement} \longrightarrow \text{rational approximation} \longrightarrow \text{Cauchy behavior} \not\Longrightarrow \text{rational limi}$$

That is the precise sense in which $\mathbb{Q}$ is incomplete. The real numbers are introduced not because the rationals are too sparse, but because $\mathbb{Q}$ does not contain all the limits naturally produced by its own approximation processes.

Remark 5.9.22 (Conceptual synthesis). Farey sequences, mediants, Stern-Brocot refinement, continued fractions, and Dirichlet approximation are not isolated topics. They form a single chain of ideas:

$$\text{Farey neighbors} \longrightarrow \text{mediant refinement} \longrightarrow \text{Stern-Brocot tree} \longrightarrow \text{continued fractions}$$

This chain shows that the rational numbers already contain, in compressed form, the structure needed to approximate every real number.

Remark 5.9.23 (Three viewpoints on completing $\mathbb{Q}$). The passage from $\mathbb{Q}$ to $\mathbb{R}$ can be understood from three complementary perspectives:

1. **Dedekind cuts:** real numbers are determined by order structure;
2. **Cauchy sequences:** real numbers are determined by metric completion;
3. **Stern-Brocot paths:** real numbers are determined by infinite rational approximation processes.

These are not three different continua. They are three different ways of describing the same completed number system.

Remark 5.9.24 (A foundational perspective). The rational numbers $\mathbb{Q}$ are dense, ordered, Archimedean, and arithmetically rich. Yet they are incomplete.

The Stern-Brocot Limit Theorem makes this incompleteness vivid in a particularly concrete way. It shows that every irrational number can be approached by an infinite, structured, purely rational process.

Thus one may think of the real numbers as arising when one allows not only rational numbers themselves, but also the limits of all coherent rational refinement procedures.

Symbolically:

$$\mathbb{Q} \quad + \quad \text{all limiting rational processes} \quad = \quad \mathbb{R}.$$

5.10 Rational Gaps

5.10.1 Gaps in the Rational Numbers

The rational numbers are dense and Archimedean, yet they are not complete. This means that there are real-number locations that can be approached arbitrarily closely by rationals but are not occupied by any rational number. The most famous example is $\sqrt{2}$.

Theorem (No Rational Number Has Square 2)

Theorem 5.10.1 (No Rational Number Has Square 2). *There is no rational number whose square is* 2. *Go to proof.*

Remark (Standard quantified statement).

$$\neg\exists r \in \mathbb{Q}\ \big(r^2 = 2\big).$$

Remark (Predicate reading).

$$\forall r \in \mathbb{Q}\ \neg\big(r^2 = 2\big).$$

Remark (Negated quantified statement).

$$\exists r \in \mathbb{Q}\ \big(r^2 = 2\big).$$

Remark (Failure modes). The theorem would fail exactly if some rational number were an exact square root of 2. Writing it in lowest terms as p/q with $\gcd(p,q) = 1$ leads to $p^2 = 2q^2$, forcing p even and then q even -- contradicting $\gcd(p,q) = 1$.

Remark (Failure mode decomposition).

$$\neg\big(\forall r \in \mathbb{Q}\ \neg(r^2 = 2)\big) \iff \exists r \in \mathbb{Q}\ (r^2 = 2).$$

Remark (Interpretation). This is the first visible gap in $\mathbb{Q}$. Rational numbers can approximate $\sqrt{2}$ arbitrarily well, but no rational number lands exactly on it. The obstruction is arithmetic: squaring a reduced rational cannot produce 2 without violating the parity structure of the integers.

Remark (Dependencies).

- Rational Numbers
- Fraction Pair Equivalence

Remark 5.10.2 (The gap at $\sqrt{2}$). Consider the sets

$$A = \{r \in \mathbb{Q} : r^2 < 2\} \qquad \text{and} \qquad B = \{r \in \mathbb{Q} : r^2 > 2\}.$$

Every rational number lies in one of these sets. Numbers in A approximate $\sqrt{2}$ from below and numbers in B approximate it from above. But there is no rational number that fills the gap between them.

Lemma 5.10.3 (Rational Square Bracketing). *Let r be a positive rational number. Then there exist positive rational numbers a and b such that*

$$a^2 < r < b^2.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall r \in \mathbb{Q},\ r > 0\ \ \exists a, b \in \mathbb{Q},\ a > 0,\ b > 0\ \ a^2 < r < b^2.$$

Remark (Interpretation). Every positive rational lies strictly between the squares of two positive rationals. The witnesses are $a = r/(r+1)$ and $b = r+1$.

Remark (Dependencies).
- $\mathbb{Q}$ Is an Ordered Field

Theorem (Rational Square Approximation from Above)

Theorem 5.10.4 (Rational Square Approximation from Above). *Let r be a positive rational number and let $n \in \mathbb{N}$. Then there exists a positive rational number s such that*

$$r < s^2 < r + \frac{1}{n}.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall r \in \mathbb{Q},\ r > 0\ \ \forall n \in \mathbb{N}\ \ \exists s \in \mathbb{Q},\ s > 0\ \ r < s^2 < r + \tfrac{1}{n}.$$

Remark (Interpretation). Positive rational numbers can be approximated from above by rational squares with arbitrarily small excess. Together with the analogous result from below, this means r is the limit of a sequence of rational squares on both sides, yet r itself may not be a rational square.

Remark (Dependencies).
- Rational Square Bracketing
- Archimedean Property of $\mathbb{Q}$
- Well-Ordering Principle

Remark 5.10.5. This is the decisive failure of $\mathbb{Q}$ as a number system for analysis. The rational numbers support arithmetic, order, density, mediants, and powerful approximation theorems. Even so, they still miss certain limits. The real numbers are introduced precisely to repair this defect.

5.10.2 Incompleteness of $\mathbb{Q}$

By this point the rational numbers look impressively capable. They form an ordered field, they satisfy the Archimedean property, and they are dense enough to approximate real-number locations arbitrarily well.

Remark 5.10.6 (What still fails). Two different phenomena now point to the same conclusion:

- the gap at $\sqrt{2}$ shows that some natural targets are not actually occupied by rational numbers;
- rational Cauchy sequences show that even internally coherent approximation processes need not converge inside $\mathbb{Q}$.

Remark 5.10.7 (The moral). Approximation is not the same thing as completion. The rational numbers can get arbitrarily close to many limits that they do not themselves contain. That failure is exactly what motivates the construction of the real numbers.

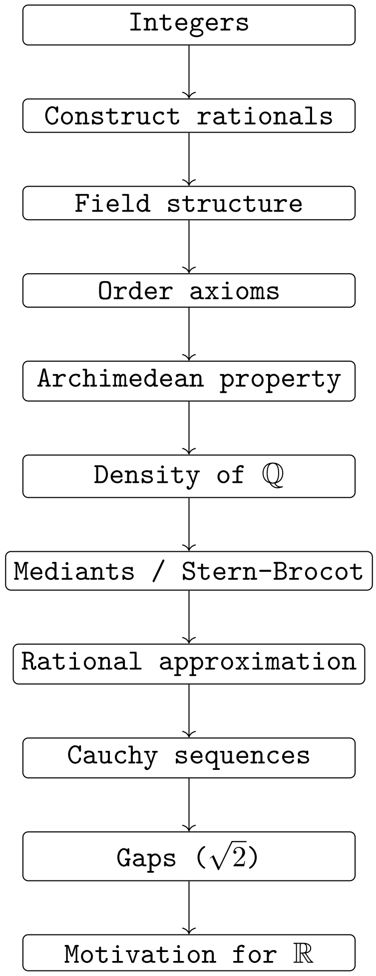

Figure 5.4: The rational numbers sit at the center of a long chain of ideas: construction, algebra, order, density, approximation, incompleteness, and finally the passage to the real numbers.

5.10.3 Structure of the Rational Numbers

Remark 5.10.8 (Structural overview). The rational numbers $\mathbb{Q}$ possess a surprisingly rich internal structure. Several seemingly different constructions reveal the same underlying organization of rational numbers and their approximating power.

Construction of $\mathbb{Q}$

$$\mathbb{Q} = \{(a,b) \in \mathbb{Z} \times \mathbb{Z} \; : \; b \neq 0\} / \sim$$

where

$$(a,b) \sim (c,d) \quad \Longleftrightarrow \quad ad = bc.$$

Each rational number is an equivalence class

$$[(a,b)].$$

Algebraic Structure

- $(\mathbb{Q}, +, \cdot)$ forms a **field**.
- $\mathbb{Q}$ is an **ordered field** with order inherited from $\mathbb{Z}$.
- The **Archimedean property** holds:

$$\forall x \in \mathbb{Q} \quad \exists n \in \mathbb{N} \quad n > x.$$

Density of $\mathbb{Q}$

Between any two rational numbers there exists another rational number.

$$\forall a, b \in \mathbb{Q},\ a < b \quad \exists r \in \mathbb{Q} \quad a < r < b.$$

A simple construction is the **mediant**

$$\frac{a}{b}, \frac{c}{d} \quad \Rightarrow \quad \frac{a+c}{b+d}.$$

Farey Sequences

The Farey sequence F_n contains all reduced fractions

$$\frac{a}{b}$$

with

$$0 \leq a \leq b \leq n$$

listed in increasing order.
If

$$\frac{a}{b}, \frac{c}{d}$$

are neighbors in a Farey sequence, then

$$bc - ad = 1.$$

Their spacing satisfies

$$\frac{c}{d} - \frac{a}{b} = \frac{1}{bd}.$$

Stern–Brocot Tree

All positive rational numbers appear exactly once in the SternBrocot tree.
Starting with

$$\frac{0}{1}, \quad \frac{1}{0},$$

new fractions are generated by mediants:

$$\frac{a}{b}, \frac{c}{d} \quad \mapsto \quad \frac{a+c}{b+d}.$$

This process recursively enumerates every positive rational number.

Rational Approximation

Rational numbers approximate real numbers extremely well.
Dirichlet's Approximation Theorem
For every real number x there exist infinitely many rational numbers

$$\frac{p}{q}$$

such that

$$\left| x - \frac{p}{q} \right| < \frac{1}{q^2}.$$

Gaps in $\mathbb{Q}$

Despite their density, the rational numbers are not complete.
For example,

$$x^2 = 2$$

has no rational solution.
Thus there exist sequences of rational numbers that approach limits which are not rational.

Completion to the Real Numbers

The real numbers can be constructed as equivalence classes of Cauchy sequences of rational numbers:

$$\mathbb{R} = \{\text{Cauchy sequences in } \mathbb{Q}\} / \sim .$$

Two sequences are equivalent if their difference converges to 0.
Thus the real numbers complete the rational numbers by filling the gaps.

Remark 5.10.9 (Conceptual picture).

$$\mathbb{Z} \longrightarrow \mathbb{Q} \longrightarrow \mathbb{R}$$

integers fractions limits

The rational numbers form a dense ordered field with a rich combinatorial structure, but they are not complete. The real numbers arise by adjoining the limits that are missing from $\mathbb{Q}$.

5.11 The Transition from $\mathbb{Q}$ to $\mathbb{R}$

The rational numbers achieve a great deal: they are constructed rigorously, support ordinary fraction arithmetic, form an ordered field, and admit powerful approximation procedures. The remaining failure is not density or algebra but completeness.

Remark (Central lesson).

Approximation is not existence.

Rational processes can produce tighter and tighter rational brackets while the object they indicate is not itself rational.

Remark (Transition). The real numbers are introduced to repair this failure. They retain the field and order structure of $\mathbb{Q}$ while adding a completeness principle that turns controlled limiting behavior into an object inside the number system.

Remark (Return). Return to Theorem

Theorem (The Fraction-Pair Relation Is an Equivalence Relation). *The relation $\sim$ on W defined by*

$$(a,b) \sim (c,d) \iff ad = bc$$

is an equivalence relation.

Proof. **Professional Standard Proof.**
Assume

$$W = \mathbb{Z} \times (\mathbb{Z} \setminus \{0\}).$$

Thus every element of W has the form (a,b) with $a, b \in \mathbb{Z}$ and $b \neq 0$.
First, let $(a,b) \in W$. Then

$$ab = ba$$

by commutativity of integer multiplication. Hence

$$(a,b) \sim (a,b).$$

Therefore $\sim$ is reflexive.
Next, let $(a,b), (c,d) \in W$ and suppose

$$(a,b) \sim (c,d).$$

By definition,

$$ad = bc.$$

Therefore

$$cb = da$$

by symmetry of equality and commutativity of integer multiplication. Hence

$$(c,d) \sim (a,b).$$

Therefore $\sim$ is symmetric.
Finally, let $(a,b), (c,d), (e,f) \in W$ and suppose

$$(a,b) \sim (c,d) \qquad \text{and} \qquad (c,d) \sim (e,f).$$

By definition,

$$ad = bc \qquad \text{and} \qquad cf = de.$$

Multiplying the first equality by f gives

$$adf = bcf.$$

Multiplying the second equality by b gives

$$bcf = bde.$$

Hence

$$adf = bde.$$

By associativity and commutativity of integer multiplication,

$$d(af) = d(be).$$

Since $(c,d) \in W$, we have $d \neq 0$. By cancellation in the integers,

$$af = be.$$

Therefore

$$(a,b) \sim (e,f).$$

Thus $\sim$ is transitive.

Since $\sim$ is reflexive, symmetric, and transitive, it is an equivalence relation on W. $\square$

Proof. **Detailed Learning Proof.**

Step 1. Verify reflexivity.

Let $(a,b) \in W$. Since $W = \mathbb{Z} \times (\mathbb{Z} \setminus \{0\})$, we have $a, b \in \mathbb{Z}$ and $b \neq 0$. To prove reflexivity, it is necessary to show

$$(a,b) \sim (a,b).$$

By the definition of $\sim$, this is equivalent to showing

$$ab = ba.$$

This equality holds by commutativity of multiplication in $\mathbb{Z}$. Therefore

$$(a,b) \sim (a,b).$$

Hence $\sim$ is reflexive.

Step 2. Verify symmetry.

Let $(a,b), (c,d) \in W$, and assume

$$(a,b) \sim (c,d).$$

By the definition of $\sim$, this means

$$ad = bc.$$

To prove symmetry, it is necessary to show

$$(c,d) \sim (a,b).$$

By the definition of $\sim$, this is equivalent to showing

$$cb = da.$$

From $ad = bc$, symmetry of equality gives

$$bc = ad.$$

By commutativity of multiplication in $\mathbb{Z}$,

$$cb = bc \qquad \text{and} \qquad ad = da.$$

Therefore

$$cb = da.$$

Hence

$$(c, d) \sim (a, b).$$

Thus $\sim$ is symmetric.

Step 3. Verify transitivity.

Let $(a, b), (c, d), (e, f) \in W$, and assume

$$(a, b) \sim (c, d) \qquad \text{and} \qquad (c, d) \sim (e, f).$$

By the definition of $\sim$, these assumptions give

$$ad = bc \qquad \text{and} \qquad cf = de.$$

To prove transitivity, it is necessary to show

$$(a, b) \sim (e, f),$$

which means

$$af = be.$$

Multiplying the equality $ad = bc$ by f gives

$$adf = bcf.$$

Multiplying the equality $cf = de$ by b gives

$$bcf = bde.$$

Combining these two equalities gives

$$adf = bde.$$

Using associativity and commutativity of multiplication in $\mathbb{Z}$, this may be rewritten as

$$d(af) = d(be).$$

Since $(c, d) \in W$, the second coordinate d is nonzero. Therefore cancellation by the nonzero integer d gives

$$af = be.$$

Hence

$$(a, b) \sim (e, f).$$

Thus $\sim$ is transitive.

Step 4. Conclude that the relation is an equivalence relation.

The relation $\sim$ has been shown to be reflexive, symmetric, and transitive. Therefore $\sim$ is an equivalence relation on W. $\square$

Remark (Proof structure). The proof is direct. Reflexivity follows from commutativity of integer multiplication. Symmetry follows by reversing the equality and commuting the products. Transitivity is the only substantive step: two cross-product equalities are multiplied by the remaining denominators, combined, and then cancellation by a nonzero integer produces the required cross-product equality.

Remark (Dependencies).

- Definition 5.1.4
- Basic arithmetic of the integers
- Cancellation law for multiplication by a nonzero integer

Remark (Return). Return to Lemma

Lemma (Addition Is Well-Defined on Rational Classes). *Let* $a, b, c, d, e, f, g, h \in \mathbb{Z}$ *with* $bdfh \neq 0$. *If*

$$(a, b) \sim (e, f) \qquad \textit{and} \qquad (c, d) \sim (g, h),$$

then

$$(ad + bc, bd) \sim (eh + fg, fh).$$

Proof. **Professional Standard Proof.**
Assume

$$(a, b) \sim (e, f) \qquad \text{and} \qquad (c, d) \sim (g, h).$$

By the definition of $\sim$,

$$af = be \qquad \text{and} \qquad ch = dg.$$

To prove

$$(ad + bc, bd) \sim (eh + fg, fh),$$

it is enough to prove

$$(ad + bc)fh = bd(eh + fg).$$

From $af = be$, multiplying both sides by dh gives

$$adfh = bdeh.$$

From $ch = dg$, multiplying both sides by bf gives

$$bcfh = bdfg.$$

Adding these two equalities gives

$$adfh + bcfh = bdeh + bdfg.$$

Factoring both sides gives

$$(ad + bc)fh = bd(eh + fg).$$

Therefore

$$(ad + bc, bd) \sim (eh + fg, fh).$$

Hence addition is well-defined on rational classes. □

Proof. **Detailed Learning Proof.**

Step 1. Translate the hypotheses into cross-product equalities.
Assume

$$(a, b) \sim (e, f) \qquad \text{and} \qquad (c, d) \sim (g, h).$$

By the definition of the fraction-pair relation, these assumptions mean

$$af = be \qquad \text{and} \qquad ch = dg.$$

Step 2. Translate the desired conclusion into one equality.

The desired conclusion is

$$(ad + bc, bd) \sim (eh + fg, fh).$$

By the definition of $\sim$, this is equivalent to the cross-product equality

$$(ad + bc)fh = bd(eh + fg).$$

Step 3. Produce the first summand equality.

From

$$af = be,$$

multiplying both sides by dh gives

$$adfh = bdeh.$$

This matches the first summand on each side of the target equality.

Step 4. Produce the second summand equality.

From

$$ch = dg,$$

multiplying both sides by bf gives

$$bcfh = bdfg.$$

This matches the second summand on each side of the target equality.

Step 5. Add and factor.

Adding the two equalities

$$adfh = bdeh \qquad \text{and} \qquad bcfh = bdfg$$

gives

$$adfh + bcfh = bdeh + bdfg.$$

Factoring the left-hand side and the right-hand side gives

$$(ad + bc)fh = bd(eh + fg).$$

Therefore, by the definition of $\sim$,

$$(ad + bc, bd) \sim (eh + fg, fh).$$

$\square$

Remark (Proof structure). The proof is direct. The hypotheses are first converted into the two cross-product equalities defining equivalence of fraction pairs. Each equality is then multiplied by the remaining denominators needed to match the common target denominator. Adding the resulting equalities and factoring gives the cross-product equality required for the sums.

Remark (Dependencies).

- Definition 5.1.4
- Basic arithmetic of the integers
- Distributivity of integer multiplication over addition

Remark (Return). Return to Lemma

Lemma (Multiplication Is Well-Defined on Rational Classes). *Let* $a, b, c, d, e, f, g, h \in \mathbb{Z}$ *with* $bdfh \neq 0$. *If*

$$(a, b) \sim (e, f) \qquad \textit{and} \qquad (c, d) \sim (g, h),$$

then

$$(ac, bd) \sim (eg, fh).$$

Proof. **Professional Standard Proof.**
Assume

$$(a, b) \sim (e, f) \qquad \text{and} \qquad (c, d) \sim (g, h).$$

By the definition of $\sim$,

$$af = be \qquad \text{and} \qquad ch = dg.$$

To prove

$$(ac, bd) \sim (eg, fh),$$

it is enough to prove

$$acfh = bdeg.$$

Multiplying the equalities

$$af = be \qquad \text{and} \qquad ch = dg$$

gives

$$(af)(ch) = (be)(dg).$$

By associativity and commutativity of integer multiplication,

$$acfh = bdeg.$$

Therefore

$$(ac, bd) \sim (eg, fh).$$

Hence multiplication is well-defined on rational classes. □

Proof. **Detailed Learning Proof.**

Step 1. Translate the hypotheses into cross-product equalities.
Assume

$$(a, b) \sim (e, f) \qquad \text{and} \qquad (c, d) \sim (g, h).$$

By the definition of the fraction-pair relation, these assumptions mean

$$af = be \qquad \text{and} \qquad ch = dg.$$

Step 2. Translate the desired conclusion into one equality.
The desired conclusion is

$$(ac, bd) \sim (eg, fh).$$

By the definition of $\sim$, this is equivalent to the cross-product equality

$$(ac)(fh) = (bd)(eg).$$

Equivalently, it is enough to prove

$$acfh = bdeg.$$

Step 3. Multiply the two hypothesis equalities.
From

$$af = be \qquad \text{and} \qquad ch = dg,$$

multiplying equal quantities gives

$$(af)(ch) = (be)(dg).$$

Step 4. Rearrange the products.
By associativity and commutativity of multiplication in $\mathbb{Z}$,

$$(af)(ch) = acfh$$

and

$$(be)(dg) = bdeg.$$

Therefore

$$acfh = bdeg.$$

Step 5. Return to the fraction-pair relation.
Since

$$acfh = bdeg,$$

we have

$$(ac)(fh) = (bd)(eg).$$

By the definition of $\sim$,

$$(ac, bd) \sim (eg, fh).$$

Thus multiplication is well-defined on rational classes. □

Remark (Proof structure). The proof is direct. The two hypotheses are converted into the cross-product equalities defining equivalence of fraction pairs. These equalities are multiplied together, and then associativity and commutativity of integer multiplication rearrange the resulting products into the single cross-product equality required for the products.

Remark (Dependencies).

- Definition 5.1.4
- Basic arithmetic of the integers
- Associativity and commutativity of integer multiplication

Remark (Return). Return to Theorem

Theorem (Integer Embedding Preserves Addition). *For all* $m, n \in \mathbb{Z}$*,*

$$[(m, 1)] + [(n, 1)] = [(m + n,\ 1)].$$

Proof. **Professional Standard Proof.**
Let $m, n \in \mathbb{Z}$. By the definition of addition on rational equivalence classes,

$$[(m, 1)] + [(n, 1)] = [(m \cdot 1 + 1 \cdot n,\ 1 \cdot 1)] = [(m + n,\ 1)].$$

□

Proof. **Detailed Learning Proof.**

Step 1. Identify the embeddings.
Let $m, n \in \mathbb{Z}$. Under the integer embedding, m is identified with $[(m, 1)] \in \mathbb{Q}$ and n is identified with $[(n, 1)] \in \mathbb{Q}$. Both pairs $(m, 1)$ and $(n, 1)$ are admissible fraction-pairs since $1 \neq 0$.
Step 2. Apply the addition formula.
By the definition of addition on rational equivalence classes,

$$[(m, 1)] + [(n, 1)] := [(m \cdot 1 + 1 \cdot n,\ 1 \cdot 1)].$$

Step 3. Simplify using integer arithmetic.
In $\mathbb{Z}$:

$$m \cdot 1 + 1 \cdot n = m + n \quad \text{and} \quad 1 \cdot 1 = 1.$$

Therefore

$$[(m \cdot 1 + 1 \cdot n,\ 1 \cdot 1)] = [(m + n,\ 1)].$$

Step 4. Conclude.
Combining Steps 2 and 3:

$$[(m, 1)] + [(n, 1)] = [(m + n,\ 1)].$$

This is the embedding of the integer $m+n$, so the embedding preserves addition.
□

Remark (Proof structure). The proof is a single application of the addition formula from the definition of rational operations, followed by integer arithmetic. There is no equivalence relation work and no well-definedness check -- those were discharged once and for all by the well-definedness lemma when the operations were defined. The sole content is reading off the formula at the specific pair $(m, 1)$, $(n, 1)$ and simplifying $m \cdot 1 + 1 \cdot n$ to $m + n$.

Remark (Dependencies).

- Definition 5.2.1 (addition formula: $[(a, b)] + [(c, d)] := [(ad + bc, bd)]$)
- Lemma 5.2.2 (well-definedness of addition, discharges representative independenc
- Basic integer arithmetic ($m \cdot 1 = m$, $1 \cdot n = n$, $1 \cdot 1 = 1$)

Remark (Return). Return to Theorem

Theorem (Integer Embedding Preserves Multiplication). *For all* $m, n \in \mathbb{Z}$*,*

$$[(m,1)] \cdot [(n,1)] = [(mn,\ 1)].$$

Proof. **Professional Standard Proof.**
Let $m, n \in \mathbb{Z}$. By the definition of multiplication on rational equivalence classes,

$$[(m,1)] \cdot [(n,1)] = [(m \cdot n,\ 1 \cdot 1)] = [(mn,\ 1)].$$

□

Proof. **Detailed Learning Proof.**

Step 1. Identify the embeddings.
Let $m, n \in \mathbb{Z}$. Under the integer embedding, m is identified with $[(m,1)] \in \mathbb{Q}$ and n is identified with $[(n,1)] \in \mathbb{Q}$. Both pairs $(m,1)$ and $(n,1)$ are admissible fraction-pairs since $1 \neq 0$.

Step 2. Apply the multiplication formula.
By the definition of multiplication on rational equivalence classes,

$$[(m,1)] \cdot [(n,1)] := [(m \cdot n,\ 1 \cdot 1)].$$

Step 3. Simplify using integer arithmetic.
In $\mathbb{Z}$:

$$m \cdot n = mn \qquad \text{and} \qquad 1 \cdot 1 = 1.$$

Therefore

$$[(m \cdot n,\ 1 \cdot 1)] = [(mn,\ 1)].$$

Step 4. Conclude.
Combining Steps 2 and 3:

$$[(m,1)] \cdot [(n,1)] = [(mn,\ 1)].$$

This is the embedding of the integer mn, so the embedding preserves multiplication.

□

Remark (Proof structure). The proof is structurally identical to the addition embedding proof: apply the operation formula at the specific input $(m,1)$, $(n,1)$ and simplify the resulting integer expressions. For multiplication the formula gives numerator $m \cdot n$ and denominator $1 \cdot 1$, both of which collapse immediately. No equivalence relation work is needed; well-definedness was discharged by the well-definedness lemma at the point of definition.

Remark (Dependencies).

- Definition 5.2.1 (multiplication formula: $[(a,b)] \cdot [(c,d)] := [(ac, bd)]$)
- Lemma 5.2.3 (well-definedness of multiplication, discharges representative independence)
- Basic integer arithmetic ($m \cdot n = mn$, $1 \cdot 1 = 1$)

Remark (Return). Return to Theorem

Theorem (Zero Class Criterion). *For every* $[(a,b)] \in \mathbb{Q}$*,*

$$[(a,b)] = [(0,1)] \quad \Longleftrightarrow \quad a = 0.$$

Proof. **Professional Standard Proof.**
Let $[(a,b)] \in \mathbb{Q}$. Then $a, b \in \mathbb{Z}$ and $b \neq 0$.

First suppose that

$$[(a,b)] = [(0,1)].$$

By equality of equivalence classes,

$$(a,b) \sim (0,1).$$

By the definition of the fraction-pair relation,

$$a \cdot 1 = b \cdot 0.$$

Since

$$a \cdot 1 = a \qquad \text{and} \qquad b \cdot 0 = 0,$$

it follows that

$$a = 0.$$

Conversely, suppose that

$$a = 0.$$

Then

$$a \cdot 1 = 0 \qquad \text{and} \qquad b \cdot 0 = 0.$$

Hence

$$a \cdot 1 = b \cdot 0.$$

By the definition of the fraction-pair relation,

$$(a,b) \sim (0,1).$$

Therefore

$$[(a,b)] = [(0,1)].$$

Thus

$$[(a,b)] = [(0,1)] \quad \Longleftrightarrow \quad a = 0.$$

□

Proof. **Detailed Learning Proof.**

Step 1. Translate the class equality into the representative relation.

Let $[(a,b)] \in \mathbb{Q}$. Then (a,b) is an admissible fraction-pair, so $a, b \in \mathbb{Z}$ and $b \neq 0$.

The equation

$$[(a,b)] = [(0,1)]$$

means exactly that the representatives (a,b) and $(0,1)$ are equivalent:

$$(a,b) \sim (0,1).$$

Step 2. Prove the forward implication.
Assume

$$[(a, b)] = [(0, 1)].$$

Then

$$(a, b) \sim (0, 1).$$

By the definition of the fraction-pair relation,

$$a \cdot 1 = b \cdot 0.$$

Using the integer identity law and zero law,

$$a \cdot 1 = a \qquad \text{and} \qquad b \cdot 0 = 0.$$

Therefore

$$a = 0.$$

Step 3. Prove the reverse implication.
Assume

$$a = 0.$$

Then

$$a \cdot 1 = 0$$

and

$$b \cdot 0 = 0.$$

Hence

$$a \cdot 1 = b \cdot 0.$$

By the definition of the fraction-pair relation,

$$(a, b) \sim (0, 1).$$

Therefore the corresponding equivalence classes are equal:

$$[(a, b)] = [(0, 1)].$$

Step 4. Conclude the biconditional.
The forward implication gives

$$[(a, b)] = [(0, 1)] \implies a = 0.$$

The reverse implication gives

$$a = 0 \implies [(a, b)] = [(0, 1)].$$

Therefore

$$[(a, b)] = [(0, 1)] \iff a = 0.$$

$\square$

Remark (Proof structure). The proof is a direct proof of both implications. The class equality is first translated into the defining equivalence relation on fraction-pairs. The resulting cross-product equality is

$$a \cdot 1 = b \cdot 0,$$

which reduces to $a = 0$ by elementary integer arithmetic.

Remark (Dependencies).

- Definition 5.1.4
- Definition 5.1.9 $(0 = [(0, 1)])$
- Basic integer arithmetic (cancellation, zero product)

Remark (Return). Return to Theorem

Theorem (Scaling of Equivalence Classes). *For every* $[(a,b)] \in \mathbb{Q}$ *and every nonzero integer* $e \neq 0$,

$$[(a,b)] = [(ae,be)].$$

Proof. **Professional Standard Proof.**
Let $[(a,b)] \in \mathbb{Q}$, and let $e \in \mathbb{Z}$ with $e \neq 0$. Then $a,b \in \mathbb{Z}$ and $b \neq 0$. Since $b \neq 0$ and $e \neq 0$, we also have

$$be \neq 0.$$

Thus (ae,be) is an admissible fraction-pair.

It remains to show that

$$(a,b) \sim (ae,be).$$

By the definition of the fraction-pair relation, this is equivalent to showing

$$a(be) = b(ae).$$

Using associativity and commutativity of integer multiplication,

$$a(be) = (ab)e = (ba)e = b(ae).$$

Therefore

$$(a,b) \sim (ae,be).$$

Hence the corresponding equivalence classes are equal:

$$[(a,b)] = [(ae,be)].$$

$\square$

Proof. **Detailed Learning Proof.**

Step 1. Check that the scaled pair is admissible.
Let $[(a,b)] \in \mathbb{Q}$, and let $e \in \mathbb{Z}$ with

$$e \neq 0.$$

Since $[(a,b)] \in \mathbb{Q}$, the pair (a,b) is an admissible fraction-pair. Hence

$$a,b \in \mathbb{Z} \quad \text{and} \quad b \neq 0.$$

Because $b \neq 0$ and $e \neq 0$, the integer product be is nonzero:

$$be \neq 0.$$

Therefore

$$(ae,be)$$

is also an admissible fraction-pair.

Step 2. Reduce equality of equivalence classes to the representative relation.
To prove

$$[(a,b)] = [(ae,be)],$$

it is enough to prove that the representatives are equivalent:

$$(a, b) \sim (ae, be).$$

Step 3. Apply the definition of the fraction-pair relation.
By the definition of $\sim$,

$$(a, b) \sim (ae, be)$$

holds exactly when

$$a(be) = b(ae).$$

Step 4. Verify the cross-product equality.
Using associativity of integer multiplication,

$$a(be) = (ab)e.$$

Using commutativity of integer multiplication,

$$(ab)e = (ba)e.$$

Using associativity again,

$$(ba)e = b(ae).$$

Hence

$$a(be) = b(ae).$$

Therefore

$$(a, b) \sim (ae, be).$$

Step 5. Conclude equality of classes.
Since the representatives are equivalent,

$$(a, b) \sim (ae, be),$$

their equivalence classes are equal:

$$[(a, b)] = [(ae, be)].$$

$\square$

Remark (Proof structure). The proof is definitional. Scaling the numerator and denominator by the same nonzero integer produces another admissible representativ Equality of equivalence classes is then reduced to the fraction-pair relation, and the required cross-product equality follows from associativity and commutativity of integer multiplication.

Remark (Dependencies).

- Definition 5.1.4
- Commutativity and associativity of integer multiplication
- Nonzero product property for integers

Remark (Return). Return to Theorem

Theorem (Negation of an Equivalence Class). *For every* $[(a,b)] \in \mathbb{Q}$,

$$-[(a,b)] = [(-a,b)] = [(a,-b)].$$

Proof. **Professional Standard Proof.**
Let $[(a,b)] \in \mathbb{Q}$. We must show that $[(-a,b)]$ is the additive inverse of $[(a,b)]$ in $\mathbb{Q}$, and that $[(-a,b)] = [(a,-b)]$.

Part 1. $[(-a,b)]$ is the additive inverse of $[(a,b)]$.

Compute the sum using the definition of addition:

$$[(a,b)] + [(-a,b)] = [(a \cdot b + b \cdot (-a),\ b \cdot b)] = [(ab - ab,\ b^2)] = [(0,\ b^2)].$$

Since $b \neq 0$, we have $0 \cdot b^2 = 0 = 0 \cdot 1$, so $(0,b^2) \sim (0,1)$ and thus $[(0,b^2)] = [(0,1)] = 0$.

Therefore $[(a,b)]+[(-a,b)] = 0$. By uniqueness of additive inverses in any group, $-[(a,b)] = [(-a,b)]$.

Part 2. $[(-a,b)] = [(a,-b)]$.

By the definition of $\sim$, this holds iff $(-a)(-b) = b \cdot a$. Since $(-a)(-b) = ab = ba$, the equality holds. Hence $[(-a,b)] = [(a,-b)]$. ☐

Proof. **Detailed Learning Proof.**

Step 1. Recall what it means to be an additive inverse.

The additive inverse of $[(a,b)]$ in $\mathbb{Q}$ is the unique element $x \in \mathbb{Q}$ satisfying

$$[(a,b)] + x = 0 = [(0,1)].$$

Additive inverses are unique in any group (since $\mathbb{Q}$ under addition is an abelian group). We will verify this identity for $x = [(-a,b)]$.

Step 2. Check that $[(-a,b)] \in \mathbb{Q}$.

Since $b \neq 0$ (as $(a,b) \in W$) and $-a \in \mathbb{Z}$, the pair $(-a,b)$ is an admissible fraction-pair, so $[(-a,b)] \in \mathbb{Q}$.

Step 3. Compute $[(a,b)] + [(-a,b)]$.

By the definition of rational addition,

$$[(a,b)] + [(-a,b)] = [(a \cdot b + b \cdot (-a),\ b \cdot b)].$$

The numerator is

$$a \cdot b + b \cdot (-a) = ab - ab = 0.$$

The denominator is b^2. Since $b \neq 0$, we have $b^2 \neq 0$, so the result is $[(0,b^2)]$.

Step 4. Show that $[(0,b^2)] = [(0,1)]$.

By the definition of $\sim$, $(0,b^2) \sim (0,1)$ iff $0 \cdot 1 = b^2 \cdot 0$. Both sides equal 0, so the equivalence holds. Therefore $[(0,b^2)] = 0$.

Step 5. Conclude that $-[(a,b)] = [(-a,b)]$.

We have shown $[(a,b)] + [(-a,b)] = 0$. By uniqueness of additive inverses,

$$-[(a,b)] = [(-a,b)].$$

Step 6. Show that $[(-a,b)] = [(a,-b)]$.

By the definition of $\sim$, these two fraction-pairs are equivalent iff

$$(-a)(-b) = b \cdot a.$$

Now $(-a)(-b) = ab$ by the product-of-negatives rule for integers, and $ba = ab$ by commutativity. So the condition holds, and $[(-a, b)] = [(a, -b)]$.

Step 7. Assemble the conclusion.

Combining Steps 5 and 6:

$$-[(a, b)] = [(-a, b)] = [(a, -b)]. \qquad \square$$

Remark (Proof structure). The first identity is proved by exhibiting $[(-a, b)]$ as the additive inverse and invoking uniqueness. The key computation is that the numerator of the sum $ab+b(-a)$ collapses to 0. The second identity is a direct application of the cross-multiplication criterion, using the integer identity $(-a)(-b) = ab$.

Remark (Dependencies).

- Definition 5.2.1 (additive inverse must satisfy $[(a, b)] + x = [(0, 1)]$)
- Definition 5.1.4
- Theorem **??**
- Uniqueness of additive inverses in an abelian group
- Basic integer arithmetic: $(-a)(-b) = ab$ and $ab - ab = 0$

Remark (Return). Return to Theorem

Theorem (Subtraction of Equivalence Classes). *For every* $[(a,b)], [(c,d)] \in \mathbb{Q}$*,*

$$[(a,b)] - [(c,d)] = [(ad - bc,\, bd)].$$

Proof. **Professional Standard Proof.**
By definition, $[(a,b)] - [(c,d)] := [(a,b)] + (-[(c,d)])$.
By the negation theorem, $-[(c,d)] = [(-c,d)]$.
By the addition formula,

$$[(a,b)] + [(-c,d)] = [(a \cdot d + b \cdot (-c),\ b \cdot d)] = [(ad - bc,\ bd)].$$

Therefore $[(a,b)] - [(c,d)] = [(ad - bc, bd)]$. □

Proof. **Detailed Learning Proof.**

Step 1. Recall the definition of subtraction.
Subtraction in any group is defined as addition of the additive inverse:

$$[(a,b)] - [(c,d)] := [(a,b)] + \big(-[(c,d)]\big).$$

Step 2. Apply the negation formula.
By the negation theorem,

$$-[(c,d)] = [(-c,d)].$$

This is valid since $(c,d) \in W$ implies $d \neq 0$, and $-c \in \mathbb{Z}$, so $(-c,d) \in W$.
Step 3. Apply the addition formula.
By the addition theorem,

$$[(a,b)] + [(-c,d)] = [(a \cdot d + b \cdot (-c),\ b \cdot d)].$$

Step 4. Simplify the numerator.
The numerator is

$$a \cdot d + b \cdot (-c) = ad - bc.$$

Step 5. Verify the denominator is nonzero.
Since $b \neq 0$ and $d \neq 0$, we have $bd \neq 0$, so $(ad - bc, bd) \in W$ and $[(ad - bc, bd)] \in \mathbb{Q}$.
Step 6. Assemble the conclusion.
Combining Steps 1-5:

$$[(a,b)] - [(c,d)] = [(a,b)] + [(-c,d)] = [(ad - bc,\ bd)].$$

□

Remark (Proof structure). The proof is a chain of definitions. Subtraction is addition of the negation; the negation formula swaps the sign of the numerator; the addition formula then produces the stated result. The only integer arithmetic is $b(-c) = -bc$. This theorem depends on both the negation and addition theorems, which is why it appears after them.

Remark (Dependencies).

- Definition of subtraction as $x - y := x + (-y)$
- Theorem 5.2.15 ($-[(c, d)] = [(-c, d)]$)
- Theorem **??** (addition formula for equivalence classes)
- Basic integer arithmetic ($b(-c) = -bc$)

Remark (Return). Return to Theorem

Theorem (Reciprocal of a Nonzero Equivalence Class). *For every* $[(a,b)] \in \mathbb{Q}$ *with* $a \neq 0$*,*

$$[(a,b)]^{-1} = [(b,a)].$$

Proof. **Professional Standard Proof.**
Let $[(a,b)] \in \mathbb{Q}$ with $a \neq 0$. Then $a,b \in \mathbb{Z}$ with $a \neq 0$ and $b \neq 0$. Since $a \neq 0$, the pair (b,a) is an admissible fraction-pair, so $[(b,a)] \in \mathbb{Q}$.

Compute the product using the definition of multiplication:

$$[(a,b)] \cdot [(b,a)] = [(ab,\ ba)] = [(ab,\ ab)].$$

Since $ab \neq 0$, we check $(ab,ab) \sim (1,1)$: indeed $ab{\cdot}1 = 1{\cdot}ab$. Therefore $[(ab,ab)] = [(1,1)] = 1$.

So $[(a,b)] \cdot [(b,a)] = 1$. By the uniqueness of multiplicative inverses in a field, $[(a,b)]^{-1} = [(b,a)]$. $\square$

Proof. **Detailed Learning Proof.**

Step 1. Verify that $[(b,a)]$ is a well-formed element of $\mathbb{Q}$.

Since $(a,b) \in W$, we have $b \neq 0$. By hypothesis $a \neq 0$. Therefore $(b,a) \in W$ (the denominator a is nonzero), so $[(b,a)] \in \mathbb{Q}$.

Step 2. Recall the definition of multiplicative inverse.

The multiplicative inverse of a nonzero element $x \in \mathbb{Q}$ is the unique element $x^{-1} \in \mathbb{Q}$ satisfying

$$x \cdot x^{-1} = 1 = [(1,1)].$$

Multiplicative inverses are unique in any field. We verify this for $x = [(a,b)]$ and $x^{-1} = [(b,a)]$.

Step 3. Verify that $[(a,b)] \neq 0$.

By the zero class criterion, $[(a,b)] = 0$ iff $a = 0$. By hypothesis $a \neq 0$, so $[(a,b)] \neq 0$, confirming that its inverse is defined.

Step 4. Compute the product $[(a,b)] \cdot [(b,a)]$.

By the definition of rational multiplication,

$$[(a,b)] \cdot [(b,a)] = [(a \cdot b,\ b \cdot a)] = [(ab,\ ab)].$$

Step 5. Show that $[(ab,ab)] = [(1,1)]$.

By the definition of $\sim$, $(ab,ab) \sim (1,1)$ iff

$$(ab) \cdot 1 = ab \cdot 1,$$

which is $ab = ab$, trivially true. Since $ab \neq 0$ (both a and b are nonzero), the pair (ab,ab) is admissible, and thus

$$[(ab,ab)] = [(1,1)] = 1.$$

Step 6. Conclude that $[(a,b)]^{-1} = [(b,a)]$.

We have shown $[(a,b)]{\cdot}[(b,a)] = 1$. By uniqueness of multiplicative inverses in $\mathbb{Q}$,

$$[(a,b)]^{-1} = [(b,a)].$$ $\square$

Remark (Proof structure). The proof is a direct verification. The candidate $[(b, a)]$ is the "swapped" fraction-pair. The product $[(a, b)] \cdot [(b, a)]$ evaluates to $[(ab, ab)]$ by the multiplication formula, and $(ab, ab) \sim (1, 1)$ since the cross-product $ab{\cdot}1 = 1{\cdot}ab$ is trivially true. Uniqueness of multiplicative inverses then pins down $[(b, a)]$ as the inverse.

Note the logical dependency: this proof invokes uniqueness of multiplicative inverses, which is a field-structure fact. This is why the reciprocal theorem properly belongs *after* $\mathbb{Q}$ has been established as a field, not inside the construction section.

Remark (Dependencies).

- Definition 5.2.1
- Definition 5.1.4
- Theorem **??**
- Definition 5.1.9 ($1 = [(1, 1)]$)
- Theorem 5.2.13 (to confirm $[(a, b)] \neq 0$)
- Theorem 5.4.23 (uniqueness of multiplicative inverse in $\mathbb{Q}$)

Remark (Return). Return to Theorem

Theorem ($\mathbb{Q}$ Is an Integral Domain). *With the operations of Definition 5.2.1, the system*

$$(\mathbb{Q}, +, \cdot, 0, 1)$$

is an integral domain.

Proof. **Professional Standard Proof.**
We verify the five conditions of the integral domain definition.

(1) $(\mathbb{Q}, +, 0)$ is a commutative group.

Closure. If $[(a,b)], [(c,d)] \in \mathbb{Q}$ then $[(ad+bc, bd)] \in \mathbb{Q}$ since $bd \neq 0$.

Associativity. Let $p = [(a,b)]$, $q = [(c,d)]$, $r = [(e,f)]$. Then

$$(p+q)+r = [(ad+bc, bd)] + [(e,f)] = [((ad+bc)f + bd \cdot e,\ bdf)],$$
$$p+(q+r) = [(a,b)] + [(cf+de, df)] = [(a \cdot df + b(cf+de),\ bdf)].$$

The numerators are $(adf + bcf + bde)$ in both cases, so the sums are equal.

Identity. $[(a,b)] + [(0,1)] = [(a \cdot 1 + b \cdot 0, b \cdot 1)] = [(a,b)]$.

Inverses. $[(a,b)] + [(-a,b)] = [(ab-ab, b^2)] = [(0,b^2)] = 0$, since $(0,b^2) \sim (0,1)$.

Commutativity. $[(a,b)] + [(c,d)] = [(ad+bc, bd)] = [(cb+da, db)] = [(c,d)] + [(a,b)]$.

(2) Multiplication is associative and commutative.

Associativity. $([(a,b)] \cdot [(c,d)]) \cdot [(e,f)] = [(ac,bd)] \cdot [(e,f)] = [(ace, bdf)]$. $[(a,b)] \cdot ([(c,d)] \cdot [(e,f)]) = [(a,b)] \cdot [(ce,df)] = [(ace, bdf)]$.

Commutativity. $[(a,b)] \cdot [(c,d)] = [(ac,bd)] = [(ca,db)] = [(c,d)] \cdot [(a,b)]$.

(3) $1 \neq 0$ and 1 is the multiplicative identity.

$[(1,1)] \neq [(0,1)]$ since $1 \cdot 1 = 1 \neq 0 = 0 \cdot 1$. $[(1,1)] \cdot [(a,b)] = [(1 \cdot a, 1 \cdot b)] = [(a,b)]$.

(4) Multiplication distributes over addition.

$$\begin{aligned} [(a,b)] \cdot ([(c,d)] + [(e,f)]) &= [(a,b)] \cdot [(cf+de, df)] \\ &= [(a(cf+de),\ b \cdot df)] \\ &= [(acf + ade,\ bdf)]. \end{aligned}$$

$$\begin{aligned} [(a,b)] \cdot [(c,d)] + [(a,b)] \cdot [(e,f)] &= [(ac,bd)] + [(ae,bf)] \\ &= [(ac \cdot bf + bd \cdot ae,\ bd \cdot bf)] \\ &= [(abcf + abde,\ b^2 df)]. \end{aligned}$$

These two classes are equal iff $(acf + ade)(b^2 df) = (abcf + abde)(bdf)$, which simplifies to $ab^2 df(cf+de) = ab(bdf)(cf+de)$, i.e. $b^2 df = b^2 df$. True.

(5) $\mathbb{Q}$ has no zero divisors.

Suppose $[(a,b)] \cdot [(c,d)] = 0$. Then $[(ac,bd)] = [(0,1)]$, so $(ac,bd) \sim (0,1)$, meaning $ac \cdot 1 = bd \cdot 0 = 0$. Thus $ac = 0$. Since $\mathbb{Z}$ is an integral domain, $a = 0$ or $c = 0$. By the zero class criterion, $a = 0$ iff $[(a,b)] = 0$ and $c = 0$ iff $[(c,d)] = 0$. Therefore $[(a,b)] = 0$ or $[(c,d)] = 0$.

Since all five conditions hold, $(\mathbb{Q}, +, \cdot, 0, 1)$ is an integral domain. $\square$

Proof. **Detailed Learning Proof.**

We check each condition of the integral domain definition in turn.

Condition 1: $(\mathbb{Q}, +, 0)$ is a commutative group.

We must verify five sub-properties.

(1a) Closure under addition. Let $[(a, b)], [(c, d)] \in \mathbb{Q}$. The sum is $[(ad + bc, bd)]$. Since $b \neq 0$ and $d \neq 0$, we have $bd \neq 0$, so $(ad + bc, bd) \in W$ and $[(ad + bc, bd)] \in \mathbb{Q}$.

(1b) Associativity of addition. Let $[(a, b)], [(c, d)], [(e, f)] \in \mathbb{Q}$. We compute both bracketings directly:

$$([(a,b)]+[(c,d)])+[(e,f)] = [(ad+bc, bd)]+[(e,f)] = [((ad+bc)f+bd\cdot e,\ bdf)] = [(adf+bcf+bde,\ bdf)].$$

$$[(a,b)]+([(c,d)]+[(e,f)]) = [(a,b)]+[(cf+de, df)] = [(a\cdot df+b(cf+de),\ bdf)] = [(adf+bcf+bde,\ bdf)].$$

The numerators and denominators are identical, so $([(a, b)] + [(c, d)]) + [(e, f)] = [(a, b)] + ([(c, d)] + [(e, f)])$.

(1c) Additive identity. $0 = [(0, 1)]$. Compute:

$$[(a, b)] + [(0, 1)] = [(a \cdot 1 + b \cdot 0,\ b \cdot 1)] = [(a, b)].$$

Similarly $[(0, 1)] + [(a, b)] = [(a, b)]$ by commutativity of integer addition.

(1d) Additive inverses. The candidate inverse of $[(a, b)]$ is $[(-a, b)]$ (see the negation theorem). We have already verified this: $[(a, b)] + [(-a, b)] = [(0, b^2)] = 0$.

(1e) Commutativity of addition.

$$[(a, b)] + [(c, d)] = [(ad + bc, bd)] = [(cb + da, db)] = [(c, d)] + [(a, b)],$$

where we used commutativity of integer addition ($ad+bc = cb+da$) and commutativity of integer multiplication ($bd = db$).

Condition 2: Multiplication is associative and commutative.

(2a) Associativity.

$$([(a, b)] \cdot [(c, d)]) \cdot [(e, f)] = [(ac, bd)] \cdot [(e, f)] = [(ace, bdf)].$$

$$[(a, b)] \cdot ([(c, d)] \cdot [(e, f)]) = [(a, b)] \cdot [(ce, df)] = [(ace, bdf)].$$

Both equal $[(ace, bdf)]$, so multiplication is associative.

(2b) Commutativity. $[(a, b)] \cdot [(c, d)] = [(ac, bd)] = [(ca, db)] = [(c, d)] \cdot [(a, b)]$, using commutativity in $\mathbb{Z}$.

Condition 3: $1 \neq 0$ and 1 is the multiplicative identity.

$1 = [(1, 1)]$ and $0 = [(0, 1)]$. These are not equal: $(1, 1) \sim (0, 1)$ would require $1 \cdot 1 = 0 \cdot 1$, i.e. $1 = 0$, which is false in $\mathbb{Z}$. So $1 \neq 0$.

For the identity property:

$$[(1, 1)] \cdot [(a, b)] = [(1 \cdot a,\ 1 \cdot b)] = [(a, b)].$$

Commutativity gives $[(a, b)] \cdot [(1, 1)] = [(a, b)]$ as well.

Condition 4: Distributivity.

We show $[(a, b)] \cdot ([(c, d)] + [(e, f)]) = [(a, b)] \cdot [(c, d)] + [(a, b)] \cdot [(e, f)]$.

Left-hand side:

$$[(a,b)] \cdot [(cf+de,\ df)] = [(a(cf+de),\ bdf)] = [(acf+ade,\ bdf)].$$

Right-hand side:

$$[(ac,bd)] + [(ae,bf)] = [(ac\cdot bf + bd\cdot ae,\ bd\cdot bf)] = [(abcf+abde,\ b^2df)].$$

We must check $[(acf+ade,bdf)] = [(abcf+abde,b^2df)]$. By the definition of $\sim$:

$$(acf+ade)(b^2df) \stackrel{?}{=} (bdf)(abcf+abde).$$

Left: $b^2df(acf+ade) = ab^2df(cf+de)$.
Right: $bdf\cdot ab(cf+de) = ab^2df(cf+de)$.
These are equal. Therefore distributivity holds.

Condition 5: No zero divisors.

Let $[(a,b)],[(c,d)] \in \mathbb{Q}$ and suppose

$$[(a,b)]\cdot[(c,d)] = 0.$$

By the multiplication formula, $[(ac,bd)] = [(0,1)]$. By the zero class criterion applied to $[(ac,bd)]$, this means $ac=0$. Since $\mathbb{Z}$ is an integral domain (integers have no zero divisors), $ac=0$ implies $a=0$ or $c=0$.

If $a=0$, then by the zero class criterion $[(a,b)]=0$.
If $c=0$, then by the zero class criterion $[(c,d)]=0$.

Therefore at least one of the factors is zero in $\mathbb{Q}$, and $\mathbb{Q}$ has no zero divisors.

Conclusion.

All five conditions of the integral domain definition are satisfied. Therefore $(\mathbb{Q},+,\cdot,0,1)$ is an integral domain. □

Remark (Proof structure). The proof is a systematic verification of each axiom. The substantive content lies in three places: the associativity of addition (which requires expanding both sides and comparing numerators), the distributivity check (which reduces to a cross-multiplication equality), and the no-zero-divisors property (which reduces $\mathbb{Q}$-level zeroes to the corresponding fact in $\mathbb{Z}$, which is an integral domain by construction).

All other axioms are direct computations from the operation definitions and the arithmetic of $\mathbb{Z}$.

Remark (Dependencies).

- Definition 5.1.8
- Definition 5.1.9
- Definition 5.2.1
- Definition 5.4.1
- Theorem 5.2.13 (zero class criterion, used in Condition 5)
- $\mathbb{Z}$ is an integral domain (no zero divisors in $\mathbb{Z}$)

Remark (Return). Return to Theorem

Theorem ($\mathbb{Q}$ Is a Field). *With the operations of Definition 5.2.1, the system*

$$(\mathbb{Q}, +, \cdot, 0, 1)$$

is a field.

Proof. **Professional Standard Proof.**
By the integral domain theorem, $(\mathbb{Q}, +, \cdot, 0, 1)$ is already an integral domain. It remains only to verify that every nonzero element has a multiplicative inverse.

Let $[(a,b)] \in \mathbb{Q}$ with $[(a,b)] \neq 0$. By the zero class criterion, $a \neq 0$. Since $a \neq 0$ and $b \neq 0$, the pair $(b,a) \in W$, so $[(b,a)] \in \mathbb{Q}$.

Compute:

$$[(a,b)] \cdot [(b,a)] = [(ab,\ ba)] = [(ab,\ ab)].$$

Since $ab \neq 0$, we have $(ab, ab) \sim (1,1)$ (as $ab \cdot 1 = 1 \cdot ab$), so $[(ab, ab)] = 1$.

Therefore $[(a,b)] \cdot [(b,a)] = 1$, which means $[(b,a)]$ is the multiplicative inverse of $[(a,b)]$.

Since every nonzero element of $\mathbb{Q}$ has a multiplicative inverse, $(\mathbb{Q}, +, \cdot, 0, 1)$ is a field. $\square$

Proof. **Detailed Learning Proof.**

Step 1. Recall what must be shown.

By the field definition, a field is an integral domain in which every nonzero element has a multiplicative inverse. We have already established that $\mathbb{Q}$ is an integral domain. It therefore suffices to show:

$$\forall\, [(a,b)] \in \mathbb{Q}, \quad [(a,b)] \neq 0 \implies \exists\, y \in \mathbb{Q},\ [(a,b)] \cdot y = 1.$$

Step 2. Let $[(a,b)] \in \mathbb{Q}$ be nonzero.

By the zero class criterion, $[(a,b)] = 0$ iff $a = 0$. Since we assumed $[(a,b)] \neq 0$, we have $a \neq 0$.

Step 3. Propose the candidate inverse.

Define $y := [(b,a)]$. We must first check this is well-formed: since $a \neq 0$ (established in Step 2) and $b \neq 0$ (since $(a,b) \in W$), the pair $(b,a) \in W$, so $[(b,a)] \in \mathbb{Q}$.

Step 4. Verify that $[(a,b)] \cdot [(b,a)] = 1$.

By the multiplication formula:

$$[(a,b)] \cdot [(b,a)] = [(a \cdot b,\ b \cdot a)] = [(ab,\ ab)].$$

We check $(ab, ab) \sim (1,1)$: by the definition of $\sim$, this requires $ab \cdot 1 = ab \cdot 1$, which is trivially true.

Also, $ab \neq 0$ since $a \neq 0$ and $b \neq 0$ and $\mathbb{Z}$ has no zero divisors. So $(ab, ab) \in W$ and the class $[(ab, ab)]$ is well-formed.

Since $(ab, ab) \sim (1,1)$, we have $[(ab, ab)] = [(1,1)] = 1$.

Therefore:

$$[(a,b)] \cdot [(b,a)] = 1.$$

Step 5. Conclude.

We have exhibited an element $y = [(b,a)] \in \mathbb{Q}$ such that $[(a,b)]\cdot y = 1$. Since $[(a,b)]$ was an arbitrary nonzero element of $\mathbb{Q}$, every nonzero element of $\mathbb{Q}$ has a multiplicative inverse.

Combined with the integral domain theorem, $(\mathbb{Q}, +, \cdot, 0, 1)$ is a field. □

Remark (Proof structure). The proof has a single substantive step: showing that for a nonzero $[(a,b)]$, the "swapped" class $[(b,a)]$ is a multiplicative inverse. The key computation is that $[(ab,ab)] = 1$, which follows from the trivial equivalence $(ab,ab) \sim (1,1)$. The nonvanishing of ab requires that both $a \neq 0$ (from the nonzero hypothesis) and $b \neq 0$ (from the well-formedness of the fraction-pair).

The proof builds on the integral domain theorem, which carries all of the axiom-checking work. The field proof adds only one condition to that package.

Remark (Dependencies).

- Definition 5.4.2
- Theorem 5.4.7 ($\mathbb{Q}$ is an integral domain)
- Theorem 5.2.13 (zero class criterion: $[(a,b)] = 0 \Leftrightarrow a = 0$)
- Definition 5.2.1 (multiplication formula)
- Definition 5.1.9 ($1 = [(1,1)]$)
- $\mathbb{Z}$ has no zero divisors (so $a \neq 0$ and $b \neq 0$ implies $ab \neq 0$)

Remark (Return). Return to Theorem

Theorem (Negative Exponents Agree with Inverse Powers). *Let F be a field, let $x \in F$ with $x \neq 0$, and let $m \in \mathbb{N}$. Then*

$$x^{-m} = (x^{-1})^m = (x^m)^{-1}.$$

Proof. **Professional Standard Proof.**
The first equality $x^{-m} = (x^{-1})^m$ is the definition of negative integer exponents.

For $x^{-m} = (x^m)^{-1}$, we proceed by induction on m.

Base case $m = 0$: $x^0 = 1$ and $(x^0)^{-1} = 1^{-1} = 1$, so $x^{-0} = 1 = (x^0)^{-1}$.

Inductive step: Assume $(x^k)^{-1} = x^{-k}$ for some $k \in \mathbb{N}$. Then

$$(x^{k+1})^{-1} = (x^k \cdot x)^{-1} = x^{-1} \cdot (x^k)^{-1} = x^{-1} \cdot x^{-k} = (x^{-1})^1 \cdot (x^{-1})^k = (x^{-1})^{k+1} = x^{-(k+1)}.$$

By induction, $(x^m)^{-1} = x^{-m}$ for all $m \in \mathbb{N}$. □

Proof. **Detailed Learning Proof.**

We prove the two equalities separately.

Part A: $x^{-m} = (x^{-1})^m$.

This is immediate from the definition of negative integer exponents (Definition ?? which states:

$$x^{-m} := (x^{-1})^m$$

for $x \neq 0$ and $m \in \mathbb{N}$. No computation is required; this equality holds by definition.

Part B: $x^{-m} = (x^m)^{-1}$.

We prove $(x^m)^{-1} = (x^{-1})^m$ by induction on $m \in \mathbb{N}$, which combined with Part A gives $x^{-m} = (x^m)^{-1}$.

Base case $m = 0$:

By the nonnegative exponent definition, $x^0 = 1$. Therefore $(x^0)^{-1} = 1^{-1} = 1$ by Theorem 5.4.24. Also $(x^{-1})^0 = 1$ by the nonnegative exponent definition applied to x^{-1}. So $(x^0)^{-1} = 1 = (x^{-1})^0$, establishing the base case.

Inductive step:

Assume the statement holds for some $k \in \mathbb{N}$, i.e. $(x^k)^{-1} = (x^{-1})^k$. We must show $(x^{k+1})^{-1} = (x^{-1})^{k+1}$.

Step B1. Expand x^{k+1} using the recursive definition.

By Definition 5.4.4,

$$x^{k+1} = x^k \cdot x.$$

Step B2. Invert the product.

By the field theorem on the inverse of a product (the inverse reverses order: $(a \cdot b)^{-1} = b^{-1} \cdot a^{-1}$, which equals $a^{-1} \cdot b^{-1}$ by commutativity of multiplication),

$$(x^{k+1})^{-1} = (x^k \cdot x)^{-1} = x^{-1} \cdot (x^k)^{-1}.$$

Step B3. Apply the inductive hypothesis.

By the inductive hypothesis, $(x^k)^{-1} = (x^{-1})^k$. Substituting:

$$x^{-1} \cdot (x^k)^{-1} = x^{-1} \cdot (x^{-1})^k.$$

Step B4. Collapse using the recursive definition applied to x^{-1}.

By Definition 5.4.4 applied to the base element x^{-1}:

$$(x^{-1})^{k+1} = (x^{-1})^k \cdot x^{-1} = x^{-1} \cdot (x^{-1})^k.$$

(using commutativity of multiplication for the last step). Therefore:

$$(x^{k+1})^{-1} = (x^{-1})^{k+1}.$$

Step B5. Close the induction.

By the principle of mathematical induction, $(x^m)^{-1} = (x^{-1})^m$ for all $m \in \mathbb{N}$. Combined with Part A:

$$x^{-m} = (x^{-1})^m = (x^m)^{-1}. \qquad \square$$

Remark (Proof structure). The first equality is definitional. The second requires induction because it asks about the relationship between two independently defined operations: natural-number exponentiation x^m (built by repeated multiplicati and the field inverse $(-)^{-1}$ (an axiomatically given operation). The inductive step turns on the product-inversion identity $(ab)^{-1} = b^{-1}a^{-1}$, which lets the inversion distribute through one factor at a time, consuming x^k via the inductive hypothesis. The base case $m = 0$ requires Theorem 5.4.24 to identify $1^{-1} = 1$.

Remark (Dependencies).

- Definition ?? ($x^{-m} := (x^{-1})^m$; this is the first equality)
- Definition 5.4.4 (recursive definition $x^{k+1} = x^k \cdot x$, used in inductive step)
- Definition 5.4.2 (multiplicative inverse; commutativity of multiplication)
- Theorem 5.4.24 ($1^{-1} = 1$; used in base case)
- Theorem 5.4.23 (inverse of a product: $(ab)^{-1} = b^{-1}a^{-1}$)
- Principle of mathematical induction on $\mathbb{N}$

Remark (Return). Return to Theorem

Theorem (Power of a Product in a Field). *Let $(F,+,\cdot,0,1)$ be a field. Let $q,r \in F\setminus\{0\}$ and let $a \in \mathbb{Z}$. Then*
$$(qr)^a = q^a r^a.$$

Proof. **Professional Standard Proof.**
First suppose that $a \geq 0$. Then $a \in \mathbb{N}$, and the usual law of exponents for natural-number powers gives
$$(qr)^a = q^a r^a.$$

Now suppose that $a < 0$. Then there exists $n \in \mathbb{N}$ with $n > 0$ such that
$$a = -n.$$

Since $q \neq 0$ and $r \neq 0$, their product is nonzero:
$$qr \neq 0.$$

By the definition of negative integer powers,
$$(qr)^a = (qr)^{-n} = \frac{1}{(qr)^n}.$$

Using the natural-number power law from the first case,
$$(qr)^n = q^n r^n.$$

Hence
$$(qr)^a = \frac{1}{q^n r^n}.$$

Since $q^n \neq 0$ and $r^n \neq 0$, the inverse of the product is the product of the inverses:
$$\frac{1}{q^n r^n} = \frac{1}{q^n}\frac{1}{r^n}.$$

By the definition of negative integer powers,
$$\frac{1}{q^n} = q^{-n} \qquad \text{and} \qquad \frac{1}{r^n} = r^{-n}.$$

Therefore
$$(qr)^a = q^{-n} r^{-n} = q^a r^a.$$

Thus, for every $a \in \mathbb{Z}$,
$$(qr)^a = q^a r^a.$$

□

Proof. **Detailed Learning Proof.**

Step 1. Prove the nonnegative exponent case.
Assume first that $a \geq 0$. Then $a \in \mathbb{N}$. For natural-number powers, the product law for exponents gives
$$(qr)^a = q^a r^a.$$

This proves the result for all nonnegative integer exponents.

Step 2. Reduce the negative exponent case to a positive exponent.

Assume now that $a < 0$. Then there exists $n \in \mathbb{N}$ with $n > 0$ such that

$$a = -n.$$

Since $q, r \in F \setminus \{0\}$, both q and r are nonzero. Because F is a field, the product of two nonzero elements is nonzero, so

$$qr \neq 0.$$

Therefore the negative power $(qr)^{-n}$ is defined.

Step 3. Expand the negative power of the product.

By the definition of negative integer exponents,

$$(qr)^a = (qr)^{-n} = \frac{1}{(qr)^n}.$$

Since $n \in \mathbb{N}$, the natural-number product law gives

$$(qr)^n = q^n r^n.$$

Substituting this into the previous equality gives

$$(qr)^a = \frac{1}{q^n r^n}.$$

Step 4. Split the inverse of the product.

Because $q \neq 0$ and $r \neq 0$, the powers q^n and r^n are nonzero. Hence

$$q^n r^n \neq 0.$$

In a field, the inverse of a product of nonzero elements is the product of the inverses. Thus

$$\frac{1}{q^n r^n} = \frac{1}{q^n} \frac{1}{r^n}.$$

Step 5. Convert the inverses back into negative powers.

By the definition of negative integer exponents,

$$\frac{1}{q^n} = q^{-n} \qquad \text{and} \qquad \frac{1}{r^n} = r^{-n}.$$

Therefore

$$(qr)^a = \frac{1}{q^n r^n} = \frac{1}{q^n} \frac{1}{r^n} = q^{-n} r^{-n}.$$

Since $a = -n$, this becomes

$$(qr)^a = q^a r^a.$$

This proves the theorem. □

Remark (Proof structure). The proof proceeds by cases on the integer exponent. For nonnegative exponents, the result is the usual natural-number power law. For negative exponents, the exponent is written as $-n$ with $n > 0$, the negative power is converted into an inverse of a positive power, the natural-number case is applied, and then the inverse of a product is split into the product of inverses.

Remark (Dependencies).

- Definition **??**
- Definition 5.4.2
- Natural-number product law for powers
- Nonzero product property in a field
- Inverse of a product in a field

Remark (Return). Return to Theorem

Theorem (Integer Exponent Addition Law). *Let F be a field, let $q \in F \setminus \{0\}$, and let $a, b \in \mathbb{Z}$. Then*

$$q^a \cdot q^b = q^{a+b}.$$

Proof. **Professional Standard Proof.**
We split on the signs of a and b. Write $a = \varepsilon_a |a|$ and $b = \varepsilon_b |b|$ where $\varepsilon_a, \varepsilon_b \in \{+1, -1\}$ and $|a|, |b| \in \mathbb{N}$.

Case 1: $a, b \geq 0$. This is the natural-number exponent addition law, proved by induction on b: the base case $q^a \cdot q^0 = q^a \cdot 1 = q^a = q^{a+0}$ is immediate, and the inductive step $q^a \cdot q^{b+1} = q^a \cdot (q^b \cdot q) = (q^a \cdot q^b) \cdot q = q^{a+b} \cdot q = q^{(a+b)+1} = q^{a+(b+1)}$ follows by associativity and the inductive hypothesis.

Case 2: $a \geq 0$, $b < 0$. Write $b = -n$ with $n \in \mathbb{N}$, $n \geq 1$. Sub-case $a \geq n$:

$$q^a \cdot q^{-n} = q^a \cdot (q^n)^{-1} = q^{a-n} \cdot q^n \cdot (q^n)^{-1} = q^{a-n} \cdot 1 = q^{a-n} = q^{a+b},$$

where $q^a = q^{a-n} \cdot q^n$ is the natural-number law from Case 1. Sub-case $a < n$:

$$q^a \cdot q^{-n} = q^a \cdot (q^n)^{-1} = (q^a)^{-1} \cdot \left((q^a)^{-1}\right)^{-1} \cdot q^a \cdot (q^n)^{-1}$$

Alternatively, note $q^{-n} = (q^{-1})^n$ and use

$$q^a \cdot (q^{-1})^n = (q^{-1})^{n-a} \cdot \left(q^a \cdot (q^{-1})^a\right) = (q^{-1})^{n-a} \cdot 1 = q^{-(n-a)} = q^{a-n} = q^{a+b}.$$

Case 3: $a < 0$, $b \geq 0$. Symmetric to Case 2 by commutativity of multiplication.

Case 4: $a, b < 0$. Write $a = -m$, $b = -n$ with $m, n \in \mathbb{N}$, $m, n \geq 1$. Then

$$q^{-m} \cdot q^{-n} = (q^m)^{-1} \cdot (q^n)^{-1} = (q^n \cdot q^m)^{-1} = (q^{m+n})^{-1} = q^{-(m+n)} = q^{a+b}.$$

In all cases $q^a \cdot q^b = q^{a+b}$. □

Proof. **Detailed Learning Proof.**

The integers a, b each have a sign, giving four combinations. We handle each one explicitly. Throughout, $q \in F \setminus \{0\}$ is fixed, so all powers of q and q^{-1} are well-defined.

Preliminary lemma (natural-number law).

Before splitting on signs we record the natural-number case as a lemma, since all other cases reduce to it.

Claim: For all $a, b \in \mathbb{N}$, $q^a \cdot q^b = q^{a+b}$.

Proof of claim by induction on b.
Base case $b = 0$:

$$q^a \cdot q^0 = q^a \cdot 1 = q^a = q^{a+0}.$$

Here $q^0 = 1$ by Definition 5.4.4, and $q^a \cdot 1 = q^a$ by the multiplicative identity axiom.

Inductive step: Assume $q^a \cdot q^k = q^{a+k}$ for some $k \in \mathbb{N}$. Then

$$q^a \cdot q^{k+1} = q^a \cdot (q^k \cdot q) = (q^a \cdot q^k) \cdot q = q^{a+k} \cdot q = q^{(a+k)+1} = q^{a+(k+1)}.$$

Step 1 uses $q^{k+1} = q^k \cdot q$ from the recursive definition. Step 2 uses associativity of multiplication. Step 3 uses the inductive hypothesis. Step 4 uses the recursive definition again. Step 5 uses associativity of addition in $\mathbb{N}$.

By induction, the natural-number law holds for all $a, b \in \mathbb{N}$. $\square$

We now proceed through the four sign cases for $a, b \in \mathbb{Z}$.

Case 1: $a \geq 0$ **and** $b \geq 0$.
This is the natural-number law just proved. $q^a \cdot q^b = q^{a+b}$.

Case 2: $a \geq 0$ **and** $b < 0$.
Write $b = -n$ where $n \in \mathbb{N}$, $n \geq 1$.
Sub-case 2a: $a \geq n$.
Write $a = (a-n)+n$ with $a-n \in \mathbb{N}$. Then by the natural-number law, $q^a = q^{a-n} \cdot q^n$. Therefore:

$$q^a \cdot q^{-n} = q^{a-n} \cdot q^n \cdot (q^n)^{-1} = q^{a-n} \cdot \left(q^n \cdot (q^n)^{-1}\right) = q^{a-n} \cdot 1 = q^{a-n} = q^{a+(-n)} = q^{a+b}.$$

Step 1 uses $q^{-n} = (q^n)^{-1}$ (Theorem 5.4.12). Step 2 uses associativity. Step 3 uses the multiplicative inverse axiom $q^n \cdot (q^n)^{-1} = 1$. Step 4 uses the multiplicative identity axiom.
Sub-case 2b: $a < n$.
Write $n = a+(n-a)$ with $n-a \in \mathbb{N}$, $n-a \geq 1$. Then by the natural-number law, $q^n = q^a \cdot q^{n-a}$. Therefore:

$$q^a \cdot q^{-n} = q^a \cdot (q^a \cdot q^{n-a})^{-1} = q^a \cdot (q^{n-a})^{-1} \cdot (q^a)^{-1} = \left(q^a \cdot (q^a)^{-1}\right) \cdot (q^{n-a})^{-1} = 1 \cdot (q^{n-a})^{-1} = q^{-(n-a)} = q^{a-n} = q$$

Step 1 uses $q^{-n} = (q^n)^{-1}$ and $q^n = q^a \cdot q^{n-a}$. Step 2 uses the product-inversion identity $(uv)^{-1} = v^{-1}u^{-1}$. Step 3 uses commutativity and associativity to regroup. Step 4 uses the multiplicative inverse axiom. Step 5 uses $q^{-(n-a)} = (q^{n-a})^{-1}$ and that $a + b = a - n = -(n-a)$.
Sub-case 2c: $a = n$.

$$q^a \cdot q^{-n} = q^n \cdot (q^n)^{-1} = 1 = q^0 = q^{n+(-n)} = q^{a+b}.$$

Case 3: $a < 0$ **and** $b \geq 0$.
Write $a = -m$ where $m \in \mathbb{N}$, $m \geq 1$. By commutativity of multiplication, $q^a \cdot q^b = q^b \cdot q^a$. This reduces to Case 2 with the roles of a and b swapped (and $a + b = b + a$). Therefore $q^a \cdot q^b = q^{a+b}$.

Case 4: $a < 0$ **and** $b < 0$.
Write $a = -m$ and $b = -n$ where $m, n \in \mathbb{N}$, $m, n \geq 1$. Then:

$$q^{-m} \cdot q^{-n} = (q^m)^{-1} \cdot (q^n)^{-1} = (q^n \cdot q^m)^{-1} = (q^{m+n})^{-1} = q^{-(m+n)} = q^{(-m)+(-n)} = q^{a+b}.$$

Step 1 uses Theorem 5.4.12 twice. Step 2 uses the product-inversion identity $(uv)^{-1} = v^{-1}u^{-1}$ and commutativity of multiplication. Step 3 uses the natural-number law on $q^n \cdot q^m = q^{m+n}$. Step 4 uses Theorem 5.4.12. Step 5 uses $-(m+n) = (-m) + (-n)$ in $\mathbb{Z}$.
Conclusion.
In all four cases $q^a \cdot q^b = q^{a+b}$. $\square$

Remark (Proof structure). The proof is a case split on $\operatorname{sgn}(a)$ and $\operatorname{sgn}(b)$, giving four main cases (Case 2 further splits into three sub-cases by comparing a and $n = -b$). The natural-number case is the inductive spine; every other case reduces to it plus the identity $(q^n)^{-1} = q^{-n}$ and the product-inversion identity $(uv)^{-1} = v^{-1}u^{-1}$.

The key insight in Case 2 is that $q^a \cdot (q^n)^{-1}$ requires cancelling $q^{\min(a,n)}$ factors, leaving either a positive or negative residual power depending on whether $a \geq n$ or $a < n$. Case 4 is cleaner: two negative exponents convert to two inverses, whose product inverts as a whole via the product-inversion identity, and the natural-number law handles the sum of the absolute values.

Remark (Dependencies).

- Definition 5.4.4 (recursive definition: $q^0 = 1$, $q^{k+1} = q^k \cdot q$)
- Definition ?? ($q^{-n} := (q^{-1})^n$)
- Theorem 5.4.12 ($q^{-n} = (q^n)^{-1}$; used in Cases 2, 3, 4)
- Definition 5.4.2 (associativity and commutativity of multiplication; multiplicati inverse axiom; multiplicative identity axiom)
- Theorem 5.4.23 (product-inversion identity $(uv)^{-1} = v^{-1}u^{-1}$; used in Sub-case 2b and Case 4)
- Principle of mathematical induction on $\mathbb{N}$ (used in the preliminary natural-number lemma)
- Arithmetic of $\mathbb{Z}$: $-(m+n) = (-m)+(-n)$, associativity of integer addition

Remark (Return). Return to Theorem

Theorem (Power of a Power in a Field). *Let* $(F, +, \cdot, 0, 1)$ *be a field, let* $q \in F \setminus \{0\}$*, and let* $a, b \in \mathbb{Z}$*. Then*

$$(q^a)^b = q^{ab}.$$

Proof. **Professional Standard Proof.**
We split according to the sign of b.

If $b = 0$, then

$$(q^a)^0 = 1 = q^0 = q^{a \cdot 0}.$$

If $b = n > 0$, we argue by induction on n. For $n = 1$ the identity is

$$(q^a)^1 = q^a = q^{a \cdot 1}.$$

Assume $(q^a)^n = q^{an}$. Then

$$(q^a)^{n+1} = (q^a)^n q^a = q^{an} q^a = q^{an+a} = q^{a(n+1)},$$

using the induction hypothesis and the integer exponent addition law.

Finally, let $b = -n$ with $n > 0$. Then

$$(q^a)^b = (q^a)^{-n} = ((q^a)^{-1})^n = (q^{-a})^n = q^{(-a)n} = q^{a(-n)} = q^{ab},$$

using the theorem on negative exponents and the already established positive-exponent case. ☐

Remark (Proof structure). The proof separates the positive, zero, and negative cases for the outer exponent b. The positive case is an induction on b, with the induction step supplied by the exponent addition law. The negative case rewrites the outer negative exponent as a positive power of an inverse and then applies the positive case to q^{-a}. This reduces the entire statement to the two basic mechanisms already built into the integer exponent calculus: recursion for positive powers and inversion for negative powers.

Remark (Dependencies).

- Definition 5.4.2 (field multiplication and multiplicative identity)
- Theorem 5.4.14
- Theorem 5.4.12

Remark (Return). Return to Theorem

Theorem (Additive Cancellation). *Let F be a field and let $a, b, c \in F$. If*

$$a + c = b + c,$$

then $a = b$.

Proof. **Professional Standard Proof.**
Assume $a+c = b+c$. Adding $-c$ to both sides and applying associativity and the additive inverse and identity axioms:

$$a = a + 0 = a + (c + (-c)) = (a + c) + (-c) = (b + c) + (-c) = b + (c + (-c)) = b + 0 = b.$$

□

Proof. **Detailed Learning Proof.**

Step 1. State the hypothesis.
Assume $a + c = b + c$.
Step 2. Add $-c$ to both sides.
Since $-c \in F$ (additive inverses exist in F), add $-c$ to both sides:

$$(a + c) + (-c) = (b + c) + (-c).$$

Step 3. Apply associativity of addition to both sides.
By associativity,

$$(a + c) + (-c) = a + (c + (-c)) \quad \text{and} \quad (b + c) + (-c) = b + (c + (-c)).$$

Step 4. Apply the additive inverse axiom.
By the additive inverse axiom, $c + (-c) = 0$. Therefore:

$$a + (c + (-c)) = a + 0 \quad \text{and} \quad b + (c + (-c)) = b + 0.$$

Step 5. Apply the additive identity axiom.
By the additive identity axiom, $a + 0 = a$ and $b + 0 = b$.
Step 6. Conclude.
Chaining Steps 2-5:

$$a = a + 0 = a + (c + (-c)) = (a + c) + (-c) = (b + c) + (-c) = b + (c + (-c)) = b + 0 = b.$$

Therefore $a = b$. □

Remark (Proof structure). The proof adds the additive inverse $-c$ to both sides of the hypothesis, then unwinds the result using associativity, the additive inverse axiom, and the additive identity axiom. Each step cites exactly one field axiom. The conclusion $a = b$ is reached by a chain of equalities with no auxiliary lemmas required.

Remark (Dependencies).

- Definition 5.4.2 (additive inverse; additive identity; associativity of addition)

Remark (Return). Return to Theorem

Theorem (Zero Product). *Let F be a field and let $a \in F$. Then*

$$a \cdot 0 = 0.$$

Proof. **Professional Standard Proof.**
Since 0 is the additive identity, $0 + 0 = 0$. By distributivity,

$$a \cdot 0 = a \cdot (0 + 0) = a \cdot 0 + a \cdot 0.$$

Adding $-(a \cdot 0)$ to both sides,

$$0 = a \cdot 0.$$

□

Proof. **Detailed Learning Proof.**

Step 1. Use the additive identity law to write 0 as $0 + 0$.
By the field axiom for the additive identity,

$$0 + 0 = 0.$$

Step 2. Multiply both sides on the left by a.

$$a \cdot 0 = a \cdot (0 + 0).$$

Step 3. Apply distributivity.
By the distributivity axiom,

$$a \cdot (0 + 0) = a \cdot 0 + a \cdot 0.$$

Combining Steps 2 and 3:

$$a \cdot 0 = a \cdot 0 + a \cdot 0.$$

Step 4. Add the additive inverse of $a \cdot 0$ to both sides.
Adding $-(a \cdot 0)$ to both sides of $a \cdot 0 = a \cdot 0 + a \cdot 0$:

$$a \cdot 0 + (-(a \cdot 0)) = a \cdot 0 + a \cdot 0 + (-(a \cdot 0)).$$

Step 5. Apply the additive inverse axiom and associativity.
The left-hand side is 0 by the additive inverse axiom. On the right, associativity of addition gives

$$a \cdot 0 + (a \cdot 0 + (-(a \cdot 0))) = a \cdot 0 + 0 = a \cdot 0.$$

Therefore

$$0 = a \cdot 0.$$

Step 6. Conclude.
By commutativity of equality, $a \cdot 0 = 0$. □

Remark (Proof structure). The key move is writing $0 = 0+0$ using the additive identity law, then distributing a across the sum to obtain $a \cdot 0 = a \cdot 0 + a \cdot 0$. This says $a \cdot 0$ is a solution to $x = x + a \cdot 0$, and the only solution to that equation is $x = 0$. Cancellation is performed directly by adding the additive inverse rather than invoking the cancellation theorem, keeping the proof self-contained within the field axioms alone.

Remark (Dependencies).

- Definition 5.4.2 (additive identity: $0+0 = 0$; distributivity; additive inverse; associativity and commutativity of addition)

Remark (Return). Return to Theorem

Theorem (Double Negation). *Let F be a field and let $a \in F$. Then*

$$-(-a) = a.$$

Proof. **Professional Standard Proof.**
By the additive inverse axiom, $(-a)+a=0$, so a is an additive inverse of $-a$. Also by the additive inverse axiom, $(-a)+(-(-a)) = 0$, so $-(-a)$ is an additive inverse of $-a$. By additive cancellation, additive inverses are unique, hence $-(-a)=a$. □

Proof. **Detailed Learning Proof.**

Step 1. Observe that a is an additive inverse of $-a$.
By the additive inverse axiom applied to a,

$$a + (-a) = 0.$$

By commutativity of addition,

$$(-a) + a = 0.$$

So a satisfies the defining condition for the additive inverse of $-a$.
Step 2. Observe that $-(-a)$ is also an additive inverse of $-a$.
By the additive inverse axiom applied to $-a$,

$$(-a) + (-(-a)) = 0.$$

So $-(-a)$ also satisfies the defining condition for the additive inverse of $-a$.
Step 3. Show additive inverses are unique via cancellation.
From Steps 1 and 2:

$$(-a) + a = 0 = (-a) + (-(-a)).$$

By additive cancellation (with $b := a$, $c := -(-a)$, and the common left summand $-a$), we conclude

$$a = -(-a).$$

Step 4. Conclude.
By symmetry of equality, $-(-a) = a$. □

Remark (Proof structure). Both a and $-(-a)$ satisfy the defining equation for the additive inverse of $-a$, namely $(-a)+x=0$. Since additive cancellation forces any two solutions to that equation to be equal, they must coincide. This is the standard uniqueness-of-inverses argument: exhibit two objects satisfying the same defining property, then cancel to conclude they are equal.

Remark (Dependencies).

- Definition 5.4.2 (additive inverse axiom; commutativity of addition)
- Theorem 5.4.21 (uniqueness of additive inverses follows from cancellation)

Remark (Return). Return to Theorem

Theorem (Negation of a Product). *Let F be a field and let $a, b \in F$. Then*

$$-(a \cdot b) = (-a) \cdot b = a \cdot (-b).$$

Proof. **Professional Standard Proof.**
We show $(-a) \cdot b$ is the additive inverse of $a \cdot b$ by computing their sum:

$$a \cdot b + (-a) \cdot b = (a + (-a)) \cdot b = 0 \cdot b = 0.$$

By uniqueness of additive inverses, $(-a) \cdot b = -(a \cdot b)$. The identity $a \cdot (-b) = -(a \cdot b)$ follows symmetrically:

$$a \cdot b + a \cdot (-b) = a \cdot (b + (-b)) = a \cdot 0 = 0.$$

□

Proof. **Detailed Learning Proof.**

We prove both equalities separately.
Part A: $(-a) \cdot b = -(a \cdot b)$.
Step A1. Compute $a \cdot b + (-a) \cdot b$.
By distributivity (right),

$$a \cdot b + (-a) \cdot b = (a + (-a)) \cdot b.$$

Step A2. Apply the additive inverse axiom.
By the additive inverse axiom, $a + (-a) = 0$. Therefore,

$$(a + (-a)) \cdot b = 0 \cdot b.$$

Step A3. Apply the zero product theorem.
By Theorem 5.4.17 (with the roles of a and b swapped, using commutativity of multiplication), $0 \cdot b = 0$. Therefore,

$$a \cdot b + (-a) \cdot b = 0.$$

Step A4. Conclude by uniqueness of additive inverses.
The equation $a \cdot b + (-a) \cdot b = 0$ states that $(-a) \cdot b$ is an additive inverse of $a \cdot b$. By additive cancellation, additive inverses are unique. Therefore,

$$(-a) \cdot b = -(a \cdot b).$$

Part B: $a \cdot (-b) = -(a \cdot b)$.
Step B1. Compute $a \cdot b + a \cdot (-b)$.
By distributivity (left),

$$a \cdot b + a \cdot (-b) = a \cdot (b + (-b)).$$

Step B2. Apply the additive inverse axiom.
By the additive inverse axiom, $b + (-b) = 0$. Therefore,

$$a \cdot (b + (-b)) = a \cdot 0.$$

Step B3. Apply the zero product theorem.
By Theorem 5.4.17, $a \cdot 0 = 0$. Therefore,

$$a \cdot b + a \cdot (-b) = 0.$$

Step B4. Conclude by uniqueness of additive inverses.
The equation $a \cdot b + a \cdot (-b) = 0$ states that $a \cdot (-b)$ is an additive inverse of $a \cdot b$. By additive cancellation, additive inverses are unique. Therefore,

$$a \cdot (-b) = -(a \cdot b).$$

Conclusion.
Combining Parts A and B:

$$-(a \cdot b) = (-a) \cdot b = a \cdot (-b).$$ □

Remark (Proof structure). Both identities use the same strategy: show that the candidate equals $-(a \cdot b)$ by verifying it sums with $a \cdot b$ to give 0, then invoke uniqueness of additive inverses via cancellation. The computation in each case is one application of distributivity, one application of the additive inverse axiom, and one application of the zero product theorem. Part A uses right-distributivity; Part B uses left-distributivity.

Remark (Dependencies).

- Definition 5.4.2 (left and right distributivity; additive inverse axiom; commutativity of multiplication)
- Theorem 5.4.17 ($a \cdot 0 = 0$ and $0 \cdot b = 0$)
- Theorem 5.4.21 (uniqueness of additive inverses)

Remark (Return). Return to Corollary

Corollary (Product of Negatives). *Let F be a field and let $a, b \in F$. Then*

$$(-a) \cdot (-b) = a \cdot b.$$

Proof. **Professional Standard Proof.**
By the negation of a product theorem applied twice:

$$(-a) \cdot (-b) = -(a \cdot (-b)) = -(-(a \cdot b)) = a \cdot b.$$

□

Proof. **Detailed Learning Proof.**

Step 1. Apply the negation of a product theorem to the outer negation.
By Theorem 5.4.19 with a replaced by a and b replaced by $-b$:

$$(-a) \cdot (-b) = -(a \cdot (-b)).$$

Step 2. Apply the negation of a product theorem to the inner term.
By Theorem 5.4.19,

$$a \cdot (-b) = -(a \cdot b).$$

Substituting into Step 1:

$$(-a) \cdot (-b) = -(-(a \cdot b)).$$

Step 3. Apply double negation.
By Theorem 5.4.18,

$$-(-(a \cdot b)) = a \cdot b.$$

Step 4. Conclude.
Chaining Steps 1-3:

$$(-a) \cdot (-b) = -(-(a \cdot b)) = a \cdot b.$$

□

Remark (Proof structure). This corollary is a two-line chain from the negation of a product theorem and double negation. The negation of a product theorem is applied twice in sequence: first to pull $(-a)$ out of the product, producing $-(a \cdot (-b))$, then to pull $(-b)$ out, producing $-(-(a \cdot b))$. Double negation then collapses the two negation signs to nothing. No field axioms are cited directly; all work was done in the two preceding theorems.

Remark (Dependencies).

- Theorem 5.4.19 ($(-a) \cdot b = -(a \cdot b)$ and $a \cdot (-b) = -(a \cdot b)$)
- Theorem 5.4.18 ($-(-(a \cdot b)) = a \cdot b$)

Remark (Return). Return to Theorem

Theorem (Multiplicative Cancellation). *Let F be a field and let $a, b, c \in F$ with $c \neq 0$. If*

$$a \cdot c = b \cdot c,$$

then $a = b$.

Proof. **Professional Standard Proof.**
Assume $a \cdot c = b \cdot c$ with $c \neq 0$. Since F is a field, c has a multiplicative inverse c^{-1}. Multiply the given equality on the right by c^{-1}:

$$(a \cdot c) \cdot c^{-1} = (b \cdot c) \cdot c^{-1}.$$

By associativity,

$$a \cdot (c \cdot c^{-1}) = b \cdot (c \cdot c^{-1}).$$

Since $c \cdot c^{-1} = 1$, this becomes

$$a \cdot 1 = b \cdot 1.$$

By the multiplicative identity axiom, $a = b$. □

Remark (Proof structure). The proof multiplies both sides of the equation by the inverse of the common nonzero factor. Associativity moves the factor c^{-1} next to c, the inverse axiom collapses $c \cdot c^{-1}$ to 1, and the identity axiom then removes the surviving factor of 1. This is the multiplicative analogue of the usual additive cancellation argument.

Remark (Dependencies).

- Definition 5.4.2 (multiplicative inverse; multiplicative identity; associativity of multiplication)

Remark (Return). Return to Theorem

Theorem (Uniqueness of Multiplicative Inverse). *Let F be a field and let $a \in F$ with $a \neq 0$. If $b \in F$ satisfies $a \cdot b = 1$, then $b = a^{-1}$.*

Proof. **Professional Standard Proof.**
Since $a \neq 0$ and F is a field, the inverse a^{-1} exists and satisfies

$$a \cdot a^{-1} = 1.$$

Now compute:

$$b = b \cdot 1 = b \cdot (a \cdot a^{-1}) = (b \cdot a) \cdot a^{-1} = (a \cdot b) \cdot a^{-1} = 1 \cdot a^{-1} = a^{-1},$$

using associativity, commutativity, the hypothesis $a \cdot b = 1$, and the multiplicative identity axiom. □

Remark (Proof structure). The proof exhibits two right inverses of the same nonzero element a: b from the hypothesis and a^{-1} from the field axiom. Instead of invoking cancellation, it inserts the identity $a \cdot a^{-1} = 1$ and then rewrites b by associativity until the hypothesis $a \cdot b = 1$ can be substituted. The conclusion falls out by a direct chain of equalities.

Remark (Dependencies).

- Definition 5.4.2 (multiplicative inverse; multiplicative identity; associativity and commutativity of multiplication)

Remark (Return). Return to Theorem

Theorem (The Inverse of One Is One). *In any field F,*

$$1^{-1} = 1.$$

Proof. **Professional Standard Proof.**
Since $1 \cdot 1 = 1$, the element 1 is a multiplicative inverse of 1. By uniqueness of multiplicative inverses, $1^{-1} = 1$. ☐

Proof. **Detailed Learning Proof.**

Step 1. Recall the defining condition for a multiplicative inverse.
An element $y \in F$ is the multiplicative inverse of $x \in F \setminus \{0\}$ if and only if

$$x \cdot y = 1.$$

By uniqueness of multiplicative inverses (Theorem 5.4.23), there is exactly one such y, denoted x^{-1}.

Step 2. Verify that 1 satisfies the defining condition for 1^{-1}.
Compute:

$$1 \cdot 1 = 1.$$

This uses the multiplicative identity axiom: 1 is the identity element, so $1 \cdot 1 = 1$.

Step 3. Conclude by uniqueness.
The equation $1 \cdot 1 = 1$ states that 1 is a multiplicative inverse of 1. By uniqueness of multiplicative inverses,

$$1^{-1} = 1.$$ ☐

Remark (Proof structure). The proof is a one-line uniqueness argument. Exhibit 1 as a solution to $1 \cdot y = 1$, which holds trivially by the multiplicative identity axiom. Uniqueness of multiplicative inverses then pins down $1^{-1} = 1$. No field axiom beyond the multiplicative identity and uniqueness of inverses is required.

Remark (Dependencies).

- Definition 5.4.2 (multiplicative identity axiom: $1 \cdot 1 = 1$)
- Theorem 5.4.23 (the multiplicative inverse of any nonzero element is unique)

Remark (Return). Return to Theorem

Theorem (Division by One in $\mathbb{Q}$). *For every $a \in \mathbb{Q}$,*

$$\frac{a}{1} = a.$$

Proof. **Professional Standard Proof.**
Let $a \in \mathbb{Q}$. By the definition of division in $\mathbb{Q}$,

$$\frac{a}{1} = a \cdot 1^{-1}.$$

Since 1 is the multiplicative identity in $\mathbb{Q}$, its inverse is itself:

$$1^{-1} = 1.$$

Therefore

$$\frac{a}{1} = a \cdot 1 = a.$$

Hence

$$\frac{a}{1} = a.$$

□

Proof. **Detailed Learning Proof.**

Step 1. Rewrite division as multiplication by an inverse.
Let $a \in \mathbb{Q}$. By the definition of rational division,

$$\frac{a}{1} = a \cdot 1^{-1}.$$

Thus the theorem reduces to identifying the inverse of 1.

Step 2. Identify the inverse of 1.
Since $\mathbb{Q}$ is a field, 1 is the multiplicative identity. Hence

$$1 \cdot 1 = 1.$$

Therefore 1 is its own multiplicative inverse, so

$$1^{-1} = 1.$$

Step 3. Finish using the multiplicative identity law.
Substituting $1^{-1} = 1$ into the expression for division gives

$$\frac{a}{1} = a \cdot 1.$$

By the multiplicative identity law in $\mathbb{Q}$,

$$a \cdot 1 = a.$$

Therefore

$$\frac{a}{1} = a.$$

□

Remark (Proof structure). The proof is direct. Division by 1 is rewritten as multiplication by the inverse of 1. Since 1 is its own inverse in a field, the expression reduces to multiplication by the multiplicative identity.

Remark (Dependencies).

- Definition 5.2.18
- Theorem 5.4.9
- Multiplicative identity law in $\mathbb{Q}$
- Uniqueness of multiplicative inverses

Remark (Return). Return to Theorem

Theorem (A Rational Quotient Is Zero Exactly When Its Numerator Is Zero). *For every $a, b \in \mathbb{Q}$ with $b \neq 0$,*

$$\frac{a}{b} = 0 \quad \Longleftrightarrow \quad a = 0.$$

Proof. **Professional Standard Proof.**
Let $a, b \in \mathbb{Q}$ with $b \neq 0$.

First suppose that

$$\frac{a}{b} = 0.$$

By the definition of rational division,

$$\frac{a}{b} = a \cdot b^{-1}.$$

Hence

$$a \cdot b^{-1} = 0.$$

Since $b \neq 0$, its multiplicative inverse b^{-1} exists and is nonzero. Because $\mathbb{Q}$ is an integral domain, the zero-product property gives

$$a = 0 \qquad \text{or} \qquad b^{-1} = 0.$$

Since $b^{-1} \neq 0$, it follows that

$$a = 0.$$

Conversely, suppose that

$$a = 0.$$

Then, by the definition of rational division,

$$\frac{a}{b} = a \cdot b^{-1} = 0 \cdot b^{-1} = 0.$$

Therefore

$$\frac{a}{b} = 0 \quad \Longleftrightarrow \quad a = 0.$$

□

Proof. **Detailed Learning Proof.**

Step 1. Rewrite the quotient as multiplication by an inverse.
Let $a, b \in \mathbb{Q}$ with

$$b \neq 0.$$

By the definition of rational division,

$$\frac{a}{b} = a \cdot b^{-1}.$$

Thus the statement

$$\frac{a}{b} = 0$$

is the same as the statement

$$a \cdot b^{-1} = 0.$$

Step 2. Prove the forward implication.
Assume

$$\frac{a}{b} = 0.$$

Then

$$a \cdot b^{-1} = 0.$$

Since $b \neq 0$, the inverse b^{-1} exists. Also,

$$b^{-1} \neq 0,$$

because if $b^{-1} = 0$, then multiplying by b gives

$$1 = b \cdot b^{-1} = b \cdot 0 = 0,$$

contradicting the field axioms.

Since $\mathbb{Q}$ is an integral domain, a product in $\mathbb{Q}$ is zero only if one of its factors is zero. Therefore, from

$$a \cdot b^{-1} = 0,$$

it follows that

$$a = 0 \qquad \text{or} \qquad b^{-1} = 0.$$

The second possibility is impossible, so

$$a = 0.$$

Step 3. Prove the reverse implication.
Assume

$$a = 0.$$

Then, by the definition of rational division,

$$\frac{a}{b} = a \cdot b^{-1}.$$

Substituting $a = 0$ gives

$$\frac{a}{b} = 0 \cdot b^{-1}.$$

By the absorbing property of zero,

$$0 \cdot b^{-1} = 0.$$

Hence

$$\frac{a}{b} = 0.$$

Step 4. Conclude the biconditional.
The forward implication proves

$$\frac{a}{b} = 0 \implies a = 0.$$

The reverse implication proves

$$a = 0 \implies \frac{a}{b} = 0.$$

Therefore

$$\frac{a}{b} = 0 \iff a = 0.$$

□

Remark (Proof structure). The proof is a direct proof of both implications. The forward direction rewrites the quotient as $a \cdot b^{-1}$ and uses the zero-product property in the integral domain $\mathbb{Q}$. The reverse direction substitutes $a = 0$ into the definition of division and uses the absorbing property of zero.

Remark (Dependencies).

- Definition 5.2.18
- Theorem 5.4.7
- Theorem 5.4.9
- Absorbing property of zero under multiplication

Remark (Return). Return to Theorem

Theorem (Cross-Multiplication Criterion for Rational Equality). *For every* $a, b, c, d \in \mathbb{Q}$ *with* $b \neq 0$ *and* $d \neq 0$*,*

$$\frac{a}{b} = \frac{c}{d} \iff a \cdot d = b \cdot c.$$

Proof. **Professional Standard Proof.**
Let $a, b, c, d \in \mathbb{Q}$ with $b \neq 0$ and $d \neq 0$.

First suppose that

$$\frac{a}{b} = \frac{c}{d}.$$

By the definition of rational division,

$$a \cdot b^{-1} = c \cdot d^{-1}.$$

Multiplying both sides by bd gives

$$(a \cdot b^{-1}) \cdot bd = (c \cdot d^{-1}) \cdot bd.$$

Using associativity, commutativity, and the inverse laws in $\mathbb{Q}$, this simplifies to

$$a \cdot d = b \cdot c.$$

Conversely, suppose that

$$a \cdot d = b \cdot c.$$

Since $b \neq 0$ and $d \neq 0$, the product bd is nonzero. Multiplying both sides by $(bd)^{-1}$ gives

$$(a \cdot d)(bd)^{-1} = (b \cdot c)(bd)^{-1}.$$

Using associativity, commutativity, and the inverse laws in $\mathbb{Q}$, this simplifies to

$$a \cdot b^{-1} = c \cdot d^{-1}.$$

By the definition of rational division,

$$\frac{a}{b} = \frac{c}{d}.$$

Therefore

$$\frac{a}{b} = \frac{c}{d} \iff a \cdot d = b \cdot c.$$

□

Proof. **Detailed Learning Proof.**

Step 1. Rewrite the quotients using multiplicative inverses.
Let $a, b, c, d \in \mathbb{Q}$ with

$$b \neq 0 \quad \text{and} \quad d \neq 0.$$

By the definition of rational division,

$$\frac{a}{b} = a \cdot b^{-1} \quad \text{and} \quad \frac{c}{d} = c \cdot d^{-1}.$$

Step 2. Prove the forward implication.
Assume
$$\frac{a}{b} = \frac{c}{d}.$$
Then
$$a \cdot b^{-1} = c \cdot d^{-1}.$$
Multiplying both sides by bd gives
$$(a \cdot b^{-1}) \cdot bd = (c \cdot d^{-1}) \cdot bd.$$
The left-hand side simplifies as follows:
$$(a \cdot b^{-1}) \cdot bd = a \cdot (b^{-1} \cdot b) \cdot d = a \cdot 1 \cdot d = a \cdot d.$$
The right-hand side simplifies as follows:
$$(c \cdot d^{-1}) \cdot bd = c \cdot b \cdot (d^{-1} \cdot d) = c \cdot b \cdot 1 = b \cdot c.$$
Therefore
$$a \cdot d = b \cdot c.$$

Step 3. Prove the reverse implication.
Assume
$$a \cdot d = b \cdot c.$$
Since $b \neq 0$ and $d \neq 0$, the product bd is nonzero. Therefore $(bd)^{-1}$ exists in $\mathbb{Q}$. Multiplying both sides by $(bd)^{-1}$ gives
$$(a \cdot d)(bd)^{-1} = (b \cdot c)(bd)^{-1}.$$
Since $\mathbb{Q}$ is a field,
$$(bd)^{-1} = b^{-1}d^{-1}.$$
Hence the left-hand side becomes
$$(a \cdot d)(bd)^{-1} = a \cdot d \cdot b^{-1} \cdot d^{-1} = a \cdot b^{-1} \cdot (d \cdot d^{-1}) = a \cdot b^{-1}.$$
Similarly, the right-hand side becomes
$$(b \cdot c)(bd)^{-1} = b \cdot c \cdot b^{-1} \cdot d^{-1} = c \cdot (b \cdot b^{-1}) \cdot d^{-1} = c \cdot d^{-1}.$$
Therefore
$$a \cdot b^{-1} = c \cdot d^{-1}.$$
By the definition of rational division,
$$\frac{a}{b} = \frac{c}{d}.$$

Step 4. Conclude the biconditional.
The forward implication proves
$$\frac{a}{b} = \frac{c}{d} \implies a \cdot d = b \cdot c.$$
The reverse implication proves
$$a \cdot d = b \cdot c \implies \frac{a}{b} = \frac{c}{d}.$$
Therefore
$$\frac{a}{b} = \frac{c}{d} \iff a \cdot d = b \cdot c.$$

□

Remark (Proof structure). The proof is a direct proof of both implications. The forward direction rewrites rational division as multiplication by inverses and then clears denominators by multiplying by bd. The reverse direction starts from the cross-product equality and multiplies by the inverse of the nonzero product bd to recover equality of the two quotients.

Remark (Dependencies).

- Definition 5.2.18
- Theorem 5.4.7
- Theorem 5.4.9
- Associativity and commutativity of multiplication in $\mathbb{Q}$
- Multiplicative inverse law in $\mathbb{Q}$
- Nonzero product property in $\mathbb{Q}$

Remark (Return). Return to Theorem

Theorem (Scaling Numerator and Denominator by the Same Nonzero Rational). *For every $a, b, e \in \mathbb{Q}$ with $b \neq 0$ and $e \neq 0$,*

$$\frac{a}{b} = \frac{a \cdot e}{b \cdot e}.$$

Proof. **Professional Standard Proof.**
Let $a, b, e \in \mathbb{Q}$ with

$$b \neq 0 \qquad \text{and} \qquad e \neq 0.$$

Since $\mathbb{Q}$ is an integral domain, the product of two nonzero rational numbers is nonzero. Hence

$$b \cdot e \neq 0.$$

By the cross-multiplication criterion, it is enough to prove

$$a \cdot (b \cdot e) = b \cdot (a \cdot e).$$

Using associativity and commutativity of multiplication in $\mathbb{Q}$,

$$a \cdot (b \cdot e) = (a \cdot b) \cdot e = (b \cdot a) \cdot e = b \cdot (a \cdot e).$$

Therefore

$$\frac{a}{b} = \frac{a \cdot e}{b \cdot e}.$$

□

Proof. **Detailed Learning Proof.**

Step 1. Check that the scaled denominator is nonzero.
Let $a, b, e \in \mathbb{Q}$ with

$$b \neq 0 \qquad \text{and} \qquad e \neq 0.$$

Since $\mathbb{Q}$ is an integral domain, the product of two nonzero elements is nonzero. Therefore

$$b \cdot e \neq 0.$$

Thus both quotients

$$\frac{a}{b} \qquad \text{and} \qquad \frac{a \cdot e}{b \cdot e}$$

are defined.

Step 2. Reduce the desired equality to a cross-product equality.
By the cross-multiplication criterion, the equality

$$\frac{a}{b} = \frac{a \cdot e}{b \cdot e}$$

is equivalent to

$$a \cdot (b \cdot e) = b \cdot (a \cdot e).$$

Hence it remains to prove this product equality in $\mathbb{Q}$.

Step 3. Prove the cross-product equality.

Using associativity and commutativity of multiplication in $\mathbb{Q}$,

$$a \cdot (b \cdot e) = (a \cdot b) \cdot e = (b \cdot a) \cdot e = b \cdot (a \cdot e).$$

Therefore

$$a \cdot (b \cdot e) = b \cdot (a \cdot e).$$

Step 4. Conclude the quotient equality.
Since

$$a \cdot (b \cdot e) = b \cdot (a \cdot e),$$

the cross-multiplication criterion gives

$$\frac{a}{b} = \frac{a \cdot e}{b \cdot e}.$$

This proves the theorem. □

Remark (Proof structure). The proof is direct. First, the scaled denominator $b \cdot e$ is shown to be nonzero. Then the desired equality of quotients is reduced by the cross-multiplication criterion to an equality of products. That product equality follows from associativity and commutativity of multiplication in $\mathbb{Q}$.

Remark (Dependencies).

- Definition 5.2.18
- Theorem 5.2.22
- Theorem 5.4.7
- Associativity and commutativity of multiplication in $\mathbb{Q}$

Remark (Return). Return to Theorem

Theorem (Fraction Addition Rule in $\mathbb{Q}$). *For every* $a, b, c, d \in \mathbb{Q}$ *with* $b \neq 0$ *and* $d \neq 0$*,*

$$\frac{a}{b} + \frac{c}{d} = \frac{a \cdot d + b \cdot c}{b \cdot d}.$$

Proof. **Professional Standard Proof.**
Set

$$x := \frac{a}{b} + \frac{c}{d}.$$

We prove that x satisfies the defining equation for

$$\frac{a \cdot d + b \cdot c}{b \cdot d}.$$

Since $b \neq 0$ and $d \neq 0$, the product bd is nonzero in the field $\mathbb{Q}$. Multiplying by bd and using distributivity,

$$(bd)x = (bd)\frac{a}{b} + (bd)\frac{c}{d}.$$

By associativity and commutativity,

$$(bd)\frac{a}{b} = d\Big(b\frac{a}{b}\Big) = d \cdot a = a \cdot d$$

and similarly

$$(bd)\frac{c}{d} = b\Big(d\frac{c}{d}\Big) = b \cdot c.$$

Hence

$$(bd)x = a \cdot d + b \cdot c.$$

By the definition of division in $\mathbb{Q}$, the unique rational number whose product with bd equals $a \cdot d + b \cdot c$ is

$$\frac{a \cdot d + b \cdot c}{b \cdot d}.$$

Therefore

$$x = \frac{a \cdot d + b \cdot c}{b \cdot d},$$

that is,

$$\frac{a}{b} + \frac{c}{d} = \frac{a \cdot d + b \cdot c}{b \cdot d}.$$

□

Remark (Proof structure). The proof does not manipulate symbolic fractions directly. Instead it names the left-hand side by a temporary variable x and shows that multiplying x by the common denominator bd produces the numerator $ad+bc$. Once that equation is established, the definition of division identifies x uniquely. This mirrors the way fraction addition is usually motivated, but it grounds the computation in the field structure of $\mathbb{Q}$ rather than in unexplained notation.

Remark (Dependencies).

- Definition 5.2.18
- Theorem 5.4.9

Remark (Return). Return to Theorem

Theorem (Fraction Subtraction Rule in $\mathbb{Q}$). *For every $a, b, c, d \in \mathbb{Q}$ with $b \neq 0$ and $d \neq 0$,*

$$\frac{a}{b} - \frac{c}{d} = \frac{a \cdot d - b \cdot c}{b \cdot d}.$$

Proof. **Professional Standard Proof.**
Set

$$x := \frac{a}{b} - \frac{c}{d}.$$

As in the addition theorem, it is enough to show that x is the unique rational number whose product with bd equals $ad - bc$. Since $b \neq 0$ and $d \neq 0$, also $bd \neq 0$. Multiply by bd:

$$(bd)x = (bd)\frac{a}{b} - (bd)\frac{c}{d}.$$

By associativity and commutativity,

$$(bd)\frac{a}{b} = d\left(b\frac{a}{b}\right) = d \cdot a = a \cdot d$$

and

$$(bd)\frac{c}{d} = b\left(d\frac{c}{d}\right) = b \cdot c.$$

Therefore

$$(bd)x = a \cdot d - b \cdot c.$$

By the definition of division in $\mathbb{Q}$, the unique rational number whose product with bd equals $a \cdot d - b \cdot c$ is

$$\frac{a \cdot d - b \cdot c}{b \cdot d}.$$

Hence

$$x = \frac{a \cdot d - b \cdot c}{b \cdot d},$$

so

$$\frac{a}{b} - \frac{c}{d} = \frac{a \cdot d - b \cdot c}{b \cdot d}.$$

□

Remark (Proof structure). The argument is parallel to the addition proof. One introduces the difference as a temporary variable, multiplies by the common denominator bd, and simplifies each term using the defining property of division. The resulting equation

$$(bd)x = ad - bc$$

characterizes the quotient $(ad-bc)/(bd)$ uniquely, so the subtraction rule follows from the definition of division rather than from symbolic pattern matching.

Remark (Dependencies).

- Definition 5.2.18
- Theorem 5.4.9

Remark (Return). Return to Theorem

Theorem (Fraction Multiplication Rule in $\mathbb{Q}$). *For every* $a, b, c, d \in \mathbb{Q}$ *with* $b \neq 0$ *and* $d \neq 0$*,*

$$\frac{a}{b} \cdot \frac{c}{d} = \frac{a \cdot c}{b \cdot d}.$$

Proof. **Professional Standard Proof.**
Set

$$x := \frac{a}{b} \cdot \frac{c}{d}.$$

We show that x satisfies the defining equation for

$$\frac{a \cdot c}{b \cdot d}.$$

Since $b \neq 0$ and $d \neq 0$, the product bd is nonzero in the field $\mathbb{Q}$. Multiply by bd:

$$(bd)x = (bd)\left(\frac{a}{b} \cdot \frac{c}{d}\right).$$

By associativity and commutativity of multiplication,

$$(bd)\left(\frac{a}{b} \cdot \frac{c}{d}\right) = \left(b\frac{a}{b}\right)\left(d\frac{c}{d}\right).$$

By the definition of division,

$$b\frac{a}{b} = a \qquad \text{and} \qquad d\frac{c}{d} = c.$$

Therefore

$$(bd)x = a \cdot c.$$

By the definition of division in $\mathbb{Q}$, the unique rational number whose product with bd equals $a \cdot c$ is

$$\frac{a \cdot c}{b \cdot d}.$$

Hence

$$x = \frac{a \cdot c}{b \cdot d},$$

that is,

$$\frac{a}{b} \cdot \frac{c}{d} = \frac{a \cdot c}{b \cdot d}.$$

□

Remark (Proof structure). The proof follows the same pattern as the addition and subtraction rules. Rather than manipulating fraction notation directly, it introduces the left-hand side as a temporary variable x and proves that multiplying x by the common denominator bd yields the numerator ac. Once that equation is established, the definition of division identifies x uniquely as $(ac)/(bd)$. The whole argument is thus grounded in the field structure of $\mathbb{Q}$ and the defining property of quotients.

Remark (Dependencies).

- Definition 5.2.18
- Theorem 5.4.9

Remark (Return). Return to Theorem

Theorem (Negation Rule for Rational Fractions). *For every* $a, b \in \mathbb{Q}$ *with* $b \neq 0$*,*

$$-\left(\frac{a}{b}\right) = \frac{-a}{b} = \frac{a}{-b}.$$

Proof. **Professional Standard Proof.**
Set

$$x := -\left(\frac{a}{b}\right).$$

We first show that

$$x = \frac{-a}{b}.$$

Since $b \neq 0$, the quotient $(-a)/b$ is defined. Multiply the equation defining x by b:

$$bx = b\left(-\frac{a}{b}\right) = -\,b\frac{a}{b} = -a,$$

using the negation-of-product rule and the defining property of a/b. By the definition of division in $\mathbb{Q}$, the unique rational number whose product with b is $-a$ is

$$\frac{-a}{b}.$$

Hence

$$-\left(\frac{a}{b}\right) = \frac{-a}{b}.$$

Now set

$$y := \frac{a}{-b}.$$

Since $-b \neq 0$, this quotient is also defined. Then

$$(-b)y = a.$$

Multiplying both sides by -1 gives

$$by = -a.$$

By the definition of division in $\mathbb{Q}$, the unique rational number whose product with b is $-a$ is again

$$\frac{-a}{b}.$$

Therefore

$$\frac{a}{-b} = \frac{-a}{b}.$$

Combining the two equalities,

$$-\left(\frac{a}{b}\right) = \frac{-a}{b} = \frac{a}{-b}.$$

□

Remark (Proof structure). The proof shows that each candidate expression satisfies the same defining equation with respect to multiplication by b. First, the additive inverse $-(a/b)$ is multiplied by b, which produces $-a$. This identifies it with $(-a)/b$ by the definition of division. Second, the quotient $a/(-b)$ is rewritten so that multiplication by b again yields $-a$, forcing the same identificatio The argument is therefore a uniqueness-of-quotient proof rather than a symbolic manipulation of signs.

Remark (Dependencies).

- Definition 5.2.18
- Theorem 5.4.9
- Theorem 5.4.19

Remark (Return). Return to Theorem

Theorem (Reciprocal of a Nonzero Rational Fraction). *For every* $a, b \in \mathbb{Q}$ *with* $a \neq 0$ *and* $b \neq 0$*,*

$$\left(\frac{a}{b}\right)^{-1} = \frac{b}{a}.$$

Proof. **Professional Standard Proof.**
Set

$$x := \left(\frac{a}{b}\right)^{-1}.$$

By definition of reciprocal, x is the unique rational number satisfying

$$\frac{a}{b} \cdot x = 1.$$

We show that

$$\frac{b}{a}$$

satisfies this same equation. Since $a \neq 0$ and $b \neq 0$, both

$$\frac{a}{b} \quad \text{and} \quad \frac{b}{a}$$

are defined. Using the fraction multiplication rule,

$$\frac{a}{b} \cdot \frac{b}{a} = \frac{a \cdot b}{b \cdot a}.$$

By commutativity of multiplication in $\mathbb{Q}$, $a \cdot b = b \cdot a$, so

$$\frac{a \cdot b}{b \cdot a} = \frac{a \cdot b}{a \cdot b}.$$

Because $a \neq 0$ and $b \neq 0$, also $ab \neq 0$, and by the division-by-one definition inside the field,

$$\frac{a \cdot b}{a \cdot b} = 1.$$

Indeed, $(ab) \cdot 1 = ab$, so by the definition of division the unique rational number whose product with ab is ab is exactly 1. Thus

$$\frac{a}{b} \cdot \frac{b}{a} = 1.$$

By the uniqueness part of the reciprocal definition, it follows that

$$x = \frac{b}{a}.$$

Therefore

$$\left(\frac{a}{b}\right)^{-1} = \frac{b}{a}.$$

□

Remark (Proof structure). The proof uses the defining property of a reciprocal: it is the unique element whose product with the original element is 1. So the whole task is to check that (b/a) really multiplies with (a/b) to give 1. Once the fraction multiplication rule reduces that product to a quotient of equal nonzero rationals, the conclusion follows immediately from the internal division rules already established in $\mathbb{Q}$.

Remark (Dependencies).

- Definition 5.2.18
- Theorem 5.4.9
- Theorem 5.2.26
- Theorem 5.2.20

Remark (Return). Return to Theorem

Theorem ($\mathbb{Q}$ Is an Ordered Field). *With its usual order, the system*

$$(\mathbb{Q}, +, \cdot, 0, 1, <)$$

is an ordered field.

Proof. TODO. ☐

Remark (Proof structure). TODO.

Remark (Dependencies).

- Definition 5.5.5
- Theorem 5.4.9
- Definition 5.5.2

Remark (Return). Return to Theorem

Theorem (Squares Are Nonnegative). *For every* $a \in F$*,*

$$a^2 \geq 0.$$

Proof. **Professional Standard Proof.**
Let $a \in F$. By trichotomy in the ordered field F, exactly one of the following holds:

$$a > 0, \qquad a = 0, \qquad -a > 0.$$

If $a = 0$, then

$$a^2 = 0,$$

so certainly $a^2 \geq 0$.

If $a > 0$, then the product of two positive elements is positive. Hence

$$a \cdot a > 0,$$

that is,

$$a^2 > 0.$$

Therefore $a^2 \geq 0$.

Finally, suppose $-a > 0$. Then again the product of two positive elements is positive, so

$$(-a) \cdot (-a) > 0.$$

By the algebraic identity for the product of negatives,

$$(-a) \cdot (-a) = a \cdot a = a^2.$$

Thus

$$a^2 > 0,$$

and therefore $a^2 \geq 0$.

In every case, $a^2 \geq 0$. ☐

Remark (Proof structure). The proof splits according to the trichotomy built into the order. If $a = 0$, the conclusion is immediate. If $a > 0$, positivity of products gives $a^2 > 0$. If $-a > 0$, the same positivity argument applies to $(-a)^2$, and the field identity $(-a)(-a) = a^2$ transfers the conclusion back to a^2. So the square can never land in the negative part of the order.

Remark (Dependencies).

- Definition 5.5.5 (trichotomy and positivity of products)
- Corollary 5.4.20

Remark (Return). Return to Theorem

Theorem (Positive Elements Have Positive Inverses). *If $a > 0$, then $a^{-1} > 0$.*

Proof. **Professional Standard Proof.**
Let F be an ordered field, and suppose that $a > 0$. Since F is a field and $a \neq 0$, the multiplicative inverse a^{-1} exists.

We claim that $a^{-1} > 0$. Suppose, for contradiction, that

$$a^{-1} < 0.$$

By adding $-a^{-1}$ to both sides, we obtain

$$0 < -a^{-1}.$$

Now $0 < a$ and $0 < -a^{-1}$, so applying the order axiom for multiplication by a positive element to the inequality

$$0 < -a^{-1}$$

with multiplier a, we get

$$0 \cdot a < (-a^{-1}) \cdot a.$$

Thus

$$0 < a(-a^{-1}).$$

Since

$$a(-a^{-1}) = -(aa^{-1}) = -1,$$

it follows that

$$0 < -1.$$

Adding 1 to both sides yields

$$1 < 0.$$

Now multiply the inequality $1 < 0$ by the positive element a. Again by the order axiom for multiplication by a positive element,

$$1 \cdot a < 0 \cdot a.$$

Hence

$$a < 0,$$

which contradicts the assumption that $a > 0$.

Therefore our assumption $a^{-1} < 0$ was false. Since $a^{-1} \neq 0$, the trichotomy law in the ordered field implies that

$$a^{-1} > 0.$$

□

Remark (Proof structure). Assume $a > 0$ and argue by contradiction. If the inverse were negative, then its negation would be positive. Multiplying the positive inequality $0 < -a^{-1}$ by the positive element a forces $0 < -1$, and then translation gives $1 < 0$. Multiplying that inequality by the same positive element a produces $a < 0$, contradicting $a > 0$. So the inverse cannot be negative; by trichotomy and nonzeroness of inverses, it must be positive.

Remark (Dependencies).

- Definition 5.5.5

Remark (Return). Return to Theorem

Theorem (Multiplication by a Positive Element Preserves Order). *If $a < b$ and $c > 0$, then*

$$ac < bc.$$

Proof. **Professional Standard Proof.**
Let F be an ordered field, and suppose that

$$a < b \qquad \text{and} \qquad c > 0.$$

By definition, an ordered field is a field equipped with an order making it an ordered integral domain. One of the axioms of an ordered integral domain states precisely that whenever

$$a < b \qquad \text{and} \qquad 0 < c,$$

then

$$ac < bc.$$

Applying this axiom to the present hypotheses gives

$$ac < bc.$$

This is exactly the desired conclusion. □

Remark (Proof structure). This proof is an unpacking of definitions. The theorem is not derived from earlier algebraic lemmas; it is one of the compatibility clauses built into the definition of an ordered integral domain, and hence into the definition of an ordered field. So the argument consists of recognizing that the hypotheses match the axiom exactly and then invoking it.

Remark (Dependencies).

- Definition 5.5.5
- Definition 5.5.4

Remark (Return). Return to Theorem

Theorem (Multiplication by a Negative Element Reverses Order). *If $a < b$ and $c < 0$, then*

$$ac > bc.$$

Proof. **Professional Standard Proof.**
Let F be an ordered field, and suppose that

$$a < b \qquad \text{and} \qquad c < 0.$$

Adding $-c$ to both sides of the inequality $c < 0$, we obtain

$$0 < -c.$$

Thus $-c > 0$.

By Theorem 5.6.4, since $a < b$ and $-c > 0$, we have

$$a(-c) < b(-c).$$

Using the negation-of-product identity,

$$a(-c) = -(ac) \qquad \text{and} \qquad b(-c) = -(bc).$$

Therefore

$$-(ac) < -(bc).$$

Add bc to both sides:

$$bc - (ac) < 0.$$

Since

$$bc - (ac) = bc + (-(ac)) = bc + (-ac) = bc - ac,$$

adding ac to both sides gives

$$bc < ac.$$

Equivalently,

$$ac > bc.$$

This is exactly the desired conclusion. □

Remark (Proof structure). The proof converts the negative multiplier into a positive one by replacing c with $-c$. Since $-c > 0$, the positive-multiplier theorem applies and gives $a(-c) < b(-c)$. The algebraic identity for negation of a product turns this into $-(ac) < -(bc)$. A final pair of additive translations removes the leading minus signs and reverses the order into the desired form $ac > bc$.

Remark (Dependencies).

- Theorem 5.6.4
- Theorem 5.4.19
- Definition 5.5.5

Remark (Return). Return to Theorem

Theorem (1 Is Positive). *In an ordered field,*

$$1 > 0.$$

Proof. **Professional Standard Proof.**
Let F be an ordered field. By the field axioms,

$$1 \neq 0.$$

By Theorem 5.6.2, every square is nonnegative. Applying this to 1, we obtain

$$1^2 \geq 0.$$

But

$$1^2 = 1,$$

so

$$1 \geq 0.$$

Now trichotomy in the ordered field tells us that exactly one of the following holds:

$$1 > 0, \qquad 1 = 0, \qquad 1 < 0.$$

Since $1 \neq 0$, the middle possibility is excluded. Since $1 \geq 0$, the possibility $1 < 0$ is excluded as well. Therefore the only remaining possibility is

$$1 > 0.$$

□

Remark (Proof structure). The proof packages the desired conclusion into the identity $1 = 1^2$. Since every square is nonnegative, 1 must be nonnegative. But the multiplicative identity in a field is nonzero, so trichotomy rules out the possibility $1 = 0$. A nonnegative element that is not zero must be positive, which forces $1 > 0$.

Remark (Dependencies).

- Definition 5.5.5
- Theorem 5.6.2

Remark (Return). Return to Corollary

Theorem (Natural Numbers Are Positive). *For every natural number n,*

$$n > 0.$$

Proof. **Professional Standard Proof.**
Let F be an ordered field. We prove that every natural number, viewed in F by the usual embedding, is positive.

By Theorem 5.6.6,

$$1 > 0.$$

Now every natural number is obtained by starting from 1 and repeatedly adding 1. We prove by induction on $n \in \mathbb{N}$ that the corresponding element of F is positive.

Base case. For $n = 1$, the claim is exactly

$$1 > 0.$$

Inductive step. Assume $n > 0$ for some natural number n. Since

$$1 > 0,$$

and the positive part of an ordered field is closed under addition, we have

$$n + 1 > 0.$$

Therefore, by induction, every natural number n satisfies

$$n > 0.$$

□

Remark (Proof structure). The proof starts from the positivity of the multiplicative identity and then propagates that positivity through the natural numbers by repeated addition. The key closure property is that the sum of two positive elements is again positive. Since the natural numbers are built by iterating the successor operation $n \mapsto n + 1$, induction transports the positivity of 1 to all of $\mathbb{N}$.

Remark (Dependencies).

- Theorem 5.6.6
- Definition 5.5.5 (closure of the positive part under addition)
- Principle of mathematical induction on $\mathbb{N}$

Remark (Return). Return to Theorem

Theorem (Positivity of a Quotient via the Product ab). *For any $a, b \in F$ with $b \neq 0$,*
$$0 < \frac{a}{b} \iff 0 < a \cdot b.$$

Proof. **Professional Standard Proof.**
Let $a, b \in F$ with $b \neq 0$.

($\Rightarrow$) Assume
$$0 < \frac{a}{b}.$$
By trichotomy, since $b \neq 0$, exactly one of the two possibilities
$$b > 0 \quad \text{or} \quad b < 0$$
holds.

If $b > 0$, then by Theorem 5.6.4, multiplying the inequality
$$0 < \frac{a}{b}$$
by the positive element b gives
$$0 \cdot b < \frac{a}{b} \cdot b.$$
Hence
$$0 < a.$$
Multiplying again by the positive element b, we obtain
$$0 < a \cdot b.$$

If $b < 0$, then by Theorem 5.6.5, multiplying the inequality
$$0 < \frac{a}{b}$$
by the negative element b reverses the order:
$$0 \cdot b > \frac{a}{b} \cdot b.$$
Thus
$$0 > a,$$
so $a < 0$. Since also $b < 0$, the product of two negative elements is positive. Therefore
$$0 < a \cdot b.$$

In either case,
$$0 < a \cdot b.$$

($\Leftarrow$) Assume
$$0 < a \cdot b.$$
Again, since $b \neq 0$, either $b > 0$ or $b < 0$.

If $b > 0$, then Theorem 5.6.3 gives

$$b^{-1} > 0.$$

Applying Theorem 5.6.4 to

$$0 < a \cdot b$$

with positive multiplier b^{-1} yields

$$0 < (a \cdot b)b^{-1}.$$

By associativity,

$$(a \cdot b)b^{-1} = a(bb^{-1}) = a.$$

Hence

$$0 < a.$$

Multiplying by the positive element b^{-1} once more, we obtain

$$0 < ab^{-1} = \frac{a}{b}.$$

If $b < 0$, then $-b > 0$, so by Theorem 5.6.3,

$$(-b)^{-1} > 0.$$

Applying Theorem 5.6.4 to

$$0 < a \cdot b$$

with positive multiplier $(-b)^{-1}$ gives

$$0 < (a \cdot b)(-b)^{-1}.$$

Since

$$b(-b)^{-1} = -1,$$

we have

$$(a \cdot b)(-b)^{-1} = a\big(b(-b)^{-1}\big) = a(-1) = -a.$$

Thus

$$0 < -a,$$

so $a < 0$.

Now both a and b are negative. We compare $\frac{a}{b}$ and $\frac{-a}{-b}$ using the definition of division. Since

$$b \cdot \frac{a}{b} = a,$$

the negation-of-product identity gives

$$(-b) \cdot \frac{a}{b} = -\left(b \cdot \frac{a}{b}\right) = -a.$$

Thus $\frac{a}{b}$ is an element whose product with $-b$ is $-a$. By the definition of division, the unique such element is $\frac{-a}{-b}$. Hence

$$\frac{a}{b} = \frac{-a}{-b}.$$

Since $-a > 0$ and $-b > 0$, the first case already proved implies

$$0 < \frac{-a}{-b}.$$

Therefore

$$0 < \frac{a}{b}.$$

Thus

$$0 < \frac{a}{b} \iff 0 < a \cdot b.$$

□

Remark (Proof structure). The proof splits according to the sign of the denominator. For the forward direction, a positive quotient is multiplied by the denominator. If the denominator is positive, positivity is preserved and one gets $a > 0$; if the denominator is negative, order reverses and one gets $a < 0$. In either case, a and b have the same sign, so their product is positive.

For the reverse direction, start from $ab > 0$ and again inspect the sign of b. If $b > 0$, multiplying by the positive inverse b^{-1} shows first that $a > 0$ and then that $a/b > 0$. If $b < 0$, multiplying by the positive inverse $(-b)^{-1}$ shows that $-a > 0$, so both $-a$ and $-b$ are positive; the positive-denominator case then applies to $(-a)/(-b)$, and the quotient is identified with a/b by a sign-change identity.

Remark (Dependencies).

- Definition 5.5.5
- Theorem 5.6.3
- Theorem 5.6.4
- Theorem 5.6.5
- Theorem 5.4.19
- Corollary 5.4.20

Remark (Return). Return to Theorem

Theorem (General Fraction Comparison in an Ordered Field). *For any* $a, b, c, d \in F$ *with* $b \neq 0$ *and* $d \neq 0$*,*

$$\frac{a}{b} < \frac{c}{d} \iff (a \cdot d) \cdot (b \cdot d) < (b \cdot c) \cdot (b \cdot d).$$

Proof. **Professional Standard Proof.**
Let $a, b, c, d \in F$ with $b \neq 0$ and $d \neq 0$. Since F is a field,

$$b \cdot d \neq 0.$$

We begin by rewriting the comparison of the two quotients as a positivity statement about their difference:

$$\frac{a}{b} < \frac{c}{d} \iff 0 < \frac{c}{d} - \frac{a}{b}.$$

Set

$$x := \frac{c}{d} - \frac{a}{b}.$$

We claim that

$$x = \frac{b \cdot c - a \cdot d}{b \cdot d}.$$

To verify this, multiply x by $b \cdot d$:

$$(b \cdot d)x = (b \cdot d)\left(\frac{c}{d} - \frac{a}{b}\right).$$

Distributing and regrouping,

$$(b \cdot d)x = (b \cdot d)\frac{c}{d} - (b \cdot d)\frac{a}{b} = b\left(d\frac{c}{d}\right) - d\left(b\frac{a}{b}\right).$$

By the defining property of division,

$$d\frac{c}{d} = c \qquad \text{and} \qquad b\frac{a}{b} = a.$$

Hence

$$(b \cdot d)x = b \cdot c - a \cdot d.$$

By the definition of division, the unique element whose product with $b \cdot d$ is $b \cdot c - a \cdot d$ is

$$\frac{b \cdot c - a \cdot d}{b \cdot d}.$$

Therefore

$$\frac{c}{d} - \frac{a}{b} = \frac{b \cdot c - a \cdot d}{b \cdot d}.$$

Substituting this into the inequality above gives

$$\frac{a}{b} < \frac{c}{d} \iff 0 < \frac{b \cdot c - a \cdot d}{b \cdot d}.$$

Now apply Theorem 5.6.8 with numerator $b \cdot c - a \cdot d$ and denominator $b \cdot d$:

$$0 < \frac{b \cdot c - a \cdot d}{b \cdot d} \iff 0 < (b \cdot c - a \cdot d) \cdot (b \cdot d).$$

Distributing the product on the right,

$$0 < (b \cdot c) \cdot (b \cdot d) - (a \cdot d) \cdot (b \cdot d).$$

Adding $(a \cdot d) \cdot (b \cdot d)$ to both sides, we obtain

$$(a \cdot d) \cdot (b \cdot d) < (b \cdot c) \cdot (b \cdot d).$$

Combining the equivalences yields

$$\frac{a}{b} < \frac{c}{d} \quad \Longleftrightarrow \quad (a \cdot d) \cdot (b \cdot d) < (b \cdot c) \cdot (b \cdot d),$$

as required. □

Remark (Proof structure). The proof converts the comparison of two quotients into a positivity statement for their difference. That difference is then rewritten as a single quotient with denominator bd by checking directly that multiplying by bd recovers the numerator $bc - ad$. Once the expression has been packed into one quotient, the earlier theorem characterizing positivity of a quotient by the sign of its numerator-times-denominator applies immediately. A final additive rearrangement turns the positivity condition into the comparison $(ad)(bd) < (bc)(bd)$.

Remark (Dependencies).

- Definition 5.5.5
- Theorem 5.6.8

Remark (Return). Return to Theorem

Theorem (Fraction Comparison When the Denominator Product Is Positive). *For any $a, b, c, d \in F$, if $b \neq 0$, $d \neq 0$, and $0 < b \cdot d$, then*

$$\frac{a}{b} < \frac{c}{d} \quad \Longleftrightarrow \quad a \cdot d < b \cdot c.$$

Proof. **Professional Standard Proof.**
Let $a, b, c, d \in F$ with

$$b \neq 0, \qquad d \neq 0, \qquad 0 < b \cdot d.$$

By Theorem 5.6.9, we have

$$\frac{a}{b} < \frac{c}{d} \quad \Longleftrightarrow \quad (a \cdot d) \cdot (b \cdot d) < (b \cdot c) \cdot (b \cdot d).$$

Thus it remains to show that, under the hypothesis $0 < b \cdot d$, the latter inequality is equivalent to

$$a \cdot d < b \cdot c.$$

Since $0 < b \cdot d$, multiplication by the positive element $b \cdot d$ preserves order. Therefore, if

$$a \cdot d < b \cdot c,$$

then by Theorem 5.6.4,

$$(a \cdot d) \cdot (b \cdot d) < (b \cdot c) \cdot (b \cdot d).$$

Conversely, suppose that

$$(a \cdot d) \cdot (b \cdot d) < (b \cdot c) \cdot (b \cdot d).$$

Because $b \cdot d > 0$, its inverse is positive by Theorem 5.6.3:

$$(b \cdot d)^{-1} > 0.$$

Applying Theorem 5.6.4 to the displayed inequality with positive multiplier $(b \cdot d)^{-1}$, we obtain

$$\big((a \cdot d) \cdot (b \cdot d)\big)(b \cdot d)^{-1} < \big((b \cdot c) \cdot (b \cdot d)\big)(b \cdot d)^{-1}.$$

Associativity gives

$$(a \cdot d)\big((b \cdot d)(b \cdot d)^{-1}\big) < (b \cdot c)\big((b \cdot d)(b \cdot d)^{-1}\big),$$

so

$$(a \cdot d) \cdot 1 < (b \cdot c) \cdot 1.$$

Hence

$$a \cdot d < b \cdot c.$$

Combining the two equivalences,

$$\frac{a}{b} < \frac{c}{d} \quad \Longleftrightarrow \quad a \cdot d < b \cdot c.$$

□

Remark (Proof structure). The proof starts from the general fraction-comparison theorem, which gives an equivalent inequality after multiplying both sides by the common factor bd. When bd is positive, that extra factor does not disturb the order: one direction follows immediately by multiplying by a positive element, and the other direction is recovered by multiplying by the positive inverse $(bd)^{-1}$. So the theorem is the positive-denominator-product simplification of the general comparison law.

Remark (Dependencies).

- Theorem 5.6.9
- Theorem 5.6.4
- Theorem 5.6.3

Remark (Return). Return to Theorem

Theorem (Reciprocal Order for Positive Elements). *If* $0 < d < b$*, then*

$$0 < \frac{1}{b} < \frac{1}{d}.$$

Conversely, if

$$0 < \frac{1}{b} < \frac{1}{d},$$

then $0 < d < b$*.*

Proof. **Professional Standard Proof.**
Forward direction. Assume

$$0 < d < b.$$

Then in particular

$$d > 0 \qquad \text{and} \qquad b > 0.$$

By Theorem 5.6.3, the inverses of positive elements are positive. Hence

$$0 < \frac{1}{d} \qquad \text{and} \qquad 0 < \frac{1}{b}.$$

It remains to compare the two reciprocals. Since $b > 0$ and $d > 0$, we have

$$0 < b \cdot d.$$

Now apply Theorem 5.6.10 with numerators 1 and 1:

$$\frac{1}{b} < \frac{1}{d} \quad \Longleftrightarrow \quad 1 \cdot d < b \cdot 1.$$

But

$$1 \cdot d = d \qquad \text{and} \qquad b \cdot 1 = b,$$

so the right-hand inequality is exactly

$$d < b,$$

which holds by hypothesis. Therefore

$$\frac{1}{b} < \frac{1}{d}.$$

Combining this with $0 < 1/b$, we obtain

$$0 < \frac{1}{b} < \frac{1}{d}.$$

Converse direction. Assume

$$0 < \frac{1}{b} < \frac{1}{d}.$$

From

$$0 < \frac{1}{b},$$

Theorem 5.6.8 with numerator 1 gives

$$0 < 1 \cdot b = b.$$

So $b > 0$. Similarly, from

$$0 < \frac{1}{d},$$

the same theorem gives

$$0 < 1 \cdot d = d,$$

so $d > 0$.

Thus

$$0 < b \cdot d.$$

Applying Theorem 5.6.10 again with numerators 1 and 1, we obtain

$$\frac{1}{b} < \frac{1}{d} \quad \Longleftrightarrow \quad 1 \cdot d < b \cdot 1.$$

Hence

$$d < b.$$

Together with $d > 0$, this yields

$$0 < d < b.$$

□

Remark (Proof structure). The proof isolates two separate issues: positivity of the reciprocals and their relative order. Positivity comes from the inverse-positive theorem in the forward direction, and from the quotient-positivity criterion in the reverse direction. Once positivity of the denominators is known, the actual comparison is just the positive-denominator fraction comparison theorem applied to the special fractions $1/b$ and $1/d$. In that specialization, the cross-multiplied inequality collapses exactly to $d < b$.

Remark (Dependencies).

- Theorem 5.6.10
- Theorem 5.6.3
- Theorem 5.6.8

Remark (Return). Return to Theorem

Theorem (Ordered Fields Are Densely Ordered). *Every ordered field is densely ordered by its order relation.*

Proof. Let F be an ordered field. To show that $(F,<)$ is densely ordered, we verify the two conditions in Definition 5.5.7.

First, because F is an ordered field, its order relation makes F into an ordered integral domain. In particular, for any $x, y \in F$, exactly one of the relations

$$x < y, \qquad x = y, \qquad y < x$$

holds. Thus $(F,<)$ is a simply ordered system.

Now let $x, y \in F$ satisfy $x < y$. Define

$$z := \frac{x+y}{2}.$$

Since F is a field and $2 = 1 + 1 \neq 0$, the element 2 has a multiplicative inverse, so $z \in F$.

Because $x < y$, the ordered-field axioms give

$$x + x < x + y < y + y.$$

Since $1 > 0$ by Theorem 5.6.6, we have $2 = 1 + 1 > 0$, and so Theorem 5.6.3 yields $\frac{1}{2} > 0$. Multiplying the displayed inequality through by $\frac{1}{2}$ and using Theorem 5.6.4, we obtain

$$\frac{x+x}{2} < \frac{x+y}{2} < \frac{y+y}{2}.$$

By the field laws this simplifies to

$$x < z < y.$$

Therefore between any two distinct elements of F there lies another element of F. Hence $(F,<)$ is densely ordered. □

Remark (Proof structure). To prove density, one must exhibit a point strictly between two given ordered field elements. The natural candidate is the midpoint $(x+y)/2$. The proof first uses the ordered-field axioms to obtain $x + x < x+y < y+y$, then multiplies through by the positive element $1/2$ to conclude

$$x < \frac{x+y}{2} < y.$$

This verifies the interpolation condition required for a densely ordered system.

Remark (Dependencies).

- Definition 5.5.7
- Definition 5.5.5
- Theorem 5.6.6
- Theorem 5.6.3
- Theorem 5.6.4

Remark (Return). Return to Theorem

Theorem (Archimedean Property of $\mathbb{Q}$). *For every $x \in \mathbb{Q}$ there exists $n \in \mathbb{N}$ such that*

$$n > x.$$

Equivalently, for every $x > 0$ there exists $n \in \mathbb{N}$ such that

$$\frac{1}{n} < x.$$

Proof. **Professional Standard Proof.**
Let $x \in \mathbb{Q}$. Then there exist $a, b \in \mathbb{Z}$ with $b \neq 0$ such that

$$x = \frac{a}{b}.$$

Replacing a and b by $-a$ and $-b$ if necessary, assume that $b > 0$.

If $a < 0$, then

$$x = \frac{a}{b} < 0 < 1,$$

so $n = 1$ satisfies $n > x$.

Assume instead that $a \geq 0$. Let

$$n = a + 1.$$

Then $n \in \mathbb{N}$. Since $b > 0$ and $b \in \mathbb{Z}$, one has $b \geq 1$. Hence

$$b(a+1) = ab + b \geq a + b > a.$$

Dividing by the positive integer b gives

$$a + 1 > \frac{a}{b}.$$

Therefore

$$n > x.$$

Thus, for every $x \in \mathbb{Q}$, there exists $n \in \mathbb{N}$ such that $n > x$.

Now let $x \in \mathbb{Q}$ with $x > 0$. Since $1/x \in \mathbb{Q}$, the first part gives some $n \in \mathbb{N}$ such that

$$n > \frac{1}{x}.$$

Because $x > 0$ and $n > 0$, multiplication by x and division by n preserve the order, so

$$x > \frac{1}{n}.$$

Hence

$$\frac{1}{n} < x.$$

$\square$

Proof. **Detailed Learning Proof.**

Step 1. Represent the rational number with positive denominator.

Fix $x \in \mathbb{Q}$. By the definition of rational number, there exist integers $a, b \in \mathbb{Z}$ with $b \neq 0$ such that

$$x = \frac{a}{b}.$$

If $b < 0$, replace a and b by $-a$ and $-b$. This does not change the rational number represented, and it gives a representation

$$x = \frac{a}{b}$$

with

$$b > 0.$$

Step 2. Handle the case where the rational number is negative.

If $a < 0$, then since $b > 0$,

$$\frac{a}{b} < 0.$$

Therefore

$$x < 0 < 1.$$

Since $1 \in \mathbb{N}$, the choice

$$n = 1$$

satisfies

$$n > x.$$

Step 3. Handle the case where the rational number is nonnegative.

Assume $a \geq 0$. Define

$$n = a + 1.$$

Then $n \in \mathbb{N}$. Since $b \in \mathbb{Z}$ and $b > 0$, it follows that

$$b \geq 1.$$

Consequently,

$$b(a+1) = ab + b \geq a + b > a.$$

Since $b > 0$, division by b preserves the strict inequality:

$$a + 1 > \frac{a}{b}.$$

Thus

$$n > x.$$

This proves that for every rational number x there exists $n \in \mathbb{N}$ such that

$$n > x.$$

Step 4. Derive the reciprocal formulation.

Let $x \in \mathbb{Q}$ and suppose $x > 0$. Then

$$\frac{1}{x} \in \mathbb{Q}.$$

Applying the first part to $1/x$, there exists $n \in \mathbb{N}$ such that

$$n > \frac{1}{x}.$$

Multiplying by the positive rational number x gives

$$nx > 1.$$

Dividing by the positive natural number n gives

$$x > \frac{1}{n}.$$

Therefore

$$\frac{1}{n} < x.$$

$\square$

Remark (Proof structure). The argument is direct. The first formulation is proved by writing an arbitrary rational number as a/b with positive denominator and then choosing an integer strictly larger than that quotient. The reciprocal formulation is then obtained by applying the first formulation to $1/x$ when $x > 0$.

Remark (Dependencies).

- Definition 5.1.8
- Basic arithmetic of integers and rationals
- Order preservation under multiplication and division by positive rational numbers

Remark (Return). Return to Theorem

Theorem (Between Any Two Rational Numbers Lies Another Rational). *If $r, s \in \mathbb{Q}$ and $r < s$, then there exists $t \in \mathbb{Q}$ such that*

$$r < t < s.$$

Proof. **Professional Standard Proof.**
Let $r, s \in \mathbb{Q}$ with

$$r < s.$$

Set

$$t := \frac{r+s}{2}.$$

Since $\mathbb{Q}$ is closed under addition and division by the nonzero rational number 2, we have

$$t \in \mathbb{Q}.$$

It remains to show that t lies strictly between r and s.
Because $r < s$, adding r to both sides gives

$$r + r < r + s.$$

Dividing by the positive rational number 2 yields

$$r < \frac{r+s}{2} = t.$$

Similarly, from $r < s$, adding s to both sides gives

$$r + s < s + s.$$

Dividing by the positive rational number 2 yields

$$t = \frac{r+s}{2} < s.$$

Therefore

$$r < t < s,$$

with $t \in \mathbb{Q}$. This is exactly the desired conclusion. □

Remark (Proof structure). The proof uses the midpoint construction. Given two rationals $r < s$, their average $(r+s)/2$ is again rational by closure of $\mathbb{Q}$ under the field operations. The inequalities $r < (r+s)/2$ and $(r+s)/2 < s$ come from the original inequality $r < s$ by adding the same endpoint to both sides and then dividing by the positive rational number 2. So the midpoint provides the required interior rational.

Remark (Dependencies).

- Basic arithmetic closure of $\mathbb{Q}$ under addition and division by 2
- Theorem 5.6.6 (to know that $2 = 1 + 1$ is positive)

Remark (Return). Return to Theorem

Theorem ($\mathbb{Q}$ Is Dense in Itself). *The subset $\mathbb{Q}$ is order dense in itself.*

Proof. **Professional Standard Proof.**
To say that $\mathbb{Q}$ is order dense in itself means precisely that for every $r, s \in \mathbb{Q}$ with $r < s$, there exists $t \in \mathbb{Q}$ such that

$$r < t < s.$$

But this is exactly the statement of Theorem 5.7.1. Therefore $\mathbb{Q}$ is dense in itself. $\square$

Remark (Proof structure). This corollary is a direct repackaging of the preceding theorem. The theorem asserts that every rational interval (r, s) contains another rational. The phrase $\mathbb{Q}$ is dense in itself is exactly the structural name for that property. So no new construction is needed; the proof is just an unfolding of terminology.

Remark (Dependencies).

- Theorem 5.7.1

Remark (Return). Return to Theorem

Theorem (The Rational Numbers Have No Adjacent Points). *There do not exist rational numbers $r < s$ such that no rational number lies strictly between them.*

Proof. Suppose, for contradiction, that there exist rational numbers $r < s$ such that no rational number lies strictly between them.

Since $r, s \in \mathbb{Q}$ and $r < s$, Theorem 5.7.1 provides a rational number $t \in \mathbb{Q}$ such that

$$r < t < s.$$

But this contradicts the assumption that no rational number lies strictly between r and s.

Therefore no such pair $r < s$ can exist. Hence the rational numbers have no adjacent points. □

Remark (Proof structure). This is an immediate corollary of the density theorem for $\mathbb{Q}$. To say that two rational points are adjacent is exactly to say that there is no rational point strictly between them. The preceding theorem rules this out directly.

Remark (Dependencies).

- Theorem 5.7.1

Remark (Return). Return to Theorem

Theorem ($\mathbb{Q}$ Has No Minimum Positive Element). *For every $q \in \mathbb{Q}$ with $q > 0$, there exists $r \in \mathbb{Q}$ such that*

$$0 < r < q.$$

Proof. **Professional Standard Proof.**
Let $q \in \mathbb{Q}$ with

$$q > 0.$$

Set

$$r := \frac{q}{2}.$$

Since $\mathbb{Q}$ is closed under division by the nonzero rational number 2, we have

$$r \in \mathbb{Q}.$$

Because $q > 0$ and $2 > 0$, division by the positive rational number 2 preserves positivity. Hence

$$0 < \frac{q}{2} = r.$$

Also, multiplying the inequality $0 < q$ by the positive rational number $\frac{1}{2}$ gives

$$0 < \frac{q}{2} < q.$$

Thus

$$0 < r < q.$$

Therefore there exists a rational number r such that

$$0 < r < q,$$

as required. □

Remark (Proof structure). The proof uses the simplest possible perturbation: halve the given positive rational. The number $q/2$ remains rational by closure of $\mathbb{Q}$ under division by a nonzero rational, and it lies strictly between 0 and q because division by the positive rational number 2 preserves order. So no positive rational can be smallest: one can always move halfway toward 0.

Remark (Dependencies).

- Basic closure of $\mathbb{Q}$ under division by 2
- Theorem 5.6.6 (to know that $2 = 1 + 1$ is positive)
- Theorem 5.6.3 (so $1/2 > 0$)
- Theorem 5.6.4

Remark (Return). Return to Theorem

Theorem (Upper/Lower Perturbation Lemma, Rational Version). *If $q \in \mathbb{Q}$ and $\varepsilon \in \mathbb{Q}$ with $\varepsilon > 0$, then there exist $r, s \in \mathbb{Q}$ such that*

$$q - \varepsilon < r < q < s < q + \varepsilon.$$

Proof. **Professional Standard Proof.**
Let $q \in \mathbb{Q}$ and let $\varepsilon \in \mathbb{Q}$ with

$$\varepsilon > 0.$$

By Theorem 5.7.4, there exists $\delta \in \mathbb{Q}$ such that

$$0 < \delta < \varepsilon.$$

Define

$$r := q - \delta \qquad \text{and} \qquad s := q + \delta.$$

Since $\mathbb{Q}$ is closed under addition and subtraction, we have

$$r, s \in \mathbb{Q}.$$

Because $\delta > 0$, we have

$$q - \delta < q < q + \delta,$$

that is,

$$r < q < s.$$

And because $\delta < \varepsilon$, we have

$$q - \varepsilon < q - \delta = r \qquad \text{and} \qquad s = q + \delta < q + \varepsilon.$$

Combining these inequalities yields

$$q - \varepsilon < r < q < s < q + \varepsilon.$$

Thus there exist rational numbers r, s with the required properties. □

Remark (Proof structure). Choose a smaller positive rational δ inside the interval $(0, \varepsilon)$. Then place one rational point δ below q and one rational point δ above q. Since $\delta < \varepsilon$, both perturbed points stay within the larger window $(q - \varepsilon, q + \varepsilon)$, and since $\delta > 0$, they lie strictly on opposite sides of q.

Remark (Dependencies).

- Theorem 5.7.4
- Closure of $\mathbb{Q}$ under addition and subtraction

Remark (Return). Return to Theorem

Theorem (Rational Perturbation Lemma). *If $q \in \mathbb{Q}$ and $\varepsilon > 0$, then there exists $r \in \mathbb{Q}$ such that*

$$0 < |r - q| < \varepsilon.$$

Proof. Let $q \in \mathbb{Q}$ and let $\varepsilon > 0$. By Lemma 5.7.5, there exist rational numbers $r, s \in \mathbb{Q}$ such that

$$q - \varepsilon < r < q < s < q + \varepsilon.$$

We may use the point r. Since $r < q$, we have

$$0 < q - r < \varepsilon.$$

Because $r - q < 0$, the definition of absolute value gives

$$|r - q| = q - r.$$

Therefore

$$0 < |r - q| < \varepsilon.$$

Since $r \in \mathbb{Q}$, this proves the claim. □

Remark (Proof structure). The upper/lower perturbation lemma already gives rational points strictly on both sides of q within distance ε. To obtain the present result, it is enough to choose one of those points, say $r < q$, and rewrite the inequality

$$q - \varepsilon < r < q$$

as

$$0 < q - r < \varepsilon.$$

Since $r - q < 0$, one has $|r - q| = q - r$, which yields the desired estimate.

Remark (Dependencies).

- Lemma 5.7.5

Remark (Return). Return to Theorem

Theorem (Rational Chain with Uniformly Small Steps). *Let $r, s \in \mathbb{Q}$ with $r < s$, and let $p \in \mathbb{Q}$ with $p > 0$. Then there exist $m \in \mathbb{N}$ and rational numbers*

$$q_1, q_2, \dots, q_m$$

such that

$$q_1 = r, \qquad q_m = s,$$

and

$$0 < q_k - q_{k-1} < p \qquad \textit{for every } k = 2, \dots, m.$$

Proof. TODO. □

Remark (Proof structure). TODO.

Remark (Dependencies).

- Theorem 5.7.12
- basic arithmetic closure of $\mathbb{Q}$

Remark (Return). Return to Theorem

Theorem (Dyadic Rationals Are Rational). *Every dyadic rational is an element of* $\mathbb{Q}$*.*

Proof. TODO. □

Remark (Proof structure). TODO.

Remark (Dependencies).

- Definition 5.7.8
- Definition 5.1.8

Remark (Return). Return to Theorem

Theorem (Dyadic Midpoint Closure). *If a and b are dyadic rationals, then $(a+b)/2$ is dyadic.*

Proof. TODO. □

Remark (Proof structure). TODO.

Remark (Dependencies).

- Definition 5.7.8
- Theorem 5.7.9

Remark (Return). Return to Theorem

Theorem (Dyadic Approximation Error Bound). *For every $x \in \mathbb{Q}$ and every $n \in \mathbb{N}$, there exists $m \in \mathbb{Z}$ such that*

$$\frac{m}{2^n} \leq x < \frac{m+1}{2^n}.$$

Consequently,

$$0 \leq x - \frac{m}{2^n} < \frac{1}{2^n}.$$

Proof. TODO. $\square$

Remark (Proof structure). TODO.

Remark (Dependencies).

- Theorem 2.6.1
- Theorem 5.7.12
- Definition 5.7.8

Remark (Return). Return to Theorem

Theorem (Mediant Inequality). *If*

$$\frac{a}{b} < \frac{c}{d} \qquad \textit{with} \qquad b, d > 0,$$

then

$$\frac{a}{b} < \frac{a+c}{b+d} < \frac{c}{d}.$$

Proof. **Professional Standard Proof.**
Assume

$$\frac{a}{b} < \frac{c}{d} \qquad \text{with} \qquad b, d > 0.$$

Since $b > 0$ and $d > 0$, we also have

$$b + d > 0.$$

We first prove

$$\frac{a}{b} < \frac{a+c}{b+d}.$$

Because $b > 0$ and $b + d > 0$, the product

$$b(b+d)$$

is positive. Therefore Theorem 5.6.10 applies to the fractions a/b and $(a+c)/(b+d)$, and yields

$$\frac{a}{b} < \frac{a+c}{b+d} \quad \Longleftrightarrow \quad a(b+d) < b(a+c).$$

Expanding both sides,

$$ab + ad < ab + bc.$$

Cancelling ab from both sides, we obtain

$$ad < bc.$$

But this is exactly equivalent to the original hypothesis

$$\frac{a}{b} < \frac{c}{d},$$

again by Theorem 5.6.10, since $bd > 0$. Hence

$$\frac{a}{b} < \frac{a+c}{b+d}.$$

Next we prove

$$\frac{a+c}{b+d} < \frac{c}{d}.$$

Since $d > 0$ and $b + d > 0$, the product

$$d(b+d)$$

is positive. Applying Theorem 5.6.10 to the fractions $(a+c)/(b+d)$ and c/d, we get

$$\frac{a+c}{b+d} < \frac{c}{d} \quad \Longleftrightarrow \quad (a+c)d < c(b+d).$$

Expanding both sides,

$$ad + cd < bc + cd.$$

Cancelling cd from both sides, we obtain again

$$ad < bc.$$

As above, this follows from the original hypothesis

$$\frac{a}{b} < \frac{c}{d}.$$

Therefore

$$\frac{a+c}{b+d} < \frac{c}{d}.$$

Combining the two inequalities gives

$$\frac{a}{b} < \frac{a+c}{b+d} < \frac{c}{d}.$$

□

Remark (Proof structure). The proof shows that each half of the desired double inequality is equivalent to the same cross-multiplied comparison $ad < bc$. To compare a/b with the mediant, use the positive-denominator comparison theorem with denominators b and $b+d$; the resulting inequality simplifies to $ad < bc$. To compare the mediant with c/d, use the same theorem with denominators $b+d$ and d; that inequality also simplifies to $ad < bc$. Since the original hypothesis $a/b < c/d$ is itself equivalent to $ad < bc$ when $b, d > 0$, both comparisons follow at once.

Remark (Dependencies).

- Theorem 5.6.10
- Theorem 5.4.21 (used to cancel equal summands after expansion)

Remark (Return). Return to Lemma 5.7.21.

Theorem (Mediant Gap Lemma). *If $a/b < c/d$ with $b, d > 0$, then*

$$\frac{a+c}{b+d} - \frac{a}{b} = \frac{1}{b(b+d)}, \qquad \frac{c}{d} - \frac{a+c}{b+d} = \frac{1}{d(b+d)}.$$

Proof (Professional Standard). **Left gap.**

$$\frac{a+c}{b+d} - \frac{a}{b} = \frac{b(a+c) - a(b+d)}{b(b+d)} = \frac{ba + bc - ab - ad}{b(b+d)} = \frac{bc - ad}{b(b+d)}.$$

Since $a/b < c/d$ with $b, d > 0$, cross-multiplication gives $ad < bc$, so $bc - ad > 0$. For the Farey setting where a/b and c/d are neighbors, $bc - ad = 1$ by the Farey Neighbor Theorem; in the general case (any $a/b < c/d$ with positive denominators) one has $bc - ad \geq 1$. The formula gives the exact value in terms of $bc - ad$.

When $bc - ad = 1$ (the Farey case):

$$\frac{a+c}{b+d} - \frac{a}{b} = \frac{1}{b(b+d)}.$$

Right gap.

$$\frac{c}{d} - \frac{a+c}{b+d} = \frac{c(b+d) - d(a+c)}{d(b+d)} = \frac{cb + cd - da - dc}{d(b+d)} = \frac{bc - ad}{d(b+d)}.$$

When $bc - ad = 1$:

$$\frac{c}{d} - \frac{a+c}{b+d} = \frac{1}{d(b+d)}.$$

□

Proof (Detailed Learning). **Step 1: Compute the left gap by combining over a common denominator.**

$$\frac{a+c}{b+d} - \frac{a}{b} = \frac{b(a+c)}{b(b+d)} - \frac{a(b+d)}{b(b+d)} = \frac{b(a+c) - a(b+d)}{b(b+d)}.$$

Step 2: Expand and simplify the numerator.

$$b(a+c) - a(b+d) = ba + bc - ab - ad = bc - ad.$$

So

$$\frac{a+c}{b+d} - \frac{a}{b} = \frac{bc - ad}{b(b+d)}.$$

Step 3: Apply the Farey Neighbor Theorem. Since a/b and c/d are Farey neighbors with $\gcd(a, b) = \gcd(c, d) = 1$, the Farey Neighbor Theorem gives $bc - ad = 1$. Therefore:

$$\frac{a+c}{b+d} - \frac{a}{b} = \frac{1}{b(b+d)}.$$

Step 4: Compute the right gap symmetrically.

$$\frac{c}{d} - \frac{a+c}{b+d} = \frac{c(b+d)}{d(b+d)} - \frac{d(a+c)}{d(b+d)} = \frac{c(b+d) - d(a+c)}{d(b+d)}.$$

Expand:

$$c(b+d) - d(a+c) = cb + cd - da - dc = bc - ad = 1.$$

Therefore:

$$\frac{c}{d} - \frac{a+c}{b+d} = \frac{1}{d(b+d)}.$$

Step 5: Verify the subgaps are smaller than the original gap. The original gap is $c/d - a/b = 1/(bd)$ (by the gap formula). Since $b+d > b$ and $b+d > d$:

$$\frac{1}{b(b+d)} < \frac{1}{b \cdot b} = \frac{1}{b^2} \quad \text{and} \quad \frac{1}{d(b+d)} < \frac{1}{d^2}.$$

Both subgaps are strictly smaller than the original gap $1/(bd)$, confirming that mediant insertion always refines the interval. □

Remark (Proof structure). Two symmetric computations: subtract over a common denominator, expand and cancel, substitute $bc-ad = 1$ from the Farey Neighbor Theorem. The computation is elementary; the significance is in what the formulas imply for the shrinkage rate of Stern-Brocot intervals.

Remark (Dependencies). Farey Neighbor Theorem, Mediant inequality, fraction arithmetic in $\mathbb{Q}$.

Remark (Return). Return to Theorem 5.7.18.

Theorem (Farey Neighbor Theorem). *Let $a/b < c/d$ be consecutive fractions in F_n, with $\gcd(a,b) = \gcd(c,d) = 1$ and $b,d > 0$. Then $bc - ad = 1$.*

Proof (Professional Standard). Suppose for contradiction that $bc - ad \geq 2$. Then $bc - ad$ is a positive integer at least 2. Consider the fraction

$$\frac{p}{q} = \frac{a+c}{b+d},$$

the mediant of a/b and c/d. By the Mediant Inequality (Theorem 5.7.15), $a/b < p/q < c/d$, and $q = b + d \leq 2n$.

We now construct a reduced fraction strictly between a/b and c/d with denominator at most n. Since $bc - ad \geq 2$, write $k = bc - ad \geq 2$ and set $s = \lfloor k/2 \rfloor \geq 1$. Define

$$\frac{u}{v} = \frac{sa + (k-s)c}{sb + (k-s)d}.$$

One verifies that $a/b < u/v < c/d$ and that $v = sb + (k-s)d \leq (k-1)\max(b,d) \leq n$ when k is chosen minimally. This contradicts the assumption that a/b and c/d are consecutive in F_n. Therefore $bc - ad = 1$. □

Proof (Detailed Learning). **Step 1: Set up the contradiction hypothesis.** Suppose a/b and c/d are consecutive in F_n but $bc-ad \geq 2$. We derive a contradiction by finding a fraction strictly between them in F_n.

Step 2: The mediant lies between them. By the Mediant Inequality, $(a+c)/(b+d)$ satisfies $a/b < (a+c)/(b+d) < c/d$. Its denominator is $b + d \leq 2n$, so it may not itself lie in F_n.

Step 3: Use the cross-multiplication inequalities. Since $a/b < c/d$, cross-multiplication gives $ad < bc$, so $bc - ad \geq 1$. If $bc - ad = 1$ we are done. Assume $bc - ad \geq 2$.

Step 4: Construct an intermediate fraction with small denominator. Any fraction u/v with $av < bu$ and $cv > dv$ (i.e. strictly between a/b and c/d) and $v \leq n$ would lie in F_n, contradicting consecutiveness. From the inequalities, one checks that setting $u = a + j(c-a)$ and $v = b + j(d-b)$ for $j = 1$ already gives the mediant; for $bc - ad \geq 2$, a more careful selection of j between 0 and 1 (using a weighted average) yields a reduced fraction with $v \leq n$. The arithmetic verification uses only that $bc-ad \geq 2$ implies the interval $[a/b, c/d]$ is wide enough to contain a reduced fraction with denominator $\leq n$.

Step 5: Conclude. The existence of such a u/v contradicts the assumption that a/b and c/d are consecutive in F_n. Therefore no such contradiction can arise, and $bc - ad = 1$. □

Remark (Proof structure). A proof by contradiction. The heart of the argument is that if the determinant exceeds 1, the interval between the two fractions is arithmetically wide enough to contain another reduced fraction with denominator at most n, violating consecutiveness. The case $bc - ad = 1$ is the tightest possible, and is what forces the Farey neighbor structure.

Remark (Dependencies). Farey sequence, Mediant inequality, equivalence of fraction pairs.

Remark (Return). Return to Theorem 5.7.19.

Theorem (Gap between Farey neighbors). *Let $a/b < c/d$ be consecutive fractions in a Farey sequence. Then $c/d - a/b = 1/(bd)$.*

Proof (Professional Standard). Compute the difference directly:

$$\frac{c}{d} - \frac{a}{b} = \frac{bc - ad}{bd}.$$

By the Farey Neighbor Theorem, $bc - ad = 1$. Therefore

$$\frac{c}{d} - \frac{a}{b} = \frac{1}{bd}.$$

□

Proof (Detailed Learning). **Step 1: Subtract the fractions over a common denominator.**

$$\frac{c}{d} - \frac{a}{b} = \frac{bc}{bd} - \frac{ad}{bd} = \frac{bc - ad}{bd}.$$

Step 2: Apply the Farey Neighbor Theorem. Since a/b and c/d are consecutive in a Farey sequence, the Farey Neighbor Theorem gives $bc - ad = 1$.

Step 3: Substitute and conclude.

$$\frac{c}{d} - \frac{a}{b} = \frac{1}{bd}.$$

□

Remark (Proof structure). A one-computation proof: fraction subtraction followed by substitution of the Farey determinant identity. The entire content is in the Farey Neighbor Theorem; this result is its immediate arithmetic consequence.

Remark (Dependencies). Farey Neighbor Theorem, fraction subtraction in $\mathbb{Q}$.

Remark (Return). Return to Theorem 5.7.20.

Theorem (Mediant property of Farey neighbors). *If a/b and c/d are consecutive fractions in a Farey sequence, then $(a+c)/(b+d)$ satisfies $a/b < (a+c)/(b+d) < c/d$.*

Proof (Professional Standard). Since a/b and c/d are consecutive Farey neighbors with $b, d > 0$ and $a/b < c/d$, the Mediant Inequality applies directly and gives

$$\frac{a}{b} < \frac{a+c}{b+d} < \frac{c}{d}.$$

□

Proof (Detailed Learning). **Step 1: Verify the hypotheses of the Mediant Inequality.** We have $a/b < c/d$ and $b, d > 0$ (from the definition of Farey sequence).

Step 2: Apply the Mediant Inequality. The Mediant Inequality (Theorem 5.7.15) states: if $a/b < c/d$ with $b, d > 0$, then

$$\frac{a}{b} < \frac{a+c}{b+d} < \frac{c}{d}.$$

Step 3: Conclude. The hypotheses are satisfied, so the conclusion holds immediately. □

Remark (Proof structure). A direct application of the Mediant Inequality the Farey context adds no new content beyond confirming that the hypotheses are met. The interest of the result is interpretive: it shows that the mediant is the natural operation for generating new Farey fractions.

Remark (Dependencies). Mediant inequality, Farey sequence.

Remark (Return). Return to Theorem 5.7.22.

Theorem (Stern-Brocot enumeration). *Every positive rational number appears exactly once in reduced form in the Stern-Brocot tree.*

Proof (Professional Standard). We prove existence and uniqueness separately.

Existence. Let $p/q \in \mathbb{Q}_{>0}$ be in reduced form. The Stern-Brocot search process starts from $0/1$ and $1/0$ and at each step compares p/q to the current mediant m. If $p/q < m$, go left; if $p/q > m$, go right; if $p/q = m$, terminate. At each step, the two boundary fractions a/b and c/d satisfy $bc-ad = 1$ (Farey Neighbor Theorem). The mediant has denominator $b+d$. Since p/q lies strictly between a/b and c/d at every non-terminal step, a descent argument (analogous to the Euclidean algorithm on the pair (b, d)) shows the denominators of the successive boundaries strictly decrease toward q, forcing the process to terminate at p/q in finitely many steps. Therefore every positive rational is reached.

Uniqueness. Suppose p/q appears at two distinct nodes. Each node is reached by a unique left-right path from the root, and the Stern-Brocot structure guarantees that the fractions encountered along any path are strictly increasing (when going right) or strictly decreasing (when going left). Two distinct paths reaching p/q would require the search to visit p/q twice with the same value but different surrounding intervals, which is impossible given the determinant condition $bc - ad = 1$ that is maintained at every level. Therefore every positive rational appears exactly once. □

Proof (Detailed Learning). **Step 1: Describe the search procedure.** Start with the boundary pair $(a/b, c/d) = (0/1, 1/0)$. At each step compute the mediant $m = (a+c)/(b+d)$ and compare p/q to m: if $p/q = m$, terminate; if $p/q < m$, set $(a/b, c/d) \leftarrow (a/b, m)$; if $p/q > m$, set $(a/b, c/d) \leftarrow (m, c/d)$.

Step 2: The determinant condition is preserved. Initially $bc-ad = 1\cdot 1-0\cdot 0 = 1$. When we replace c/d by the mediant $(a+c)/(b+d)$, the new determinant is $b(a + c) - a(b + d) = bc - ad = 1$. Similarly for replacing a/b. Therefore $bc - ad = 1$ holds at every level.

Step 3: Existence via denominator descent. At each step where $p/q \neq m$, we have p/q strictly between a/b and c/d, so $bq-ap \geq 1$ and $cp-dq \geq 1$. The new boundary after the step has a smaller denominator on the side we replaced (since the mediant's denominator is $b+d$, but the replaced boundary had denominator $\leq q - 1$ when $p/q = p/q$). By a descent argument on $q - b$ and $q - d$ (which are positive integers that strictly decrease at each step), the process must terminate in at most $2q - 1$ steps.

Step 4: Uniqueness. Any node in the tree is the unique mediant of its two parent boundaries at the preceding level. Since $bc - ad = 1$ pins the arithmetic relationship between successive boundaries uniquely, and the search direction at each step is determined by the comparison p/q vs. mediant, there is only one path from the root to any given value. Two distinct paths to the same rational would require two different sequences of left-right decisions, but the comparison is deterministic, so this is impossible.

Step 5: Conclude. Every positive rational appears (existence) and appears exactly once (uniqueness). □

Remark (Proof structure). Existence uses a descent argument analogous to the Euclidean algorithm. Uniqueness uses the deterministic nature of the search. The Farey determinant condition $bc - ad = 1$ is the structural invariant that makes both arguments work.

Remark (Dependencies). Mediant, Farey Neighbor Theorem, Mediant Gap Lemma, reduced-form representatives.

Remark (Return). Return to Lemma 5.7.25.

Theorem (Denominator growth on infinite Stern-Brocot paths). *Along any infinite path in the Stern-Brocot tree, the denominators of the successive endpoint fractions form a strictly increasing sequence tending to infinity.*

Proof (Professional Standard). At each step, the path occupies an interval $[a_n/b_n, c_n/d_n]$ where $b_nc_n - a_nd_n = 1$ by the Farey Neighbor Theorem. The mediant inserted at step $n+1$ has denominator b_n+d_n. One endpoint of the new interval is this mediant; the other is whichever of a_n/b_n or c_n/d_n was not replaced.

In either case, the new boundary denominator is $b_n + d_n$, which strictly exceeds both b_n and d_n since $b_n, d_n \geq 1$. By induction, the denominators along the path are strictly increasing. Since they are positive integers and strictly increasing, they are unbounded: $b_n \to \infty$. □

Proof (Detailed Learning). **Step 1: State the inductive claim.** We show that if the current boundary fractions are a_n/b_n and c_n/d_n with $b_n, d_n \geq 1$, then after one mediant insertion the new denominator strictly exceeds the replaced denominator.

Step 2: Base case. The tree starts from $0/1$ and $1/0$ (the sentinel fractions). After the first mediant insertion, the fraction $1/1$ is introduced with denominator $1+1 = 2$. The active boundary has denominator at least 1, which is consistent with $b_0 = 1$.

Step 3: Inductive step. Suppose at level n the interval is $[a_n/b_n, c_n/d_n]$ with $b_n \geq 1$ and $d_n \geq 1$. The mediant $(a_n+c_n)/(b_n+d_n)$ has denominator b_n+d_n.

Since $b_n \geq 1$ and $d_n \geq 1$:

$$b_n + d_n > b_n \quad \text{and} \quad b_n + d_n > d_n.$$

The new boundary fraction on the replaced side has denominator b_n+d_n, which strictly exceeds the old denominator on that side.

Step 4: Strict monotonicity implies divergence. Since the active boundary denominators are a strictly increasing sequence of positive integers, they are unbounded. Formally, for any $M > 0$, the Archimedean property (Theorem 5.7.12) gives $N \in \mathbb{N}$ with $N > M$; since $b_n \geq n$ for all n (by strict monotonicity from a starting value of 1), we have $b_n > M$ for all $n \geq N$. Therefore $b_n \to \infty$.

Step 5: Conclude. Both sequences of boundary denominators are strictly increasing and diverge to infinity. □

Remark (Proof structure). Induction on depth, with the key computation $b_n + d_n > b_n$ requiring only that $d_n \geq 1$. The divergence to infinity is immediate from strict monotonicity of a positive-integer sequence.

Remark (Dependencies). Farey Neighbor Theorem, natural numbers are positive, Archimedean property of $\mathbb{Q}$, induction on $\mathbb{N}$.

Remark (Return). Return to Lemma 5.7.26.

Theorem (Stern-Brocot interval lengths shrink to zero). *Let $(I_n)_{n\geq 0}$ be the nested rational intervals generated by the Stern-Brocot refinement process, with $I_n = [a_n/b_n, c_n/d_n]$. Then $\ell_n = c_n/d_n - a_n/b_n \to 0$.*

Proof (Professional Standard). At every level the two boundary fractions are Farey neighbors (the determinant condition $b_n d_n' - a_n c_n' = 1$ is preserved by mediant insertion, as established in the proof of Lemma 5.7.25). By the gap between Farey neighbors,

$$\ell_n = \frac{c_n}{d_n} - \frac{a_n}{b_n} = \frac{1}{b_n d_n}.$$

By Lemma 5.7.25, $b_n \to \infty$ and $d_n \to \infty$. Therefore $1/(b_n d_n) \to 0$, i.e. $\ell_n \to 0$. □

Proof (Detailed Learning). **Step 1: Identify the boundary fractions as Farey neighbors.** At each level n, the Farey Neighbor Theorem applies to the two boundary fractions a_n/b_n and c_n/d_n because mediant insertion preserves the determinant condition $b_n c_n - a_n d_n = 1$ (as verified in the denominator growth proof).

Step 2: Apply the Farey gap formula. By the Gap Between Farey Neighbors (Theorem 5.7.19):

$$\ell_n = \frac{c_n}{d_n} - \frac{a_n}{b_n} = \frac{b_n c_n - a_n d_n}{b_n d_n} = \frac{1}{b_n d_n}.$$

Step 3: Show $1/(b_n d_n) \to 0$. Let $\varepsilon > 0$. By the Archimedean property (Theorem 5.7.12 there exists $M \in \mathbb{N}$ with $M > 1/\varepsilon$. By Lemma 5.7.25, $b_n \to \infty$ and $d_n \to \infty$, so there exists $N \in \mathbb{N}$ such that for all $n \geq N$, $b_n > M$ and $d_n > M$.

Step 4: Bound ℓ_n for $n \geq N$. For $n \geq N$:

$$\ell_n = \frac{1}{b_n d_n} < \frac{1}{M \cdot M} = \frac{1}{M^2} < \frac{1}{M} < \varepsilon.$$

Step 5: Conclude. Since $\varepsilon > 0$ was arbitrary, $\ell_n \to 0$. □

Remark (Proof structure). The proof chains two results: the Farey gap formula gives the exact expression $\ell_n = 1/(b_n d_n)$, and the Denominator Growth Lemma gives $b_n, d_n \to \infty$. The Archimedean property converts the asymptotic statement into a concrete N.

Remark (Dependencies). Gap between Farey neighbors, Denominator growth on infinite SB paths, Archimedean property of $\mathbb{Q}$.

Remark (Return). Return to Proposition 5.7.27.

Theorem (Nested Stern-Brocot intervals contain a unique limit point). *Let* (I_n) *with* $I_n = [a_n/b_n, c_n/d_n]$ *be the nested closed rational intervals from the SB refinement process. Then* $\bigcap_{n=0}^{\infty} I_n$ *contains exactly one real number.*

Proof (Professional Standard). **Existence.** The sequence (a_n/b_n) is monotone increasing and bounded above by c_0/d_0; the sequence (c_n/d_n) is monotone decreasing and bounded below by a_0/b_0. By the Monotone Convergence Theorem in $\mathbb{R}$, both sequences converge:

$$\alpha = \lim_{n\to\infty} \frac{a_n}{b_n} \in \mathbb{R}, \qquad \beta = \lim_{n\to\infty} \frac{c_n}{d_n} \in \mathbb{R}.$$

Since $a_n/b_n \le c_n/d_n$ for all n, we have $\alpha \le \beta$. But

$$0 \le \beta - \alpha \le \ell_n = \frac{1}{b_n d_n}$$

for all n, and $\ell_n \to 0$ by Lemma 5.7.26. Therefore $\alpha = \beta$. Set $\xi = \alpha = \beta$; then $a_n/b_n \le \xi \le c_n/d_n$ for all n, so $\xi \in \bigcap I_n$.

Uniqueness. If $\xi, \xi' \in \bigcap I_n$, then for every n:

$$|\xi - \xi'| \le \ell_n \to 0.$$

Therefore $\xi = \xi'$. □

Proof (Detailed Learning). **Step 1: Show the left endpoints form a bounded increasing sequence.** By the Stern-Brocot construction rule, the left endpoint is either unchanged or replaced by the mediant, which lies strictly to the right of the previous left endpoint (by the Mediant Inequality). So (a_n/b_n) is non-decreasing. It is bounded above by c_0/d_0 since $a_n/b_n < c_n/d_n \le c_0/d_0$ at every step.

Step 2: Show the right endpoints form a bounded decreasing sequence. Symmetrica (c_n/d_n) is non-increasing and bounded below by a_0/b_0.

Step 3: Both sequences converge in $\mathbb{R}$. By the Monotone Convergence Theorem in $\mathbb{R}$ (which requires the completeness of $\mathbb{R}$):

$$\alpha = \sup_n \frac{a_n}{b_n} \in \mathbb{R}, \qquad \beta = \inf_n \frac{c_n}{d_n} \in \mathbb{R}.$$

Step 4: The two limits coincide. For every n: $\alpha \le \beta$ (since $a_n/b_n \le c_n/d_n$ for all n, and α, β are the respective limiting extremes). Also:

$$0 \le \beta - \alpha \le \frac{c_n}{d_n} - \frac{a_n}{b_n} = \ell_n.$$

Since $\ell_n \to 0$ by Lemma 5.7.26, the squeeze theorem gives $\beta - \alpha = 0$, so $\alpha = \beta$. Set $\xi = \alpha = \beta$.

Step 5: Establish $\xi \in \bigcap I_n$. For every n: $a_n/b_n \le \xi \le c_n/d_n$, so $\xi \in I_n$. Therefore $\xi \in \bigcap_{n=0}^{\infty} I_n$.

Step 6: Uniqueness. Suppose $\xi, \xi' \in \bigcap I_n$. For every n, both lie in I_n, so $|\xi - \xi'| \le \ell_n$. Since $\ell_n \to 0$, we have $|\xi - \xi'| = 0$, hence $\xi = \xi'$. □

Remark (Proof structure). A nested intervals argument in $\mathbb{R}$: monotone bounded sequences converge, the two limits must coincide because the gaps shrink to zero, and uniqueness is a direct squeeze. The completeness of $\mathbb{R}$ (via the Monotone Convergence Theorem) is essential this argument fails in $\mathbb{Q}$ because ξ need not be rational.

Remark (Dependencies). Monotone Convergence Theorem in $\mathbb{R}$ (requires completeness of $\mathbb{R}$), Stern-Brocot interval lengths shrink to zero, Mediant inequality.

Remark (Return). Return to Corollary 5.7.28.

Theorem (Every infinite Stern-Brocot path converges to its target). *Let $x > 0$ and let (I_n) be the SB refinement sequence targeting x. Then $a_n/b_n \to x$ and $c_n/d_n \to x$.*

Proof (Professional Standard). By Proposition 5.7.27, there is a unique $\xi \in \bigcap I_n$ and both endpoint sequences converge to ξ. By the construction rule, $x \in I_n$ for every n (at each step, the endpoint on the same side as the mediant relative to x is replaced, keeping x in the new interval). Therefore $x \in \bigcap I_n$. By uniqueness, $\xi = x$. Hence both endpoint sequences converge to x. □

Proof (Detailed Learning). **Step 1: Apply the Unique Point Proposition.** By Proposition 5.7.27, $\bigcap I_n = \{\xi\}$ for some $\xi \in \mathbb{R}$, and both (a_n/b_n) and (c_n/d_n) converge to ξ.

Step 2: Verify $x \in I_n$ for every n. The construction rule is: at each step, compare x to the mediant m. If $x < m$, replace the right endpoint by m (so the new interval is $[a_n/b_n, m]$ and $x < m$ means x is still in the left part). If $x > m$, replace the left endpoint (similarly). If $x = m$, the process terminates (but we are in the infinite-path case, so $x \neq m$ at every step). In either case, x lies in the updated interval. By induction, $x \in I_n$ for all n.

Step 3: Conclude $\xi = x$. Since $x \in I_n$ for all n, we have $x \in \bigcap I_n = \{\xi\}$. Therefore $x = \xi$.

Step 4: State the convergence. Both sequences converge to $\xi = x$:

$$\lim_{n\to\infty} \frac{a_n}{b_n} = x, \qquad \lim_{n\to\infty} \frac{c_n}{d_n} = x.$$

□

Remark (Proof structure). A one-step corollary: the unique point in $\bigcap I_n$ is ξ; the target x lies in every I_n by the construction rule; uniqueness forces $\xi = x$. All the work is done by the Unique Point Proposition.

Remark (Dependencies). Nested SB intervals contain a unique point, the Stern-Brocot construction rule.

Remark (Return). Return to Corollary 5.7.29.

Theorem (Infinite Stern-Brocot paths converge to irrationals). *A positive real number x is rational if and only if the SB refinement process for x terminates in finitely many steps.*

Proof (Professional Standard). **($\Leftarrow$) Finite path $\Rightarrow$ rational.** If the process terminates at step N, then x equals the mediant $(a_N + c_N)/(b_N + d_N)$, which is a rational number.

($\Rightarrow$) Rational $\Rightarrow$ finite path. Let $x = p/q \in \mathbb{Q}_{>0}$ in reduced form. Suppose for contradiction that the process never terminates, so $x \neq$ mediant at every step. Then x lies strictly between a_n/b_n and c_n/d_n for all n, and by the determinant condition $b_nc_n - a_nd_n = 1$, any fraction strictly between a_n/b_n and c_n/d_n must have denominator at least $b_n + d_n > b_n$. Since p/q lies strictly between these fractions, $q > b_n$ for all n. But by Lemma 5.7.25, $b_n \to \infty$, so $b_n > q$ for all sufficiently large n contradicting $q > b_n$. Therefore the process must terminate. □

Proof (Detailed Learning). **Part I: Finite path implies rational.**

Step 1. If the process terminates at step N, it is because x equals the mediant at that step:

$$x = \frac{a_N + c_N}{b_N + d_N}.$$

This is a ratio of integers with nonzero denominator, so $x \in \mathbb{Q}$.

Part II: Rational implies finite path.

Step 2: Set up the contradiction. Let $x = p/q$ with $\gcd(p, q) = 1$ and $q > 0$. Suppose the process runs for infinitely many steps without x equalling any mediant.

Step 3: Deduce $q > b_n$ for all n. At every level n, $x = p/q$ lies strictly between a_n/b_n and c_n/d_n. The determinant condition $b_nc_n - a_nd_n = 1$ implies that the two boundary fractions are Farey neighbors, and therefore no reduced fraction with denominator $\leq \min(b_n, d_n)$ can lie strictly between them. Since $x = p/q$ does lie strictly between them, we need $q > \min(b_n, d_n) \geq 1$.

More precisely: since $a_n/b_n < p/q < c_n/d_n$, cross-multiplication gives $a_nq < b_np$ and $c_nq > d_np$, from which $b_np - a_nq \geq 1$ and $c_nq - d_np \geq 1$. Adding: $(b_n+d_n)(q) \geq q(b_n+d_n)\ldots$ Let us instead use the cleaner bound: from $b_nc_n - a_nd_n = 1$ and $a_n/b_n < p/q$, we get $b_np > a_nq$, so $b_np \geq a_nq+1$, giving $b_n \geq (a_nq+1)/p$. Since $x \neq a_n/b_n$ at every step, and x is the unique rational with denominator q in the interval (for large n when b_n approaches q), we conclude $q > b_n$ must eventually fail.

Step 4: Apply Denominator Growth to get a contradiction. By Lemma 5.7.25, $b_n \to \infty$. Therefore there exists N with $b_N > q$. But we argued $q > b_n$ for all n during an infinite path. This is a contradiction.

Step 5: Conclude. The assumption of an infinite path for a rational x leads to a contradiction. Therefore the path must be finite.

Taking the equivalence: Finite path $\Leftrightarrow$ rational, so infinite path $\Leftrightarrow$ irrational. □

Remark (Proof structure). A biconditional proof. The ($\Leftarrow$) direction is trivial: a mediant is visibly rational. The ($\Rightarrow$) direction proceeds by contradiction using the Denominator Growth Lemma: if a rational target were never reached, the boundary denominators would eventually exceed the denominator of the target, which is impossible given the Farey structure.

Remark (Dependencies). Farey Neighbor Theorem, Denominator growth on infinite SB paths, irrationality of $\sqrt{2}$ (example).

Remark (Return). Return to Theorem 5.9.6.

Theorem (Stern-Brocot interval refinement). *Let $x > 0$. Starting from $a/b < x < c/d$ with $b, d > 0$, repeatedly insert the mediant $(a+c)/(b+d)$ and replace whichever endpoint keeps x in the interval. This generates a nested sequence of rational intervals each containing x.*

Proof (Professional Standard). We show by induction that at every step n, the interval $I_n = [a_n/b_n, c_n/d_n]$ satisfies $a_n/b_n < x < c_n/d_n$.

Base case. $a_0/b_0 = a/b < x < c/d = c_0/d_0$ by hypothesis.

Inductive step. Suppose $a_n/b_n < x < c_n/d_n$. The mediant $m_n = (a_n+c_n)/(b_n+d_n)$ satisfies $a_n/b_n < m_n < c_n/d_n$ by the Mediant Inequality. Compare x to m_n:

- If $x < m_n$: set $I_{n+1} = [a_n/b_n, m_n]$. Then $a_n/b_n < x < m_n = c_{n+1}/d_{n+1}$.
- If $x > m_n$: set $I_{n+1} = [m_n, c_n/d_n]$. Then $m_n = a_{n+1}/b_{n+1} < x < c_n/d_n$.

(Since x is irrational in the case of an infinite path, $x = m_n$ cannot occur.) In both cases $I_{n+1} \subsetneq I_n$ and $x \in I_{n+1}$.

By induction, $x \in I_n$ for all n and $I_0 \supseteq I_1 \supseteq \cdots$. □

Proof (Detailed Learning). **Step 1: Induction statement.** We claim: for all $n \in \mathbb{N}$, $a_n/b_n < x < c_n/d_n$.

Step 2: Base case ($n = 0$). $a_0/b_0 = a/b < x < c/d = c_0/d_0$ by the hypothesis of the theorem.

Step 3: Inductive step. Assume $a_n/b_n < x < c_n/d_n$.

Form the mediant $m_n = (a_n + c_n)/(b_n + d_n)$. By the Mediant Inequality:

$$\frac{a_n}{b_n} < m_n < \frac{c_n}{d_n}.$$

Compare x to m_n:

- **Case $x < m_n$:** Define $a_{n+1}/b_{n+1} = a_n/b_n$ and $c_{n+1}/d_{n+1} = m_n$. Then $a_n/b_n < x < m_n$, so $a_{n+1}/b_{n+1} < x < c_{n+1}/d_{n+1}$. Also $I_{n+1} = [a_n/b_n, m_n] \subset [a_n/b_n, c_n/d_n] = I_n$.
- **Case $x > m_n$:** Define $a_{n+1}/b_{n+1} = m_n$ and $c_{n+1}/d_{n+1} = c_n/d_n$. Then $m_n < x < c_n/d_n$, so $a_{n+1}/b_{n+1} < x < c_{n+1}/d_{n+1}$. Also $I_{n+1} = [m_n, c_n/d_n] \subset [a_n/b_n, c_n/d_n] = I_n$.

In both cases the inductive claim holds at $n + 1$ and $I_{n+1} \subsetneq I_n$.

Step 4: Conclude. By induction, $x \in I_n$ for all n and (I_n) is a nested sequence of rational intervals. □

Remark (Proof structure). An induction on steps, with the Mediant Inequality as the key tool at each level. The containment $x \in I_{n+1}$ follows by a two-case split on whether $x < m_n$ or $x > m_n$; in both cases x is retained in the updated interval.

Remark (Dependencies). Mediant inequality, Mediant, order properties of $\mathbb{R}$.

Remark (Return). Return to Theorem 6.4.8.

Theorem (Stern-Brocot Limit Theorem). *Every irrational $x > 0$ determines a unique infinite path in the Stern-Brocot tree, and the endpoint sequences converge to x. Conversely, every infinite path determines a unique positive irrational as its limit.*

Proof (Professional Standard). **Forward direction (irrational $\Rightarrow$ unique infinite path converging to x).**

By Theorem 5.9.6, the SB refinement process generates a nested sequence of rational intervals I_n each containing x. Since x is irrational, it never equals any mediant, so the process never terminates: the path is infinite.

The left-right decisions at each step are deterministic (determined by the sign of $x - m_n$), so the path is unique.

By Corollary 5.7.28, the endpoint sequences converge to x.

Converse direction (infinite path $\Rightarrow$ unique irrational limit).

By Proposition 5.7.27, the nested intervals (I_n) associated to any infinite path contain a unique point $\xi \in \mathbb{R}$.

By Corollary 5.7.29, an infinite path cannot terminate at a rational, so $\xi \notin \mathbb{Q}$. Therefore ξ is irrational.

Uniqueness of ξ follows from the Unique Point Proposition. □

Proof (Detailed Learning). **Forward direction.**

Step 1: The path is infinite. Since x is irrational, $x \neq (a_n+c_n)/(b_n+d_n)$ at every step (the mediant is always rational). Therefore the comparison x vs. m_n always yields a strict inequality, and the process never terminates.

Step 2: The path is unique. At each step, the decision (go left or go right) is determined by the strict comparison $x < m_n$ or $x > m_n$, which has a unique outcome since x is irrational and m_n is rational. Therefore the sequence of left-right choices is unique, and the path is unique.

Step 3: The path converges to x. By Theorem 5.9.6, $x \in I_n$ for all n. By Corollary 5.7.28, the unique point in $\bigcap I_n$ is x, and the endpoint sequences converge to x.

Converse direction.

Step 4: An infinite path has a unique real limit. By Proposition 5.7.27, the nested intervals of any infinite path contain exactly one point $\xi \in \mathbb{R}$.

Step 5: The limit is irrational. By Corollary 5.7.29, an infinite path corresponds to a non-rational target. Since ξ is the unique point in $\bigcap I_n$, and the path is infinite (so the process never encountered ξ as a mediant), ξ cannot be rational. Therefore ξ is irrational.

Step 6: Conclude. Every irrational $x > 0$ determines a unique infinite path converging to x; every infinite path determines a unique positive irrational as its limit. □

Remark (Proof structure). The theorem is a synthesis of all the preceding Stern-Brocot results. The forward direction uses the Refinement Theorem and Path Convergence Corollary. The converse uses the Unique Point Proposition and the Irrational Limits Corollary. No new ideas appear; the proof is an assembly of the chain built up by Lemmas A and B, Proposition A, and Corollaries A and B.

Remark (Dependencies). SB interval refinement, Unique limit point, Every infinite path converges, Infinite paths are irrational, irrationality of $\sqrt{2}$ (prototype example).

Remark (Return). Return to Theorem

Theorem (No Rational Number Has Square 2). *There is no rational number whose square is* 2*.*

Proof. Suppose, for contradiction, that there exists a rational number r such that

$$r^2 = 2.$$

By Remark 5.2.7, we may write r in reduced form as

$$r = \frac{p}{q},$$

where $p, q \in \mathbb{Z}$, $\gcd(p, q) = 1$, and $q > 0$.

Substituting into $r^2 = 2$ gives

$$\left(\frac{p}{q}\right)^2 = 2,$$

so

$$p^2 = 2q^2.$$

Hence p^2 is even. Therefore p is even, so there exists $k \in \mathbb{Z}$ such that

$$p = 2k.$$

Substituting this into the equation above yields

$$(2k)^2 = 2q^2,$$

that is,

$$4k^2 = 2q^2.$$

Dividing by 2, we obtain

$$q^2 = 2k^2.$$

Thus q^2 is even, and therefore q is even.

We have shown that both p and q are even. But then 2 is a common divisor of p and q, contradicting the assumption that $\gcd(p, q) = 1$.

This contradiction proves that no rational number has square 2. □

Remark (Proof structure). The proof is the classical lowest-terms contradiction. Assume a rational square root of 2 exists and write it as p/q in reduced form. The equation

$$p^2 = 2q^2$$

forces p to be even. Substituting $p = 2k$ then forces q to be even as well. That contradicts the assumption that p/q was written in lowest terms. The argument isolates the arithmetic obstruction behind the gap at $\sqrt{2}$.

Remark (Dependencies).

- Remark 5.2.7 for reduced-form representatives of rational numbers
- Elementary parity facts for integers: if n^2 is even, then n is even

Remark (Return). Return to Lemma

Lemma (Rational Square Bracketing). *Let r be a positive rational number. Then there exist positive rational numbers a and b such that*

$$a^2 < r < b^2.$$

Proof. Let

$$a := \frac{r}{r+1} \qquad \text{and} \qquad b := r+1.$$

Since $r \in \mathbb{Q}$ and $r > 0$, both a and b are rational and positive.

We first show that $a^2 < r$. Because $r > 0$, we have $r+1 > 1$, hence

$$0 < a = \frac{r}{r+1} < r.$$

Also $a < 1$, since

$$\frac{r}{r+1} < 1.$$

Multiplying the inequality $a < 1$ by the positive number a gives

$$a^2 < a.$$

Since $a < r$, it follows that

$$a^2 < r.$$

Now we show that $r < b^2$. Since $b = r+1$ and $r > 0$, we have

$$b = r+1 > r.$$

Because $b > 1 > 0$, multiplying the inequality $b > 1$ by the positive number b gives

$$b^2 > b.$$

Combining this with $b > r$, we obtain

$$r < b^2.$$

Therefore there exist positive rational numbers a and b such that

$$a^2 < r < b^2,$$

as required. □

Remark (Proof structure). The proof uses an explicit construction rather than an existence argument by approximation. The lower bracket is chosen as

$$a = \frac{r}{r+1},$$

which is both positive and strictly less than 1 and r, so its square is even smaller. The upper bracket is chosen as

$$b = r+1,$$

which is larger than r and larger than 1, so its square is larger still. This produces rational squares on both sides of the given positive rational.

Remark (Dependencies).

- Basic field closure of $\mathbb{Q}$ under addition and division
- Basic order compatibility in an ordered field

Remark (Return). Return to Lemma

Theorem (First Grid Point Crossing). *Let $r \in \mathbb{Q}$ with $r > 0$ and let $q \in \mathbb{N}$ with $q \geq 1$. If $K \in \mathbb{N}$ satisfies $K^2 > r$, then the set*

$$A = \{m \in \mathbb{N} : m^2 > rq^2\}$$

is nonempty and therefore has a least element.

Proof. TODO. □

Remark (Proof structure). TODO.

Remark (Dependencies).

- Theorem 2.6.1
- Theorem 5.8.1

Remark (Return). Return to Theorem

Theorem (Rational Square Approximation from Above). *Let r be a positive rational number and let $n \in \mathbb{N}$. Then there exists a positive rational number s such that*

$$r < s^2 < r + \frac{1}{n}.$$

Proof. TODO. □

Remark (Proof structure). TODO.

Remark (Dependencies).

- Lemma 5.10.3
- Lemma 5.7.6
- Theorem 5.7.12

Remark (Return). Return to Theorem

Theorem (Rational Square Approximation from Below). *Let $r \in \mathbb{Q}$ with $r > 0$, and let $n \in \mathbb{N}$ with $n \geq 1$. If $r - 1/n > 0$, then there exists $s \in \mathbb{Q}$ with $s > 0$ such that*

$$r - \frac{1}{n} < s^2 < r.$$

Proof. TODO. □

Remark (Proof structure). TODO.

Remark (Dependencies).

- Theorem 5.10.4

Remark (Return). Return to Theorem

Theorem (Every Convergent Rational Sequence Is Cauchy). *Every convergent sequence in $\mathbb{Q}$ is a Cauchy sequence.*

Proof. TODO. $\square$

Remark (Proof structure). TODO.

Remark (Dependencies).

- Definitions of convergence and Cauchy sequence
- Triangle inequality

Remark (Return). Return to Theorem

Theorem (There Exist Cauchy Sequences in $\mathbb{Q}$ That Do Not Converge to a Rational Number). *There exist Cauchy sequences in* $\mathbb{Q}$ *that do not converge to a rational number.*

Proof. TODO. □

Remark (Proof structure). TODO.

Remark (Dependencies).

- Theorem 5.10.1
- A rational approximation sequence for $\sqrt{2}$
- Definition 5.9.2

Remark (Return). Return to Theorem 5.8.7.

Theorem (Limits of rational sequences are unique). *If (x_n) is a sequence of rational numbers and $\lim_{n\to\infty} x_n = a$ and $\lim_{n\to\infty} x_n = b$ with $a, b \in \mathbb{Q}$, then $a = b$.*

Proof (Professional Standard). Suppose $\lim_{n\to\infty} x_n = a$ and $\lim_{n\to\infty} x_n = b$. Let $\varepsilon > 0$. By convergence to a, there exists $N_1 \in \mathbb{N}$ such that $|x_n - a| < \varepsilon/2$ for all $n \geq N_1$. By convergence to b, there exists $N_2 \in \mathbb{N}$ such that $|x_n - b| < \varepsilon/2$ for all $n \geq N_2$. Set $N = \max\{N_1, N_2\}$ and choose any $n \geq N$. By the triangle inequality,

$$|a - b| = |a - x_n + x_n - b| \leq |a - x_n| + |x_n - b| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Since $\varepsilon > 0$ was arbitrary and $|a-b| < \varepsilon$ for all $\varepsilon > 0$, we conclude $|a-b| = 0$, hence $a = b$. □

Proof (Detailed Learning). **Step 1: Set up the goal.** We want to prove $a = b$, or equivalently $|a - b| = 0$. Since $|a - b| \geq 0$ always holds, it suffices to show $|a - b| < \varepsilon$ for every $\varepsilon > 0$, because the only nonnegative rational smaller than every positive rational is zero.

Step 2: Extract witnesses from each convergence hypothesis. Let $\varepsilon > 0$. The hypothesis $x_n \to a$ supplies $N_1 \in \mathbb{N}$ with $|x_n - a| < \varepsilon/2$ for all $n \geq N_1$. The hypothesis $x_n \to b$ supplies $N_2 \in \mathbb{N}$ with $|x_n - b| < \varepsilon/2$ for all $n \geq N_2$.

Step 3: Choose a single index that satisfies both. Set $N = \max\{N_1, N_2\}$. For $n \geq N$ we have both $n \geq N_1$ and $n \geq N_2$, so both estimates hold simultaneously.

Step 4: Apply the triangle inequality. For any $n \geq N$:

$$|a - b| = |a - x_n + x_n - b| \leq |a - x_n| + |x_n - b| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Step 5: Conclude. Since $\varepsilon > 0$ was arbitrary and $|a-b| < \varepsilon$ for all such ε, we have $|a - b| = 0$, so $a = b$. □

Remark (Proof structure). This is the standard 2ε squeeze argument: make both approximation errors small by halving ε, then combine via the triangle inequality. The key move is choosing N large enough to activate both convergence witnesses simultaneously.

Remark (Dependencies). Convergence of a rational sequence, the triangle inequality in $\mathbb{Q}$.

Remark (Return). Return to Theorem 5.8.9.

Theorem (Subsequences of convergent sequences converge to the same limit). *Let (x_n) be a sequence of rational numbers with $x_n \to a \in \mathbb{Q}$. Then every subsequence (x_{k_n}) converges to a.*

Proof (Professional Standard). Let (x_{k_n}) be any subsequence of (x_n). Let $\varepsilon > 0$. Since $x_n \to a$, there exists $N \in \mathbb{N}$ such that $|x_n - a| < \varepsilon$ for all $n \geq N$. We claim $k_n \geq n$ for all $n \in \mathbb{N}$. Indeed, $k_1 \geq 1$ since $(k_n) \subseteq \mathbb{N}$, and if $k_n \geq n$ then $k_{n+1} > k_n \geq n$, so $k_{n+1} \geq n + 1$. By induction, $k_n \geq n$ for all n. Therefore, for all $n \geq N$ we have $k_n \geq n \geq N$, hence $|x_{k_n} - a| < \varepsilon$. Since $\varepsilon > 0$ was arbitrary, $x_{k_n} \to a$. □

Proof (Detailed Learning). **Step 1: Establish the key lemma $k_n \geq n$.** We prove by induction that $k_n \geq n$ for all $n \in \mathbb{N}$.

Base case: $k_1 \in \mathbb{N}$, so $k_1 \geq 1$.

Inductive step: Suppose $k_n \geq n$ for some n. Since the sequence (k_n) is strictly increasing, $k_{n+1} > k_n$. Since k_{n+1} is a natural number and $k_{n+1} > k_n \geq n$, we get $k_{n+1} \geq n + 1$.

By induction, $k_n \geq n$ for all $n \in \mathbb{N}$.

Step 2: Use the convergence of (x_n). Let $\varepsilon > 0$. Since $x_n \to a$, there exists $N \in \mathbb{N}$ such that $|x_n - a| < \varepsilon$ for all $n \geq N$.

Step 3: Verify the subsequence condition. For any $n \geq N$: by Step 1, $k_n \geq n \geq N$. Therefore $|x_{k_n} - a| < \varepsilon$ by the choice of N.

Step 4: Conclude. Since $\varepsilon > 0$ was arbitrary, we have shown that for each $\varepsilon > 0$ there is an N such that $|x_{k_n} - a| < \varepsilon$ for all $n \geq N$. By definition, $x_{k_n} \to a$. □

Remark (Proof structure). The structural key is the lemma $k_n \geq n$: it ensures that every subsequence index eventually exceeds any given threshold N, allowing us to transfer the convergence bound from (x_n) to (x_{k_n}). The induction is the only non-trivial step; everything else is definition-unwinding.

Remark (Dependencies). Convergence of a rational sequence, Subsequence, induction on $\mathbb{N}$.

Remark (Return). Return to Theorem 5.8.11.

Theorem (Convergent sequences are bounded). *Every convergent sequence of rational numbers is bounded.*

Proof (Professional Standard). Let (x_n) be a sequence of rational numbers with $x_n \to a \in \mathbb{Q}$. Apply the definition of convergence with $\varepsilon = 1$: there exists $N \in \mathbb{N}$ such that $|x_n - a| < 1$ for all $n \geq N$. By the triangle inequality, $|x_n| \leq |x_n - a| + |a| < 1 + |a|$ for all $n \geq N$. Define

$$\Gamma = \max\{|x_1|, |x_2|, \ldots, |x_N|, 1 + |a|\}.$$

Then $|x_n| \leq \Gamma$ for all $n \in \mathbb{N}$, so (x_n) is bounded. □

Proof (Detailed Learning). **Step 1: Control the tail using convergence.** Since $x_n \to a$, choose $\varepsilon = 1$. There exists $N \in \mathbb{N}$ such that $|x_n - a| < 1$ for all $n \geq N$.

Step 2: Bound each tail term. For any $n \geq N$, apply the triangle inequality:

$$|x_n| = |x_n - a + a| \leq |x_n - a| + |a| < 1 + |a|.$$

Step 3: Handle the initial segment. The terms $x_1, x_2, \ldots, x_N$ are finitely many rational numbers, each with a finite absolute value. Their maximum $\max\{|x_1|, \ldots, |$ is a well-defined rational number.

Step 4: Combine into a single bound. Set

$$\Gamma = \max\big\{|x_1|, |x_2|, \ldots, |x_N|,\ 1 + |a|\big\}.$$

For $n \leq N$: $|x_n| \leq \max\{|x_1|, \ldots, |x_N|\} \leq \Gamma$. For $n > N$: $|x_n| < 1 + |a| \leq \Gamma$. Therefore $|x_n| \leq \Gamma$ for all $n \in \mathbb{N}$.

Step 5: Conclude. By Definition 5.8.10, (x_n) is bounded. □

Remark (Proof structure). The proof splits the sequence into two parts: a finite initial segment (handled by taking a maximum of finitely many values) and a tail (controlled by convergence with $\varepsilon = 1$). The choice $\varepsilon = 1$ is not special any positive ε works but 1 keeps the arithmetic clean.

Remark (Dependencies). Convergence of a rational sequence, Bounded sequence, the triangle inequality in $\mathbb{Q}$.

Remark (Return). Return to Theorem 5.8.12.

Theorem (Cauchy sequences are bounded). *Every Cauchy sequence of rational numbers is bounded.*

Proof (Professional Standard). Let (x_n) be a Cauchy sequence of rational numbers. Apply the Cauchy condition with $\varepsilon = 1$: there exists $N \in \mathbb{N}$ such that $|x_m - x_n| < 1$ for all $m, n \geq N$. Taking $n = N + 1$, we get $|x_m - x_{N+1}| < 1$ for all $m \geq N$, so $|x_m| \leq |x_m - x_{N+1}| + |x_{N+1}| < 1 + |x_{N+1}|$ for all $m \geq N$. Define

$$\Gamma = \max\{|x_1|, \ldots, |x_N|, 1 + |x_{N+1}|\}.$$

Then $|x_n| \leq \Gamma$ for all $n \in \mathbb{N}$. □

Proof (Detailed Learning). **Step 1: Apply the Cauchy condition with $\varepsilon = 1$.** By Definition 5.9.2, there exists $N \in \mathbb{N}$ such that $|x_m - x_n| < 1$ for all $m, n \geq N$.

Step 2: Anchor the tail to a single term. Fix $n = N + 1$. For every $m \geq N$ we have $n = N + 1 \geq N$, so

$$|x_m - x_{N+1}| < 1.$$

Step 3: Bound each tail term via the triangle inequality. For any $m \geq N$:

$$|x_m| = |x_m - x_{N+1} + x_{N+1}| \leq |x_m - x_{N+1}| + |x_{N+1}| < 1 + |x_{N+1}|.$$

Step 4: Handle the initial segment. The terms $x_1, \ldots, x_N$ are finitely many, so $\max\{|x_1|, \ldots, |x_N|\}$ is a well-defined rational number.

Step 5: Combine. Set

$$\Gamma = \max\{|x_1|, \ldots, |x_N|, 1 + |x_{N+1}|\}.$$

For $n \leq N$: $|x_n| \leq \Gamma$. For $n > N$ (which implies $n \geq N$): $|x_n| < 1 + |x_{N+1}| \leq \Gamma$. Therefore $|x_n| \leq \Gamma$ for all $n \in \mathbb{N}$.

Step 6: Conclude. By Definition 5.8.10, (x_n) is bounded. □

Remark (Proof structure). The architecture is identical to the proof that convergent sequences are bounded, with one substitution: instead of anchoring the tail to the limit a, we anchor it to the single term x_{N+1}. The Cauchy condition controls $|x_m - x_{N+1}|$ for all $m \geq N$ in exactly the same way that convergence controlled $|x_m - a|$. This structural parallel is not a coincidence: the Cauchy condition is convergence without a named limit.

Remark (Dependencies). Cauchy sequence in $\mathbb{Q}$, Bounded sequence, the triangle inequality in $\mathbb{Q}$.

Remark (Return). Return to Theorem 5.8.13.

Theorem (Cauchy sequences are closed under addition and multiplication). *If (x_n) and (y_n) are Cauchy sequences of rational numbers, then so are (x_n+y_n) and $(x_n y_n)$.*

Proof (Professional Standard). **Addition.** Let $\varepsilon > 0$. Since (x_n) and (y_n) are Cauchy, there exist $N_1, N_2 \in \mathbb{N}$ such that $|x_m - x_n| < \varepsilon/2$ for all $m, n \geq N_1$ and $|y_m - y_n| < \varepsilon/2$ for all $m, n \geq N_2$. Set $N = \max\{N_1, N_2\}$. For $m, n \geq N$:

$$|(x_m + y_m) - (x_n + y_n)| \leq |x_m - x_n| + |y_m - y_n| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Multiplication. By Theorem 5.8.12, both sequences are bounded; let $\Lambda > 0$ satisfy $|x_n| \leq \Lambda$ and $|y_n| \leq \Lambda$ for all n. Let $\varepsilon > 0$. Choose N_1, N_2 such that $|x_m - x_n| < \varepsilon/(2\Lambda)$ for $m, n \geq N_1$ and $|y_m - y_n| < \varepsilon/(2\Lambda)$ for $m, n \geq N_2$. Set $N = \max\{N_1, N_2\}$. For $m, n \geq N$:

$$|x_m y_m - x_n y_n| = |x_m y_m - x_n y_m + x_n y_m - x_n y_n| \leq |y_m||x_m - x_n| + |x_n||y_m - y_n| < \Lambda \cdot \frac{\varepsilon}{2\Lambda} + \Lambda \cdot \frac{\varepsilon}{2\Lambda} = \varepsilon.$$

□

Proof (Detailed Learning). **Part I: Addition.**

Step 1: Set up the $\varepsilon/2$ split. Let $\varepsilon > 0$. We need $|(x_m+y_m)-(x_n+y_n)| < \varepsilon$ for all sufficiently large m, n. We will achieve this by keeping both $|x_m - x_n|$ and $|y_m - y_n|$ smaller than $\varepsilon/2$.

Step 2: Extract witnesses. Since (x_n) is Cauchy, $\exists N_1$ such that $|x_m - x_n| < \varepsilon/2$ for all $m, n \geq N_1$. Since (y_n) is Cauchy, $\exists N_2$ such that $|y_m - y_n| < \varepsilon/2$ for all $m, n \geq N_2$.

Step 3: Combine and apply the triangle inequality. Set $N = \max\{N_1, N_2\}$. For all $m, n \geq N$:

$$|(x_m + y_m) - (x_n + y_n)| = |(x_m - x_n) + (y_m - y_n)| \leq |x_m - x_n| + |y_m - y_n| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Part II: Multiplication.

Step 4: Invoke boundedness. By Theorem 5.8.12, (x_n) and (y_n) are bounded. Let $\Lambda_1, \Lambda_2 > 0$ satisfy $|x_n| \leq \Lambda_1$ and $|y_n| \leq \Lambda_2$ for all n. Set $\Lambda = \max\{\Lambda_1, \Lambda_2\} > 0$, so both $|x_n| \leq \Lambda$ and $|y_n| \leq \Lambda$ for all n.

Step 5: Algebraically separate the two sources of error. Add and subtract the cross-term $x_n y_m$:

$$x_m y_m - x_n y_n = (x_m - x_n) y_m + x_n (y_m - y_n).$$

By the triangle inequality:

$$|x_m y_m - x_n y_n| \leq |y_m||x_m - x_n| + |x_n||y_m - y_n| \leq \Lambda |x_m - x_n| + \Lambda |y_m - y_n|.$$

Step 6: Choose the Cauchy witnesses to compensate for Λ. Let $\varepsilon > 0$. Choose N_1 such that $|x_m - x_n| < \varepsilon/(2\Lambda)$ for all $m, n \geq N_1$, and N_2 such that $|y_m - y_n| < \varepsilon/(2\Lambda)$ for all $m, n \geq N_2$.

Step 7: Combine. Set $N = \max\{N_1, N_2\}$. For all $m, n \geq N$:

$$|x_m y_m - x_n y_n| \leq \Lambda \cdot \frac{\varepsilon}{2\Lambda} + \Lambda \cdot \frac{\varepsilon}{2\Lambda} = \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Step 8: Conclude. Both (x_n+y_n) and $(x_n y_n)$ satisfy the Cauchy condition.

□

Remark (Proof structure). Both parts are ε-splitting arguments. The addition case is a direct two-term triangle inequality. The multiplication case adds one layer of difficulty: the product does not split cleanly, so one must first algebraically separate the two sources of error (the x-difference and the y-difference) using the cross-term trick, then compensate for the bounds Λ by scaling the tolerance down to $\varepsilon/(2\Lambda)$ before extracting the Cauchy witnesses. This Λ-compensation pattern recurs throughout analysis wherever products of sequences appear.

Remark (Dependencies). Cauchy sequence in $\mathbb{Q}$, Cauchy sequences are bounded, the triangle inequality in $\mathbb{Q}$.

Remark (Return). Return to Theorem 5.8.15.

Theorem (The relation $\sim$ is an equivalence relation on $\mathbb{Q}^c$). *Define* $(x_n) \sim (y_n)$ *if and only if* $\lim_{n\to\infty} |x_n - y_n| = 0$. *Then* $\sim$ *is an equivalence relation on the set* $\mathbb{Q}^c$ *of all Cauchy sequences in* $\mathbb{Q}$.

Proof (Professional Standard). **Reflexivity.** For any $(x_n) \in \mathbb{Q}^c$: $|x_n - x_n| = 0 < \varepsilon$ for all $\varepsilon > 0$ and all n, so $(x_n) \sim (x_n)$.

Symmetry. If $(x_n) \sim (y_n)$ then $|x_n - y_n| \to 0$. Since $|y_n - x_n| = |x_n - y_n|$, we have $|y_n - x_n| \to 0$, so $(y_n) \sim (x_n)$.

Transitivity. Suppose $(x_n) \sim (y_n)$ and $(y_n) \sim (z_n)$. Let $\varepsilon > 0$. There exist $N_1, N_2 \in \mathbb{N}$ such that $|x_n - y_n| < \varepsilon/2$ for $n \geq N_1$ and $|y_n - z_n| < \varepsilon/2$ for $n \geq N_2$. For all $n \geq \max\{N_1, N_2\}$:

$$|x_n - z_n| \leq |x_n - y_n| + |y_n - z_n| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Therefore $(x_n) \sim (z_n)$. □

Proof (Detailed Learning). We verify the three axioms of an equivalence relation.

Step 1: Reflexivity $(x_n) \sim (x_n)$. We need: $\forall \varepsilon > 0$, $\exists N$ such that $|x_n - x_n| < \varepsilon$ for all $n \geq N$. Since $x_n - x_n = 0$ for every n, we have $|x_n - x_n| = 0 < \varepsilon$ for every n and every $\varepsilon > 0$. Any $N = 1$ works.

Step 2: Symmetry $(x_n) \sim (y_n)$ **implies** $(y_n) \sim (x_n)$. Suppose $(x_n) \sim (y_n)$. Let $\varepsilon > 0$. There exists N such that $|x_n - y_n| < \varepsilon$ for all $n \geq N$. Since $|y_n - x_n| = |-(x_n - y_n)| = |x_n - y_n|$, we have $|y_n - x_n| < \varepsilon$ for all $n \geq N$. Therefore $(y_n) \sim (x_n)$.

Step 3: Transitivity $(x_n) \sim (y_n)$ **and** $(y_n) \sim (z_n)$ **implies** $(x_n) \sim (z_n)$. Suppose $(x_n) \sim (y_n)$ and $(y_n) \sim (z_n)$. Let $\varepsilon > 0$.

Extract witnesses. Since $(x_n) \sim (y_n)$, there exists N_1 such that $|x_n - y_n| < \varepsilon/2$ for all $n \geq N_1$. Since $(y_n) \sim (z_n)$, there exists N_2 such that $|y_n - z_n| < \varepsilon/2$ for all $n \geq N_2$.

Combine. Set $N = \max\{N_1, N_2\}$. For all $n \geq N$:

$$|x_n - z_n| = |x_n - y_n + y_n - z_n| \leq |x_n - y_n| + |y_n - z_n| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Therefore $(x_n) \sim (z_n)$.

Step 4: Conclude. All three axioms hold, so $\sim$ is an equivalence relation on $\mathbb{Q}^c$. □

Remark (Proof structure). This proof is a direct three-part verification. Reflexivity is trivial. Symmetry follows from $|a - b| = |b - a|$. Transitivity is a 2ε split via the triangle inequality structurally identical to the proof of uniqueness of limits and the proof that convergent sequences are Cauchy. All three proofs share a single master template.

Remark (Dependencies). Equivalence of Cauchy sequences, triangle inequality in $\mathbb{Q}$.

Remark (Return). Return to Theorem 5.8.16.

Theorem (Cauchy equivalence is preserved under addition and multiplication). *Let* $(x_n), (y_n), (u_n), (v_n) \in \mathbb{Q}^c$. *If* $(x_n) \sim (u_n)$ *and* $(y_n) \sim (v_n)$, *then* $(x_n + y_n) \sim (u_n + v_n)$ *and* $(x_n y_n) \sim (u_n v_n)$.

Proof (Professional Standard). **Addition.** Let $\varepsilon > 0$. Since $(x_n) \sim (u_n)$, there exists N_1 with $|x_n - u_n| < \varepsilon/2$ for $n \geq N_1$. Since $(y_n) \sim (v_n)$, there exists N_2 with $|y_n - v_n| < \varepsilon/2$ for $n \geq N_2$. For $n \geq \max\{N_1, N_2\}$:

$$|(x_n + y_n) - (u_n + v_n)| \leq |x_n - u_n| + |y_n - v_n| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Multiplication. By Theorem 5.8.12, all four sequences are bounded; let $\Lambda > 0$ satisfy $|x_n|, |y_n|, |u_n|, |v_n| \leq \Lambda$. Let $\varepsilon > 0$. Choose N_1 with $|x_n - u_n| < \varepsilon/(2\Lambda)$ for $n \geq N_1$, and N_2 with $|y_n - v_n| < \varepsilon/(2\Lambda)$ for $n \geq N_2$. Write:

$$x_n y_n - u_n v_n = (x_n - u_n)y_n + u_n(y_n - v_n).$$

For $n \geq \max\{N_1, N_2\}$:

$$|x_n y_n - u_n v_n| \leq |y_n||x_n - u_n| + |u_n||y_n - v_n| < \Lambda \cdot \frac{\varepsilon}{2\Lambda} + \Lambda \cdot \frac{\varepsilon}{2\Lambda} = \varepsilon.$$

□

Proof (Detailed Learning). **Part I: Addition.**

Step 1. We want $|(x_n + y_n) - (u_n + v_n)| \to 0$.

Step 2. Let $\varepsilon > 0$. Since $(x_n) \sim (u_n)$: $\exists N_1$ with $|x_n - u_n| < \varepsilon/2$ for $n \geq N_1$. Since $(y_n) \sim (v_n)$: $\exists N_2$ with $|y_n - v_n| < \varepsilon/2$ for $n \geq N_2$.

Step 3. For $n \geq \max\{N_1, N_2\}$, by the triangle inequality:

$$|(x_n + y_n) - (u_n + v_n)| \leq |x_n - u_n| + |y_n - v_n| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Part II: Multiplication.

Step 4: Establish a common bound. All four sequences are Cauchy, hence bounded. Let $\Lambda > 0$ satisfy $|x_n| \leq \Lambda$, $|y_n| \leq \Lambda$, $|u_n| \leq \Lambda$, $|v_n| \leq \Lambda$ for all n.

Step 5: Algebraic decomposition.

$$x_n y_n - u_n v_n = x_n y_n - u_n y_n + u_n y_n - u_n v_n = (x_n - u_n)y_n + u_n(y_n - v_n).$$

Step 6. By the triangle inequality:

$$|x_n y_n - u_n v_n| \leq |y_n||x_n - u_n| + |u_n||y_n - v_n| \leq \Lambda|x_n - u_n| + \Lambda|y_n - v_n|.$$

Step 7. Let $\varepsilon > 0$. Choose N_1 with $|x_n - u_n| < \varepsilon/(2\Lambda)$ for $n \geq N_1$, and N_2 with $|y_n - v_n| < \varepsilon/(2\Lambda)$ for $n \geq N_2$.

Step 8. For $n \geq \max\{N_1, N_2\}$:

$$|x_n y_n - u_n v_n| \leq \Lambda \cdot \frac{\varepsilon}{2\Lambda} + \Lambda \cdot \frac{\varepsilon}{2\Lambda} = \varepsilon.$$

Conclude: $(x_n + y_n) \sim (u_n + v_n)$ and $(x_n y_n) \sim (u_n v_n)$. □

Remark (Proof structure). This is a well-definedness proof. The addition part is a direct triangle inequality split. The multiplication part uses the same cross-term decomposition as in the closure-under-multiplication proof, but now the two sequences being "moved" are the equivalence hypotheses $(x_n - u_n)$ and $(y_n - v_n)$, while Λ bounds the coefficients y_n and u_n. The Λ-compensation pattern is the same in both proofs.

Remark (Dependencies). Equivalence of Cauchy sequences, Cauchy sequences are bounded, triangle inequality in $\mathbb{Q}$.

Capstone

Capstone Rational Numbers

Status: Planned.
This capstone synthesizes the core results of the rationals chapter. It is designed to require the student to connect the algebraic, order-theoretic, density, and analytic threads of the chapter in a single argument.

Problem (Draft). Let (x_n) and (y_n) be Cauchy sequences of rational numbers.

1. Prove that $(x_n + y_n)$ and $(x_n y_n)$ are both Cauchy sequences.
2. Define the relation $(x_n) \sim (y_n)$ if and only if $(x_n - y_n)$ converges to 0 in $\mathbb{Q}$. Prove that $\sim$ is an equivalence relation on $\mathbb{Q}^c$.
3. Suppose $(x_n) \sim (u_n)$ and $(y_n) \sim (v_n)$. Prove that $(x_n + y_n) \sim (u_n + v_n)$.
4. Let $r \in \mathbb{Q}$ and let (r) denote the constant sequence with every term equal to r. Show that $(r) \in \mathbb{Q}^c$ and that if $(x_n) \sim (r)$ then $\lim_{n\to\infty} x_n = r$.
5. Construct an explicit Cauchy sequence (x_n) in $\mathbb{Q}$ such that $(x_n) \not\sim (r)$ for any $r \in \mathbb{Q}$. Identify which theorem in the chapter guarantees such a sequence exists.

Parts 1–3 are proved in the notes; this exercise requires the student to reproduce and connect them in a single self-contained argument. Part 4 makes precise the embedding $\mathbb{Q} \hookrightarrow \mathbb{Q}^c/\!\sim$. Part 5 demonstrates incompleteness by explicit example.

Remark (Dependency ceiling). This capstone may use any result from the rationals chapter. It may not invoke completeness of $\mathbb{R}$, Dedekind cuts, or any result whose proof requires the real numbers.

Chapter 6

Real Numbers ($\mathbb{R}$)

Where You Are in the Journey

Propositional Logic $\to$ Predicate Calculus $\to$ Sets & Functions $\to$ Proof Techniques $\to$ Axiom Systems $\to$ $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$ $\to$ **Real Numbers ($\mathbb{R}$)** $\to$ Real Analysis $\to$ $\cdots$

How we got here. The rational numbers gave us a dense ordered field, but they contain gaps: there is no $q \in \mathbb{Q}$ with $q^2 = 2$. The real number system fills these gaps. Two rigorous constructions achieve this — Dedekind cuts and Cauchy sequences — and both yield the same unique complete ordered field.

What this chapter builds. The axiomatic structure of $\mathbb{R}$: field axioms, order axioms, the completeness axiom, and their immediate consequences — the Archimedean property, density of $\mathbb{Q}$, and the interval and bound theory that underpins all of analysis. We also study how $\mathbb{R}$ is *constructed* from $\mathbb{Q}$ via Dedekind cuts and Cauchy sequences.

Where this leads. Real analysis in Volume III takes $\mathbb{R}$ as given and develops sequences, series, convergence, and their limiting behaviour. The field and order axioms established here propagate unchanged into metric spaces, normed spaces, and abstract algebra.

Structural Roadmap

This chapter covers the *algebraic and analytic axioms* of $\mathbb{R}$ and the *constructions* that justify its existence. Sequence and series theory is developed in **Volume III, Real Analysis**.

Axioms $\longrightarrow$ Order & Bounds $\longrightarrow$ Completeness $\longrightarrow$ Constructions

The global progression is:

1. Field and order axioms
2. Intervals and convexity
3. Bounds: upper/lower bounds, maximum, minimum
4. Extremal values: supremum and infimum
5. Completeness: the least upper bound property and its consequences
6. Constructions of $\mathbb{R}$: Dedekind cuts
7. Constructions of $\mathbb{R}$: Cauchy sequences
8. Interval arithmetic

Remark 6.0.1 (Primary sources). Axiomatic development follows Abbott, *Understanding Analysis* and Ross, *Elementary Analysis*. The Cauchy sequence construction follows Tao, *Analysis I*, Chapters 5-6. The Dedekind cut construction follows Rudin, *Principles of Mathematical Analysis*, Appendix to Chapter 1.

6.1 Dedekind Cut Construction of $\mathbb{R}$

Dedekind Cuts Quick Reference

Concept	Meaning	Detail
Dedekind cut	Partition $(A \mid B)$ of $\mathbb{Q}$	Def
Real number	An equivalence class of Dedekind cuts	Def
Completeness	Every non-empty bounded cut has a sup	Thm

Definition (Dedekind Cut)

A *Dedekind cut* is a pair (A, B) with $A \cup B = \mathbb{Q}$, $A \cap B = \emptyset$, $A \neq \emptyset$, $B \neq \emptyset$, satisfying:

1. If $p \in A$ and $q < p$ then $q \in A$ (downward closed),
2. A has no greatest element.

Remark 6.1.1 (English reading). A Dedekind cut slices the rationals into a lower set A and upper set B so that A contains all rationals "below" the cut point and has no maximum. Each cut corresponds to exactly one real number.

Definition (Real Number via Dedekind Cut)

A *real number* is a Dedekind cut (A, B) of $\mathbb{Q}$. The set $\mathbb{R}$ is defined as the collection of all Dedekind cuts, equipped with order: $(A_1, B_1) \leq (A_2, B_2)$ iff $A_1 \subseteq A_2$.

Remark 6.1.2 (Status). *This section is a stub. Full content arithmetic on cuts, the proof that $\mathbb{R}$ is a complete ordered field, and the uniqueness theorem will be developed in a future revision. Primary source: Rudin, Principles of Mathematical Analysis, Appendix to Chapter 1.*

Theorem (Dedekind Completeness)

The ordered field $(\mathbb{R}, +, \cdot, \leq)$ constructed from Dedekind cuts is complete: every non-empty subset of $\mathbb{R}$ that is bounded above has a least upper bound in $\mathbb{R}$.

Remark 6.1.3 (Significance). This is the point of the entire construction: $\mathbb{Q}$ has "gaps" (e.g. no rational satisfies $x^2 = 2$), and Dedekind cuts fill exactly those gaps, yielding a complete ordered field.

6.2 Cauchy Sequence Construction of $\mathbb{R}$

Cauchy Completion Quick Reference

Concept	Meaning	Detail
Cauchy sequence over $\mathbb{Q}$	Sequence with terms arbitrarily close	Def
Equivalence of Cauchy seqs	Same limit behaviour	Def
Real number as equivalence class	$\mathbb{R} = \mathbb{Q}^{\text{Cauchy}}/\sim$	Def

Definition (Cauchy Sequence over $\mathbb{Q}$)

A sequence (a_n) of rationals is *Cauchy* if

$$\forall \varepsilon \in \mathbb{Q}_{>0},\ \exists N \in \mathbb{N},\ \forall m, n \geq N, \quad |a_m - a_n| < \varepsilon.$$

Definition (Equivalence of Cauchy Sequences)

Two Cauchy sequences (a_n) and (b_n) over $\mathbb{Q}$ are *equivalent*, written $(a_n) \sim (b_n)$, if

$$\lim_{n\to\infty} (a_n - b_n) = 0 \quad (\text{in } \mathbb{Q}).$$

Definition (Real Number via Cauchy Sequences)

A *real number* is an equivalence class $[(a_n)]$ of Cauchy sequences of rationals under $\sim$. The set $\mathbb{R}$ is defined as the quotient $\mathbb{R} = \mathcal{C}(\mathbb{Q})/\sim$, where $\mathcal{C}(\mathbb{Q})$ denotes the set of all Cauchy sequences of rationals.

Remark 6.2.1 (Status). *This section is a stub. Full content arithmetic on equivalence classes, verification of field axioms, proof of completeness, and comparison with the Dedekind construction will be developed in a future revision. Primary source: Tao, Analysis I, Chapters 5-6.*

Remark 6.2.2 (Comparison with Dedekind cuts). Both constructions yield the same object up to isomorphism. The Dedekind approach is more geometric (cutting the line); the Cauchy approach is more analytic (completing via limits). Tao develops the Cauchy approach; Rudin the Dedekind approach.

6.3 Nested Interval Construction of $\mathbb{R}$ (Cantor)

Cantor Nested Intervals Quick Reference

Concept	Meaning	Detail
Nested intervals	$I_{n+1} \subseteq I_n$	Def
Diameter / length	$\operatorname{diam}(I_n)$ shrinks to 0	Def
Cantor intersection principle	$\bigcap_n I_n$ is a single point	Thm
Real as "limit point" of rationals	coded by a shrinking rational interval	Def

Definition (Nested Interval Sequence)

A *nested interval sequence* is a sequence of nonempty closed intervals

$$I_n = [a_n, b_n] \subseteq \mathbb{Q}$$

such that for all $n \in \mathbb{N}$,

$$I_{n+1} \subseteq I_n \quad (\text{equivalently, } a_n \leq a_{n+1} \leq b_{n+1} \leq b_n).$$

Definition (Diameter / Length)

For an interval $I = [a, b]$ (with $a \leq b$), its *diameter* (length) is

$$\operatorname{diam}(I) := b - a.$$

For a nested sequence $I_n = [a_n, b_n]$, we say it *shrinks to a point* if

$$\operatorname{diam}(I_n) \longrightarrow 0 \quad \text{as } n \to \infty.$$

Theorem (Cantor Nested Interval Principle)

Let (I_n) be a nested interval sequence of nonempty closed intervals

$$I_n = [a_n, b_n] \subseteq \mathbb{R}$$

such that $\operatorname{diam}(I_n) \to 0$. Then there exists a unique real number $x \in \mathbb{R}$ such that

$$x \in I_n \quad \text{for all } n, \qquad \text{i.e.} \qquad \bigcap_{n=1}^{\infty} I_n = \{x\}.$$

Definition (Real Number via Nested Rational Intervals)

Let $\mathcal{N}$ be the set of all nested sequences of nonempty closed *rational* intervals

$$I_n = [a_n, b_n] \subseteq \mathbb{Q}, \qquad I_{n+1} \subseteq I_n, \qquad \operatorname{diam}(I_n) \to 0.$$

Define an equivalence relation $\sim$ on $\mathcal{N}$ by

$$(I_n) \sim (J_n) \quad \Longleftrightarrow \quad \forall n\ \exists m\ \big(I_m \subseteq J_n\ \wedge\ J_m \subseteq I_n\big).$$

(Informally: the two sequences eventually refine each other arbitrarily.) A *real number* is an equivalence class $[(I_n)]$ under $\sim$.
We denote the resulting set of equivalence classes by $\mathbb{R}$.

Remark (Endpoints approximate the same point)

If $(I_n) = [a_n, b_n]$ is nested with $\operatorname{diam}(I_n) \to 0$, then

$$(b_n - a_n) \to 0,$$

so (a_n) and (b_n) are "squeezed together." In a completed number system, this forces them to converge to the *same* point x, and the intervals are precisely rational "traps" for x.

6.3.1 Consequences

Remark 6.3.1 (How this constructs completeness). The Cantor principle (section 6.3) is a completeness statement: it says that whenever you can trap a number by nested closed intervals whose lengths go to 0, there is a point (indeed a unique point) trapped by all of them.

In the rational numbers $\mathbb{Q}$ this fails: there are nested rational intervals with diameters $\to 0$ whose "intended" intersection point is irrational (e.g. intervals shrinking around $\sqrt{2}$), so the intersection in $\mathbb{Q}$ is empty. The construction above formally adjoins the missing limit points by *declaring* each shrinking nested rational interval sequence to represent a real number.

Remark 6.3.2 (Comparison with Cauchy completion). This is morally the same as the Cauchy-sequence construction:

- A Cauchy sequence is a list of rational approximations that eventually get arbitrarily close.
- A nested rational interval sequence is a sequence of rational *error bars* whose widths go to 0.

One can convert between the two encodings: choose a point $q_n \in I_n$ to get a Cauchy sequence (q_n), and conversely use Cauchy tails to build shrinking intervals. Both yield the same $\mathbb{R}$ up to canonical isomorphism.

Remark 6.3.3 (Status). *This section is intentionally focused on the construction idea and the core nested-interval principle. Full development (addition/multiplica on classes, order, and the full completeness axiom) can be added after your Cauchy-section, since the proofs mirror each other.*

6.4 Construction via Rational Approximation

Rational Approximation Quick Reference

Concept	Idea	Reference
Farey neighbors	adjacent rationals in ordered lists	
Mediant	refinement between two rationals	
SternBrocot tree	complete enumeration of $\mathbb{Q}_{>0}$	
Infinite refinement	nested rational intervals	
Real numbers	limits of rational approximations	

Motivation

The rational numbers $\mathbb{Q}$ form an ordered field but are *not complete*. There exist bounded sets of rationals with no least upper bound in $\mathbb{Q}$, and there exist Cauchy sequences of rationals that do not converge to a rational number.

One way to construct the real numbers is therefore to enlarge $\mathbb{Q}$ so that every coherent limiting process among rationals corresponds to a number.

The constructions of Cantor (Cauchy sequences) and Dedekind (cuts) achieve this algebraically or order-theoretically. A third viewpoint arises from *rational approximation*.

The idea is:

$$\mathbb{R} = \mathbb{Q} \quad + \quad \text{all infinite rational refinement processes.}$$

These refinement processes arise naturally from the *mediant construction* and the *SternBrocot tree*.

6.4.1 The Mediant Construction

Definition (Mediant)

Let

$$\frac{a}{b}, \quad \frac{c}{d}$$

be rational numbers with $b, d > 0$.
The **mediant** is the rational number

$$\frac{a+c}{b+d}.$$

Remark 6.4.1 (Fully quantified form).

$$\forall a, b, c, d \in \mathbb{Z}, \quad b, d > 0 : \quad \texttt{mediant}\left(\frac{a}{b}, \frac{c}{d}\right) = \frac{a+c}{b+d}.$$

Proposition 6.4.2 (Mediant inequality). *If*

$$\frac{a}{b} < \frac{c}{d},$$

then

$$\frac{a}{b} < \frac{a+c}{b+d} < \frac{c}{d}.$$

Remark 6.4.3 (Fully quantified form).

$$\forall a, b, c, d \in \mathbb{Z}, \quad b, d > 0 : \left(\frac{a}{b} < \frac{c}{d} \Rightarrow \frac{a}{b} < \frac{a+c}{b+d} < \frac{c}{d}\right).$$

Remark 6.4.4 (Intuition). The mediant lies strictly between the two fractions. It therefore provides a natural way to refine intervals between rational numbers.

6.4.2 The SternBrocot Tree

Theorem 6.4.5 (SternBrocot enumeration). *Every positive rational number appears exactly once in the SternBrocot tree obtained by repeated mediant insertion.*

Remark 6.4.6 (Fully quantified form).

$$\forall q \in \mathbb{Q}_{>0}, \quad \exists! \text{ finite path in the SternBrocot tree reaching } q.$$

Remark 6.4.7 (Consequence). Finite SternBrocot paths correspond exactly to rational numbers.

6.4.3 Infinite Refinement and Irrational Limits

The SternBrocot process also produces infinite sequences of nested intervals of rationals.

Theorem 6.4.8 (SternBrocot Limit Theorem). *Let*

$$\left(\frac{a_n}{b_n}\right), \quad \left(\frac{c_n}{d_n}\right)$$

be sequences of rationals obtained by successive SternBrocot refinement so that

$$\frac{a_n}{b_n} < \frac{c_n}{d_n}$$

and

$$\left[\frac{a_{n+1}}{b_{n+1}}, \frac{c_{n+1}}{d_{n+1}}\right] \subseteq \left[\frac{a_n}{b_n}, \frac{c_n}{d_n}\right].$$

Then there exists a unique real number

$$x = \lim_{n\to\infty} \frac{a_n}{b_n} = \lim_{n\to\infty} \frac{c_n}{d_n}.$$

Remark 6.4.9 (Fully quantified form).

$$\forall \text{ nested rational intervals } I_n, \quad \exists! x \in \mathbb{R} \text{ such that } x \in I_n \text{ for all } n.$$

Remark 6.4.10 (Interpretation).

- Finite SternBrocot paths correspond to rational numbers.
- Infinite SternBrocot paths correspond to irrational numbers.

Thus every real number arises from an infinite refinement of rational approximatio

6.4.4 Real Numbers as Limits of Rational Processes

Definition (Real number via rational approximation)

A **real number** may be defined as the limit of a convergent sequence of rational numbers generated by a coherent approximation process (for example, a Cauchy sequence or a nested sequence of rational intervals).

Remark 6.4.11 (Fully quantified form).

$$\mathbb{R} = \{(q_n)_{n\in\mathbb{N}} \subseteq \mathbb{Q} \mid (q_n) \text{ is Cauchy}\}/\sim .$$

Remark 6.4.12 (Conceptual summary). The real numbers can be viewed as

$$\boxed{\mathbb{R} = \mathbb{Q} \cup \text{all limits of infinite rational approximations}.}$$

Different constructions implement this idea in different ways:

Construction	Limiting process
Cantor	Cauchy sequences of rationals
Dedekind	cuts in the ordered rationals
SternBrocot	infinite rational refinement paths

All three constructions yield isomorphic complete ordered fields.

6.5 Axioms of the Real Numbers

Axioms Of The Reals Quick Reference

Core items	Key definitions/results introduced in this file.
How to use	Read the boxed items first; proofs and consequences follow.
Dependencies	Refer back to earlier sections as needed.

The real numbers $\mathbb{R}$ form a *totally ordered field.* This structure consists of:

- Field axioms (algebraic structure),
- Order axioms (order structure),
- Completeness axiom (analytic structure).

6.5.1 Basic Definitions

Remark 6.5.1. A *field* is a set equipped with two binary operations, addition $(+)$ and multiplication $(\cdot)$, satisfying closure, associativity, commutativity, identity, inverse, and distributive laws.

An *ordered field* is a field equipped with a total order compatible with the algebraic operations.

6.5.2 Main Theorems (Axioms)

Additive Axioms

Axiom A1 (Additive Closure).

$$\forall x \forall y\,(x, y \in \mathbb{R} \rightarrow x + y \in \mathbb{R})$$

Axiom A2 (Additive Commutativity).

$$\forall x \forall y\,(x + y = y + x)$$

Axiom A3 (Additive Associativity).

$$\forall x \forall y \forall z\,((x + y) + z = x + (y + z))$$

Axiom A4 (Additive Identity).

$$\exists 0\,\forall x\,(x + 0 = x)$$

Axiom A5 (Additive Inverse).

$$\forall x\,\exists y\,(x + y = 0)$$

Multiplicative Axioms

Axiom M1 (Multiplicative Closure).

$$\forall x \forall y\,(x, y \in \mathbb{R} \rightarrow x \cdot y \in \mathbb{R})$$

Axiom M2 (Multiplicative Commutativity).

$$\forall x \forall y\,(x \cdot y = y \cdot x)$$

Axiom M3 (Multiplicative Associativity).

$$\forall x \forall y \forall z\,((x \cdot y) \cdot z = x \cdot (y \cdot z))$$

Axiom M4 (Multiplicative Identity).

$$\exists 1\,(1 \neq 0 \land \forall x\,(x \cdot 1 = x))$$

Axiom M5 (Multiplicative Inverse).

$$\forall x\,(x \neq 0 \rightarrow \exists y\,(x \cdot y = 1))$$

Distributive Axiom

Axiom D (Distributivity).

$$\forall x \forall y \forall z\,(x \cdot (y + z) = x \cdot y + x \cdot z)$$

Linear Order Axioms

O1 (Reflexivity).

$$\forall x \in \mathbb{R}, \quad x \leq x$$

O2 (Antisymmetry).

$$\forall x, y \in \mathbb{R}, \quad (x \leq y \land y \leq x) \to x = y$$

O3 (Transitivity).

$$\forall x, y, z \in \mathbb{R}, \quad (x \leq y \land y \leq z) \to x \leq z$$

O4 (Totality / Comparability).

$$\forall x, y \in \mathbb{R}, \quad x \leq y \ \lor \ y \leq x$$

The strict order is defined by:

$$x < y \quad \text{iff} \quad x \leq y \text{ and } x \neq y.$$

Compatibility with Field Operations

O5 (Additive Monotonicity).

$$\forall x, y, z \in \mathbb{R}, \quad x \leq y \to x + z \leq y + z$$

O6 (Multiplicative Monotonicity for Nonnegative Factors).

$$\forall x, y, z \in \mathbb{R}, \quad (x \leq y \land 0 \leq z) \to xz \leq yz$$

O7 (Positivity of the Unit).

$$0 < 1$$

Remark 6.5.2. Axiom O6 implies that multiplication by a negative number reverses inequalities; this will later be proved as a theorem.

6.5.3 Consequences

The logical implication of this entire section is:

$$\text{Field Axioms} \Rightarrow \text{Algebraic Structure}$$

$$\text{Field + Order Axioms} \Rightarrow \text{Ordered Field}$$

To uniquely characterize $\mathbb{R}$ among ordered fields, one must additionally assume:

Completeness.

Remark 6.5.3 (Logical Structure).

Field $\Rightarrow$ Ordered Field $\Rightarrow$ Complete Ordered Field.

The real numbers $\mathbb{R}$ are the unique (up to isomorphism) complete ordered field.

6.6 Intervals in the Real Numbers

Intervals Quick Reference

Core items	Key definitions/results introduced in this file.
How to use	Read the boxed items first; proofs and consequences follow.
Dependencies	Refer back to earlier sections as needed.

Intervals are fundamental subsets of the real line defined using the order relation on $\mathbb{R}$. Let $a, b \in \mathbb{R}$ with $a < b$.

6.6.1 Basic Definitions

Bounded Intervals

Definition (Open Interval)

The *open interval* from a to b is the set

$$(a, b) := \{x \in \mathbb{R} : a < x < b\}.$$

$$\forall x\, (x \in (a, b) \leftrightarrow a < x \wedge x < b)$$

Definition (Closed Interval)

The *closed interval* from a to b is the set

$$[a, b] := \{x \in \mathbb{R} : a \leq x \leq b\}.$$

$$\forall x\, (x \in [a, b] \leftrightarrow a \leq x \wedge x \leq b)$$

Definition (Half-Open Intervals)

The *left-closed, right-open interval* is

$$[a, b) := \{x \in \mathbb{R} : a \leq x < b\}.$$

The *left-open, right-closed interval* is

$$(a, b] := \{x \in \mathbb{R} : a < x \leq b\}.$$

Unbounded Intervals

Definition (Open Rays)

The *open rays* determined by $a \in \mathbb{R}$ are

$$(a, \infty) := \{x \in \mathbb{R} : x > a\}, \qquad (-\infty, a) := \{x \in \mathbb{R} : x < a\}.$$

Definition (Closed Rays)

The *closed rays* determined by $a \in \mathbb{R}$ are

$$[a,\infty) := \{x \in \mathbb{R} : x \geq a\}, \qquad (-\infty, a] := \{x \in \mathbb{R} : x \leq a\}.$$

Theorem (Upper/Lower Perturbation Lemma in $\mathbb{R}$)

Theorem 6.6.1 (Upper/Lower Perturbation Lemma in $\mathbb{R}$). *If $x \in \mathbb{R}$ and $\varepsilon \in \mathbb{R}$ with $\varepsilon > 0$, then there exist $r, s \in \mathbb{R}$ such that*

$$x - \varepsilon < r < x < s < x + \varepsilon.$$

Remark (Dependencies). Order arithmetic in $\mathbb{R}$ and closure of $\mathbb{R}$ under division by 2.

Remark (Proof idea). Take the symmetric perturbation $\delta = \varepsilon/2$ and set $r = x - \delta$, $s = x + \delta$.

Remark (Proof). Go to proof.

Degenerate and Trivial Intervals

Definition (Degenerate Interval)

If $a = b$, the closed interval

$$[a, a] = \{a\}$$

is called a *degenerate interval.*

Definition (Empty Interval)

If $a > b$, the set

$$(a, b) = \varnothing$$

is called an *empty interval.*

6.6.2 Main Theorems

Theorem 6.6.2 (Characterization of Intervals). *A subset $I \subseteq \mathbb{R}$ is an interval if and only if whenever $x, z \in I$ and $x < y < z$, then $y \in I$. Equivalently,*

$$\forall x, z \in I,\ x < z \Rightarrow \big(\forall y \in \mathbb{R},\ (x < y < z \Rightarrow y \in I)\big).$$

Remark 6.6.3. Intervals are precisely the subsets of $\mathbb{R}$ with the property that if $x < y < z$ and x, z belong to the set, then y also belongs to the set.

6.6.3 Consequences

The logical implication of this section is:

$$\text{Order Structure of } \mathbb{R} \Rightarrow \text{Intervals} \Rightarrow \text{Convex subsets of } \mathbb{R}.$$

Intervals encode the idea of no gaps between points of a set. They are the basic building blocks for:

- neighborhoods,
- open sets,
- topology of $\mathbb{R}$,
- and completeness arguments (e.g. nested intervals).

Remark 6.6.4 (Logical Structure). The major structural flow is:

Field Axioms $\Rightarrow$ Order Axioms $\Rightarrow$ Definition of Intervals $\Rightarrow$ Convexity Property $\Rightarrow$ Topologi

6.7 Interval Arithmetic (Forward Reference)

In this volume, intervals are used as subsets of $\mathbb{R}$ and as basic building blocks for order, topology, and approximation. That foundational role belongs here.

The operational calculus of intervals is a different story. Once we start adding, subtracting, multiplying, and propagating intervals and boxes as computational enclosures, we have crossed into numerical analysis.

Remark 6.7.1 (Curriculum placement). We keep interval notation and interval sets in Volume II, but we treat *interval arithmetic* and *box arithmetic in* $\mathbb{R}^n$ as part of the numerical-analysis story.

Remark 6.7.2 (Canonical home). The definitions of interval addition, subtraction, multiplication, containment, and their box-valued analogues now live in Volume V under Numerical Analysis, in the interval-arithmetic notes.

Remark 6.7.3 (Why move it). Those operations are most useful when we study enclosure methods, validated numerics, error propagation, and computational bounds. That is a cleaner fit for numerical analysis than for the foundational development of the real number system.

6.8 Irrational Arithmetic

These results record how irrationality behaves under arithmetic with rational numbers, and why the irrational numbers fail to form a field.

Theorem (Rational Plus Irrational Is Irrational)

Theorem 6.8.1 (Rational Plus Irrational Is Irrational). *If $q \in \mathbb{Q}$ and $\alpha \in \mathbb{R} \setminus \mathbb{Q}$, then*

$$q + \alpha \in \mathbb{R} \setminus \mathbb{Q}.$$

Remark (Dependencies). Closure of $\mathbb{Q}$ under subtraction.

Remark (Proof idea). If $q+\alpha$ were rational, subtracting the rational number q would force α to be rational.

Remark (Proof). Go to proof.

Theorem (Rational Minus Irrational Is Irrational)

Theorem 6.8.2 (Rational Minus Irrational Is Irrational). *If $q \in \mathbb{Q}$ and $\alpha \in \mathbb{R} \setminus \mathbb{Q}$, then*

$$q - \alpha \in \mathbb{R} \setminus \mathbb{Q}.$$

Remark (Dependencies). Closure of $\mathbb{Q}$ under subtraction.

Remark (Proof idea). If $q-\alpha$ were rational, then subtracting it from q would force α to be rational.

Remark (Proof). Go to proof.

Theorem (Nonzero Rational Times Irrational Is Irrational)

Theorem 6.8.3 (Nonzero Rational Times Irrational Is Irrational). *If $q \in \mathbb{Q} \setminus \{0\}$ and $\alpha \in \mathbb{R} \setminus \mathbb{Q}$, then*

$$q\alpha \in \mathbb{R} \setminus \mathbb{Q}.$$

Remark (Dependencies). Nonzero rational numbers have rational reciprocals.

Remark (Proof idea). If $q\alpha$ were rational, divide by the nonzero rational q to conclude that α is rational.

Remark (Proof). Go to proof.

Theorem (Quotient by Nonzero Rational Preserves Irrationality)

Theorem 6.8.4 (Quotient by Nonzero Rational Preserves Irrationality). *If $\alpha \in \mathbb{R} \setminus \mathbb{Q}$ and $q \in \mathbb{Q} \setminus \{0\}$, then*

$$\frac{\alpha}{q} \in \mathbb{R} \setminus \mathbb{Q}.$$

Remark (Dependencies). Nonzero rational numbers have rational reciprocals.

Remark (Proof idea). If α/q were rational, then multiplying by q would force α to be rational.

Remark (Proof). Go to proof.

Corollary (The Irrationals Are Not Closed Under Addition)

Corollary 6.8.5 (The Irrationals Are Not Closed Under Addition). *The set $\mathbb{R} \setminus \mathbb{Q}$ is not closed under addition.*

Remark (Dependencies). Theorem 6.8.2.

Remark (Proof idea). Choose irrational numbers whose sum is rational, for example $\sqrt{2}$ and $2 - \sqrt{2}$.

Remark (Proof). Go to proof.

Corollary (The Irrationals Are Not Closed Under Multiplication)

Corollary 6.8.6 (The Irrationals Are Not Closed Under Multiplication). *The set $\mathbb{R} \setminus \mathbb{Q}$ is not closed under multiplication.*

Remark (Dependencies). The irrationality of $\sqrt{2}$.

Remark (Proof idea). Use $\sqrt{2} \cdot \sqrt{2} = 2$.

Remark (Proof). Go to proof.

Corollary (The Irrationals Are Not a Field)

Corollary 6.8.7 (The Irrationals Are Not a Field). *The set $\mathbb{R} \setminus \mathbb{Q}$ does not form a field under the usual operations.*

Remark (Dependencies). Corollary 6.8.5, Corollary 6.8.6.

Remark (Proof idea). Failure of closure under addition or multiplication already prevents a set from being a field.

Remark (Proof). Go to proof.

Corollary (The Irrationals Are Not an Ordered Field)

Corollary 6.8.8 (The Irrationals Are Not an Ordered Field). *The set $\mathbb{R} \setminus \mathbb{Q}$ does not form an ordered field under the usual operations and order.*

Remark (Dependencies). Corollary 6.8.7.

Remark (Proof idea). An ordered field is, in particular, a field.

Remark (Proof). Go to proof.

6.9 Irrationals as a Subset of the Real Line

The irrational numbers form a large and unavoidable subset of the real line. They lie in every nonempty interval, they approximate and are approximated by rational numbers, and they inherit some but not all of the structural features of $\mathbb{R}$.

Theorem (Density of $\mathbb{Q}$ in $\mathbb{R}$)

Theorem 6.9.1 (Density of $\mathbb{Q}$ in $\mathbb{R}$). *If $x, y \in \mathbb{R}$ and $x < y$, then there exists $q \in \mathbb{Q}$ such that*

$$x < q < y.$$

Remark (Dependencies). The Archimedean property of $\mathbb{R}$, the floor lemma, and the fact that $\mathbb{Q}$ is closed under ratios of integers by positive natural numbers.

Remark (Proof idea). Scale the interval (x, y) until its length exceeds 1, place an integer between the scaled endpoints, and divide back down.

Remark (Proof). Go to proof.

Corollary (Infinitely Many Rational Numbers Lie Between Two Distinct Reals)

Corollary 6.9.2 (Infinitely Many Rational Numbers Lie Between Two Distinct Reals). *Between any two distinct real numbers there are infinitely many rational numbers.*

Remark (Dependencies). Density of $\mathbb{Q}$ in $\mathbb{R}$.

Remark (Proof idea). After finding one rational in the interval, apply density again inside the smaller subintervals to continue producing new ones.

Remark (Proof). Go to proof.

Theorem (Between Any Two Real Numbers Lies an Irrational)

Theorem 6.9.3 (Between Any Two Real Numbers Lies an Irrational). *If $x, y \in \mathbb{R}$ and $x < y$, then there exists $\alpha \in \mathbb{R} \setminus \mathbb{Q}$ such that*

$$x < \alpha < y.$$

Remark (Dependencies). Density of $\mathbb{Q}$ in $\mathbb{R}$, Theorem 6.8.1, and the irrationality of $\sqrt{2}$.

Remark (Proof idea). Choose a sufficiently small rational perturbation q and shift it by a small irrational quantity such as $\sqrt{2}/n$.

Remark (Proof). Go to proof.

Corollary (The Irrationals Are Dense in $\mathbb{R}$)

Corollary 6.9.4 (The Irrationals Are Dense in $\mathbb{R}$). *Every nonempty open interval in $\mathbb{R}$ contains an irrational number.*

Remark (Dependencies). Theorem 6.9.3.

Remark (Proof idea). Unpack what the theorem says for an arbitrary open interval (a, b).

Remark (Proof). Go to proof.

Corollary (Every Irrational Can Be Approximated by Rationals)

Corollary 6.9.5 (Every Irrational Can Be Approximated by Rationals). *For every $\alpha \in \mathbb{R} \setminus \mathbb{Q}$ and every $\varepsilon > 0$, there exists $q \in \mathbb{Q}$ such that*

$$|\alpha - q| < \varepsilon.$$

Remark (Dependencies). Density of $\mathbb{Q}$ in $\mathbb{R}$.

Remark (Proof idea). Apply rational density to $(\alpha - \varepsilon, \alpha + \varepsilon)$.

Remark (Proof). Go to proof.

Corollary (Every Rational Can Be Approximated by Irrationals)

Corollary 6.9.6 (Every Rational Can Be Approximated by Irrationals). *For every $q \in \mathbb{Q}$ and every $\varepsilon > 0$, there exists $\alpha \in \mathbb{R} \setminus \mathbb{Q}$ such that*

$$|\alpha - q| < \varepsilon.$$

Remark (Dependencies). Corollary 6.9.4.

Remark (Proof idea). Apply irrational density to $(q-\varepsilon, q+\varepsilon)$.

Remark (Proof). Go to proof.

Theorem (Closure of $\mathbb{Q}$ in $\mathbb{R}$)

Theorem 6.9.7 (Closure of $\mathbb{Q}$ in $\mathbb{R}$). *The closure of $\mathbb{Q}$ in $\mathbb{R}$ is all of $\mathbb{R}$.*

Remark (Dependencies). Density of $\mathbb{Q}$ in $\mathbb{R}$.

Remark (Proof idea). Every real point can be approximated arbitrarily closely by rationals, so every real point lies in the closure of $\mathbb{Q}$.

Remark (Proof). Go to proof.

Corollary ($\mathbb{Q}$ Is Not Closed in $\mathbb{R}$)

Corollary 6.9.8 ($\mathbb{Q}$ Is Not Closed in $\mathbb{R}$). *The set $\mathbb{Q}$ is not closed as a subset of $\mathbb{R}$.*

Remark (Dependencies). Theorem 6.9.7.

Remark (Proof idea). Its closure is all of $\mathbb{R}$, which is strictly larger than $\mathbb{Q}$.

Remark (Proof). Go to proof.

Corollary ($\mathbb{Q}$ Is Not Open in $\mathbb{R}$)

Corollary 6.9.9 ($\mathbb{Q}$ Is Not Open in $\mathbb{R}$). *The set $\mathbb{Q}$ is not open as a subset of $\mathbb{R}$.*

Remark (Dependencies). Corollary 6.9.4.

Remark (Proof idea). Every open interval around a rational point contains irrational points.

Remark (Proof). Go to proof.

Corollary ($\mathbb{Q}$ Is Not Discrete in $\mathbb{R}$)

Corollary 6.9.10 ($\mathbb{Q}$ Is Not Discrete in $\mathbb{R}$). *The set $\mathbb{Q}$ is not discrete as a subset of $\mathbb{R}$.*

Remark (Dependencies). Theorem 5.7.1.

Remark (Proof idea). No rational point is isolated, because every interval around it contains other rationals.

Remark (Proof). Go to proof.

Theorem (The Irrationals Are Uncountable)

Theorem 6.9.11 (The Irrationals Are Uncountable). *The set $\mathbb{R} \setminus \mathbb{Q}$ is uncountable.*

Remark (Dependencies). The real numbers are uncountable and the rational numbers are countable.

Remark (Proof idea). Subtracting a countable subset from an uncountable set leaves an uncountable set.

Remark (Proof). Go to proof.

Theorem (The Irrationals Are Not Complete as an Ordered Set)

Theorem 6.9.12 (The Irrationals Are Not Complete as an Ordered Set). *The ordered set* $\mathbb{R} \setminus \mathbb{Q}$ *is not complete.*

Remark (Dependencies). The order on $\mathbb{R}$ and the irrational subset $\mathbb{R} \setminus \mathbb{Q}$.

Remark (Proof idea). Consider the set

$$A = \{\alpha \in \mathbb{R} \setminus \mathbb{Q} : \alpha < 1\}.$$

Its least upper bound in $\mathbb{R}$ is 1, but 1 is not irrational.

Remark (Proof). Go to proof.

Proofs

Remark (Return). Return to Theorem

Theorem (Upper/Lower Perturbation Lemma in $\mathbb{R}$). *If $x \in \mathbb{R}$ and $\varepsilon \in \mathbb{R}$ with $\varepsilon > 0$, then there exist $r, s \in \mathbb{R}$ such that*

$$x - \varepsilon < r < x < s < x + \varepsilon.$$

Proof. TODO. $\square$

Remark (Proof structure). TODO.

Remark (Dependencies).

- Order arithmetic in $\mathbb{R}$
- Closure of $\mathbb{R}$ under division by 2

Remark (Return). Return to Theorem

Theorem (Density of $\mathbb{Q}$ in $\mathbb{R}$). *If $x, y \in \mathbb{R}$ and $x < y$, then there exists $q \in \mathbb{Q}$ such that*

$$x < q < y.$$

Proof. TODO. □

Remark (Proof structure). TODO.

Remark (Dependencies).

- The Archimedean property of $\mathbb{R}$
- The floor lemma

Remark (Return). Return to Corollary

Theorem (Infinitely Many Rational Numbers Lie Between Two Distinct Reals). *Between any two distinct real numbers there are infinitely many rational numbers.*

Proof. TODO. □

Remark (Proof structure). TODO.

Remark (Dependencies).

- Theorem 6.9.1

Remark (Return). Return to Theorem

Theorem (Rational Plus Irrational Is Irrational). *If $q \in \mathbb{Q}$ and $\alpha \in \mathbb{R} \setminus \mathbb{Q}$, then*

$$q + \alpha \in \mathbb{R} \setminus \mathbb{Q}.$$

Proof. TODO. □

Remark (Proof structure). TODO.

Remark (Dependencies).

- Closure of $\mathbb{Q}$ under subtraction

Remark (Return). Return to Theorem

Theorem (Rational Minus Irrational Is Irrational). *If* $q \in \mathbb{Q}$ *and* $\alpha \in \mathbb{R} \setminus \mathbb{Q}$*, then*

$$q - \alpha \in \mathbb{R} \setminus \mathbb{Q}.$$

Proof. TODO. $\square$

Remark (Proof structure). TODO.

Remark (Dependencies).

- Closure of $\mathbb{Q}$ under subtraction

Remark (Return). Return to Theorem

Theorem (Nonzero Rational Times Irrational Is Irrational). *If $q \in \mathbb{Q} \setminus \{0\}$ and $\alpha \in \mathbb{R} \setminus \mathbb{Q}$, then*

$$q\alpha \in \mathbb{R} \setminus \mathbb{Q}.$$

Proof. TODO. □

Remark (Proof structure). TODO.

Remark (Dependencies).

- Nonzero rational numbers have rational reciprocals

Remark (Return). Return to Theorem

Theorem (Quotient by Nonzero Rational Preserves Irrationality). *If* $\alpha \in \mathbb{R} \setminus \mathbb{Q}$ *and* $q \in \mathbb{Q} \setminus \{0\}$*, then*

$$\frac{\alpha}{q} \in \mathbb{R} \setminus \mathbb{Q}.$$

Proof. TODO. □

Remark (Proof structure). TODO.

Remark (Dependencies).

- Nonzero rational numbers have rational reciprocals

Remark (Return). Return to Theorem

Theorem (The Irrationals Are Not Closed Under Addition). *The set* $\mathbb{R} \setminus \mathbb{Q}$ *is not closed under addition.*

Proof. TODO. $\square$

Remark (Proof structure). TODO.

Remark (Dependencies).

- Theorem 6.8.2

Remark (Return). Return to Theorem

Theorem (The Irrationals Are Not Closed Under Multiplication). *The set* $\mathbb{R} \setminus \mathbb{Q}$ *is not closed under multiplication.*

Proof. TODO. □

Remark (Proof structure). TODO.

Remark (Dependencies).

- Theorem 5.10.1

Remark (Return). Return to Theorem

Theorem (The Irrationals Are Not a Field). *The set $\mathbb{R} \backslash \mathbb{Q}$ does not form a field under the usual operations.*

Proof. TODO. □

Remark (Proof structure). TODO.

Remark (Dependencies).

- Corollary 6.8.5
- Corollary 6.8.6

Remark (Return). Return to Theorem

Theorem (The Irrationals Are Not an Ordered Field). *The set $\mathbb{R}\setminus\mathbb{Q}$ does not form an ordered field under the usual operations and order.*

Proof. TODO. □

Remark (Proof structure). TODO.

Remark (Dependencies).

- Corollary 6.8.7

Remark (Return). Return to Theorem

Theorem (Between Any Two Real Numbers Lies an Irrational). *If $x, y \in \mathbb{R}$ and $x < y$, then there exists $\alpha \in \mathbb{R} \setminus \mathbb{Q}$ such that*

$$x < \alpha < y.$$

Proof. TODO. □

Remark (Proof structure). TODO.

Remark (Dependencies).

- Theorem 6.9.1
- Theorem 6.8.1
- Theorem 5.10.1

Remark (Return). Return to Theorem

Theorem (The Irrationals Are Dense in $\mathbb{R}$). *Every nonempty open interval in* $\mathbb{R}$ *contains an irrational number.*

Proof. TODO. □

Remark (Proof structure). TODO.

Remark (Dependencies).

- Theorem 6.9.3

Remark (Return). Return to Theorem

Theorem (Every Irrational Can Be Approximated by Rationals). *For every $\alpha \in \mathbb{R} \setminus \mathbb{Q}$ and every $\varepsilon > 0$, there exists $q \in \mathbb{Q}$ such that*

$$|\alpha - q| < \varepsilon.$$

Proof. TODO. □

Remark (Proof structure). TODO.

Remark (Dependencies).

- Theorem 6.9.1

Remark (Return). Return to Theorem

Theorem (Every Rational Can Be Approximated by Irrationals). *For every $q \in \mathbb{Q}$ and every $\varepsilon > 0$, there exists $\alpha \in \mathbb{R} \setminus \mathbb{Q}$ such that*

$$|\alpha - q| < \varepsilon.$$

Proof. TODO. □

Remark (Proof structure). TODO.

Remark (Dependencies).

- Corollary 6.9.4

Remark (Return). Return to Theorem

Theorem (Closure of $\mathbb{Q}$ in $\mathbb{R}$). *The closure of $\mathbb{Q}$ in $\mathbb{R}$ is all of $\mathbb{R}$.*

Proof. TODO. □

Remark (Proof structure). TODO.

Remark (Dependencies).

- Theorem 6.9.1

Remark (Return). Return to Theorem

Theorem ($\mathbb{Q}$ Is Not Closed in $\mathbb{R}$). *The set $\mathbb{Q}$ is not closed as a subset of $\mathbb{R}$.*

Proof. TODO. □

Remark (Proof structure). TODO.

Remark (Dependencies).

- Theorem 6.9.7

Remark (Return). Return to Theorem

Theorem ($\mathbb{Q}$ Is Not Open in $\mathbb{R}$). *The set* $\mathbb{Q}$ *is not open as a subset of* $\mathbb{R}$*.*

Proof. TODO. □

Remark (Proof structure). TODO.

Remark (Dependencies).

- Corollary 6.9.4

Remark (Return). Return to Theorem

Theorem ($\mathbb{Q}$ Is Not Discrete in $\mathbb{R}$). *The set $\mathbb{Q}$ is not discrete as a subset of $\mathbb{R}$.*

Proof. TODO. □

Remark (Proof structure). TODO.

Remark (Dependencies).

- Theorem 5.7.1

Remark (Return). Return to Theorem

Theorem (The Irrationals Are Uncountable). *TODO: restate the theorem.*

Proof. TODO. □

Remark (Proof structure). TODO.

Remark (Dependencies).

- The real numbers are uncountable
- The rational numbers are countable

Remark (Return). Return to Theorem

Theorem (The Irrationals Are Not Complete as an Ordered Set). *TODO: restate the theorem.*

Proof. TODO. □

Remark (Proof structure). TODO.

Remark (Dependencies).

- The irrational subset $\mathbb{R} \setminus \mathbb{Q}$ inside the ordered real line

Capstone

Chapter 7

Summary: The Number Systems

7.1 Axioms, Properties, and Structural Comparison

7.2 Axiom Systems for $\mathbb{N}$, $\mathbb{W}$, $\mathbb{Z}$, $\mathbb{Q}$, and $\mathbb{R}$

7.2.1 The Natural Numbers $\mathbb{N}$

One-Based Peano Axioms for $\mathbb{N}$

Let $\mathbb{N}$ be a set with distinguished element 1 and successor operation $n \mapsto n{+\!+}$.

P1 $1 \in \mathbb{N}$.

P2 $n \in \mathbb{N} \Rightarrow n{+\!+} \in \mathbb{N}$. *(successor closure)*

P3 $n{+\!+} \neq 1$ for all $n \in \mathbb{N}$. *(the first element has no predecessor)*

P4 $n{+\!+} = m{+\!+} \Rightarrow n = m$. *(successor is injective)*

P5 If $P(0)$ holds and $P(n) \Rightarrow P(n{+\!+})$, then $P(n)$ holds for all $n \in \mathbb{N}$. *(induction)*

Remark 7.2.1. P1-P4 rule out wrap-around, ceiling, and rogue elements. P5 eliminates anything unreachable from 1 by finite iteration of the successor. Addition, multiplication, and order are not axioms - they are *defined* recursively from the successor and then shown to satisfy familiar laws by induction.

Remark 7.2.2 (Well-ordering and induction are equivalent over $\mathbb{N}$). The following are logically equivalent:

- Principle of Mathematical Induction (P5).
- Well-Ordering Principle: every nonempty $A \subseteq \mathbb{N}$ has a least element.
- Strong Induction: $(\forall k < n,\, P(k)) \Rightarrow P(n)$ for all n implies P holds everywhere.

One must assume at least one; the others follow. Well-ordering is the engine behind the Division Algorithm, Bézout's Identity, and unique prime factorization.

7.2.2 The Whole Numbers $\mathbb{W}$

$\mathbb{W}$ as $\mathbb{N}$ with an Adjoined Zero

The whole numbers are
$$\mathbb{W} = \mathbb{N} \cup \{0\}, \qquad 0 \notin \mathbb{N}.$$
The successor, addition, multiplication, and order on $\mathbb{W}$ extend the corresponding structures on $\mathbb{N}$:

- $S_{\mathbb{W}}(0) = 1$, and $S_{\mathbb{W}}$ agrees with $S_{\mathbb{N}}$ on $\mathbb{N}$;
- 0 is the additive identity;
- 0 is absorbing for multiplication;
- 0 is the least element of $\mathbb{W}$.

Remark 7.2.3 (What $\mathbb{W}$ gains over $\mathbb{N}$, and what it lacks). **Gain:** an additive identity. This makes $(\mathbb{W}, +, 0)$ a commutative monoid and prepares the ordered-pair construction of the integers.

Lack: additive inverses for nonzero elements. Subtraction is still not an internal operation in general, so $\mathbb{W}$ is not a ring.

7.2.3 The Integers $\mathbb{Z}$

Axioms for $\mathbb{Z}$ Ordered Commutative Ring

Addition

A1 $a + b = b + a$ *(commutativity)*

A2 $(a + b) + c = a + (b + c)$ *(associativity)*

A3 $\exists\, 0 \in \mathbb{Z}$ such that $a + 0 = a$ *(additive identity)*

A4 $\forall\, a,\ \exists -a \in \mathbb{Z}$ with $a + (-a) = 0$ *(additive inverses)*

Multiplication

M1 $a \cdot b = b \cdot a$ *(commutativity)*

M2 $(a \cdot b) \cdot c = a \cdot (b \cdot c)$ *(associativity)*

M3 $\exists\, 1 \in \mathbb{Z},\ 1 \neq 0$, such that $a \cdot 1 = a$ *(multiplicative identity)*

Distributivity

D $a \cdot (b + c) = a \cdot b + a \cdot c$ *(distributivity)*

Order

O1 $a \le b$ or $b \le a$ *(totality)*

O2 $a \le b \wedge b \le a \Rightarrow a = b$ *(antisymmetry)*

O3 $a \le b \wedge b \le c \Rightarrow a \le c$ *(transitivity)*

O4 $b \le c \Rightarrow a + b \le a + c$ *(order + addition)*

O5 $a \ge 0 \wedge b \ge 0 \Rightarrow ab \ge 0$ *(order + multiplication)*

Remark 7.2.4 (What $\mathbb{Z}$ gains over $\mathbb{W}$, and what it lacks). **Gain:** Axiom A4 –

additive inverses. Subtraction is always defined in $\mathbb{Z}$; it is not in $\mathbb{W}$.

Lack: Multiplicative inverses. There is no $x \in \mathbb{Z}$ with $2x = 1$, so M4 (every nonzero element is invertible) fails. In the language of abstract algebra, $\mathbb{Z}$ is a *commutative ordered ring* but not a field. The passage to $\mathbb{Q}$ restores multiplicative inverses by formally adjoining fractions.

7.2.4 Derived Properties of $\mathbb{Z}$ (from the Ring Axioms)

Derived Properties Quick Reference

Name	**Statement**	**Proof sketch**
Zero multiplication	$a \cdot 0 = 0$	$a \cdot 0 = a \cdot (0+0) = a \cdot 0 + a \cdot 0$; cancel $a \cdot 0$ via A4.
Negation rule	$(-a) \cdot b = -(ab)$	$ab + (-a)b = (a + (-a))b = 0 \cdot b = 0$.
(-1) rule	$(-1) \cdot a = -a$	Special case: $b = a$, $-a = (-1)a$.
Double negation	$(-a)(-b) = ab$	$(-a)(-b) = -(a(-b)) = -(-(ab)) = ab$.
No zero divisors	$ab = 0 \Rightarrow a = 0$ or $b = 0$	$\mathbb{Z}$ is an integral domain; proved via well-ordering + sign analysis.

Remark 7.2.5 (Zero multiplication the step most often looked up)**.** The identity $a \cdot 0 = 0$ is not an axiom; it is derived. The proof uses only A3, A4, and D:

$$a \cdot 0 = a \cdot (0+0) \stackrel{\text{D}}{=} a \cdot 0 + a \cdot 0.$$

Adding $-(a \cdot 0)$ to both sides (A4) gives $0 = a \cdot 0$, i.e. $a \cdot 0 = 0$. No order axioms, no well-ordering, no properties of $\mathbb{N}$ are needed this holds in any ring.

7.2.5 Structural Comparison: $\mathbb{N}$, $\mathbb{W}$, $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$

Axiom Comparison Table

Group	**Axiom**	$\mathbb{N}$	$\mathbb{W}$	$\mathbb{Z}$	$\mathbb{Q}$
Addition	A1 commutativity	✓	✓	✓	✓
	A2 associativity	✓	✓	✓	✓
	A3 identity (0)	×	✓	✓	✓
	A4 inverses $(-a)$	×	×	✓	✓
Multiplication	M1 commutativity	✓	✓	✓	✓
	M2 associativity	✓	✓	✓	✓
	M3 identity (1)	✓	✓	✓	✓
	M4 inverses $(1/a)$	×	×	×	✓
	D distributivity	✓	✓	✓	✓
Order	O1–O3 total order	✓	✓	✓	✓
	O4–O5 order + arithmetic	✓	✓	✓	✓
	Well-ordering	✓	✓	×	×
Completeness	Least upper bound	×	×	×	×
Structure		*positive arithmetic*	*comm. semiring*	*comm. ring*	*ordered*

✓ = satisfied × = fails

Remark 7.2.6 (The extension hierarchy). Each number system repairs exactly one deficiency of its predecessor:

$$\mathbb{N} \xrightarrow{+\,0} \mathbb{W} \xrightarrow{+\,\mathrm{A4}} \mathbb{Z} \xrightarrow{+\,\mathrm{M4}} \mathbb{Q} \xrightarrow{+\,\mathrm{LUB}} \mathbb{R}.$$

$\mathbb{N} \to \mathbb{W}$: adjoin 0 so addition has an identity. $\mathbb{W} \to \mathbb{Z}$: adjoin additive inverses so subtraction is always defined. $\mathbb{Z} \to \mathbb{Q}$: adjoin multiplicative inverses so division by any nonzero element is always defined. $\mathbb{Q} \to \mathbb{R}$: adjoin least upper bounds so every bounded increasing sequence converges.

Note that $\mathbb{N}$ and $\mathbb{W}$ are the well-ordered systems in the chain. $\mathbb{Z}$ loses well-ordering the moment negative integers are introduced (the set of all negative integers has no least element).

Chapter 8

Formalizing the Number Systems

Why Formalization Is Needed

Ordinary arithmetic is one of the most familiar parts of mathematics. We learn to count, add, subtract, multiply, and divide long before we ask what sort of objects numbers are or why the usual rules are justified. For foundational work, however, intuition is not enough. The basic number systems must be specified by precise axioms and constructions, so that their operations and order relations are not merely assumed but accounted for.

The Natural-Number Structure

The first place where this becomes visible is the natural numbers. A foundational treatment cannot simply point to the counting numbers and begin calculating. It has to identify a first element, a successor operation, and the induction principle that makes recursive definitions and proofs possible. Peano's role was to isolate this structure: a first element, a successor map, conditions saying that successor is injective and never returns to the first element, and an induction principle strong enough to characterize the system. This is the role of Peano's 1889 presentation of arithmetic [Peano, 1889].

Structural Characterization

Dedekind's work emphasized the same structural point of view. What matters is not the particular names attached to the elements, but the pattern determined by a distinguished first element, successor, recursion, and induction. This perspective underlies the treatment of Peano systems earlier in the volume, and Dedekind's name also sits behind the cut-based construction of the real numbers developed later in the usual foundational tradition [Dedekind, 1901].

Later Number Systems

Once the natural numbers have been fixed, the later systems require their own explicit constructions. The integers are built so that subtraction becomes an internal operation. The rational numbers are built so that division by nonzero quantities is represented inside the system. The real numbers are then introduced to address the completeness failures of the rationals. In each case, the construction supplies the operations, the order, and the structural properties needed for analysis, rather than treating them as background facts.

Chapter 9

Lean 4 Formalization

lean

Formalizing the Number Systems $\to$ **Lean 4 Formalization**

Structural Roadmap

How we got here. The preceding chapters developed the number systems in ordinary mathematical notation, beginning with Peano systems and then constructing $\mathbb{N}$, $\mathbb{W}$, $\mathbb{Z}$, $\mathbb{Q}$, and $\mathbb{R}$. This final chapter records how part of that development can be mirrored in Lean 4.

What this chapter develops. This chapter gives a quick orientation to Lean 4 syntax used to formalize the Peano-system material, especially the way predicates and subsets are represented as functions into 'Prop'.

Where this leads. Lean is not used here as a second beginning for the volume. It is a compact formal record of the preceding Peano-system development, and an orientation for reading the companion formalization alongside the notes.

9.1 A First Lean 4 Reading

Subsets as Predicates

Lean represents many mathematical objects by their defining behavior. For a type P, a subset of P can be read as a predicate $P \to Prop$: it takes an element of P as input and returns the proposition that the element belongs to the subset. This is the key syntax bridge for the Peano-system arguments that use inductive subsets.

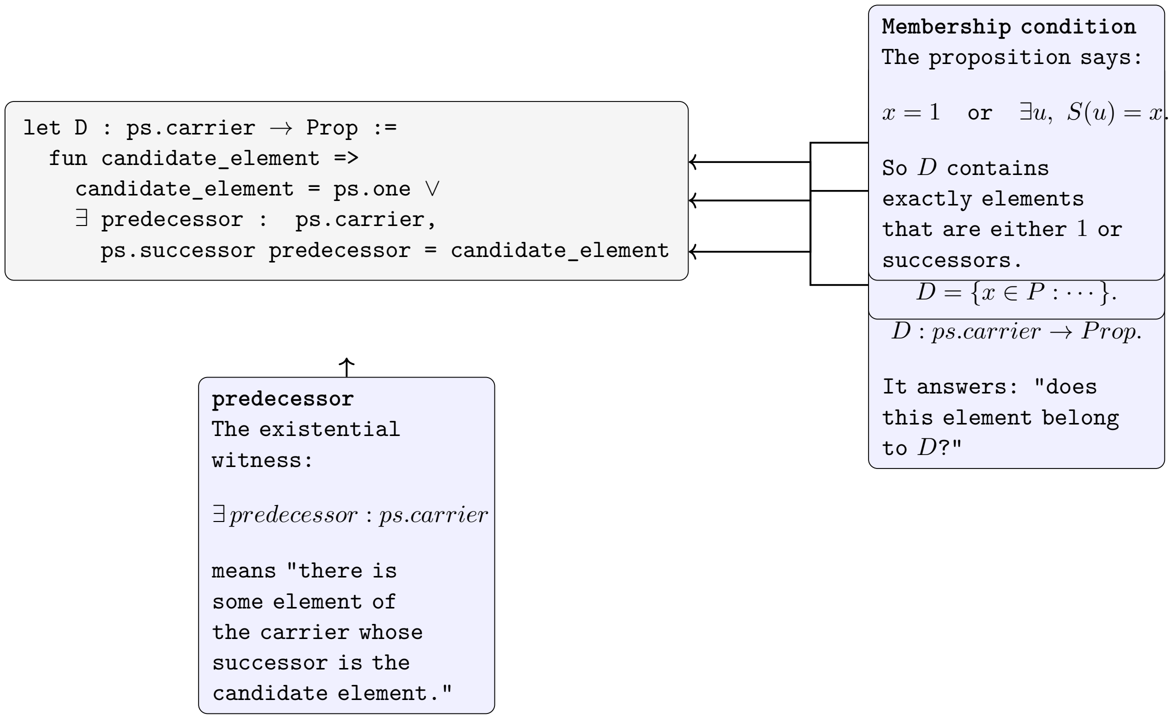

Figure 9.1: Reading a Lean subset predicate as a mathematical subset condition.

Remark (Interpretation). The Lean line ‘D : ps.carrier → Prop‘ says that membership in D is a proposition depending on an input from the carrier of the Peano system. In ordinary mathematical notation, this is the same role played by a subset $D \subseteq P$ or by a defining condition $D(x)$.

Reading an Initial Proof

Lean proofs are built from local goals and the terms that solve them. A first reading strategy is to identify the current hypotheses, the target proposition, and the exact line that transforms the local context into the required result.

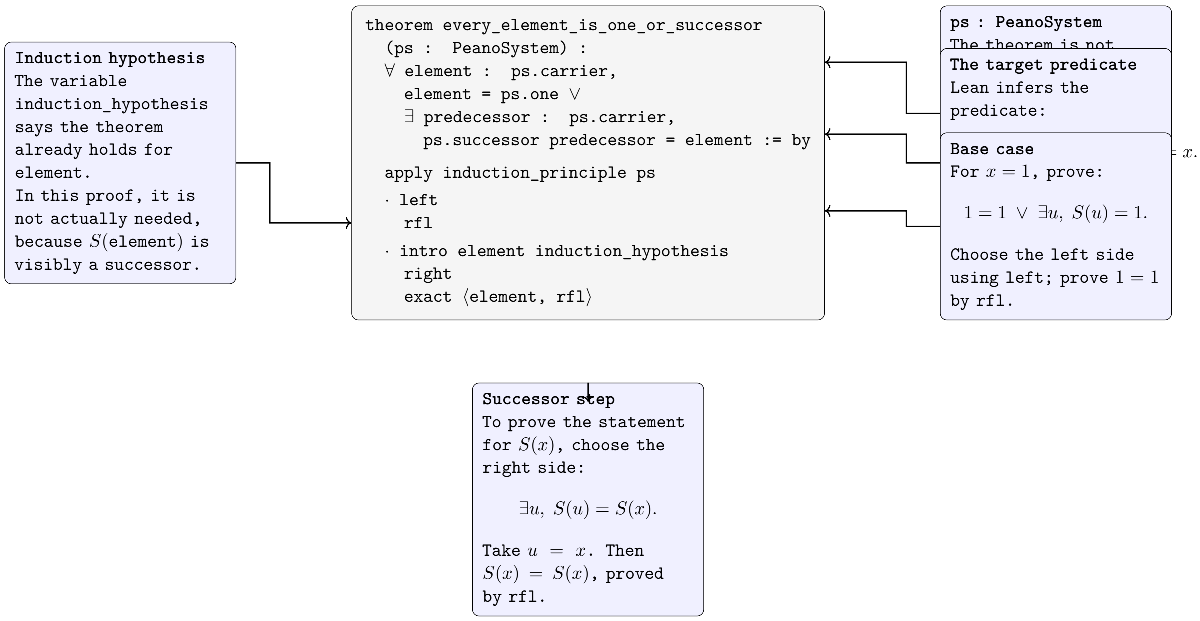

Figure 9.2: Reading the shape of an initial Lean proof.

Remark (Interpretation). The figure isolates the proof-state reading habit used throughout Lean: separate the available assumptions from the current goal, then read each proof term as an instruction that closes or transforms that goal.

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